

Imperial College London

Doctoral Thesis

Imperial College London

Department of Physics

Energy system modelling for pathways to
achieving sustainable electricity access

Author:
Hamish Beath

Supervisors:
Prof. Jenny Nelson, Dr Ajay
Gambhir & Dr Sheridan Few

Abstract

Inadequate access to electricity impedes improvements in human living standards. Cheaper low-carbon technologies enable electricity access using decentralised off-grid systems; however, the financial sustainability of such systems remains challenging. Nevertheless, renewables-based off-grid systems may be the cheapest and most reliable electricity access option for some users. In this thesis, I develop an existing energy system model and present a new geospatial model, both of which are open-source. Using these tools, I investigate economic challenges facing off-grid electricity access. Firstly, I use energy systems modelling to explore electricity supply options for healthcare and domestic use in India. I find that the grid is unreliable and that solar-based mini-grids are normally cheaper than diesel-based ones. Adding healthcare-based demand to domestic demand reduces the cost of mini-grid electricity and enhances financial sustainability. Secondly, I extend an existing energy systems model to simulate hybrid solar mini-grids with dispatched diesel generation. I then apply this model to a humanitarian setting and demonstrate the potential for reducing costs by optimising the operational strategy. Additionally, I explore the impacts of adding new consumers to the system, demonstrating that whilst surplus solar electricity limits additional costs at lower levels, at higher levels fuel costs may be a financial burden. Finally, I present a new geospatial tool for investigating least-cost electricity access pathways. I use this tool to examine pathways to universal household electricity access in Sub-Saharan Africa under different demand, reliability, and carbon price scenarios. The results suggest that higher demand levels necessitate greater investment in grid extension and that applying a financial penalty for unmet demand or a carbon price increases the share of off-grid solar used in least-cost pathways. Together, these investigations quantify the extent to which low-carbon off-grid systems can become more economically attractive in the drive towards increased electricity access.

Copyright statement

The copyright of this thesis rests with the author. Unless otherwise indicated its contents are licensed under a Creative Commons Attribution-Non Commercial 4.0 International Licence (CC BY-NC). Under this licence, you may copy and redistribute the material in any medium or format. You may also create and distribute modified versions of the work. This is on the condition that: you credit the author and do not use it, or any derivative works, for a commercial purpose. When reusing or sharing this work, ensure you make the licence terms clear to others by naming the licence and linking to the licence text. Where a work has been adapted, you should indicate that the work has been changed and describe those changes. Please seek permission from the copyright holder for uses of this work that are not included in this licence or permitted under UK Copyright Law

Statement of Originality

I hereby declare that all the work presented in this thesis is my own unless explicitly stated and referenced.

Acknowledgements

I want to acknowledge the unwavering support of my supervisors Prof Jenny Nelson, Dr Ajay Gambhir and Dr Sheridan Few. They have been patient and understanding and have made my PhD interesting and enjoyable and I am grateful to have had the opportunity to work with them. I also want to thank Dr Philip Sandwell and Dr Shivika Mittal for their valuable input and support. Finally, I want to thank my parents for their kindness and encouragement throughout my PhD.

Contents

1	Introduction	12
1.1	Overview	12
1.2	Research Questions Addressed and Approach	13
1.3	Scope of this Thesis	14
1.4	Thesis Structure	14
2	Literature Review	16
2.1	Overview	16
2.2	Access to Electricity: defining and measuring the problem	16
2.2.1	Summary	18
2.3	Progress towards SDG7 and future projections	19
2.3.1	Summary	20
2.4	The Gains from Improving Electricity Access	20
2.4.1	Livelihoods	20
2.4.2	Health	21
2.4.3	Education	22
2.4.4	Quality of Life	23
2.4.5	Summary	23
2.5	Means of Improving Access: Technology Choices	23
2.5.1	System Types	23
2.5.2	Factors Impacting Technology Choice	24
2.5.3	Solar PV	26
2.5.4	Diesel Generation	28
2.5.5	Battery Storage	29
2.5.6	Trends in costs of off-grid systems and future projections	30
2.5.7	Economic Challenges and Solutions for Off-grid Systems	31
2.5.8	Summary	32
2.6	Electricity Access & Climate Change	32

2.6.1	Summary	33
2.7	Modelling tools for Electricity Access Research	34
2.7.1	Modelling tools for Off-grid Energy System Design	34
2.7.2	Geospatial Modelling for Electricity Access	37
2.7.3	Summary	39
2.8	Discussion of Literature and Research Gaps	39
2.9	Summary	42
3	Methods	43
3.1	Overview	43
3.2	The CLOVER Energy System Model	44
3.2.1	Model Structure	47
3.2.2	System Simulation	47
3.2.3	Model Modifications	49
3.2.4	System Optimisation	50
3.2.5	Summary	51
3.3	Inclusion of Dispatched Generation in CLOVER	52
3.3.1	Summary	57
3.4	Least-cost geospatial electrification model	57
3.4.1	High-Level Description of Model	57
3.4.2	Mathematical Description of Model	58
3.4.3	Summary	62
3.5	Summary of Methodological Contributions	62
4	Improving Mini-grid Economics via Anchor Loads	64
4.1	Overview	64
4.2	Background	65
4.3	Methods	66
4.3.1	Modelling Domestic and Health Centre Demand	68
4.3.2	Modelling Framework and Scenario Description	70
4.3.3	Comparison with the National Grid	70
4.4	Results of Analysis	72
4.4.1	Demand Profiles	72
4.4.2	National Grid Unmet Energy Estimations	73
4.4.3	Cost and Emissions Implications of Health Centre Anchor Loads	74
4.4.4	Sensitivity of Results to Increased Energy Demand Met	76

4.4.5	Cost and Emissions of 100% Reliable Hybrid PV, Battery and Diesel Systems	81
4.5	Discussion of Findings	81
4.6	Summary	83
5	Improving the Operational Performance of Hybrid Mini-grids	85
5.1	Overview	85
5.2	Background	86
5.2.1	Operating hybrid mini-grids	86
5.2.2	Energy in Displacement Settings	87
5.3	Case Study: Rwanda and Mahama Refugee Camp	89
5.4	Methods	91
5.4.1	Demand Data	92
5.4.2	Modelling diesel generation	93
5.4.3	Assessing the impacts of connecting business customers	95
5.5	Results of Analysis	95
5.5.1	Input data and load profiles	95
5.5.2	Investigation of diesel operation strategies	98
5.5.3	Impact of connecting business customers	102
5.6	Conclusions	107
5.6.1	Designing hybrid systems to mitigate diesel usage	107
5.6.2	Supporting business customers	108
5.6.3	Limitations of the study	109
5.7	Summary	110
6	Assessing Pathways to Universal Electricity Access using Geospatial Modelling	111
6.1	Overview	111
6.2	Background	112
6.3	Methods	113
6.3.1	Electricity Demand	115
6.3.2	Off-grid Energy System Modelling	116
6.3.3	Geospatial Data	117
6.3.4	Calculating Population Trajectories	117
6.3.5	Least-cost Electrification Modelling	118
6.4	Results of Analysis	121
6.4.1	Baseline scenario	121
6.4.2	Pathways to universal household access	122

6.4.3	Variation in Reliability of Electricity Supply	124
6.4.4	Inclusion of Carbon Pricing Scheme	126
6.4.5	Sensitivity Analysis	129
6.5	Discussion and Conclusions	130
6.6	Summary	133
7	Conclusions	135
7.1	Overview	135
7.2	Key Results of Thesis	136
7.3	Implications of Research	137
7.4	Limitations of Work	139
7.5	Future Work	141
7.6	Thesis Summary and Final Considerations	142
A	Supplementary Tables and Information Chapter 4	143
B	Supplementary Tables, Figures, and Information Chapter 5	146
B.1	Constructing load profiles for Mahama Camp	146
B.2	Mathematical Description of Appraisal Metrics	148
B.3	Model Inputs	149
C	Supplementary Tables, Figures, and Information Chapter 6	151

List of Figures

2.1	ESMAP Multi-tier Tracking Framework	18
2.2	Population without access to electricity 2012-2022	19
2.3	Diagram outlining different electrification options in different geographical contexts	25
3.1	Overview of methods sections and relevant results chapters	44
3.2	Structure of the CLOVER energy system model	46
3.3	Graphical representation of the CLOVER optimisation process	51
3.4	Flowchart of the updated CLOVER energy system function to accommodate dispatched diesel generation	53
4.1	Outline of the methodological process for Chapter 4	67
4.2	Electricity demand profiles for domestic and healthcare uses in Chapter 4	72
4.3	Average hourly availability across three national grid monitoring sites examined in Chapter 4	74
4.4	Scatter plot of cost and emissions of all systems and demand scenarios at 95% of electricity demand met, Chapter 4	76
4.5	Scatter plot of cost and emissions of all systems and demand scenarios at 100% of electricity demand met, Chapter 4	78
4.6	Bar plots comparing the LCUE values for each demand scenario, system type and reliability level in Chapter 4	80
5.1	Simplified system diagram of the electricity system at Mahama Refugee Camp	90
5.2	A flow diagram containing different methodological components for Chapter 5	91
5.3	Monitored energy demand of the Mahama camp system	96
5.4	Monitored night energy demand remaining until and since sunrise Mahama Camp	97
5.5	Fuel savings and battery lifetime consumed of different operating strategies at Mahama Refugee Camp	100
5.6	The hourly state of charge of battery storage for the original strategy and the alternative strategy on the system at Mahama Camp	103
5.7	Probability of use of the diesel generator under original and alternative strategies at Mahama Camp	104

5.8	Costs and revenues arising from different changes to the system at Mahama Camp	107
6.1	Flow diagram for methods used in Chapter 6	114
6.2	Estimated spatial distribution of change in population without access in Sub-Saharan Africa between 2020 and 2030	122
6.3	Estimated spatial distribution of population without access in Sub-Saharan Africa in 2020 and 2030	123
6.4	Spatially distributed least-cost electrification method for the Ref_central, Rel_penalty_0.5 and Ctax_median scenarios	124
6.5	Mapped country-specific factors influencing shares of off-grid PV	127
6.6	Stacked horizontal bars of each electrification mode by country (Tier 3)	128
6.7	Figure showing sensitivity analysis for carbon pricing and unmet demand penalty values in Chapter 6	130
B.1	Load profiles of business types and scenarios for Chapter 5	148
C.1	Greyscale Version: Estimated spatial distribution of population without access in Sub-Saharan Africa in 2020 and 2030	152
C.2	Greyscale version: Spatially distributed least-cost electrification method for the Ref_central, Rel_penalty_0.5 and Ctax_median scenarios	155
C.3	Greyscale version: stacked horizontal bars of each electrification mode by country (Tier 3)	156
C.4	Greyscale version: figure showing sensitivity analysis for carbon pricing and unmet demand penalty values in Chapter 6	157

List of Tables

2.1	Summary of off-grid energy system models and their characteristics	36
3.1	Example technical CLOVER inputs for dispatched diesel modelling	52
4.1	Power ratings and data sources of the devices at the PHC and Pathlab	69
4.2	Electricity demand scenarios used in Chapter 4	70
4.3	The locations and their characteristics used for the grid analysis in Chapter 4 . . .	71
4.4	Electricity demand data for the five demand scenarios in Chapter 4	73
4.5	Data describing the energy demand met by different grid sites in Chapter 4	73
4.6	Results summary for PV and Battery systems optimising for cost with 95% of energy demand met in Chapter 4	75
4.7	Results summary for diesel systems optimising for cost with 95% of energy demand met in Chapter 4	75
4.8	Results summary for PV & Battery systems optimising for cost with 100% energy demand met Chapter 4	77
4.9	Results summary for diesel-only systems optimising for cost with 100% of energy demand met, Chapter 4	79
4.10	Results summary for hybrid systems optimising for cost with 100% of energy demand met, Chapter 4	81
5.1	Breakdown of the businesses in the marketplace in Mahama Camp	93
5.2	Fuel costs, fuel GHG emissions and technical performance of system configurations at Mahama Camp	105
5.3	Tariff structures for different business numbers and percentage difference between fuel costs and revenues for the system at Mahama Camp	106
6.1	Multi-tier framework annual household electricity consumption (Tiers 1-4	115
6.2	Main scenarios for analysis in Chapter 6	121
6.3	Estimated population without access to electricity in all study countries 2020, 2030 & 2035	122
6.4	Total investment, emissions and reliability metrics for selected scenarios in Chapter 6	125

6.5	AR6 C1 C2 Carbon Prices used for carbon pricing	129
A.1	Grid of probabilities used for a single device in Chapter 4	143
A.2	Technical Inputs used for Chapter 4 Analysis	144
A.3	Emissions Inputs used for Chapter 4 Analysis	144
A.4	Financial Inputs used for Chapter 4 Analysis	145
B.1	Model inputs used for the system technical specification in Chapter 5	150
B.2	Model inputs used for the economic assessment in Chapter 5	150
B.3	Model inputs used for the environmental assessment for the analysis in Chapter 5.	150
C.1	Model inputs used for technical specification of off-grid systems in Chapter 6	151
C.2	Model inputs used for the economic and GHG assessment of off-grid systems Chapter 6	152
C.3	Model inputs for least-cost electrification model in Chapter 6	152
C.4	Country-level inputs used for least-cost electrification modelling in Chapter 6	153
C.5	Table of country substitutions used in Chapter 6	154

Chapter 1

Introduction

1.1 Overview

Access to electricity remains a significant global problem. An estimated 770 million people remain without any electricity supply, the majority of whom reside in Sub-Saharan Africa. Progress over the last decade in reducing this number has been marked: in 2012 approximately 1.3 billion people lived without access. However, this progress has stalled in the last three years, due in part to the Covid-19 pandemic and additional impacts from the conflict in Ukraine [1, 2]. The number of people with an electricity connection does not capture the poor service reliability experienced by many who have a connection: globally, as many as 3.5 billion may live with unreliable power [3]. This causes costs to both households [4] and businesses [5] and results in a significant drag on economic growth [6]. Achieving universal electricity access, a key component of United Nations (UN) Sustainable Development Goal (SDG) 7 - affordable, reliable, sustainable, and modern energy for all by 2030 - is going to be extremely difficult given the anticipated population growth in low-access countries, and the number of people still without power [1, 7]. An urgent acceleration in the rate of progress is needed if we are to come close to achieving universal household access in the next decade. Failing to address the lack of access to electricity will continue to prevent gains on a number of other SDGs [8], including improved healthcare (SDG3) [9, 10], educational outcomes (SDG4) [11] and economic opportunities (SDGs 1, 8) [12].

The optimal approach to connect people - by national grid extension, stand-alone systems for single consumers, or mini-grids serving clusters of consumers - varies on the context. Electricity demand, population density, and distance to existing national grid infrastructure are crucial factors. Regarding the use of renewables-based off-grid systems to meet SDG7, arguments for greater emphasis can be made on cost, reliability or environmental grounds [13, 14]. PV-based off-grid systems have become markedly cheaper, benefiting from increased deployment of PV globally and mass production of batteries for electric vehicles. Capital costs for mini-grids fell by 65-85% in the decade up to 2019, and are expected to fall further to 2030 [15]. The International Energy Agency (IEA) estimates that decentralised options such as solar photovoltaics (PV) -based mini-grids and stand-alone systems may be the least-cost option for much as half of the population needing access [1]. Additionally, off-grid solutions may offer consumers higher reliability and more predictable power than grid infrastructure in some contexts, which may encourage consumer preferences towards such systems [14, 16]. Finally, renewables-based off-grid systems have the potential to provide equivalent energy services with reduced greenhouse gas (GHG) emissions when com-

pared with grid or fossil fuel-based off-grid alternatives [17, 18]. Despite these benefits, in many low-access countries, off-grid solutions are omitted from national electrification strategies [19].

However, there are still economic challenges facing the deployment of off-grid systems to boost electricity access. For electricity supply companies and other organisations building off-grid systems such as NGOs, high average expenditure and low average revenues per user mean that in many contexts projects may not be economically viable without grant funding or subsidies [16, 20]. Such financial support mechanisms are not widely available, thus deployment is limited. Additionally, there is uncertainty regarding the electricity demand of newly connected users. It is difficult to estimate latent electricity demand for households or businesses and therefore off-grid systems can be incorrectly sized, increasing their costs or leading to supply deficits [21–23]. Work is needed to help improve the economics of off-grid systems so that deployment can be accelerated, in turn increasing the number of people who will gain access each year.

Large numbers of people need to be connected to electricity supplies in low-income countries, often with limited resources available to bodies responsible for electricity planning and access [19]. Historically, the extension of national grids has taken precedence; however, the reduced cost of low-carbon, off-grid electricity systems has widened the options available for providing electricity access. Knowing where to emphasise the deployment of different available electricity solutions to use resources efficiently can help to maximise benefits. This challenge is further complicated when considering additional factors such as the electricity demand level [24], or the relative reliability of electricity supply of different options. Poor national grid reliability is a challenge that is common in low-income countries and should therefore be factored in when considering technology choice [25, 26]. Research that can help to explore the costs of each technology spatially and under different scenarios can help electricity planners make the best use of limited resources. This in turn can help ensure that the maximum number of people can benefit from investment in additional infrastructure.

This thesis uses energy system modelling and geospatial modelling to ask research questions which address the challenges described above. With two detailed case studies, I present ways in which renewables-based off-grid systems can provide reduced costs or improved service over alternative electricity access options. Further, within these studies, I present ways in which off-grid renewables-based systems can have a potentially improved economic outlook, by diversifying the electricity consumer base (Chapters 4 & 5) and optimising the way off-grid systems are operated (Chapter 5). Additionally, covering 43 countries in Sub-Saharan Africa, I present a study that assesses the use of different electricity access technologies under different demand, carbon pricing and reliability conditions (Chapter 6).

1.2 Research Questions Addressed and Approach

This thesis addresses a range of research questions relating to the use of off-grid electricity supply options in low-income countries. The research questions in this thesis are unified in that they each address the costs of low-carbon electricity access supply technologies and use energy system modelling methods to do so. To improve on many existing studies, where possible, real demand and cost data were used. The geographies considered and the specifics of the case studies were to an extent defined by research opportunities that provided availability of high-quality data. The main research questions are as follows:

1. How does combining domestic and healthcare electricity demands improve the cost profile for solar PV-based mini-grids in India? (Chapter 4)
2. Does the grid provide reliable electricity for domestic and healthcare electricity demands in rural India? (Chapter 4)
3. How can hybrid diesel, PV and battery off-grid systems be operated more efficiently to reduce fuel usage in humanitarian settings in Rwanda? (Chapter 5)
4. How does connecting additional electricity consumers create trade-offs and revenue opportunities within hybrid mini-grids in humanitarian settings in Rwanda? (Chapter 5)
5. To what extent do electricity demand levels impact the least-cost pathways to universal household electricity access in Sub-Saharan Africa? (Chapter 6)
6. How does adding a financial penalty to units of unmet electricity demand, or applying a carbon price impact least-cost pathways to universal household electricity access in Sub-Saharan Africa? (Chapter 6)

In this thesis, my approach to addressing the research questions outlined above is to use and develop open-source modelling tools, details of which are outlined in Chapter 3. Additionally, where possible I use data which is case-study or country-specific to improve the accuracy and usefulness of the findings. I first use an existing open-source modelling tool to produce the results provided in Chapter 4. I use site-specific grid reliability, electricity demand and cost data to strengthen the study. Secondly, I adapt the model used for Chapter 4 and add new functionality to be able to produce the results in Chapter 5. In Chapter 5 I also use site-specific electricity demand and cost data. Finally, for the results in Chapter 6, I develop a new geospatial data model and use energy system modelling alongside energy demand modelling to prepare country-level model inputs.

1.3 Scope of this Thesis

The work presented in this thesis uses modelling techniques to address research questions around technology cost and performance over time, and under different conditions. Electricity access is a complex challenge that must be looked at from a variety of perspectives. On the one hand, it is important to ensure that a given technology performs well, and can be supplied at a cost low enough to permit the largest number of people to benefit from it. However, on the other hand, to successfully connect households and businesses to electricity systems, practitioners must be aware of many factors not addressed in this thesis. These include, but are not limited to, social structures or cultural factors within communities, levels of local education and training, and laws or regulations in the respective jurisdiction. Using modelling as its primary method, this thesis considers questions relating to technologies and their cost and performance over time under different conditions. The reader should be mindful of this limitation of scope when interpreting the findings of this thesis.

1.4 Thesis Structure

This thesis is structured as follows. Chapter 2 gives a review of the relevant literature and concepts for the research presented in the subsequent chapters. Chapter 3 gives descriptions of the modelling

tools used and developed to produce the research in this thesis. Each result chapter has a specific methods section, but the methods chapter seeks to explain key parts of the models in more detail. Chapter 4 contains the study of health centres within mini-grids in Northern India. Chapter 5 details the study looking at how hybrid mini-grids are operated to reduce fuel usage in displacement settings in Rwanda. Chapter 6 outlines the study examining least-cost pathways to universal household electricity access in 43 countries in Sub-Saharan Africa under differing demand, carbon price and unmet demand penalty scenarios. Finally, Chapter 7 gives conclusions, limitations and a thesis summary.

Chapter 2

Literature Review

2.1 Overview

In this section, I give context to the research that is presented in this thesis. This literature review is not exhaustive, but it describes key concepts and the existing, relevant work in the field. The main numbered sections are as follows:

- Section 2.2 outlines the literature on defining and measuring electricity access.
- Section 2.3 describes progress to, and future projections for reaching universal electricity access.
- Section 2.4 outlines why electricity access is important and considers its impacts on livelihoods, health, education and quality of life.
- Section 2.5 gives an overview of the technology choices available for electricity access, trends in cost with future projections and the economic challenges for off-grid systems.
- Section 2.6 provides a brief overview of electricity access and climate change, considering climate change mitigation impacts and carbon pricing.
- Section 2.7 covers the modelling tools available to answer research questions relating to electricity access; this includes tools for off-grid system design and geospatial modelling tools for electricity access planning.
- Section 2.8 discusses the literature, places it in context with my research and highlights gaps that I address.

Within each of the results Chapters 4, 5 & 6, there are also short detailed literature summaries in the 'Background' parts of the Chapter. These aim to guide the reader and focus on studies that are most similar to the study found in the respective chapters.

2.2 Access to Electricity: defining and measuring the problem

Access to electricity is far from straightforward to define and measure, with no single unifying definition of what electricity access constitutes. The way access is defined is important, as it may

dictate national electrification plans or system design and consequently the outcomes for those who still need access to electricity.

Access to electricity is easiest to measure in a binary way: echoing the approach to electrification historically taken in high-income countries. A common metric applied in low-income countries is whether national grids have reached a given population centre [27]. A key weakness of only considering national grid extension is that it fails to capture the increasing numbers of connections from off-grid sources. However, even when including off-grid sources as the IEA and others do, defining access in terms of a connection or not fails to capture important dynamics [28].

Another definition is whether a connection is able to provide access to a bundle of basic energy services: multiple light bulbs, task lighting, phone charging and a radio. This has alternatively been defined as 100 kWh of electrical energy per capita per year, or enough power to charge a mobile phone and give 5 hours of light every day in the year. This fails to take into account important characteristics such as affordability and reliability. However, the 100kWh definition is often used due to the poor availability of data in low-access regions [28]. Due to the very low level of power provision that this definition of access gives, it has been argued that the threshold should instead be redefined as an extreme energy poverty line, with further higher-defined thresholds as levels approach those similar to the energy services available in the developed world [29].

Other factors beyond physical access are critical: among populations that have access to modern electricity access, households may not be able to afford the upfront connection costs, the unit costs, or the appliances that make electricity useful [30]. Some alternative approaches attempt to take these factors into account. Recognising the need to track energy access in a more comprehensive way, the Sustainable Energy for All (SE4ALL) initiative launched the Global Tracking Framework (GTF) in 2013. A framework was proposed that included 5 tiers of access, with corresponding quantitative measures for power capacity as well as total and evening power availability. In addition, binary affordability, legality and quality (voltage) indicators were proposed [31]. Building on the GTF, in 2015 The World Bank Energy Sector Management Assistance Program (ESMAP) launched the Multi-Tier Tracking Framework (MTF) (see Figure 2.1), which is more comprehensive than the candidate GTF and includes criteria for reliability, power quality, affordability, legality and safety that must be met to reach the higher tiers of access. In addition to the MTF for household usage, frameworks detailing tiers for productive use of electricity, community institutions and street-lighting are included [32].

Although the MTF is a more comprehensive way of assessing access to electricity, it is not free of criticism. Its complexity has been noted as an issue for its use in practice for effective measurement; additionally, it has been described as too prescriptive to be accepted at the national energy planning level [33]. In a study testing the MTF in a Bangladeshi context, criticism of the algorithm for tier assessment was raised in regard to the increasingly complex nature of energy access, and the innovation in super-efficient DC devices for off-grid purposes [34].

Further frameworks exist to understand access to electricity and energy poverty. The Energy Supply Index (ESI) does so also through 5 levels, with level 1 describing users as having access only to third-party charging, and level 5 describing users as having a reliable AC connection comparable to those in high-income countries [35]. This framework is reliant on qualitative data and is good for in-depth region-specific studies; however, it is less suited to larger assessments across multiple locations that may require more standardised data [33]. In addition to the ESI, Practical Action, an NGO, has a broader framework called Total Energy Access (TEA), encompassing further criteria. Total Energy Access is achieved when households, enterprises and community facilities

			Tier 0	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Attributes	1. Peak Capacity	Power	No Electricity	Min 3 watts	Min 50 watts	Min 200 watts	Min 800 watts	Min 2 kW
		Daily Capacity		Min 12 Wh	Min 200 Wh	Min 1.0 kWh	Min 3.4 kWh	Min 8.2 kWh
	2. Duration	Hrs/Day	< 4 hrs	Min 4 hrs		Min 8 hrs	Min 16 hrs	Min 23 hrs
		Hrs/Evening	< 2 hrs	Min 1 hr	Min 2 hrs	Min 2 hrs	Min 4 hrs	
	3. Reliability						Max 14 outages/week	Max 3 outages/week totalling <2hs
	4. Quality					Voltage problems do not affect the usage of desired appliances		
	5. Affordability					Cost of standard a consumption package of 365 kWh per annum is less than 5% of household income		
	6. Legality					Bill is paid to the utility / prepaid card seller / authorized representative		
	7. Health & Safety					Absence of both past accidents and perception of high risk in the future		

Figure 2.1: Multi-tier tracking framework, adapted from [32]

have adequate access to all the energy supplies and services that are required to enable human, social and economic development. The TEA takes account of both qualitative and quantitative measures and is again suitable for in-depth community studies [36].

The above frameworks rely on data from government electrification authorities, companies or survey data collected by other parties. An alternative way to assess electricity access is through night-light data [37]. The existence of lighting at night and the concentration or brightness of light, when assessed on satellite imagery [38], can be used to determine whether a given population has access. Validating this data with survey data can give insights into its accuracy [39]. Combining night-light data with other geospatial data such as population and settlement information can give further insights into populations living without access, such as their approximate energy access tier, as defined by the MTF. A remote sensing approach for measuring access has benefits as it applies a standardised methodology across entire countries or continents. However, there are specific community-level aspects that cannot be captured, and the limitations of these data must be acknowledged with their usage [40].

2.2.1 Summary

This section has described the different approaches to measuring and defining electricity access. This is an important consideration as the method of measuring access may impact the outcomes achieved. A binary definition of electricity access will fail to capture important characteristics of supply whereas considering electricity access at different demand levels (tiers), that factor in energy service delivered, may be more useful for comparing different electrification strategies across geographies. The research presented in the coming chapters seeks to incorporate a nuanced way of considering electricity access and where appropriate refers to the MTF tiers of access and the reliability of different solutions.

2.3 Progress towards SDG7 and future projections

Achieving universal electricity access, a key part of SDG 7, target 7.1 - to ensure universal access to affordable, reliable and modern energy services, by 2030 - will be difficult due to the large numbers of people still without access, and the expected growth in population in the coming years in areas that have no or poor infrastructure [1, 7]. Notable progress in reaching universal access to electricity has been achieved in the last decade. The total number of people without access has fallen by 40%, from roughly 1.3 billion in 2012 (see Figure 2.2), to around 770 million today [1]. In Africa, 115 million people gained access between 2014-9 [41]. However, progress has stalled recently; the number of people without access stopped declining during the Covid-19 pandemic and the International Energy Agency (IEA) expects the number to increase for the first time in decades in 2022 given the hardship caused by the war in Ukraine [2].

In addition to those without any access, poor reliability of supply is a significant problem. Globally, as many as 3.5 billion may live without a reliable electricity connection [3]. Unreliable power results in an economic cost to households [4, 42], and businesses [5, 43] with evidence to suggest higher customer satisfaction for, and expressing a willingness to pay more for, reliable electricity [14]. In Africa, where grid reliability is poor, between 1995 and 2007, lack of reliable electricity is thought to have weighed on economic growth by as much as 2% of GDP per year [6].

Looking forwards to 2030, the target year for SDG7, the outlook for meeting target 7.1 is negative: the IEA's stated policies' scenario expects 670 million people globally to remain without any power; with 85% of them residing in Sub-Saharan Africa [1]. For Sub-Saharan Africa, if the pace of progress is maintained (roughly 22 million people gaining access per year), then the share of the population with access by 2030 will only reach around 62%. To achieve universal access in Africa, as many as 75 million people will need to gain access per year to reach universal access [41]. Given current progress and low levels of access in many landlocked countries in Sub-Saharan Africa, it is possible that universal access may not be reached even by 2050 [44].

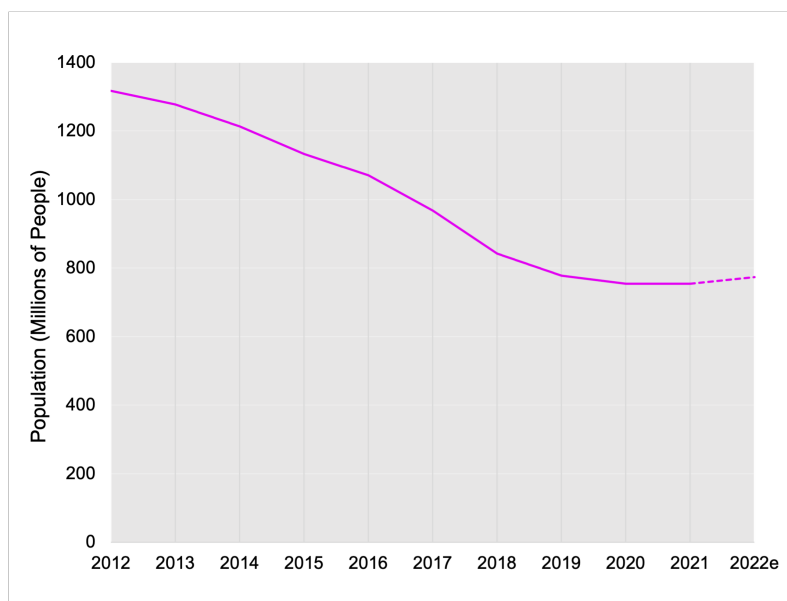


Figure 2.2: Population without access 2012-2022, data from the IEA [2]. Until 2020 there were consistent declines in the number of people without access, with that trend reversing in 2021.

2.3.1 Summary

Although there has been progress towards reaching universal access in the last decade, this has stalled since 2020 with more than 700 million people having no access; additionally, billions of people live with unreliable access. Future projections show that we are not on course to meet SDG7, with 100s of millions of people still without access in 2030. Rapid action is needed to change course.

2.4 The Gains from Improving Electricity Access

Improving access to electricity (SDG7) has been shown to impact other sustainable development goals [8]: such as reduced poverty (SDG1) and improved economic opportunity (SDG8); better healthcare provision (SDG3); and, improved educational facilities (SDG4). Additionally, access to electricity can improve quality of life in ways that are harder to measure: e.g. access to entertainment via television and access to the internet.

2.4.1 Livelihoods

Electricity may allow for stimulation and improvement of income-generating activities, cause the creation and strengthening of local markets or alternatively have an impact on the household economy [12]. The evidence available suggests that there are regional differences in the socioeconomic effects of access to electricity [45]: with more Asian case studies seeing measurable benefits, and with studies from African countries finding weaker socioeconomic impacts [46]. Links between electricity access and economic benefit are not always found. This may be due to the fact that additional supporting interventions to ensure positive impacts are sometimes absent.

Numerous case studies do support the relationship between access to electricity and the growth of enterprises or enhancement of existing income-generating activities. For example, Gibson and Olivia (2008) report that electricity boosts income and non-farm rural employment in Indonesia [47]; whilst the impact of a Nepalese hydroelectric project was reported to have caused the emergence of numerous small industries in the area served [48]. An analysis of Vietnamese panel data demonstrated considerable uplift in per capita income following electricity access. However, households in higher income brackets benefited considerably more than those in lower ones, highlighting that gains from access may be unevenly distributed [49]. Despite most evidence originating from Asia; there is evidence from Kenya indicating that electricity access boosts the rate of micro-enterprise creation, raises incomes and increases the proportion of customers from outside communities for businesses [50]. Further, solar PV systems were found to create new income-generating opportunities and allowed micro-businesses to generate more income in rural Ethiopia [51].

The amount of electrical energy available is also important, with higher per capita availability likely to have more impact on socioeconomic outcomes. Whilst better quality of light from small DC solar-home-systems (SHSs) has the potential to improve reported quality of life, whether it is sufficient to provide the direct, measurable economic benefits, is less clear [52]. Moreover, the expected causal relationship between the provision of electricity and improvements in socioeconomic outcomes is not always present. A randomised control trial providing solar power in India reported no improvements in any socioeconomic indicators: including effects on the household economy (saving and expenditures), business creation or lighting for studying [53]. Further, a review of the

implementation of 30 small-scale energy projects across the world concluded that the provision of energy services does not automatically result in productive use of energy and therefore economic development [54]. The absence of intended effects of electrification, such as improvements to the household economy may be due to the lack of complementary development interventions that maximise the gains from electrification [54, 55]. Furthermore, an integrated approach, with capacity building to encourage development of economic activity alongside access to power has been proven as a successful strategy in improving intended benefits [56].

Finally, it is important to highlight the significant challenges in measuring the impact of interventions, with self-selection bias (in those that opt to connect) and heterogeneity between the control and treatment groups between and within communities. Bensch, Kluve and Peters (2011) highlight this, and attempt to tackle such problems in a study in Rwanda. Once accounting for regional differences between the control and treatment group, measurable socioeconomic benefits were less evident [57].

2.4.2 Health

It is estimated that approximately one billion people globally rely on health centres that have no access to power, undermining the quality of care they can benefit from [58]; with almost 60% of primary health centres in low- and middle-income countries (LMICs) having inadequate energy services [59]. In Sub-Saharan Africa, it is estimated that more than 55 000 health facilities of various sizes have no access to electricity at all, [60]; however, data on healthcare facilities and their access to power is of extremely poor quality, so the true numbers may be even greater [61].

Access to electricity for health centres can affect a broad range of metrics that can improve quality of care, from extending hours of service by providing lighting, to powering essential medical equipment, whilst health centres with electricity are better able to retain the healthcare staff needed to provide care [62]. Evidence from India suggests that access to reliable electricity for health centres increased the availability and functionality of equipment and improved health outcomes for children and pregnant women [9]. Additionally, a systematic review of evidence from a range of low-income countries found that electrification is generally associated with improved health outcomes, such as lower mortality and disease prevalence [63]. Reliability is key: cutting electricity supplies to life-supporting equipment such as ventilators or incubators carries the risk of causing death. The risk of mortality in Ghanaian health facilities was shown to increase by 43% on days when the duration of power outages exceeded 2hrs, demonstrating the importance of not only providing an electricity supply but additionally of ensuring its reliability [64]. Hypoxemia, low blood oxygen, can lead to preventable mortality; and it is high in healthcare facilities with a lack of reliable electricity [65]. It is estimated that less than 2/3 of hospitals providing surgical care in a study of 21 low-income countries have reliable electricity, either from the grid or backup generators, making the provision of care more difficult.

It is estimated that 1.7 million children die each year from diseases preventable by vaccines or immunisations [66]. There are complex challenges in delivering vaccines and immunisations to those that need them: consistent power for the cold chain is critical in preserving their efficacy [62, 67]. A comprehensive study of health facilities in Uganda and Ghana found that electricity access facilitated the proper storage of vaccines [68]. Given the increased prevalence of the spread of infectious diseases, and in light of the recent Covid-19 pandemic, the importance of reliable electricity for cold storage is increasingly salient [69]

The health implications of poor air quality are also important to consider when understanding the benefits of electricity access. Exposure to indoor air pollution in low-income countries has been implicated in causing a range of diseases, including cancers, tuberculosis and increased infant mortality rates [70]. Indoor air pollution from fuel-based lighting and the resultant damaging health effects has received relatively little attention in the literature, with most authors focusing on the harmful effects of cooking fuels [71]. Burning kerosene for lighting is however a notable contributing factor to air pollution, with exposure to particulate matter and toxic gases causing adverse health impacts [72]. The association of indoor air pollution and kerosene lighting has been demonstrated in rural Uganda, with open-wick kerosene lamps used in living rooms having the worst impact [73]. Moreover, using a hurricane lamp instead of an open-wick style one has been found to reduce exposure to both PM_{2.5} and PM₁₀. Better still however, is the use of electric lighting, with an 80% reduction found in the mass of inhaled PM_{2.5} over using an open-wick kerosene lantern [74].

Switching to lighting from electricity sources such as solar photovoltaics has the potential to displace kerosene use, and therefore reduce exposure to pollution and improve health outcomes [75]. A trial in Kenya found that the provision of solar lanterns displaced the use of kerosene by 90%, and showed marked reductions in the exposure of PM_{2.5} to both adults and school pupils [76]. Displacement of kerosene and reduced pollution can then cause gains in health outcomes, e.g. in the reduction of acute respiratory infections in young children [77].

Finally, electrification can impact health outcomes in indirect ways. Access to power can allow people to receive health information. In Bangladesh, power allowing people to watch television was shown to boost health outcomes due to improved access to relevant information, for example on the importance of immunisations [78].

2.4.3 Education

Globally, as many as 200 million children are attending schools that do not have an electricity connection [79]. Electricity access for schools can lead to a range of benefits. These include being able to teach classes in the morning or evening when natural light levels are lower, the ability to use television and computers in learning, and higher staff retention [80]. Additionally, electricity access for households can have an impact: having light in hours of darkness can allow children to study for longer, thereby improving attainment [11, 81].

Regarding measurable impacts, household electricity access has been shown to lead to additional years of formal and higher education completed in the Philippines [82]; and regression analysis undertaken with data from Assam, India found that the electrification rate of households was the third most significant variable in determining levels of literacy in those aged 6 and above, behind road density and sex ratio [83]. In addition, econometric analysis of panel data from Vietnam from 2002-5 determined that household connection to the grid has positive effects on both enrolment and completion rates for those aged 5-18 years [49]. In Brazil, the Luz para Todos (Light for all) programme has provided electricity to over 15 million people (3.3 million households), with education being the component of the human development index to be most impacted by the provision of electricity [84].

2.4.4 Quality of Life

Access to electricity can positively impact people in ways that are harder to measure quantitatively such as leisure time, social connectedness and access to entertainment. This sits separately from economic measures such as income growth.

A review of evidence from South Africa, Brazil, China and India found wide-ranging impacts on quality of life from electricity access: increased leisure time, use of electronic devices such as televisions for information and entertainment, and use of lighting at night extending usable hours of the day. These benefits were found to be greater for women than for men, in part due to the higher proportion of domestic labour women typically undertake [85]. Electricity access has also been found to increase levels of reported social connectedness: in South Asia studies have shown that solar home systems can contribute to strengthening connectedness between household members and other members of their community [86]. Although hard to measure objectively, there is evidence to suggest that electricity access improves reported levels of happiness: in South Africa, community residents with electricity access were found to be happier [87], and in Indonesia, a positive effect on happiness was also found, with the impact being greatest in poorest regions [88].

2.4.5 Summary

This part of the literature review has summarised how electricity access can positively impact human development outcomes. In relation to income growth, and the development of income-generating activities, the evidence is somewhat mixed. The evidence suggests that factors such as the level of electricity access (in terms of electricity supplied), and accompanying capacity-strengthening programmes, have an impact on the level of benefit. The evidence around health, education and quality of life is less equivocal; with the evidence suggesting that electricity access improves outcomes.

2.5 Means of Improving Access: Technology Choices

To connect people who don't have access to electricity, there are a variety of technology options available; both in terms of the system configuration and the generation means. This section will describe off-grid electricity system types and the factors impacting the selection of these differing means. Further, this section will describe some generation technologies that are relevant to this thesis (solar PV, diesel and battery storage) and their usage in off-grid contexts.

2.5.1 System Types

There are three main options for improving electricity access, regardless of generation technology used: 1) the extension of and connection to national electricity grids; 2) the construction of isolated grids with their own generation capacity connecting multiple customers; and 3) the use of stand-alone systems that power single consumers with their own generating capacity.

On-grid or grid extension solutions are defined as a connection to a local network with connectivity to the transmission network, drawing power normally from large centralised power stations mixed with some distributed generation [89].

Regarding isolated grids, there is ambiguity amongst practitioners with regard to the ter-

minology that should be used; for example, with what constitutes a mini-grid as opposed to a micro-grid [90]. Local, national and international bodies use their own different metrics when defining isolated grids; such as, size of generating capacity, grid connectedness, or voltage levels. This can create difficulties in the comparison of systems when considering academic studies, policy documents or reports [91]. Some definitions distinguish micro-grids and mini-grids by whether or not they have a connection to the national grid [92]. Most, however, focus on the scale either at the generation capacity level, the number of nodes connected, or both [93]. Nano-grids, a more recent term, are broadly speaking the smallest type of grid system, with at least one power sink [94]. In addition, solutions can be categorised by both their level of isolation as well as their capacity. DfID (2013) define the first category of system as being grid-connected or proximate, with a capacity of $>1\text{MW}$, also likely to provide power that would be equivalent to a functioning grid. The second is defined as being isolated, normally with AC power of 1kW - 1MW and would provide an intermediate service to the grid. Finally, the third category is defined as being very isolated, under 10kW , often DC power, and providing basic services such as lighting [90]. It would be logical to assume under these definitions that category 1 would refer to a mini-grid, category 2 to a micro-grid and category 3 to a nano-grid.

The term mini-grid has emerged as the most commonly used in the literature and by practitioners to describe isolated grids as an energy access solution and for clarity, I will use this term henceforth in this thesis. Hybrid mini-grids typically refer to mini-grids that incorporate some combination of generation technologies; normally renewables (e.g. Solar PV) with storage capacity and back-up or dispatchable generation such as diesel generation [15].

Stand-alone systems have only one consumer of the generated capacity but can be of varying sizes. Both small systems for a single household, such as a solar home system of a few watts, and a much larger solution powering a single commercial off-taker would be considered a stand-alone system [95].

2.5.2 Factors Impacting Technology Choice

Different system types are appropriate for different contexts. The suitability of a given solution will depend on a range of factors, such as the cost of different options in a given context, the reliability of national grids, policy-making and socioeconomic considerations

The relative cost of taking a grid or off-grid approach will depend largely on the population density and distribution in a given area and the proximity to existing national grid lines. In areas far away from the grid where households are isolated from each other, the costs of distribution lines linking households in a mini-grid or micro-grid are likely to outweigh the benefits of any economies of scale [96].

Figure 2.3 illustrates that as the distance from the grid increases, and population density decreases, the cost of grid extension increases above the point at which power can be provided with mini-grids and stand-alone systems. However, the costs of different approaches vary beyond the above factors; with country context and demand levels also playing a role [98]. A number of studies have used geospatial modelling to map countries and regions based on the expected least-cost approach; this is covered in Section 2.7.2.

A further consideration for grid versus off-grid approaches may be based on reliability. Electricity grids in low-access countries often provide poor levels of service [25], necessitating the costly and often polluting use of backup off-grid generation means [26]. In some contexts, when grid

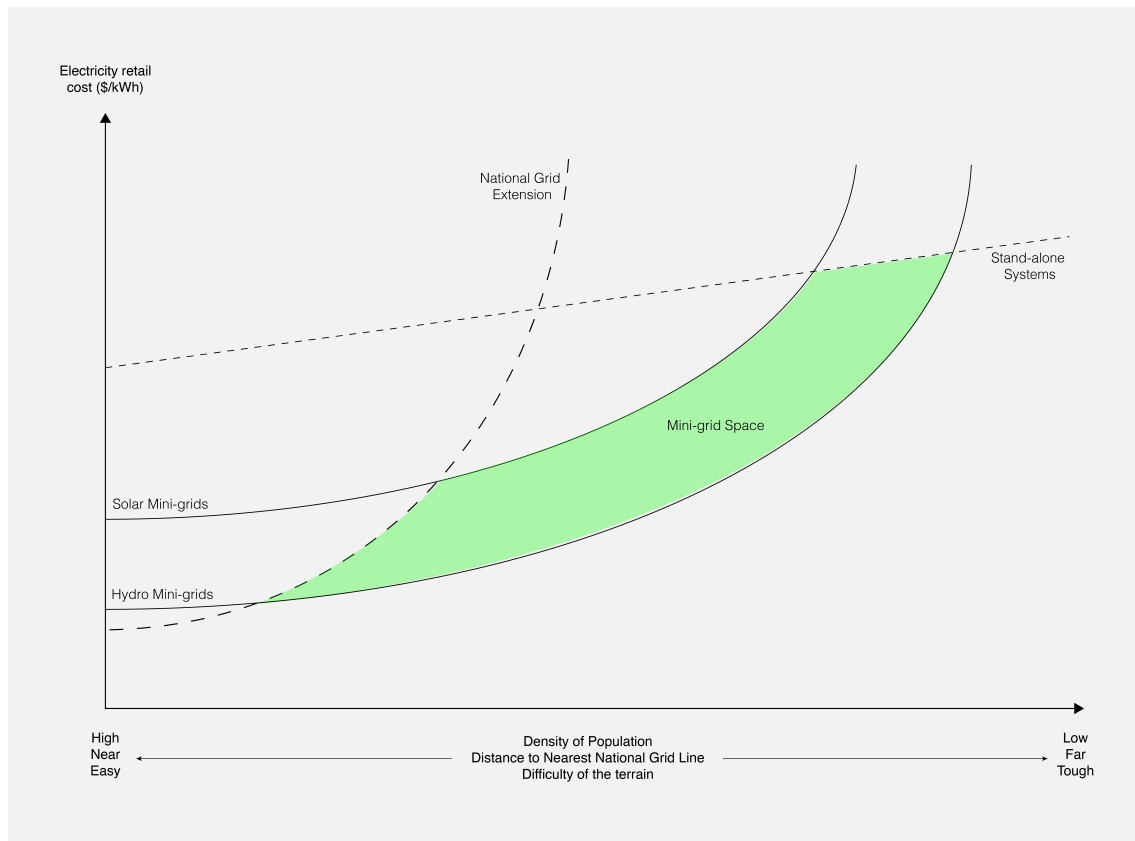


Figure 2.3: Displays different optimal systems in different geographical contexts, adapted from EU (2014) [97]

reliability is low, consumers have expressed willingness to pay more for reliable electricity [5], and there are examples of consumers choosing to pay more for reliable off-grid electricity providers rather than a grid connection that is unreliable [14]. Poor reliability of supply can discourage consumers from obtaining a connection or investing in appliances to increase demand [99].

Policies and electrification plans can impact adopted pathways to access. Many governments omit off-grid solutions from electrification plans, so regardless of what may be the least-cost approach, only grid extension is considered [19]. Other countries, e.g. Rwanda, include off-grid solutions such as mini-grids in their electrification plans by designating zones where these technologies will be implemented[100]. Other policy considerations, such as energy deregulation may also play a role in determining whether off-grid or on-grid approaches are facilitated [101]

Finally, there may be socioeconomic considerations when determining system types. For example, mini-grids rather than grid extension may be beneficial for gender equality, although it depends on the ownership model and level of community engagement [102] Off-grid systems, due to their smaller nature and that generating capacity is located at the household or community level, better facilitate ownership of assets by consumers and participation in the development of electricity projects [56], and thus an off-grid approach may be taken to reap these benefits even if costs are higher.

2.5.3 Solar PV

Since the development of PV modules with high enough efficiency for use as generators and their use in the space programme for powering satellites in the 1950s, solar PV has expanded in its deployment significantly and is now a mature technology [103]. Growth in global deployment has been rapid in the last decade, increasing from 40 gigawatts (GW) of capacity in 2010, to 900GW by the end of 2021 [104]. The weighted average efficiency of crystalline Silicon (c-Si) modules deployed has also increased in recent years: increasing from 17% to 20% in roughly the last decade [105, 106].

As solar PV has increased in capacity, its cost has fallen dramatically, with an experience curve showing that over the last 41 years, a 25% reduction in price for each doubling of capacity has taken place [105]. At the utility scale, solar PV is already the cheapest form of electricity generation in many parts of the world: the global weighted average levelised cost of electricity (LCOE) for capacity commissioned in 2021 was just \$0.048/kWh, falling on average 17% year on year since 2010 [105, 107]. For comparison, the same figure for on-shore wind was \$0.033/kWh, with fossil fuel generation such as gas and coal sitting in a range of roughly \$0.05-0.18/kWh [107].

PV's projected growth in capacity has been consistently underestimated. For example, in 2004, the IEA estimated the total installed capacity of PV would be just 29 GW in 2020; whereas the actual number was 25 times that [104, 108]. In 2010, it was predicted that only 2% of electricity demand would be met by PV in 2035; yet solar PV currently represents around 3.6% of world electricity generation [104, 109]. As costs fall further, this share is expected to increase substantially, possibly providing as much as 38% of global electricity generation by 2050 [110]. However, in the medium term, projected growth in PV is lower than it needs to be to achieve required decarbonisation targets: under the IEA's 2050 Net Zero scenario, annual growth in PV deployment up to 2026 needs to be 80% higher than the IEA project it to be [111].

PV's emissions factor of electricity generated will depend on the capacity factor, the panel type, and the configuration of the system. A review and harmonisation of life-cycle emissions studies was carried out by researchers at the National Renewable Energy Laboratory (NREL). They examined the results of the life-cycle emissions of ground-mounted and rooftop c-Si solar PV and found a median emissions intensity of 45g CO₂e/kWh, with the interquartile range of 39-49 grams of CO₂e/kWh [112]. The location of the PV determines the capacity factor and therefore the unit emissions; for example within Europe, PV located in Iceland is estimated to produce electricity at an emissions intensity of 89 grams of CO₂e/kWh versus a value of 38g CO₂e/kWh for Cyprus [113].

For off-grid applications, the emissions intensity of electricity values for PV may be much higher due to significant shares of electricity generated not being used. Sandwell, Chan, et al., performed modelling scenarios of the use of different PV technologies in Indian mini-grids, using the metric of grams CO₂e/kWh used, as in a mini-grid some generated power will be dumped. In this context c-Si PV was found to have an emissions intensity of 487g CO₂e/kWh used; other technologies such as organic photovoltaics (OPV) had lower emissions intensities due to their less energy-intensive manufacturing process [114]. A study of solar home systems in use in Sub-Saharan Africa found lower figures, with emissions intensity of electricity estimated to be in the range 140-180g CO₂e/kWh [115].

Silicon PV modules are energy intensive to produce, hence if they are under-utilised the emissions intensity of electricity can be high. Where modules are manufactured matters due to the different carbon intensities of electricity grids. For example, modules produced in Germany will

typically have a lower embedded carbon than modules produced in China [116, 117]. Around 60% of the electricity used to produce PV modules globally is coal-fired due to production being concentrated in coal-heavy regions in China. However, due to efficiencies in production and an increasing share of low-carbon electricity being used, the embedded carbon in PV modules is falling [117]. Recent data suggest the embedded carbon in silicon-based PV modules may be as low as 380kgCO₂e/kWp, whereas values above 1000kgCO₂e/kWp were typically found a decade ago [116, 118].

Solar PV for Electricity Access

Solar PV is a practical option in an off-grid context and historically, before PV became a mature technology, a greater proportion of installed solar PV was off-grid than on-grid [91]. It has the benefit of being modular and can be applied to provide power for very small applications such as solar lanterns or SHSs, or alternatively within larger off-grid systems such as mini-grids. Furthermore, it has the benefit of relatively fast deployment, with shorter project lead times than most other generation technologies [119]. Additionally, solar radiation is typically high in parts of the world with low levels of electricity access: large parts of sub-Saharan Africa and India have irradiation exceeding 2000 kWh/m²/annum [120]. The use of PV in off-grid contexts has grown in the last decade, from an estimated 1.3GW of installed off-grid capacity in 2011, to 7.4GW in 2021; however, this increase is dwarfed by the increase seen in utility-scale PV over the same time period. The off-grid PV capacity is made up of approximately 1/3 each of mini-grids, industrial users and solar home systems [104].

Solar home systems and other small solar products provide entry-level electricity (MTF tier 2 or below) to an estimated 490 million people; these products are often bought on credit, and allow consumers to pay monthly in small instalments [121]. They often utilise super-efficient DC devices that enable greater levels of energy service at lower power ratings with small panels and batteries, reducing their overall cost [122]. Countries differ in their market maturity for these products: Bangladesh represents a market that is saturated: it has had huge success in improving electricity access through the subsidisation and financing of solar home systems for millions of households [123]. Uganda and Kenya represent maturing markets, with multiple companies expanding rapidly and SHSs commonplace. Countries with large numbers of people without access such as the Democratic Republic of Congo and Madagascar have relatively nascent markets for solar products [121].

For solar-based mini-grids, the reach is lower in terms of the number of people connected; yet the level of access provided is typically higher than that of solar products. It is estimated that there are approximately 50,000 mini-grids operating in 130 countries, serving around 50 million people [16]. These are of varying sizes and use a mix of technologies. Solar mini-grids comprise a relatively small fraction: and are estimated to provide power for about 3.5 million people, approximately half of whom are in Africa [124]; although this number is expected to increase significantly over the next decade [16]. Countries with relatively high use of solar-based mini-grids include Indonesia, Cameroon and Nepal [124]. Section 2.5.6 will explore cost reductions and future trends in the growth of off-grid PV systems.

2.5.4 Diesel Generation

Diesel generation is deployed widely at a varying scale around the world, from systems at around 1kW size for a stand-alone system, to those at MW scale that can power municipal grids, or that may provide electricity for national electricity grids [125]. The lower upfront capital costs, reliability, ease of maintenance and network of suppliers and supply chains for parts have meant that the technology has become widely used for off-grid systems and backup generation; chosen over technologies with cheaper running costs, but higher capital costs [126, 127].

The global installed capacity of diesel generators is estimated to be between 350 - 500 GW, with the majority of them being grid-connected. They provide approximately 9% of electricity consumed in Sub-Saharan Africa and 2% in South Asia. The figure is over 40% for western Africa and Nigeria has one of the highest fuel usages for diesel generators in the world. This fuel use is estimated to have a global annual cost of between 30 and 50 billion US dollars a year, which corresponds to an average cost per kWh of \$0.30/kWh, on fuel alone [128].

The cost of diesel at a given location will affect the competitiveness of diesel as a technology in comparison with other options such as PV. Diesel costs are likely to be linked to transportation costs, with cheaper overall fuel costs found near population centres and ports [127]. In addition, fuel subsidies are an important factor in the choice of generation technology, and influence decision-making in many low-access countries. In Colombia for example - where diesel mini-grids are widely used - fuel subsidies paid by the central government increase the competitiveness of the technology compared to others [129]. In Africa, the role of fuel subsidies has been demonstrated to be crucial in making off-grid diesel cost advantageous over PV in some countries [127].

Diesel generation is heavily polluting, both from a GHG emissions perspective and in its contribution to local air quality. The GHG emissions intensity of diesel power generation will depend on the context but is high compared to other technologies. The emissions intensity for diesel in India is taken to be 1056 grams of CO₂-eq/kWh [114, 130]. An alternative estimate used by the United Nations Development Programme in a baseline assessment for sub-Saharan Africa is 800 grams CO₂-eq/kWh [131]. With regard to localised emissions, diesel is heavily polluting. A study of the Indian telecom network found that high levels of pollutants damaging to health such as particulate matter and nitrous oxide resulted from off-grid diesel generation for telecom towers [132].

Diesel for Electricity Access

There is estimated to be over 100GW of off-grid diesel capacity in Sub-Saharan Africa alone, with 17 countries having more than their installed total grid generation. In Africa, Nigeria has by far the highest capacity of off-grid diesel [133]; and has a chronically unreliable grid. 80% of those with a grid connection have another form of electricity backup, normally diesel. Nigeria is the largest importer of diesel generators and spends \$22 Billion annually on backup diesel generation [89] Pacific Small Island Developing States (PSIDS) also rely heavily on diesel generation for mini-grids. Large proportions of national GDP are spent on imported oil products in PSIDS, and they have some of the highest costs of electricity in the world for consumers, hampering GDP growth. A move towards a renewables-based island power system could see long-term savings in places such as Tuvalu, and additionally, increase energy independence [134].

Despite falling costs of renewables and the significant attention paid to technologies such as solar and battery storage, a large proportion of off-grid power systems are still reliant on diesel generation, and this will continue for some years to come [135]. Data from BNEF shows that

whilst small-scale diesel generator imports to low-access nations have declined slightly since the late 2000s, the annual figure is relatively stable at around 10GW per year, predominantly to East Asia and Africa [136].

2.5.5 Battery Storage

Battery storage is increasingly playing an important role in both on-grid and off-grid electricity systems. To be able to make greater use of electricity from variable renewables such as PV, battery storage can play a vital role: storing electricity when it is in excess and discharging it when there is a deficit.

Battery storage to support electricity grids is growing rapidly, with year-on-year growth of the annual amount added of 60% from 2020 to 2021, and with total installed capacity estimated to be 16GW, the majority of which uses lithium-ion chemistry [137]. From 1990-2010, the annual cost reductions for batteries were relatively modest, decreasing at a rate 50% lower than that of solar PV [138, 139]. Between 2010-2018, the cost reduction per annum for li-ion batteries has accelerated and has averaged 20% per annum [139]. Battery pack costs have fallen by a total of 24% since 2018; although between 2021 and 2022 lithium-ion battery prices increased by 7% in real terms, to around \$150/kWh, highlighting their exposure to global commodity prices and supply chain challenges [140]. Despite recent increases, the growth in electric vehicles (EVs), and the associated mass production of lithium-ion batteries has pushed down the cost of batteries for both the automotive industry and stationary applications; BNEF expects the cost to fall below \$100/kWh in 2026 and could fall below \$60/kWh by 2030 [140]. However, a review of academic work that took differing approaches e.g. asking experts or using learning rates, was less bullish and predicted prices below \$70/kWh by 2050 [141].

Battery Storage for Electricity Access

There are a variety of battery chemistries, however, for off-grid applications, the most commonly used are either lead acid or lithium-ion [142]; this section will therefore focus only on these two types of batteries.

Lead-acid batteries have until recently been more commonly used in off-grid contexts due to their lower cost and better availability. Lead acid battery technology has been in existence for over 150 years. They are cheap and have relatively stable prices, typically less than <\$200/kWh for off-grid applications [143]. They have a relatively low energy density however and do not respond well to deep discharging. They typically have a lifetime of around 500-2000 cycles and an energy density of 50-110Wh/litre. Lead-acid batteries can only be discharged at low C-rates or the lifetime will be seriously reduced [144, 145]. Lead-acid batteries are found in off-grid applications at the household level (SHSs) and also in mini-grids. Although lithium batteries are expected to displace lead-acid in some contexts; there is continued anticipated future use for larger off-grid applications e.g. used in mini-grids when combined with lithium-ion batteries to form hybrid battery storage [145].

The lithium-ion chemistries most commonly used for off-grid applications are lithium iron phosphate (LFP) and nickel manganese cobalt (NMC). Historically they have been more expensive than lead-acid batteries; but as highlighted above, lithium-ion batteries are reducing in cost, thus their usage in off-grid contexts is increasing [145]. They also have technical advantages over lead-acid: achieving a greater number of cycles (up to 3000 cycles, although this depends on how they are used), and a higher energy density typically exceeding 150Wh/L, with continued improvements

as delivered capacity expands and drives innovation. Lithium-ion batteries can also be discharged at higher C-rates than lead acid batteries and experience less degradation [144, 145]. Lithium-ion batteries dominate in small solar products and are expected to be used in increasing shares in off-grid systems at larger sizes e.g. mini-grids in the future [145].

A life-cycle assessment of different lithium-ion chemistries found there to be embodied carbon of 200–225 kgCO₂e per kWh of useable capacity. The carbon intensity of electricity stored in batteries will depend on how intensively the batteries are cycled. When cycled once a day it adds 40–80g CO₂e/kWh depending on the chemistry [146]. Lead-acid batteries typically add higher amounts of carbon per unit of energy stored due to their lower energy density [147].

2.5.6 Trends in costs of off-grid systems and future projections

As outlined above, key components such as battery storage and PV modules have seen significant cost reductions in recent years; with further projected declines to come. However, it is important to consider the trends in the cost of off-grid systems, not just components, as there are other factors impacting cost.

Regarding the cost structure of a typical PV and battery mini-grid the batteries are a significant cost: typically comprising 1/3rd of the capital expenditure, with the PV modules accounting for 1/5th [144]. Mini-grids have seen sharp declines in cost in recent years: between 2010 and 2018 mini-grid capital costs fell by more than 50%. The current tariff charged by mini-grids is around \$0.45 per kWh, higher than the average charged by the grid in SSA of around \$0.27. The LCOE for mini-grids may be reduced by around 60% by 2030, allowing mini-grid operators to charge tariffs competitive with the grid [148]. Since 2018, ESMAP estimates that for the best-in-class mini-grids, the LCOE has fallen by 30%, due in part to reductions in component costs, but also due to sector-specific innovations: better planning and advanced geospatial analysis of sites, faster construction and economies of scale [16].

Within the solar-home-system market, there has been huge growth in smaller systems providing power in DC (18–136 watts), leading to lower installed costs; however, for larger AC systems capable of providing more energy services, there is a higher cost per watt; due to the cost of inverters, other balance of system costs and the much smaller market size [119]. SHSs are often sold on credit with appliances from the same company: therefore the cost of the overall system includes the appliances, many of which are low-power DC devices. In recent years, R&D and economies of scale have led to lower-cost devices for use within such systems. This has led to whole-system cost reductions that are even greater, once also factoring in lower PV and storage costs [121].

Looking ahead to 2030 and beyond, there are predictions of significant growth in the use of off-grid systems from international organisations. The IEA expects that around a third of people without access will connect via off-grid technologies [1]. ESMAP (2022) anticipates that with the right support as many as half a billion people could get electricity access via solar mini-grids by 2030. This will require financing, innovation and good policy support and is by no means guaranteed [1, 16]. Future growth in mini-grids is expected to be driven by hybrid mini-grids, consisting of PV, storage and some back-up diesel generation, as they often present the lowest-cost configurations [15]. SHSs are expected to continue to play a critical role in expanding access to some of the poorest households up to 2030. Penetration into markets with low levels of SHS use, which may be in unstable environments such as conflict zones, will depend on de-risking financing initiatives to ensure all households can obtain systems [121].

2.5.7 Economic Challenges and Solutions for Off-grid Systems

Challenges for Off-grid Electricity Access

There are notable economic challenges to the delivery of off-grid electricity access. Despite the cost reductions discussed above, it is extremely difficult to make the revenues of off-grid projects such as mini-grids meet the expenditure needed to build and maintain them. Vulcan, a mini-grid developer, published their findings from their investment in 10 solar mini-grids in Kenya, with sizes ranging from 1.5-6.8 kWp of PV capacity. They found that in the present conditions, at most sites the mini-grids were not viable without subsidy, due to high expenditure and low demand and revenues. Key recommendations were to reduce capital spending, induce higher demand (by as much as 20-40 % in y2-5), and improve the availability of subsidy programmes (at the rate of \$500/customer) to support construction costs. This demonstrates the discrepancy between investment and returns facing developers, and the problems for enterprises operating in the space [22]. Reinforcing this, the International Finance Corporation (IFC) released a summary of findings based on interactions and data provided by 20 mini-grid companies worldwide. The findings further demonstrate the challenging environment for companies operating in the space: mean revenues per user were just \$7 per month, with the mean capital investment per user totalling \$920. Currently, the cost base for developers is too high when compared to income, and work must be done to reduce capital and operating expenditure, as well as finding ways to increase revenue by broadening service provision, and availability of power [20]

A huge uncertainty facing organisations building mini-grids is in estimating the demand for electricity. Given that customers on newly built systems are often gaining access to electricity for the first time, there is little or no data to use to estimate demand. This can lead to either under or over-sizing of systems and is a technical and financial challenge [149]. If demand is underestimated, the system may be undersized, resulting in unmet energy demand and dissatisfied users. If demand is overestimated, it will result in excessive capital costs and insufficient revenues [21, 23]. Electricity demand modelling can aid in sizing systems, helping to address this issue. This is discussed further below, in Section 2.7.

Solutions for Off-grid Electrification

If mini-grid operators are able to reduce tariffs charged, it can have a dramatic impact on electricity demand and incentivise consumers to obtain a connection. One study found that reducing tariffs saw an increase in demand over the long term and had a particularly strong effect on the poorest consumers [150]. One way mini-grid operators can reduce tariffs for consumers is by diversifying their consumer base, and having an 'anchor load.' The anchor-business-community (ABC) model, comprises a base fee-paying demand that secures revenue against the backdrop of uncertain take-up or payment of the community. Additionally, it incorporates business users as a complementary load to community users and effectively cross-subsidises domestic users that may have a low ability to pay [151]. This model, using a consistent anchor load, is proving successful in some parts of India, with telecom towers acting as a larger consistent consumer, allowing mini-grids to be viable for companies such as OMC Power [152].

Another important consideration to encourage demand is allowing consumers to pay a subscription service to a company that maintains the systems and appliances. This is often the case for solar home systems but may be applied to mini-grids too. The approach ensures the provision of services paid for e.g. lighting, TV or mobile phone charging, rather than electricity units. With

a smart meter, operators can cut off consumers when there is non-payment and, by bundling devices, it incentivises consumers to pay and to acquire greater levels of energy service [153, 154]. This opens up access to power to a broader consumer base, as consumers do not need to have the financial resources to buy the equipment (SHS) or pay a connection fee (to mini-grids) upfront. The rapid spread of mobile connectivity and mobile money has allowed companies such as BBOXX to monitor systems to assist with maintenance and to enforce payment of energy services through remote deactivation [154]. However, although this model has proven successful for some operators, the financing of appliances to consumers may not work in all contexts. A study that provided consumer financing for devices on mini-grids in Nigeria found that the financing of household devices was not effective at increasing demand. However, for productive uses, there was a sustained increase in demand, again highlighting the importance of incorporating and supporting non-domestic loads within mini-grids to improve their economic viability [155].

2.5.8 Summary

This section has described the different solutions available for electricity access, including the grid and different off-grid system types. Further, it describes key generating technologies and their costs and trends for general and off-grid usage, with a particular focus on PV, diesel and batteries. Rapid cost reductions have meant that PV and battery-based off-grid systems are increasingly used to meet electricity access challenges. Technologies such as lithium-ion batteries are improving, and reducing in cost, offering exciting potential for use in off-grid contexts. However, this section has also highlighted the challenges faced by off-grid systems: sizing systems is difficult due to uncertain demand, and there is often a significant gap between revenues and expenditure. Exploring innovative delivery models and diversifying demand types can help to mitigate these issues.

2.6 Electricity Access & Climate Change

This section will briefly cover the interrelationship between electricity access and climate change, both in terms of the climate impacts of expanded electricity access and how climate policy in the form of carbon pricing may play a role in expanding the use of low-carbon technologies.

Global Emissions Impacts

There is a range of studies attempting to quantify the impact on global emissions of reaching universal access. The IEA's estimates suggest that the impact would be negligible in global terms, with their Energy for All scenario resulting in just 70 megatonnes (Mt) of CO₂e added by 2030 [89]. Another estimate, an integrated assessment model (IAM) approach, estimated that reaching universal access would add 39Mt per year by 2030 [156], although demand levels per capita or household are not clear in this study. A different, earlier, IAM-based study estimated that emissions would be 2-4% over the baseline in 2030 to achieve total electrification; however, this analysis placed a large emphasis on grid-based fossil fuel expansion, assuming 420kWh per household per annum [157]. Finally, the demand level is important for impact. An econometrics-based study found that annual per capita demand levels of 100kWh and 2000kWh yielded cumulative additional emissions of 44 and 183Gt by 2100 respectively [158]. The numbers from these studies vary widely. The methods and assumptions used, demand levels and access modes will impact outcomes. There have been no

studies that properly quantify the potential emissions savings from taking an off-grid low-carbon intensity versus a grid and carbon-intensive route to universal electricity access. It is important to highlight that historically, areas with low or no electricity access have made almost no contribution to climate change [159].

The concept of "locking in" carbon-intensive technologies can be applied broadly across energy technologies and not just for electricity access. Adopting carbon-intensive technologies early on can create inertia that may make it harder to decarbonise [160]. Considering the lowest-cost decarbonisation pathways in the long term, it is important that investment decisions are made with technology lock-in in mind [161]. Although the research described above estimates that meeting universal access to electricity will have a relatively low impact on global emissions, this could change over the very long term. If countries become locked into carbon-intensive electricity access solutions, emissions may grow over time as electricity demand and economic growth occur over the very long term [162].

Carbon Pricing

There are many policy levers that can be used to address climate change, including subsidies for low-carbon technologies or regulations stipulating emissions limits on certain technologies. Carbon pricing, whereby, a charge is levied for each unit (per tonne) of carbon emitted can incentivise shifts away from polluting technologies. Carbon pricing offers the potential to have an impact on electricity access and the uptake of lower-carbon technologies: if a technology used for expanding electricity access is liable for a carbon tax, or a carbon credit available for it, it may incentivise or disincentivise to the point where a different technology is used. Voluntary carbon markets may have an impact on technology uptake and provide climate finance that can encourage infrastructure expansion [163]. At present these mechanisms are relatively untapped but have exciting potential for the future. Regions like Sub-Saharan Africa currently only produce a tiny fraction of the carbon credits that they could, for things like clean electricity access expansion [164]. One example of voluntary carbon markets creating credits for electricity access is the D-REC initiative. It acknowledges the lower GHG emissions from off-grid PV systems and provides financial incentives in the form of carbon credits for off-grid low-carbon projects, increasing their uptake [165]. Understanding how carbon pricing could influence the relative cost and shares of different access modes is important for forward planning. There has been some limited analysis using carbon pricing looking at electricity generation shares for Sub-Saharan Africa; however, no work that examines how carbon pricing might impact the split between different modes e.g. grid versus off-grid [156].

2.6.1 Summary

This section has briefly outlined some of the research regarding electricity access and climate change. Estimates of emissions impacts from universal electricity access vary, with the impact expected to be relatively small. The methods used and the demand level impact the outcomes. It is important to note that areas with poor electricity access have historically made negligible contributions to climate change. Carbon pricing offers exciting potential to encourage the uptake of low-carbon electricity access technologies. However, more work is needed to assess the impact at a larger scale.

2.7 Modelling tools for Electricity Access Research

This section will discuss how modelling tools can be used to address research questions relating to electricity access. A broad range of tools and approaches are available that have been applied to a wide range of contexts: both at the community scale [166], and at the region [24, 60, 127] and country level [167, 168]. This section will focus on two types of modelling tools that are used and developed in this thesis. The first is simulation and optimisation tools that are used for off-grid energy system design, e.g. for a single household or a community; and the second is geospatial tools that can help inform electricity planning for regions or countries.

2.7.1 Modelling tools for Off-grid Energy System Design

This section will outline the models that are designed for off-grid energy system design (see Table 2.1). Whilst this review is not explicitly exhaustive, it is the result of a thorough search in the academic literature for models that are in scope for addressing research related to the design of off-grid energy systems. Some models are excluded as they are no longer maintained and compatible with current operating systems (HOMER legacy [169], and Hybrid2 [170]). Table 2.1 provides a summary of key characteristics of the models discussed in this section.

The most commonly used software tool to size off-grid electricity systems is the Hybrid Optimisation Model for Electric Renewables (HOMER) modelling tool, developed by NREL, and launched for free public use in 2004 [171, 172]. The software was designed to undertake three main functions: simulation, optimisation and sensitivity analysis. The simulation function models the ability of a given system configuration to meet the demand for each hourly time-step, whilst optimisation and sensitivity analysis model a range of configurations to give the lowest life-cycle cost, and test the effects of uncertainty and changes to the inputs, respectively [173]. Following initial development, at NREL, HOMER was commercialised in 2009 and is now run by a private company. A paid licence is required for usage, with additional paid bolt-on modules such as the availability of advanced storage parameters or the ability to write dispatched generation protocols in MATLAB. There are three versions of the software available today: 1) HOMER Pro, designed for modelling generator islands or mini-grids; 2) HOMER Grid, intended for modelling behind-the-meter systems such as PV; and more recently 3) HOMER front, for modelling utility-scale storage [174].

There have been hundreds of papers published using the HOMER software since its inception. A review of 91 papers using HOMER by Bahramara, Moghaddam and Haghifam (2016) found that it was used to investigate a wide range of technology configurations, with PV the most commonly used, and system sizes ranging from 0.6kW to as large as 2.2GW. The review cited wind speed, solar resources, fuel and component prices and electricity demand as factors contributing most to uncertainty in the modelling process [166]. Studies in which HOMER is used for mini-grids that deploy solar PV in an energy access context for low-income countries have a range of notable characteristics. Many highly cited older studies using HOMER are set up based on hypothetical systems in locations that are not well specified [175–178] and do not utilise monitored demand data or load profile construction based on surveys [175–177, 179]. Additionally, whilst financial modelling is given over a multi-year time frame, it was unclear in many studies whether or how technical system optimisation had been undertaken [175–185]. This is particularly important to consider as the standard version of HOMER only optimises the technology over a 1-year time frame. HOMER has been used as part of more detailed studies that undertake other additional

components of analysis such as detailed load profile generation via surveys [186, 187] or external load models [180]; or that have conducted detailed business model analysis [186].

iHoga is another software tool, with arguably some more advanced functionality than HOMER. This started as free research software but now requires a paid license to use it to conduct research. The model is written in the programming language C++ and has features such as temperature-dependent battery degradation and the ability to model dispatched diesel generation within hybrid mini-grids [188, 189]. There have been numerous published studies using iHoga for a variety of applications of off-grid systems in different contexts [190–192]. However, while it is possible to understand the structure in published work the code is not open-source.

Although there is widespread use of commercial software such as HOMER for electricity access research, there are problems with relying on software for scientific research that does not have open-source code, such as producing research that has a lack of transparency and reproducibility [193]. There are increasing calls within the energy modelling community to use open-source models to ensure improved transparency and scientific reproducibility and foster collaboration between parties [194]. Openness can be brought into each stage of the modelling process, from data choice and processing, model formulation, processing of results and interpretation to allow better quality research and increased productivity due to collaborative efforts in developing and maintaining code [193].

To address these problems, researchers at the Reiner Lemoine Institute designed the micrOgridS open-source modelling tool to be able to simulate off-grid systems, have an adjustable time resolution (15 minute -1 hourly time-steps), and have multi-objective optimisation functionality [195]. The micrOgridS tool was developed using the open energy modelling framework (oemof). Oemof is an open-source library to model energy supply systems in the programming language Python [196]. Comparisons with HOMER show the micrOgridS model solving with a lower overall cost whilst still meeting energy demand [195]. The micrOgridS model does in its present form facilitate the modelling of dispatched generation with user-defined strategies. Offgridders is another open-source software tool, also built using the Oemof framework and also similar to micrOgridS; however, Offgridders has the functionality to simulate mini-grids interconnected with reliable or unreliable grids [197]. Studies on Nepal [198] and the Phillipines [199] have used the model to see how off-grid systems can be used to improve access. The model has a one-year time horizon and therefore is unsuitable for simulating systems over their entire lifetime. Whilst back-up diesel can be modelled within a hybrid system comprising variable renewables, it does not have the functionality to model dispatched generation. Another freely available, open-source tool that can be applied to off-grid electrification research is NREL's System Adviser Model (SAM). It includes functionality to consider a wide range of solar energy technologies, battery storage, wind generation and biomass. The model is predominantly designed for grid-connected renewables, and can only be used to simulate systems rather than optimise for a given context [200]. The model can accept weather input data in varying resolutions, from 1 minute to 1 hour [201].

The CLOVER (Continuous Lifetime Optimisation of Variable Renewables) model is a further open-source modelling tool designed to simulate and optimise renewables-based off-grid systems [202]. It uses an hourly resolution and can include a grid connection of varying reliability, back-up diesel generation, storage and renewables generation such as PV. It is designed to simulate systems over a multi-year time horizon and has the capability to add new capacity in response to increasing demand or degrading components [203–205]. The CLOVER model is used and developed in my thesis and I will give a detailed description in the methods chapter (Chapter 3).

Table 2.1: Characteristics of energy system models used for off-grid electricity research.

Model	Open-source	OS	Language	Free	Demand Module	Temporal Resolution	Multi-year	Supply Technologies	Functionality	Main Limitation
Homer Pro		Windows		X		Adjustable (1-60 min)	Paid add-on	Solar PV, Wind, Diesel, Biomass, Hydro, CHP, Hydrogen, Batteries, Flywheel, Grid	Simulation, Optimisation, Sensitivity Analysis	Not open-source
iHoga		Windows	C++	X		Adjustable (1-60 min)	X	Solar PV, Wind, Diesel, Hydro, Fuel cell, Electrolyzer, Batteries	Simulation, Optimisation	Not open-source
MicroGridS	X	Any	Python (Oemof)	X		Adjustable (15-60 min)		Solar, PV, Diesel, Batteries	Simulation, Optimisation	Not multi-year
Offgridders	X	Any	Python (Oemof)	X	X	1 hr		Solar, PV, Diesel, Batteries, Grid	Simulation, Optimisation, Sensitivity Analysis	Not multi-year
SAM	X	Windows	C++	X		Adjustable (1-60 min)	X	Solar PV, Wind, Diesel, Biomass, Marine, Fuel cells, Geothermal	Simulation	No optimisation
CLOVER	X	Any	Python	X	X	1 hr	X	Solar PV, Diesel, Batteries, Grid	Simulation, Optimisation	1 Hr resolution

Electricity Demand Simulation

As discussed above, an area of uncertainty is understanding the electricity demand of consumers, especially if they are gaining access to electricity for the first time. In response, tools have been developed to provide a better understanding and simulate load which in turn can aid in off-grid system design.

Mandelli, Brivio, et al., describe a stochastic model in MATLAB called LoadProGen that builds up the co-incidence behaviour of appliances within a system to calculate a load profile and the theoretical maximum power peak an off-grid electricity system would need to cope with. The authors combine this with a techno-economic optimisation model that simulates systems on an hourly time-step basis to determine if a given system can meet the load. The model simulates a number of system configurations to find the lowest cost [21].

Boait et al., present a decision support tool called ESCoBox, the core component of which predicts peak and average electricity demand from a population of electrical devices. The tool takes survey data, in the form of a probability that a given device within a system will be either on or off at each hour of the day to generate aggregate consumer electricity demand using a Monte Carlo method [206, 207]. The CLOVER model electricity load module uses a similar method to ESCoBox to generate a demand profile, taking input data based on surveys in the form of the probability a device will be in use at a given time. The CLOVER model adds further layers of complexity, as it takes into account seasonal variation and growth in the number of households with a connection within the project time horizon. The model additionally incorporates device ownership with diffusion rates within the community, simulating economic development. This load generation process creates an electricity demand profile that is then fed into the other parts of the model that simulate or optimise off-grid energy systems [202, 205].

The CREST (Centre for Renewable Energy Systems Technology) demand model is a more sophisticated and detailed tool for generating demand profiles at the household or community level [208]. It has typically been used for modelling energy demand in high-income countries; however, there is an adapted version more suited to low-income resource-constrained contexts relevant to electricity access research [204, 209]. It produces outputs at a high resolution that are climate and temperature-dependent based on the input location used. It is free and open-source [210].

2.7.2 Geospatial Modelling for Electricity Access

Geospatial models are mathematical and digital structures that represent things spatially, typically on Earth. These models use geospatial data, e.g. remote sensing data from satellites or geospatial data sets compiled in other ways, to represent how various aspects of phenomena may change over space in relation to time or other variables. Geospatial models allow multiple layers of data to be used to give location-based insights into research questions. Energy resources and demand vary spatially and over time, and geospatial tools are a powerful way to ensure these dynamics are captured [211]. In the context of electricity access, geospatial data and models can allow project developers, planners and researchers to ask questions that would be difficult to address at scale using other approaches. Geospatial tools can speed up analysis that would take longer through other methods e.g. manual surveys or data collection regarding possible mini-grid connections, and enable analysis to be conducted over larger areas [16].

Least-cost electricity access modelling

Szabo et al. (2011) provide one of the first examples of constructing a geospatial data model to compare the costs of different electricity access options for a large area. They map the costs of using solar PV systems off-grid, diesel off-grid systems or grid extension for all of Africa and estimate the shares of the population that would be met by each. Their inputs include solar resources; grid infrastructure maps and distance-adjusted diesel fuel costs. This analysis was done for a 2010 snapshot and did not have a representation of electricity demand or different off-grid systems (e.g. stand alone vs. mini-grids). However, it provided an important methodological precursor to other more sophisticated work. It also demonstrated the importance of fuel costs in determining different zones for different electricity access solutions, based on what is the least-cost option [127]

Following this work, a more sophisticated geospatial tool for electricity access was created that followed similar principles. The open source, spatial electrification tool (OnSSET) has been applied to Sub-Saharan Africa as a whole, and to case studies focusing on specific countries [167, 168, 212]. The model uses a 1km² resolution and establishes the cost of each technology at set electricity demand levels (e.g. MTF tier 1). The model is open-source, modular and intended to be adapted depending on usage requirements [24]. Due to the model's detailed spatial resolution (approx. 26 million cells for all of SSA), running it requires high levels of computational work. The model is best suited to snapshot analysis, although more recently has been updated to incorporate time intervals. Further, it has been updated with a new grid-extension algorithm to better estimate the route and cost of grid extension [167].

The Reference Electrification Model (REM) is a geospatial model that supports both larger-scale electrification planning and small-scale electrification projects. The model uses clustering and optimisation algorithms designed for electrification planning and produces the lowest-cost design. When data on the position of buildings is not available, the model uses satellite imagery and processing algorithms to approximate the position and size of buildings. The model can be used in two modes: Local REM, which can give detailed plans for mini-grid design for planners or companies; and, Large-scale REM, which is designed to give insights into the best approach for electrification for a given (larger) area, although it is still done on a building-by-building basis [213]. Given the model operates at the building level, it is best suited to looking at individual settlements or administrative areas rather than larger assessments looking at countries or regions.

Solar Resource Modelling

Another set of geospatial tools that produce a key input commonly needed in electricity access modelling is a representation of renewable generating resources. Typically for off-grid systems, this would be the solar resource. In the absence of monitored data from a specific site, there are tools that can provide the estimated output of a solar installation for certain locations on the planet using remote sensing techniques [214].

A commonly used tool for this purpose is PVGIS. PVGIS is a tool created by the European Union with coverage of Europe, and large parts of Africa and Asia [215]. It has recently expanded coverage to include North America. It uses historical satellite data and modelling techniques to estimate the output of a solar array of a user-defined configuration at certain points. It also has the capability to suggest the orientation and tilt angles of solar installations for a given location in its coverage [216].

Another tool is [Renewables.ninja](https://renewables.ninja/). This has global coverage, can provide estimates for both solar

and wind output and has an application programming interface (API) that can connect directly to energy system models such as CLOVER. The Renewables.ninja tool is based on outputs of validation and bias-correction of reanalysis data for Europe but provides coverage globally [217, 218]. The renewables.ninja tool is used in this thesis and is discussed further in the next chapter.

2.7.3 Summary

This section has described different modelling tools that can be used to address electricity access research questions, both at the single system level and at the region or country level. Research on modelling off-grid systems is dominated by HOMER, which is not open source. A range of open-source tools exist, including the CLOVER model, which is applied and developed in this thesis. Geospatial tools facilitate research addressing electricity access planning. Existing geospatial tools such as OnSSET are powerful and have been applied to a range of research questions with a spatial nature.

2.8 Discussion of Literature and Research Gaps

The sections above aim to give an overview of the different parts of the literature that this thesis relies on, or contributes to. A range of literature gaps have been identified, both in terms of research topics and the methods used. I will first discuss the key concepts covered above and why they are important for the research. Afterwards, I will go on to highlight the gaps in the literature that this thesis works within.

Firstly, it is important to highlight from existing research and data that electricity access is far from a solved problem. This underpins, in part, the rationale for further research in this area and the motivation behind this thesis. Although there has been marked progress toward universal electricity access over the last decade, significant numbers of people remain without electricity and will remain without it past 2030 without strong action to accelerate growth in added connections. This moreover strengthens the justification for this research. The challenge is particularly acute in Sub-Saharan Africa, where the majority of people who will need a connection in the next decade reside. Failure to address the gap between the current progress and the progress required may see people still living without access even as late as 2050. This thesis will therefore include research that focuses on regions likely to have the greatest need.

Next, serving as further justification for the work presented; there is a significant body of research that emphasises the need to solve SDG7 to be able to address other development challenges (SDGs). Electricity access is vitally important for the provision of proper healthcare (SDG3), ensuring good educational opportunities (SDG4) and facilitating growth in business activity and income-generating activities (SDGs 1 & 8). Boosting access to electricity will therefore unlock gains on a number of fronts relating to human living standards. The literature demonstrates that electricity access is an important challenge humanity faces, and further research on this topic is essential. Whilst beyond the scope of this thesis, there is a clear need in the literature for further work that looks at the impacts of electricity on livelihoods; particularly in Sub-Saharan Africa. The evidence in the literature outlining the links between electricity access and improvements in human living standards provides an additional basis for research (such as in this thesis) that seeks to unlock further growth in connections.

Given the nature of the research in this thesis, it is important to understand how electricity

access is defined in the literature. This is to ensure the work sits within the context of existing definitions and can be understood more broadly. What is clearly shown is the need to consider demand level, as well as other metrics such as reliability and affordability in definitions of access. The simple measure of whether an individual, business or household has a connection, masks the amount of electricity available or consumed. It also doesn't capture the reliability of supply. As the literature describes, more nuanced ways of defining access, such as the ESMAP MTF multi-tier framework, address these issues and provide a definition that gives access at different levels, including multiple criteria. When conducting research on access to electricity, using definitions that provide information about the level and quality of access is important as it can impact the outcomes for final users. For example, a household may be defined as having a connection to the grid, but a poor level of service means that in practice most of the time there is no connection. Referring to more complex definitions when thinking about access can ensure that desired outcomes are reached. For researchers, it is important to ensure that work on electricity access takes these factors into account and where appropriate considers factors such as demand levels and reliability in the analysis of off-grid systems and pathways to access. Given these points, the work presented in this thesis will attempt to use a definition of electricity demand that fully appreciates the demand level (in reference to the MTF where appropriate) and also considers the reliability of supply. In doing so, it is hoped the research will better consider human needs and outcomes.

It is important to consider the range of options available to address electricity challenges, and the approaches and definitions used in the literature. It is clear that although historically, electrification was via national grid extension, or for remote communities, diesel generators were used, off-grid solutions powered by modular low-carbon technologies are an increasingly used option. This is vitally important for the research in this thesis. The costs of technologies such as solar PV and lithium-ion batteries have reduced significantly over the last decade, making them even more accessible and the least-cost option for increasing numbers of people. There has also been innovation that is specific to off-grid systems, reducing the installed cost of both mini-grids and stand-alone systems and thus making these technology choices even more competitive. Cost is a key determinant of technology choice, and this is well discussed in the literature; however, there are other determinants of technology choice, such as reliability. Reliability is less explored in the literature, although there is evidence of an economic burden to households and businesses from poor reliability. Users may choose to use other means of access e.g. mini-grids or backup diesel when national grids are unreliable. Carbon pricing too, may impact technology choices in future, either through taxes paid by higher-carbon generators or through the supply of carbon credits for lower-carbon ones. The future impacts of carbon pricing on electricity access pathways have not been well explored. This provides a technology foundation for examining electricity access: solar and battery systems, but compared with incumbent technologies (grid and diesel). Additionally, there are clear avenues for further research (as discussed further below).

The literature demonstrates that despite the significant reductions in the cost of off-grid systems that have enabled growth, as described above, mini-grid operators face economic challenges in many contexts. This hampers their growth in deployment. Uncertainty of electricity demand and low levels of electricity demand from domestic users create an uncertain investment environment without necessary support schemes. Addressing these issues by considering additional components of demand from customers who can generate more revenue, and by thinking about ways to minimise the cost of off-grid systems, by exploring different system configurations and reliability levels is crucial. The literature details how incorporating other electricity demand users with a stable, large supply has the potential to make mini-grids more viable for domestic customers. However, these

issues are not well explored in the academic literature. A key weakness is the lack of accurate usage data and cost data representative of real systems that can make the work more applicable. Additionally, Work is needed to explore how off-grid systems should be designed under different demand and reliability conditions, and how they can be managed to minimise costs. To ensure off-grid projects can succeed and to stimulate investment and de-risk projects, this research gap (more below) must be addressed. The research in this thesis will therefore try to address these issues using data from existing mini-grid operators where possible.

A final and crucial area of literature covered is the methods for addressing electricity access research. It is clear from existing work using and developing models that can be applied to relevant research questions that modelling tools are a powerful method for this topic. For modelling single systems, there is a range of options available, but the research landscape is dominated by HOMER, a paid-for software tool with a range of add-ons available. Whilst HOMER has broad functionality, it is not open-source, raising issues of transparency and reproducibility. Further, whilst the user can modify settings, the closed-source nature of the model means it cannot be adapted to the needs of specific research questions or case studies. Given trends towards open models and data in energy research, having alternatives that are open-source and freely available is better for the transparency and potentially the accuracy of research. There are some good, open-source alternatives available, but need further development of features to ensure they can be applied to the broadest range of settings.

In addition to designing specific systems and communities, models can help to facilitate the planning of energy resources or for exploring research questions across countries or regions. These questions may relate to the appropriateness of different technology options (cost, reliability etc.) as outlined above. These models, typically using geospatial data, are increasingly used for electricity access research and can be harnessed using a growing amount of relevant, publicly available geospatial data, such as the estimated number of people without access to electricity in a given place. Although there are open-source tools available, e.g., OnSSET, which has been used for a range of research questions and has many powerful features, including a complex representation of grid extension. However, it is computationally intensive, which reduces the ease of running many scenarios. Further work to develop tools that have geospatial representations of electricity access challenges is needed to ensure researchers can answer different types of research questions relating to the challenges outlined above. Additionally, having available tools with reduced computational demands can better enable scenario-based research for a broader range of researchers. The research in this thesis will seek to contribute methodologically to this area.

Research Gaps this Thesis will Address

In this thesis, I present work that gives detail on the cost and performance of mini-grids, which is detailed in Chapter 4. I present a detailed study looking at mini-grids for domestic and healthcare demands in Northern India. This research directly addresses the challenge of mini-grid economics, providing a range of system types, reliability levels and electricity demand scenarios with different users. In doing so, I also aim to address the research gap around technology choice and proper comparisons of grid versus off-grid systems at different reliability levels. Chapter 4 uses cost and monitored usage data from the site, addressing the issue of a lack of real-world focused studies in the academic literature that address issues of off-grid system economics. This work uses an open-source energy system model, CLOVER, rather than commercially available software such as HOMER.

Further, in Chapter 5, I present further work relevant to the economics of mini-grids, with a detailed study of a hybrid mini-grid system built and operated in a refugee camp in Rwanda. This work addresses how adding solar and battery capacity to existing diesel capacity may be able to reduce lifetime costs. Further, it explores how different operating strategies for diesel generators alongside the batteries can potentially further reduce costs. This work aims to inform operators of similar contexts on how to improve their cost profiles, and therefore aid the financial sustainability of systems. The work in Chapter 5 also uses monitored demand data and cost data from the installed system, addressing the lack of real-world focused studies on mini-grid economics. An additional component of this work, which also addresses financial sustainability, is that it investigates electricity demand scenarios of adding extra user groups and the technical and financial implications of doing so. This work uses a version of the open-source CLOVER model that I adapted for this case study. The changes made add functionality not available in other open-source modelling tools designed for electricity access research.

In Chapter 6, I present work that considers technology choice for universal household electrification in Sub-Saharan Africa. This work uses a scenario-based approach, using the ESMAP MTF tiers of access for different electricity demand levels, and includes scenarios with both carbon pricing and that address the reliability of different electricity access technologies. This work builds on existing geospatial analysis to investigate universal household access, but addresses two specific areas that have not been addressed properly in the literature (reliability and carbon pricing). This work provides insights into the economic viability of off-grid electricity access technologies in different spatial contexts: distance from the grid, country-specific factors and population density. To produce the research in Chapter 6, I develop an open-source geospatial modelling tool, that is used in combination with energy system modelling and energy demand modelling to create required inputs, addressing the need for further available tools with different characteristics in this area of research.

As described above, the work in this thesis both uses and develops open-source modelling tools for electricity access research, specifically addressing the need for additional attention in this area. In Chapter 3, I give explanations as to the way I structured these new modelling tools, and how core calculations are handled. The purpose of this is to contribute to the literature on modelling tools for electricity access and serve as a reference for the reader as to how the results in the subsequent three results chapters were produced.

2.9 Summary

This chapter has outlined the relevant literature, giving context for this thesis on a range of topics: the current state of electricity access, defining access, the impacts on other SDGs, technology choices and the challenges they face, climate change and electricity access and finally modelling tools for electricity access research. This chapter has also detailed where the gaps in the literature exist, and how the different chapters presented in this thesis will address them.

Chapter 3

Methods

3.1 Overview

In this thesis, I develop modelling tools to be used to address electricity access research questions. This chapter describes the tools that I use and develop to produce the results found in the subsequent chapters. Figure 3.1 outlines the methods presented here and the corresponding results chapter. I will focus on presenting the structure of the core parts of the models in this chapter, with the individual methodology sections within the results chapters presenting a complete picture of how the results were generated. Within the methods sections of the results chapters, I will refer to individual sections of the following methods chapter for model-specific details where relevant.

The methods and models presented in this chapter focus on the modelling of solar PV-based off-grid systems (with or comparing with diesel generation). Whilst the methods could be applied to other small-scale generation types e.g. micro-wind or micro-hydro, they are not covered here as existing research has shown that they are not economical, or as feasible due to resource availability when compared to solar PV [24, 219].

The first part of this thesis used the CLOVER energy systems model in its original form [203, 205]. This involved gathering input data and configuring the inputs and scenarios to answer the research questions but did not require major changes to the model. Some minor changes were made, and they are included in this chapter. I present a general overview and description of the core parts of the simulation and optimisation modules of the CLOVER model in Section 3.2. The purpose of this is to both clarify how the results in Chapters 4 and 6 were produced and to provide background to the following methods section 3.3, relevant to the results in Chapter 5.

The second component of this thesis involved substantial modifications to the energy system module of the CLOVER model. These modifications were completed to provide the functionality to model dispatched diesel generation configured for battery cycle charging. The changes made are presented in Section 3.3. The adapted model is applied to a specific case study of a hybrid mini-grid in a humanitarian context, the results of which are presented in Chapter 5; however these changes have broader uses and advance the functionality of the CLOVER model.

The final component of this research develops a new geospatial data model, used in conjunction with energy system modelling and energy demand modelling. This model was developed to be able to explore pathways to universal electricity access in Sub-Saharan Africa under different electricity demand, supply reliability and carbon pricing scenarios. In Section 3.4 I present the

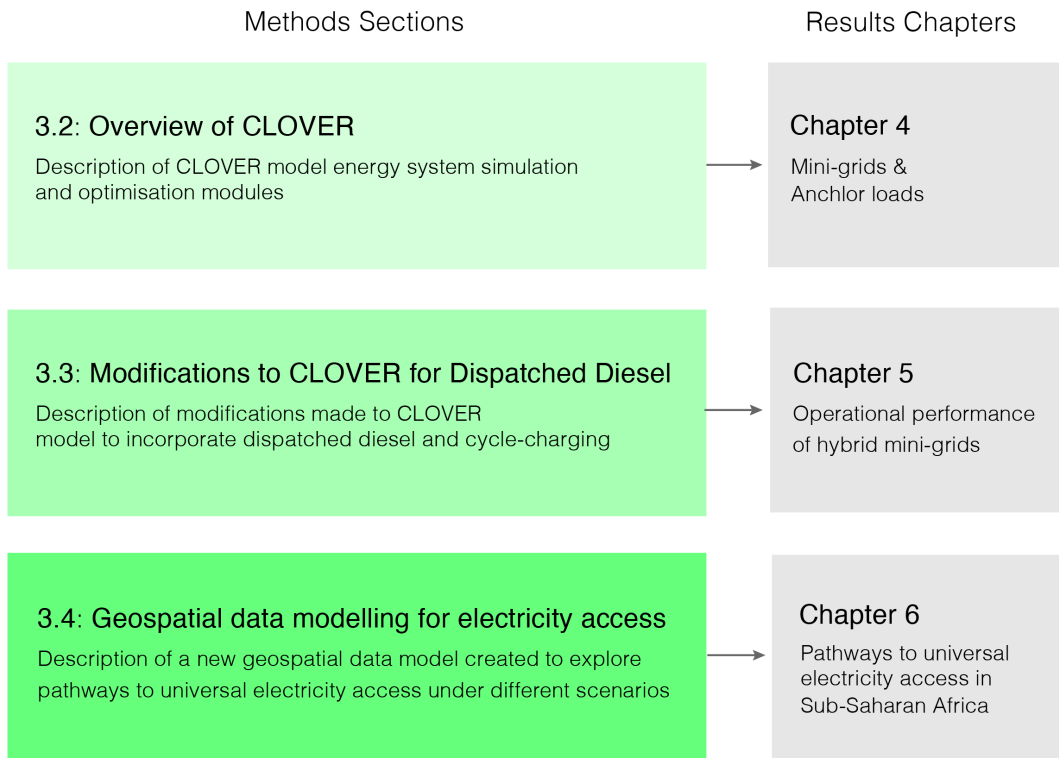


Figure 3.1: Describing the sections of the methods chapter and the corresponding results chapters the methodology parts relate to.

structure of the least-cost component of the geospatial data model, giving clarity on how outputs for different scenarios are created. The framework developed is applied to specific questions presented in Chapter 6, although it may also be applied to other regions or questions provided there is data availability.

3.2 The CLOVER Energy System Model

The CLOVER model is a software tool, which is designed to both optimise and simulate community or household scale off-grid electricity systems. It has an hourly resolution and can be used to simulate demand and supply to give insights into how technologies or system configurations may perform in a given context. CLOVER can suggest the optimally-sized system given the selected optimisation objective (normally cost), that meets a given sufficiency threshold (normally reliability). CLOVER provides technical performance outputs, both in an hourly format and as summaries, including metrics such as total emissions, hourly unmet demand and dumped electricity [202]. The CLOVER model was originally conceptualised by Sandwell et al. (2017) [203] and a detailed mathematical description of all parts of the original model can be found in Sandwell (2017) [205]. There have been several published works that use the model in different contexts [18, 204, 220–222].

I both use and adapt the CLOVER model for the work in this thesis. In this first section of the

methods, I present an overview of the CLOVER model as it was used for the analysis presented in Chapter 4. In this instance, the model was used with only some minor modifications made; these are described in section 3.2.3. In Section 3.3 I present work that details more substantial modifications to the model, that were undertaken to complete the results presented in Chapter 5.

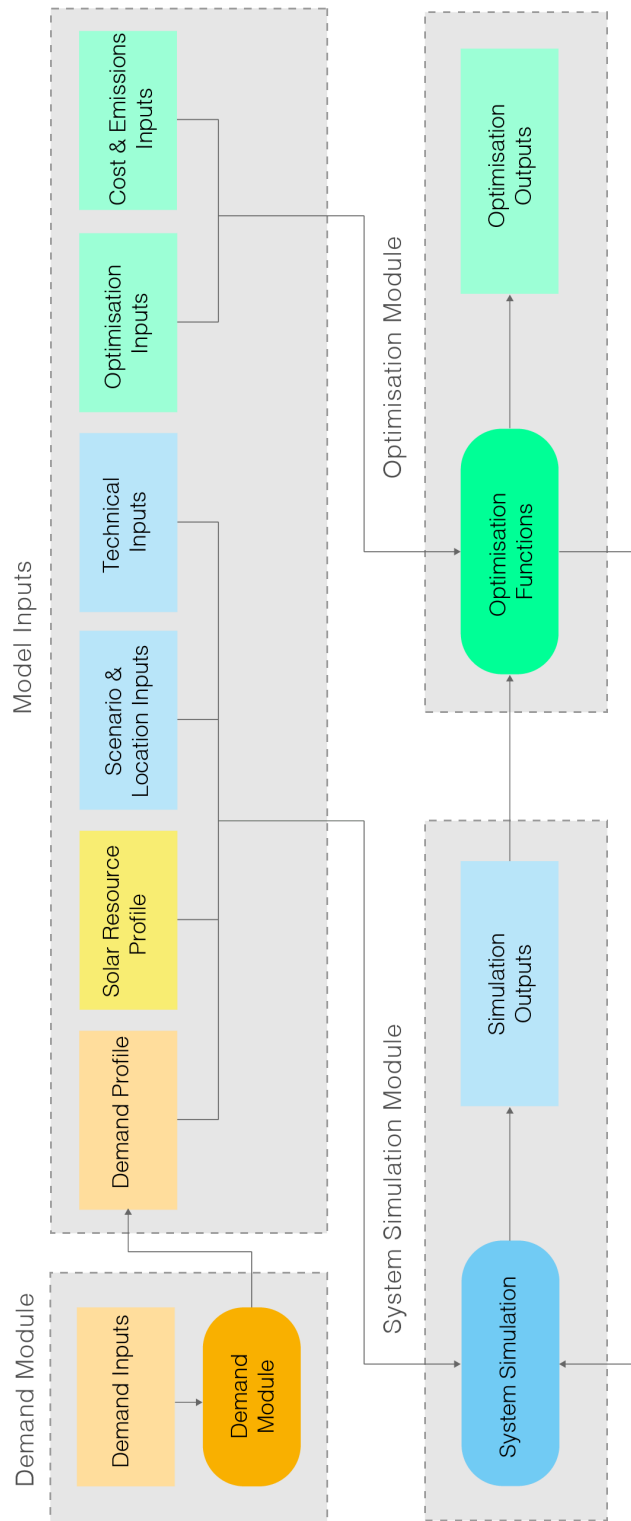


Figure 3.2: Figure showing a high-level outline of the structure of the CLOVER model. The demand module is optional, and if the user already has their own demand profile in an hourly format from other sources it can be used instead. The model can be run either to simulate a system of a specific configuration or to optimise, whereby the model runs the simulation function for multiple system configurations, assesses the outputs and selects the optimum based on user-specified criteria.

3.2.1 Model Structure

The core structure of the CLOVER model is made up of three key modules: demand, simulation and optimisation (see Figure 3.2). The model can operate either through user-generated demand profiles via the demand module, or by hourly load profiles from monitored data or other generated load profiles. A load profile may need to be generated from other data, e.g., from surveys, if monitored data is unavailable. The load profile forms an input, along with solar resources, scenario and location inputs, and technical inputs required to run the energy system simulation module. The energy system simulation module can be run to assess the performance of a given system and provides hourly simulation outputs. Additional inputs are required for the optimisation module, including cost and emissions information about the system components. The optimisation module repeatedly runs the system simulation module, modifying the system size and using its appraisal functions to assess the cost, emissions and reliability. From the system configurations run by the model, the optimum is selected, and this is provided as an output along with the appraisal outputs.

3.2.2 System Simulation

CLOVER simulates energy systems that comprise any combination of solar PV, energy storage and diesel generation over a multi-year time horizon. It can also incorporate power supplied from a national grid network with a defined availability profile, although this is not discussed here as it is not used in this thesis. The load module, which uses the hourly device in use probabilities, employs stochastic modelling to produce hourly electricity demand profiles at the household or community level. This is used in Chapter 4 to generate health centre load profiles and is explained in Section 4.3.1. When simulating, the CLOVER model first tries to meet electricity demand in a given hour with available energy from solar PV generation, then battery storage. In any hour of the simulation, if there is an oversupply then that energy is dumped, if there is a shortfall it will result in unmet demand unless diesel generation is also utilised. This is described in more detail below.

Solar PV Profile

CLOVER uses the coordinates of the system location, the user-specified PV capacity, and its orientation (tilt and azimuth) to extract hourly solar generation data from the Renewables.ninja API. Renewables.ninja uses reanalysis data from the MERRA-2 dataset [217] and provides the estimated hourly solar generation profile for the system location, $E^{PV}(t)$ (in kWh).

Energy Balance Profile

Once the solar PV profile has been extracted, the model then creates an energy balance profile for every hour, t , for the duration of the simulation. In any given time step it uses the energy demand in kWh $E^L(t)$ and the available energy from the PV array $E^{PV}(t)$ to calculate the energy balance in kWh $E^{Bal}(t)$:

$$E^{Bal}(t) = E^{PV}(t) - E^L(t) \quad (3.1)$$

Hourly Simulation

Next, the model uses the hourly energy balance profile $E^{Bal}(t)$ to simulate how it will operate, by the hour, over its lifetime. The energy that flows in and out of the battery capacity (in kWh), B , in any given hour t , referred to as $E_{Bat}^{Flw}(t)$ and in kWh, is limited by the charging and discharging C-rates, c_{in} and c_{out} . C rates are given as a fraction of the battery capacity charged or discharged in an hour. For any given hour, the battery energy flow $E_{Bat}^{Flw}(t)$ in kWh is therefore given by

$$E_{Bat}^{Flw}(t) = \begin{cases} B c_{in} & \text{if } E^{Bal}(t) > B c_{in} \\ -B c_{out} & \text{if } E^{Bal}(t) < -B c_{out} \\ E^{Bal}(t) & \text{otherwise} \end{cases} \quad (3.2)$$

The battery energy flow E_{Bat}^{Flw} is added to the amount of energy in the batteries (in kWh) for the previous hour $E^B(t-1)$ to give $E^B(t)$, ensuring that the battery energy level stays within the minimum and maximum storage levels (% depth of battery discharge) S_{min} and S_{max} and is given by

$$E^B(t) = \begin{cases} S_{min}B & \text{if } E^B(t-1) + E_{Bat}^{Flw}(t) < S_{min}B \\ S_{max}B & \text{if } E^B(t-1) + E_{Bat}^{Flw}(t) > S_{max}B \\ E^B(t-1) + E_{Bat}^{Flw}(t) & \text{Otherwise} \end{cases} \quad (3.3)$$

Any energy that cannot be stored due to the energy in the battery $E^B(t)$ or state of charge, (% of battery capacity) $S(t)$, exceeding the maximum S_{max} is dumped and the unmet energy for the hour $E^U(t)$ is zero.

If $E^B(t) < S_{min}B$ then unmet energy (in kWh) for the time step $E^U(t)$ is greater than zero:

$$E^U(t) = S_{min}B - E^B(t) \quad (3.4)$$

Blackouts

In hours with unmet energy, the model records a blackout in a binary blackout profile, $K(t) = 1$. For hours where unmet energy is zero, $K(t) = 0$. Following this simulation function, the blackout rate as a % of all hours in the simulation, K_r , is calculated:

$$K_r = \frac{\sum_{t=0}^{t_{max}} K(t)}{t_{max}} \quad (3.5)$$

Back-up Diesel Generation

Here, backup diesel generation functionality is described. The use of dispatched diesel for cycle-charging is described in the next section 3.2.4 of this chapter and is a key methodological development presented in this thesis.

When running the model with back-up diesel configured, it automatically sizes the diesel generator (at a 1kW resolution) according to the maximum demand that needs to be met. The model will use the system blackout rate K_r and a user-defined threshold of the fraction of permitted blackout hours K_{Thr} to meet unmet demand E^U with diesel energy E^D in kWh. When the blackout rate, K_r , is greater than 0 and the permitted $K_{Thr} = 0$ (i.e. the system must meet all demand), in all hours where $K(t) = 1$, diesel backup is used to meet unmet demand, $E^D(t) = E^U(t)$.

When $K_r > 0$ and $K_{Thr} > 0$, (i.e. some blackouts are permitted) the model fills in blackout hours that have the greatest unmet energy demand, $E^U(t)$, first until the blackout rate K_r for the simulation has reached the desired blackout threshold K_{Thr} . The logic of this is that a system operator would run the diesel generator in times when demand is highest and the generator would run at a higher capacity factor. To do this, it assigns a binary value to the required number of hours to reach the blackout threshold K_{Thr} using a profile $\mu(t)$: where demand is to be met $\mu(t) = 1$, and for the remaining hours, $\mu(t) = 0$ meaning demand is not to be met. Diesel energy thus fills in the unmet energy for the required hours:

$$E^D(t) = (E^U(t))\mu(t) \quad (3.6)$$

When simulating diesel-only systems you can include no PV or battery storage capacity in the model and therefore the diesel backup acts as the only source of generation meeting the resulting unmet energy from the initial parts of the system simulation.

The model allows the user to define the technical characteristics of an energy system in detail, including conversion and transmission efficiencies, the degradation of PV capacity, and the degradation of battery storage capacity as a function of time and energy throughput, over the lifetime of the system. Examples of the technical inputs used (for Chapter 4) can be found in Appendix A, Table A.2.

3.2.3 Model Modifications

For the results presented in Chapter 4, there were two minor modifications made to the above model. These were a) automatic battery replacement and b) hybrid AC and DC distribution.

Battery Replacement

Given the simulated time horizon of 20 years, it is a likely requirement during the project life for the storage capacity to be replaced if it degrades beyond a certain point. To do so I added a condition in the model, the storage replacement capacity B_{rc} given in kWh of the remaining battery capacity following degradation. For each hour, the value of the usable storage capacity $B_u(t)$ (in kWh) is calculated from the installed storage capacity B and the factor by which the storage has degraded $B_{deg}(t)$:

$$B_u(t) = B(B_{deg}(t)) \quad (3.7)$$

If $B_u(t) \leq B_{rc}$ then $B_u(t) = B$ and this time step in the simulation is recorded so that the discounted costs and associated emissions are taken into account when calculated for a given system in the optimisation process described below.

Hybrid AC and DC Distribution

For the use case described in Chapter 4, it was necessary to be able to model a system that used both AC and DC distribution, in a hybrid configuration. Mini-grid systems sometimes will have DC load e.g. lighting that runs directly from the system whilst also having some AC loads supplying for example larger productive use consumers. I adapted the CLOVER model to be able

to incorporate both AC and DC loads simultaneously. To do this, I modified the system inputs to be able to specify different parts of the load as AC or DC; e.g. domestic = DC and productive = AC. Then, updated the model so that it applies different conversion factors to these parts of the load before the system simulation commences.

3.2.4 System Optimisation

The CLOVER model can be used to optimise off-grid systems. The user must define an optimisation criterion e.g. unit cost or emissions intensity of electricity. In this section, I will define two optimisation criteria used in this thesis: levelised cost of used electricity (LCUE) and emissions intensity of used electricity (EIUE) and give an overview of how the CLOVER model performs optimisation.

Optimisation Criteria

A key metric for economic assessment used in the CLOVER model is the levelised cost of used electricity (LCUE) in \$/kWh [203]. This metric explicitly accounts for only the useful energy consumed by the users, relevant in situations with over-generation or shortfalls in demand satisfaction, as is often the case with off-grid systems. The LCUE will be greater than or equal to the levelised cost of generated electricity (LCOE), a more commonly used metric. As off-grid PV systems often dump a significant fraction of the electricity generated due to an imperfect match between supply and demand, LCUE is therefore a more appropriate metric in this context.

The LCUE is the discounted sum of all capital investment I_n , operation and maintenance M_n , and fuel costs F_n (each in \$) in every year n , divided by the sum of the discounted energy E_n in kWh that is used in each year:

$$LCUE = \frac{\sum_{n=1}^N \frac{I_n + M_n + F_n}{(1+r)^n}}{\sum_{n=1}^N \frac{E_n}{(1+r)^n}} \quad (3.8)$$

The same approach applies to the emissions intensity of used electricity (EIUE) given in kg of CO₂e per kWh. Electricity generated that is dumped because it cannot be stored or used is not considered. The emissions intensity is the sum of embedded emissions in kg of CO₂e, ξ_n^Q , in installed equipment and operating emissions ξ_n^O from the electricity system in every year over the lifetime of the project (in kg CO₂e) divided by the sum of all energy used in kWh:

$$EIUE = \frac{\sum_{n=1}^N \xi_n^Q + \xi_n^O}{\sum_{n=1}^N E_n} \quad (3.9)$$

Optimisation Process

CLOVER uses a form of exhaustive search optimisation, described fully in Sandwell (2017) to identify the combination of component sizes that meet the requirements that the user has defined [205]. The model performs simulations for systems with incrementally increasing PV and storage sizes to find the optimum system defined as that which minimises the optimisation criterion, e.g. the LCUE or EIUE as described above. It steps through a predefined set of PV and battery sizes but has a mechanism to ensure when an "edge case" occurs, the model will simulate further systems

to ensure the optimum is found.

The sizes of PV and battery units considered by the model are defined by the user. For example, this could be 0.3 kWp to represent a realistic PV module size, and 1 kWh for storage, representing increments at which storage may be installed. The hourly output of the resulting simulations combined with financial and cost data (see Appendix Tables A.4 and A.3) are used to compute the LCUE and EIUE of the systems. For a system to be considered as a potential optimum, either with the lowest LCUE or EIUE from the set of simulated systems, it must meet a defined sufficiency threshold. The sufficiency threshold is typically reliability, e.g. the fraction of unmet energy that is permitted over the simulation period. Figure 3.3 shows a graphical representation of the relationship between these thresholds and the optimal system configuration for an example demand scenario.

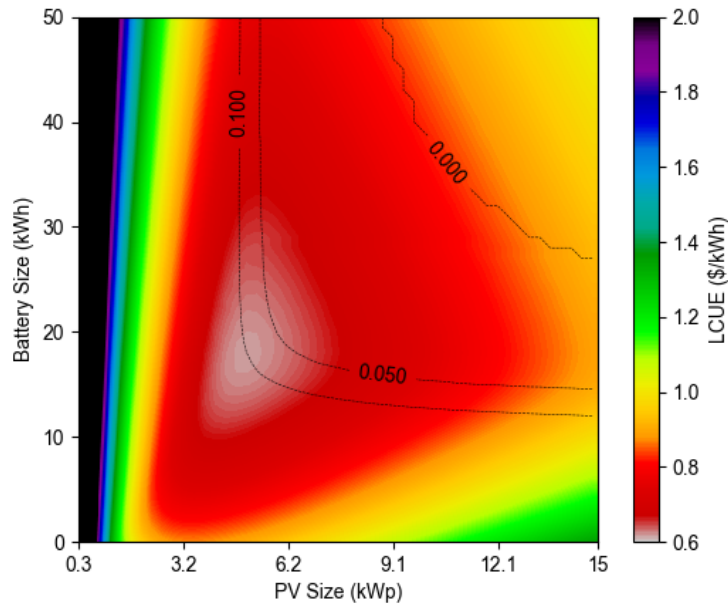


Figure 3.3: Showing the LCUE (\$/kWh) of 2450 PV (0.3 – 15kWp) and battery systems (0-50kWh) considered for a demand scenario (Domestic + Pathlab) as described in Chapter 4. The heatmap is formed from the LCUE (\$/kWh) of each system configuration simulated, with the contour lines marking the unmet energy thresholds for 10%, 5% and 0%; and the sufficient systems. The lowest LCUE systems are at or around 10% unmet energy, with a significant jump in both storage and PV capacity required between 5% and 0% unmet energy. If the sufficiency threshold is set to 5% of unmet energy, only the tested systems above that contour line would be considered.

3.2.5 Summary

This section has given a description of how the system simulation and optimisation parts of the CLOVER model work, and some minor modifications that were made to this version. The model was used to generate the outputs described in Chapter 4 relating to the integration of anchor loads into mini-grids of differing configurations. The next section will describe significant changes made to the system simulation part of CLOVER to improve functionality and its usefulness.

3.3 Inclusion of Dispatched Generation in CLOVER

This part of the methods chapter will describe changes I made to the model to accommodate dispatched generation with 'cycle charging' functionality, whereby the diesel generator can be used to charge the batteries in a hybrid system. These changes were important for the application of the model to answer research questions relating to a hybrid mini-grid in a humanitarian setting (Chapter 5). However, the changes broaden the usefulness of the CLOVER model for other users as it is common for hybrid diesel, PV and storage mini-grids to utilise dispatch strategies and cycle-charging.

The version of CLOVER described above accommodates diesel generation, but only in a simplistic way, as back-up; however, hybrid mini-grids may have diesel generators that are programmed to run at certain times, or when the battery reaches a certain level of charge. Additionally, diesel generators may be run to both charge the battery (cycle-charging) and or meet demand (load-following). The ability to simulate systems with this functionality can allow users to investigate different dispatch strategies such as time of use or state of charge levels for the batteries that may determine when the generator runs.

Existing modelling tools have various functions as described in the previous chapter in Section 2.7. It is possible to use HOMER with the MATLAB Simulink add-on to model hybrid mini-grid systems with dispatched diesel generation [174], and the tool iHoga also has dispatched generation functionality [189]. However, neither of these tools is free to use for research purposes and neither is open-source and customisable.

I adapted the CLOVER model to introduce functionality, to allow the modelling of a diesel generator using a cycle-charging (CC) dispatch strategy, in which the specific SOC set points determine whether the generator will run can vary in different parts of the day (see Figure 3.4). The following subsections mathematically outline the general operation of the model and the new capabilities integrated for the research presented in Chapter 5. The accompanying code is available on Github¹. For clarity, this description does not include transmission and conversion efficiencies, or the degradation of batteries or solar panels, but these are accounted for in the model code and details of example inputs for these values can be found in Appendix B, Table B.1.

Determining the inclusion of the diesel generator

To determine whether a diesel generator, (the size of which is size predefined by the user) is used in any given hour, t , the model checks whether the battery SOC $S(t)$ (as a percentage) is within a specified range (see Table 3.1), which varies according to the time of day, and whether the diesel generator was used in the previous hour $t - 1$.

Table 3.1: Example technical inputs which varied between simulations.

Item	Value	Unit
Diesel hours ($D_{on}(t) = 1$)	18:00-00:00	Time of day
Diesel hours switch-on SOC (S_{on}^D)	65	%
Diesel hours switch-off SOC (S_{off}^D)	95	%
Quiet hours switch-on SOC (S_{on}^Q)	45	%
Quiet hours switch-off SOC (S_{off}^Q)	65	%

The SOC set points $S_{on}(t)$ and $S_{off}(t)$ represent the battery levels (% of charged capacity) at

¹<https://github.com/hamishbeath/CLOVER>

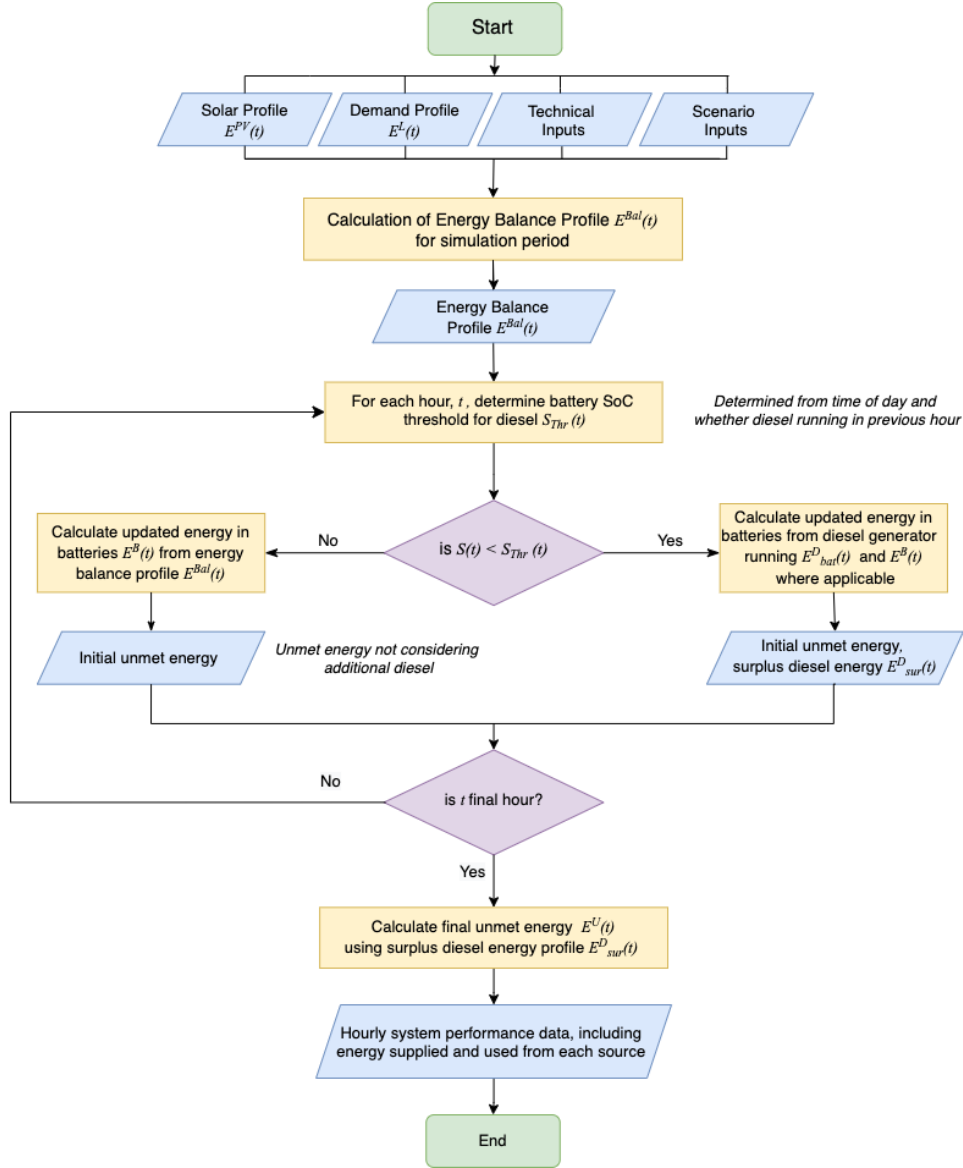


Figure 3.4: A flowchart outlining the updated energy system module with dispatched diesel functionality used in the energy system modelling.

which the generator will turn on and off and can be different depending on the hour of the day. During certain times the generator may be programmed to turn on at a higher state of charge, and therefore may be more likely to run than outside of those times. I refer to “diesel” hours (when the state of charge threshold is higher than usual) and “quiet” hours (all other times) to distinguish these two periods. Each has the relevant on and off set points S_{on}^D , S_{off}^D and S_{on}^Q , S_{off}^Q respectively.

Using the times of day set for “diesel” and “quiet” hours (see Table 3.1), CLOVER generates a binary profile $D_{on}(t)$ for the entire simulation. The profile $D_{on}(t)$ is used to attribute the relevant set points to each hour t . The minimum permitted state of charge, below which the diesel generator must turn on, is given by $S_{on}(t)$ where

$$S_{on}(t) = \begin{cases} S_{on}^D & \text{if } D_{on}(t) = 1 \\ S_{on}^Q & \text{otherwise} \end{cases} \quad (3.10)$$

and likewise the switch-off state of charge $S_{off}(t)$, which must be exceeded if the generator was

running in the previous hour in order to charge the battery adequately, is given by

$$S_{off}(t) = \begin{cases} S_{off}^D & \text{if } D_{on}(t) = 1 \\ S_{off}^Q & \text{otherwise} \end{cases} \quad (3.11)$$

The battery SOC threshold, below which the diesel generator will run, is $S_{Thr}(t)$. During the hourly simulation (see Figure 2), to determine which value – $S_{on}(t)$ or $S_{off}(t)$ – is the appropriate one for $S_{Thr}(t)$, the model checks whether the diesel generator was running in the previous hour by evaluating the value of diesel energy in kWh used by the batteries in the most recent time-step, $E_{bat}^D(t-1)$. The SOC threshold $S_{Thr}(t)$ is therefore defined by

$$S_{Thr}(t) = \begin{cases} S_{off}(t) & \text{if } E_{bat}^D(t-1) > 0 \\ S_{on}(t) & \text{otherwise} \end{cases} \quad (3.12)$$

Simulation process for hours when diesel not running

When the diesel generator does not run, the energy system simulation module operates similarly to how it did before, described above in Section 3.2.2. However, back-up diesel cannot be used when dispatched diesel is selected by the user. The following section describes the modelling process for hours when the generator does not run.

For hours where $S(t) > S_{Thr}(t)$, then the diesel generator will not run. The energy balance, $E^{Bal}(t)$, then represents the potential flow of energy (in kWh) to (if positive) or from (if negative) the battery capacity, B . This flow is constrained by the charging and discharging C-rates, c_{in} and c_{out} respectively, which are evaluated over the hour-long period for each time-step. The model first establishes the battery energy flow for the hour, $E_{Bat}^{Flw}(t)$, under the conditions

$$E_{Bat}^{Flw}(t) = \begin{cases} B c_{in} & \text{if } E^{Bal}(t) > B c_{in} \\ -B c_{out} & \text{if } E^{Bal}(t) < -B c_{out} \\ E^{Bal}(t) & \text{otherwise} \end{cases} \quad (3.13)$$

The battery energy flow E_{Bat}^{Flw} in kWh is added to the amount of energy in the batteries for the previous hour $E^B(t-1)$ to give $E^B(t)$, ensuring that the battery energy level stays within the minimum and maximum storage levels S_{min} and S_{max} (% SOC thresholds) and is given by

$$E^B(t) = \begin{cases} S_{min}B & \text{if } E^B(t-1) + E_{Bat}^{Flw}(t) < S_{min}B \\ S_{max}B & \text{if } E^B(t-1) + E_{Bat}^{Flw}(t) > S_{max}B \\ E^B(t-1) + E_{Bat}^{Flw}(t) & \text{Otherwise} \end{cases} \quad (3.14)$$

Any energy that cannot be stored due to the energy in the battery $E^B(t)$ or state of charge $S(t)$ exceeding the maximum S_{max} is dumped. The calculation of unmet energy is described later.

The battery energy supplied in kWh, $E_{Sup}^B(t)$, is calculated, with non-zero values recorded for when the energy balance for the hour $E^{Bal}(t)$ is negative, i.e. when the battery is discharging energy, given by

$$E_{Sup}^B(t) = \begin{cases} E^B(t-1) - E^{Bal}(t) & \text{if } E^{Bal}(t) < 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.15)$$

Simulation process for hours when diesel is running

When $S(t) < S_{Thr}(t)$ the diesel generator will run. The amount of energy generated by the diesel generator (in kWh) to charge the batteries depends on three factors: the empty capacity remaining in the batteries (in kWh), $E^{ec}(t)$; the maximum diesel energy that can go from the diesel generator into the batteries, E_{max}^{DBat} (in kWh), and the minimum load factor of the generator (as a percentage), Γ_{min} .

The empty capacity $E^{ec}(t)$ is calculated using the total storage capacity B (in kWh), the state of charge in the given hour $S(t)$, and the maximum state of charge permitted for the batteries S_{max} , such that

$$E^{ec}(t) = B (S_{max} - S(t)) \quad (3.16)$$

Depending on the system configuration the maximum diesel energy that can go into the batteries in one hour, E_{max}^{DBat} , is constrained either by the maximum diesel generator output (E_{max}^D in kWh, the energy provided by the total diesel capacity D (in kW) running for one hour) or the input battery C-rate, $B c_{in}$, whichever is smaller, such that

$$E_{max}^{DBat} = \min \{ B c_{in}, E_{max}^D \} \quad (3.17)$$

The minimum load factor (as a percentage) Γ_{min} , and the maximum energy E_{max}^D that installed diesel capacity D can produce in one hour determine the minimum possible output of the generator E_{min}^D via

$$E_{min}^D = \Gamma_{min} E_{max}^D \quad (3.18)$$

Given the above, the amount of energy supplied from the generator and used to charge the batteries $E_{bat}^D(t)$ is given by

$$E_{bat}^D(t) = \min \{ E^{ec}(t), E_{max}^{DBat} \} \quad (3.19)$$

When the empty capacity $E^{ec}(t) < E_{min}^D$ some diesel energy generated is not used by the batteries and is either dumped or used to meet unmet demand. The value of the energy used to charge the batteries from the diesel generator, $E_{Bat}^D(t)$, is added to the amount of energy in the batteries from the previous hour, $E^B(t-1)$, to give an initial value for the amount of energy in kWh in the batteries for the current hour, $E_I^B(t)$, following the diesel charging.

When the maximum diesel energy that can go into the batteries is limited by the battery C-rate then any surplus solar energy is dumped. The initial battery energy $E_I^B(t)$ is confirmed as the final battery energy level for the hour, $E^B(t)$. When battery charging is instead limited by the maximum diesel output then it may be possible to charge the batteries further. Not yet considering S_{max} , the model calculates the energy that could be taken from the PV to the batteries in kWh, $E_{Bat}^{PV}(t)$, via

$$E_{Bat}^{PV}(t) = \min \{ E^{Ba}(t), B c_{in} - E_{max}^D \} \quad (3.20)$$

The PV energy that can go into the battery, $E_{Bat}^{PV}(t)$, is then added to the initial battery energy, $E_I^B(t)$, to evaluate the final value for battery energy, $E^B(t)$. This is calculated considering the

additional energy from the PV panels, $E_{Bat}^{PV}(t)$, and the maximum state of charge, S_{max} , of the battery via

$$E^B(t) = \min \{E_I^B(t) + E_{Bat}^{PV}(t), B S_{max}\} \quad (3.21)$$

Any fraction of $E^{Bal}(t)$ that it is not possible to store in the batteries due to either the maximum input permitted by the C-rate c_{in} or the maximum state of charge S_{max} being exceeded, is dumped.

Given that the batteries are being charged in hours when the diesel generator is running, they do not supply energy and the model records the energy in kWh supplied from the battery for the hour, $E_{Sup}^B(t)$, as zero.

Calculating energy met by each source

The energy (in kWh) that meets demand can come from solar directly, $E_{dir}^{PV}(t)$, the battery storage (itself supplied by PV or diesel), $E_{Sup}^B(t)$, or from surplus diesel generation, $E_{sur}^D(t)$, during hours when the diesel generator is used to charge the batteries. $E_{dir}^{PV}(t)$ is calculated using the load profile $E^L(t)$ and the PV generation $E^{PV}(t)$, and is given by

$$E_{dir}^{PV}(t) = \min \{E^{PV}(t), E^L(t)\} \quad (3.22)$$

The surplus diesel available is calculated using the maximum diesel output, E_{max}^D , and the diesel energy already used to charge the battery, $E_{Bat}^D(t)$, via

$$E_{sur}^D(t) = \begin{cases} E_{max}^D(t) - E_{Bat}^D(t) & \text{if } E_{Bat}^D(t) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.23)$$

The unmet energy in kWh, $E^U(t)$, is the load energy $E^L(t)$ that remains when all available energy resources have been considered for each hour and is calculated after the simulation is complete. The total available energy in kWh is given by

$$E^{Tot}(t) = E_{dir}^{PV}(t) + E_{sup}^B(t) + E_{sur}^D(t) \quad (3.24)$$

and so the unmet energy profile $E^U(t)$ is given by

$$E^U(t) = \max \{E^L(t) - E^{Tot}(t), 0\} \quad (3.25)$$

and is reported as positive by convention.

Calculating diesel energy generated

To accurately estimate diesel fuel consumption, a profile of total diesel energy generated, $E_{Gen}^D(t)$ (in kWh), is needed. This differs from the diesel energy used when the required energy from diesel would result in a load factor below the minimum load factor, Γ_{min} , of the diesel generating capacity, D , and its resultant minimum diesel output E_{min}^D . E_{Gen}^D is therefore given by

$$E_{Gen}^D(t) = \begin{cases} \max \{D \Gamma_{min}, E_{Bat}^D(t) + E_{sup}^D(t)\} & \text{if } E_{Bat}^D(t) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (3.26)$$

where $E_{Bat}^D(t)$ is diesel energy already used to charge the battery, $E_{sup}^D(t)$ is the surplus diesel energy (in kWh) that was used to meet demand, and with the second case governing when the diesel generator is not in use. Once the simulation model has completed, the hourly profiles generated are used for the system analysis and appraisal.

3.3.1 Summary

This section has given a detailed description of how I adapted the energy system module of the CLOVER model. The changes to the model outlined in this section incorporate dispatched generation with cycle-charging functionality, and with the capability to define strategies based on times and battery states of charge. This updated version of the model is used to produce the results presented in Chapter 5; however, they represent a new feature within the CLOVER model that could be applied to other contexts.

3.4 Least-cost geospatial electrification model

This section presents a least-cost electricity access geospatial model that I developed to produce the results in Chapter 6. As discussed in the literature review (Section 2.7.2), geospatial tools offer researchers the opportunity to answer research questions with a country-wide or regional focus and with a spatial aspect. They facilitate the use of extensive geospatial data sets with wide coverage and have been developed through sophisticated remote sensing techniques [40]. Existing tools designed to examine least cost electrification, such as the OnSSET [24, 167] and REM [213] models, have advanced features and have been applied for a range of questions. However, for the research questions addressed in this thesis, neither is easily applicable due to model design and computational intensity. For the results in Chapter 6 of this thesis, I therefore developed a new simple geospatial data model, that can be used easily in conjunction with other modelling tools I deploy; and is suitable for a scenario-based approach that addresses questions that relate to electricity demand, reliability, and carbon pricing. The code is available on [Github](#).

3.4.1 High-Level Description of Model

The model is written in Python and processes inputs and provides outputs in CSV files. The geographical area of study is split into 'cells', each with its own unique ID number and location information such as the country or district it sits within and the coordinates of the cell boundaries. The model is designed with an annual temporal resolution, and a flexible spatial resolution (depending on input data) and can be used to assess what the least cost mode of electrification is in each gridded cell under different scenarios. The model is designed to compare the cost of grid extension with off-grid modes of access: stand-alone systems and mini-grids. These off-grid modes could be powered by any generation means, however; for the usage in Chapter 6 the model is configured to compare diesel and PV-based systems and the description below will cover those systems only.

The model requires a range of inputs and geospatial data layers before completing the cost-minimisation exercise. Firstly, the model requires a geospatial layer that describes the number of people who will gain access per year in each map cell. This can be created by using relevant geospatial data sets such as population, population without access and population growth. A

description of the creation of an example of this input is given in the methods section of Chapter 6 in section 6.3.4. The population inputs - a matrix of numbers of people gaining access by year and by map cell - define the simulation period covered. Using household size, which may vary by country, region or district, allows the model to know the number of households that need access in each map cell per year of the simulation. Additionally, the model requires a vector layer detailing the existence of national grid infrastructure and energy demand estimates per household. Cost inputs for each technology and system type are also required: these can vary at the country or district level. Further description of the preparation of these inputs can be found in section 6.3.1 of Chapter 6.

3.4.2 Mathematical Description of Model

Here, I present a description of the calculations, which the model performs to estimate the emissions and costs of each electricity access mode for each map cell.

National Grid Costs

Grid extension costs, for a given cell ϕ ; are defined by the centroid of the cell's distance from the nearest grid line G^{dist} (in km), multiplied by the grid extension cost, C^{km} (in \$ per km),

$$C_G^{ext}(\phi) = C^{km} (G^{dist}(\phi)) \quad (3.27)$$

Grid capacity costs, C_G^{cap} (\$/kW) for a given cell ϕ are calculated using an estimation of the additional capacity required for each cell $G^{cap}(\phi)$ (in kW). The additional capacity per cell $G^{cap}(\phi)$ is calculated using the weighted average grid capacity factor G_y (%) for the country κ of the given cell (ϕ), and the average hourly electricity demand, E^L (in kWh), in the final year, N , in the selected cell ϕ , taking into account the assumed reliability of the national grid in rural areas, in each country, $G^R(\kappa)$. The additional grid capacity that would be required in each cell $G^{cap}(\phi)$ is therefore given by

$$G^{cap}(\phi) = \left(\frac{1}{G_y(\kappa)} \right) (E_N^L(\phi)) (G^R(\kappa)) \quad (3.28)$$

The grid capacity investment requirements (in 2022 USD) for the cell, $C_G^{cap}(\phi)$, are given by

$$C_G^{cap}(\phi) = (G^{cap}(\phi)) C_G^I(\kappa) \quad (3.29)$$

where $C_G^I(\kappa)$ is the assumed grid capacity cost (in \$/kW) for the country the cell is within. The total grid costs for each cell $C_G^{tot}(\phi)$ are the sum of the above costs in all years, each discounted to the year of construction n , at discount rate r :

$$C_G^{tot}(\phi) = \sum_{n=1}^N \frac{(C_G^{ext}(\phi) + C_G^{cap}(\phi))}{(1+r)^n} \quad (3.30)$$

The year of construction is determined by the population inputs and is covered in more detail below. Considering all map cells in the simulation, the model outputs a column vector of total costs, $C_G^{tot}(\Phi)$ to be used later in the model.

National Grid Emissions

The emissions from each source are used to provide emissions totals, for each scenario. They are also used to calculate carbon tax amounts added for scenarios where carbon pricing is included. Emissions are tracked for each map cell, ϕ , and by year, n . The total grid emissions by cell (kgCO₂e), and for a given year, $\xi_{G,n}(\phi)$, comprise firstly, embedded emissions in grid infrastructure for the cell and year. The year the grid is extended to a given cell ϕ , is determined by the user-defined population model (see Chapter 6, Section 6.3.4), which provides the year that the first household within a non-connected cell gains access. The model stores a binary matrix for all cells and years, based on the population model, with 1, denoting the year of the first connection and thus the building of the required infrastructure, $\Theta_n(\phi)$. So, grid extension emissions are only factored for a given year, n , when $\Theta_n(\phi) = 1$. Therefore,

$$\xi_{G,n}^{ext}(\phi) = \begin{cases} \xi^{km}(G^{dist}(\phi)) & \text{if } \Theta_n(\phi) = 1 \\ 0 & \text{otherwise} \end{cases} \quad (3.31)$$

where ξ^{km} is the embedded emissions (kgCO₂e) per km of grid extension, and $G^{dist}(\phi)$ is the distance from the grid of the given cell ϕ . The second component of national grid emissions is from electricity generation ξ_G^{Gen} , which is dependent on the generation mix for the country κ ; and the associated emissions intensity of electricity generated, $\xi_G^O(\kappa)$, (in kgCO₂e/kWh) and the electricity demand E^L (in kWh) in a given year, n :

$$\xi_{G,n}^{Gen}(\phi) = \xi_G^O(\kappa)(E_n^L(\phi)) \quad (3.32)$$

The total grid emissions per cell (kgCO₂e), in a given year is $\xi_{G,n}^{tot}(\phi)$, is the extension infrastructure emissions $\xi_{G,n}^{ext}(\phi)$ added to the $\xi_{G,n}^{Gen}(\phi)$ generation emissions. The model stores a matrix of the emissions occurring in each cell, and each year of the simulation, used for the calculation of carbon taxation amounts. Additionally, it sums emissions over all years in each cell to provide a column vector of total emissions from the grid in each cell, $\xi_G^{tot}(\Phi)$ (in (kgCO₂e).

Whilst embedded emissions for grid extension infrastructure are included in the model, embedded emissions in grid-scale generating capacity are omitted. This is due to the complexities of including this infrastructure, and the data requirements of doing so. The omission of these emissions will underrepresent the emissions from the grid and may create a positive bias towards grid infrastructure for scenarios considering carbon pricing.

Off-grid Costs

For off-grid systems, system costs and emissions can be defined at the region, country or cell level. For the use case of the model described later in Chapter 6, cost and emissions outputs were provided by the CLOVER model at the country level for different off-grid system types. The following description will assume off-grid system inputs at the country level, however; these could vary at the individual cell level. The model is configured to compare off-grid systems powered by diesel or by solar PV; at the household (stand-alone-system) and community level (mini-grids) with national grids. It could be easily adapted to compare additional off-grid means of access.

Although mini-grids and stand-alone systems each powered by PV or diesel will have their system-specific per-household investment cost inputs, the process for calculating respective investment costs for all off-grid systems is the same, and here each of the off-grid systems is given as a

type, j . The total costs for an off-grid system C_{og} , of type j in a given map cell $C_{og}^j(\phi)$ is defined by the country level investment, $C_{og}^I(\kappa)$ and OPEX values, $C_{og}^O(\kappa)$; and the number of households added in a given cell $H(\phi)$ in each year n of the simulation; discounted respectively. For a given year n , cell ϕ and off-grid type j , the investment cost (in 2022 USD) is defined by

$$C_{og,n}^{I,j}(\phi) = (C_{og}^{I,j}(\kappa)) (H_n(\phi)) \quad (3.33)$$

For operational expenditure (OPEX) (in 2022 USD) e.g. fuel or new components, the per-household amount for the off-grid type j , is simultaneously factored in for years after the construction year n , up to the final year N . In the model, the OPEX costs are added in and discounted to be in the present value of the year of construction n :

$$C_{og,n}^{O,j}(\phi) = \sum_{x=n}^N \frac{(C_{og}^{O,j}(\kappa)) (H_n(\phi))}{(1+r)^{x-n}} \quad (3.34)$$

where x is any year between the construction year n , and the end of the simulation N . The total cost in a given cell and for an off-grid type, $C_{og}^j(\phi)$ is the sum of the OPEX costs and investment costs occurring in all years and discounted to give the present value,

$$C_{og}^j(\phi) = \sum_{n=1}^N \frac{C_{og,n}^{I,j} + C_{og,n}^{O,j}(\phi)}{(1+r)^n} \quad (3.35)$$

The model stores the costs for each cell ϕ and off-grid type j summed across all years as column vectors, $C_{og}^j(\Phi)$, to be used to assess the least-cost mode for each map cell.

Off-grid Emissions

Off-grid emissions follow a similar pattern to costs, explained above, without discounting. However, emissions are tracked on an annual basis, as this is necessary for carbon pricing, which is explained further below. The infrastructure emissions, embedded in the initial capacity installed ξ_{og}^{inf} (kgCO₂e/kWp/kW or kWh), for a given cell ϕ , year, n and off-grid type, j is given by

$$\xi_{og,n}^{inf,j}(\phi) = (\xi_{og,n}^{inf,j}(\kappa)) (H_n(\phi)) \quad (3.36)$$

where $\xi_{og,n}^{inf,j}(\kappa)$ is the infrastructure emissions per household in the given country κ and for the off-grid type, j . The OPEX emissions for the given year n , are given by

$$\xi_{og,n}^{O,j}(\phi) = (\xi_{og}^{O,j}(\kappa)) (H_n^{tot}(\phi)) \quad (3.37)$$

whereby H_n^{tot} is the total accumulated number of households connected in the grid cell in the given year n . The total emissions for an off-grid system type, ξ_{og}^j in a cell, ϕ for a specific year, n is the infrastructure emissions $\xi_{og,n}^{inf,j}(\phi)$ added to the $\xi_{og,n}^{O,j}(\phi)$ OPEX emissions. The output of this process is a matrix of cells covering the geographic area, and years, with respective emissions occurring in each cell in each year of the simulation. This is used for calculating carbon tax (see below).

Similar to costs, the embedded emissions in off-grid infrastructure are added in the year of construction. Given infrastructure such as solar panels may last many years beyond the model time horizon, this may mean there is an overrepresentation for off-grid infrastructure emissions.

This will therefore have a negative bias towards off-grid infrastructure in scenarios considering a carbon price.

Inclusion of Carbon Tax

The model can accommodate a single carbon price, or dynamic carbon pricing, varying by year. For the results produced in Chapter 6, the carbon price varies by year. To calculate the carbon tax values for a given electrification mode, m , and in each cell, ϕ , the model uses the matrices of emissions in each cell, produced in each year (as outlined above). For a given year, n , there is a column vector of cells Φ from the respective matrix containing emissions ξ (kgCO₂e), from each electrification mode m . The carbon tax value for the given year as defined by the user, C_n^{tax} , (in 2022 USD / tonne) is multiplied by the emissions from the relevant mode, m , and discounted. The sum of this across all years gives the total carbon tax amount for the mode, $C^{tax,m}(\Phi)$:

$$C^{tax,m}(\Phi) = \sum_{n=1}^N \frac{(C_n^{tax}) \xi_n^m(\Phi)}{(1+r)^n} \quad (3.38)$$

The net present value of the carbon tax amount, in a column vector of all, $C^{tax,m}(\Phi)$, is included in total costs for each mode, m , for scenarios with a carbon tax.

Inclusion of Unmet Demand Penalty

The model can apply an unmet demand penalty (in \$ per kWh unmet) across all electrification modes, m , with the level varying between modes e.g. off-grid versus grid; and in each country, κ . It does so using the reliability, R (% of demand met) for each mode m in each country κ , and the energy demand, E^L (in kWh) in the relevant year n , and cell ϕ such that

$$C^{rel}(\phi) = \sum_{n=1}^N \frac{C^{rel} (1 - R^m(\kappa)) (E_n^D(\phi))}{(1+r)^n} \quad (3.39)$$

where the C^{rel} is the financial value of the unmet demand penalty. The total amounts per cell, $C^{rel}(\phi)$, form a column vector $C^{rel}(\Phi)$ to be added to total costs for each mode m , for scenarios including an unmet demand penalty.

Least-Cost Electrification by cell

Once total net present costs, in each cell for all years and for each mode m are calculated, the model then selects the least-cost option for each grid cell based on the total net present cost, factoring in either carbon pricing or unmet demand penalties if specified for the scenario. Prior to this, it uses the population density (persons per cell) to determine whether each cell is most suitable for off-grid electrification. There is a user-defined population density threshold P^{Thr} , below which, for the given cell, the model will compare the total cost for stand-alone-systems, $C_{SA}(\phi)$, in favour of the cost of mini-grid systems, $C_{MG}(\phi)$. This is the case for both PV and diesel-powered off-grid systems, such that

$$C_{og}(\phi) = \begin{cases} C_{MG}(\phi) & \text{if } P_N^{Den}(\phi) > P^{Thr} \\ C_{SA}(\phi) & \text{otherwise} \end{cases} \quad (3.40)$$

where $P_N^{Den}(\phi)$ is the population density of people getting access in each cell, in the final year of the simulation, N , and $C_{og}(\phi)$ is the selected off-grid system type (stand-alone or mini-grid), based on the population density in the cell. Once establishing whether a cell, ϕ is best suited to stand-alone systems or mini-grids, the next step is to compare the costs off-grid modes of each generation type, PV, C_{og}^{PV} , and diesel, C_{og}^D for each cell with the cost of the national grid, C_G :

$$C(\phi) = \min \{C_{og}^{PV}(\phi), C_{og}^D, C_G(\phi)\} \quad (3.41)$$

Model Outputs

The model outputs a column vector and the cost and respective selected electrification mode for each cell. Outputs from the model are provided in a by-cell output, with data such as emissions, cost and electricity access mode along with the cell location data used for mapping outputs; total scenario appraisal outputs giving the percentage of the population met by each mode and investment and emissions via each mode, and the same appraisal but by-country included in the respective study.

3.4.3 Summary

In this section, I have described how the least-cost electrification geospatial data model functions, factoring in the costs and emissions for each grid-cell, for a given scenario. This model produces the results in Chapter 6, focusing on universal household electricity access in Sub-Saharan Africa. However, with different data inputs, it could be applied to a single country or a different region.

3.5 Summary of Methodological Contributions

One of the key contributions of this thesis is the development of open-source modelling tools to address electricity access research questions. This chapter has described these contributions, with further context regarding their usefulness given in the respective results Chapters 5 and 6.

The first part of this chapter (Section 3.2) gave an overview of the CLOVER energy system model, designed for modelling low-carbon community-scale electricity systems. I made minor modifications to the model to be able to undertake the research in Chapter 4; however, the novelty arises from the application to the case study and research questions rather than model development. This is described in detail in the next chapter.

The second part of this chapter (Section 3.3) described changes made to the CLOVER energy systems model to include the functionality for the inclusion of dispatched diesel generation within hybrid mini-grid systems. These changes allow users to model diesel generators configured for cycle-charging (diesel generator charges batteries as well as meets electricity demand) and to use different settings at different times of the day. The inclusion of this functionality in an open-source energy systems model is a key methodological contribution within this thesis. This version of the model is applied to a case study described in Chapter 5 and could be further used to model other electricity systems with a similar configuration.

The final part of this chapter, Section 3.4 described the structure of the geospatial least-cost electrification model that I built to produce the results I will outline in Chapter 6. This model can be applied to any country or region where the necessary data inputs are available. The model

can be used to assess the least-cost mode of electricity access across a given area. The model can be used to consider research questions relating to dynamic electricity demand, different levels of reliability between countries or regions, and the application of carbon prices or a penalty for unmet demand. This tool, which could be applied to future research questions beyond those which I address in Chapter 6, is the second main methodological contribution within this thesis.

Chapter 4

Improving Mini-grid Economics via Anchor Loads

Acknowledgement of Contributions

The study was designed and completed in conjunction with M Hauser. Supervision and guidance were given by A Gambhir, S Few, P Sandwell, C Chambon and J Nelson. Written contributions to the paper resulting from this work were given by M Hauser to all sections and by A Gambhir and C Chambon for the introduction and discussion sections. Comments and edits on the paper manuscript were given by M Hauser, A Gambhir, C Chambon, S Few, P Sandwell and J Nelson. The CLOVER model version for this study was designed and written by P Sandwell.

Previous Presentation of this work

Hamish Beath, Muriel Hauser, Philip Sandwell, Ajay Gambhir, Sheridan Few, Clementine L. Chambon, Jenny Nelson, The cost and emissions advantages of incorporating anchor loads into solar mini-grids in India, Renewable and Sustainable Energy Transition, Volume 1, 2021, 100003, doi.org/10.1016/j.rset.2021.100003.

4.1 Overview

As discussed in Chapter 2, renewables-based mini-grids have the potential to improve electricity access for certain consumers at lower cost, and in some instances at higher reliability than available national grids in rural areas. Further, in Chapter 2 I highlighted the economic hurdles faced by mini-grids described in the literature: these systems have a challenging cost-to-revenue ratio, hindering their implementation and there is a need to increase demand and revenues to improve the economic viability. Combining residential loads with an anchor load, a relatively large non-domestic user, can help to improve mini-grid economics.

Using energy modelling and measured electricity demand data from India, in this chapter, I assess the cost and emissions advantages of integrating health clinics as anchor loads within domestic solar-based mini-grids. For comparison, I also assess the ability of the national grid to meet my demand scenarios using monitored Indian grid data from locations close to the mini-grid

site. I apply a scenario-based approach, using separate domestic and health centre load demand profiles, and both types of electricity demand profiles in combination. I do this by meeting two levels of reliability (% of electricity demand met), 95% and 100%; and assess this by comparing systems using PV and batteries, diesel, and hybrid PV, battery and diesel generation.

The work in this chapter contributes novelty in that is the first techno-economic assessment of rural health facilities acting as anchor loads in domestic mini-grids. Furthermore, and differing from existing research looking at anchor loads, the work presented in this chapter exploits high-resolution monitored demand data and detailed cost data from an existing domestic solar mini-grid site. Further novelty arises from the use of granular, monitored grid availability data from close to the case-study site to estimate the ability of the national grid to meet various demand scenarios and thus give a thorough comparison of different electrification options. Additionally, it highlights the issue of national grid intermittency in making such comparisons.

4.2 Background

Anchor loads are larger, consistent electricity consumers with greater capacity to make reliable payments, reducing the financial risk of connecting smaller consumers with low ability to pay. Anchor loads can also perform important social and economic functions for a community, which makes this model additionally attractive [151]. In India, telecommunication towers are the anchor load in some mini-grids with the surrounding community using surplus electricity [223]. Using hypothetical demand data, techno-economic analysis has found a 48% reduction in the unit cost of electricity when anchor loads constitute 30% of the load in PV and battery mini-grids [224]. Another techno-economic study, using estimated demand and willingness to pay data, suggests that a business model for a mini-grid operator using PV and batteries with anchor loads offers improved viability, but still requires subsidies [151]. Rural health centres acting as an anchor load may provide additional benefits since, as well as improving the viability of off-grid systems, they enable the supply of electricity for healthcare needs [62, 225]. Other options for additional loads in rural India are often agricultural and seasonal, meaning they cannot provide consistent demand and payment year-round [226]

Access to electricity for rural, unserved health facilities has been demonstrated to improve patient outcomes [9] and therefore enables further progress towards SDG3: ensuring healthy lives and promoting well-being for all [227]. Electricity can extend service hours through lighting, allow the use of life-saving electrical medical equipment and improve staff recruitment and retention [62]. Additionally, electricity is needed for maintaining the cold chain that allows for storing blood and temperature-sensitive medicines such as vaccines [228, 229]. A lack of electricity supply has serious consequences for healthcare provision in low-income countries. For example, at night in the absence of electric lighting, practitioners are forced to perform surgery or child delivery using candles or mobile phone torches, significantly increasing patient risk [230, 231]. The reliability of any supply is of vital importance. Good quality electricity supplies result in health centres using energy services more widely in their care provision, improving outcomes [10]. Further, blackouts and voltage drops – that cause life-support equipment such as ventilators to cease functioning – lead to a higher incidence of patient mortality [64].

In India, the first line of health care is provided by primary health centres (PHCs); and it is estimated that roughly 50% of primary health centres (PHCs) still have no reliable electricity access, with a further 38 million people dependent on a PHC with no electricity at all [58]. In

India's most populous state Uttar Pradesh, where the research in this chapter is focused, access is amongst the lowest across states: 30% of PHCs have no access to electricity at all, and only 10% have a reliable supply [232].

Moreover, although India has made rapid progress with connecting households to the national grid many 'connected' villages exist, where all or most households do not have a physical grid connection or connections are so unreliable that households are connecting to mini-grids to get a more reliable service [14, 233]. Mini-grid operators such as OMC Power are building mini-grids in Uttar Pradesh in communities that are grid-connected because the grid is not providing a reliable service, especially at peak times [16]. The mini-grid used as the case study in this chapter exists within a community that has a grid connection, with households having a mixture of grid and mini-grid connections. The fact that consumers are relying on off-grid connections demonstrates the importance of reliability for households too, not just health facilities.

4.3 Methods

To investigate the use of health centres acting as anchor loads in mini-grids, I created 5 different electricity demand profiles consisting of data from two health centres and a rural village located in Northern India. These demand profiles were used in energy system modelling to find the least-cost and least-emission electricity system configurations. In addition, I created three national grid electricity profiles and assessed their ability to meet the demand of the 5 demand profiles. The methodological process is shown in Figure 4.1 and is covered in more detail in the following sections:

- Section 4.3.1: Description of the construction of five hourly energy demand profiles incorporating domestic and health centre demand used for this study. I use a mixture of monitored demand data and energy use survey data.
- Section 4.3.3 Explanation of the use of monitored grid data to select and construct three hourly national grid profiles with varying levels of availability. I use the demand profiles and grid availability profiles to assess the ability of the grid to meet this hypothetical demand.
- Section 4.3.2 Using the demand profiles and additional financial, emissions and technical data, I use energy system modelling to find the cost- and emissions- optimal mini-grid system sizes capable of supplying 95% and 100% of energy demand to each demand scenario.

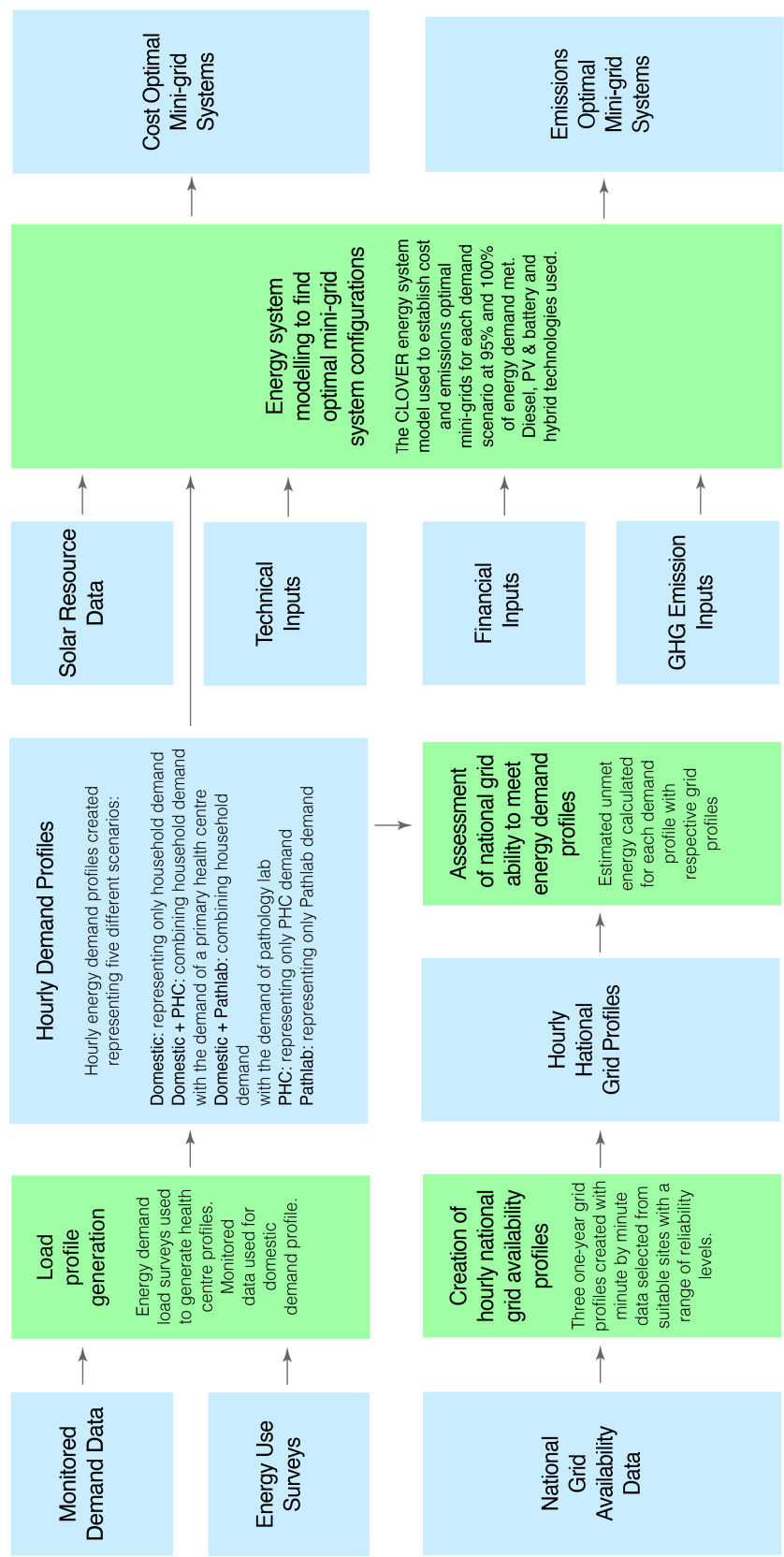


Figure 4.1: Showing outline of the methodological process. Inputs and outputs are shown in blue, with processes in green. Figure Adapted from Beath et al.(2021) [18].

4.3.1 Modelling Domestic and Health Centre Demand

Domestic Demand Profiles

For the domestic demand profile, I used electricity usage data monitored from households connected to a direct current (DC) solar and battery mini-grid in the village of Sarvantara located in the district of Bahraich, Uttar Pradesh, India. The system has been operational since June 2017 and supplies approximately half of the 100 households in the village with electricity for USB phone charging, LED lighting and fans; providing electricity at around Tier 2, as outlined by the ESMAP MTF [32]. The national grid is present in the village with the mini-grid operating alongside it.

At the time of data collection, 45 households were connected to the mini-grid, 29 of which were being monitored by smart meters (BBOXX Ltd, UK), providing minute-by-minute load data which I aggregated to give hourly electricity demand for one year from 1.12.2017 – 30.11.2018. To represent the load of all households connected at the time, I synthesised 16 additional household profiles from the existing data of the 29 monitored households to represent the total community size of 45 households. I assumed the data followed a normal distribution and used the hourly mean and standard deviation of the existing data to produce randomised values for every hour in the year, to be used for the additional household profiles. Where gaps in the data sets existed for other households, such as periods where data were missing due to meter faults, these were synthesised using the same procedure.

Healthcare Facility Demand Profiles

For the health centre demand profiles, I used data from two facilities: a pathology lab located 50 km from the village of Sarvantara, and a primary health centre, located in the state of Maharashtra. The data for the two health centres were gathered via correspondence with local organisations Oorja Development Solutions & Gram Oorja. These organisations carried out quantitative load surveys that detailed existing electrical devices in use with their daily usage pattern and respective power rating. This was supplemented with data from the literature (see below). The usage hours were converted into probability profiles that indicate the likelihood that a given device would be in use at a given hour of the day, to meet the model input requirements.

The PHC used for this study is typical of those in rural India in terms of services offered, and with opening hours from 9:00 to 17:00. The devices and their power ratings are shown in Table 4.1. The pathology lab (Pathlab) is a larger facility than the PHC but is also located in a rural area. The main activity performed in the Pathlab is the analysis of medical samples of patients, which is why it has a greater range of devices in use (see Table 4.1). It represents a health centre that is larger and has a higher electricity consumption than the PHC but is still critical to a rural community. It has opening hours of 7:00 to 20:00 when patients can arrive to give samples and receive results.

The CLOVER model load module was used to develop the device-level data into an actual load profile. One of the required inputs for CLOVER is a utilisation profile of each device. For every month of the year, the hourly usage probability for one representative day had to be created for each device. The quantitative surveys from external organisations working with the healthcare institutions contained information on the present electrical devices, their wattage and daily average usage profiles. For most devices, the probability was calculated by dividing their usage hours by the number of opening hours of the health centre as it was not always clear from the load survey

Table 4.1: Power ratings and data sources of the devices at the PHC and Pathlab

Device	Wattage [W]	Source
PHC		
Lights	15	[62]
Fans	65	[62]
ILR	100	[234]
Refrigerator	200	[62]
Freezer	25	[235]
Nebulizer	100	Survey data
Suction Machine	145	[62]
Shadowless Lamp	150	[236]
Foetal Monitor	2.25	[62]
Pathlab		
Analyser	110	
Blood Mixer	50	
Cell Counter	5	
Centrifuge	138	
Exhaust Fan	50	
Fans	40	
Fridge	200	Survey data
Incubator	750	
Laptop	60	
LED	10	
Microscope	20	
Nebuliser	80	
Spectrophotometer	110	
Printer	300	
X-ray Machine	600	[237]

which exact hours of the day the devices were used. For some of the devices, such as the ILR, freezer and fridge, a probability of 0.25 was assumed for all 24h. This assumption is justified as these devices do not constantly draw power to cool their interiors but rather on an interval basis. To see an example probability profile used as an input into the demand profile process, see Table A.1 in Appendix A.

The X-ray machine was reported to be used for one hour with a power rating of 3 kW. However, this data point was adjusted because an X-ray machine normally has a high-power requirement but only for a very short period, it would be inaccurate to assume that it operates at its peak power of 3 kW for one full hour. Hence, I approximated a more representative usage profile through data available in the literature. According to COCIR (2017), an X-ray machine has three different operating modes, namely, an off-/low-power mode, standby mode and scan mode. Depending on the number of patients per day, the energy consumed during the scanning mode can be negligible as it accounts for only a few seconds to minutes per day and is exceeded by the total energy consumption of the standby mode [237]. As in this case, the X-ray machine is in the scanning mode less than 10% per day (1h per 13h), it can be assumed that the standby mode is the prevailing contributor to its overall daily energy consumption [238].

Given all the required probability inputs for each device and health facility, the total load was modelled. CLOVER computes the total load profile based on the load of each device and its probability of usage. By using a Monte Carlo simulation, a randomised device utilisation pattern was created. Based on a binomial distribution, the devices are randomly assigned to be on or off in

each hour. This allows the creation of usage patterns that differ each day and ultimately produces a more realistic demand profile. A detailed mathematical description of the load module within the CLOVER model can be found in Sandwell (2017) [205]. To be consistent with the domestic load profile construction, I assume a constant load over the 20 years without growth in demand.

4.3.2 Modelling Framework and Scenario Description

I used the open-source energy model CLOVER Model [202] to optimise mini-grid systems with the minimum unit cost and emissions intensity of electricity for each scenario over a system lifetime of 20 years. CLOVER was developed by Sandwell et al. (2017) [203], written in Python and is freely available on [Github](#). A high-level mathematical description of the energy system and optimisation modules of the version of the model used in this chapter is provided in Section 3.2 of the methods chapter, along with details of the modifications made to the model for this study.

I investigated the use of electricity supply systems using the following technologies:

- PV and battery storage
- Diesel generation
- A hybrid system using PV, batteries and diesel

I modelled two levels of energy demand met: for meeting 95% of energy demand I compared PV and battery systems with diesel systems, and for meeting 100% of demanded energy I compared all three system types. I chose to use 95% of demanded energy met as the base case as this is the highest level of uptime of the grid locations examined in this study in the region (see 4.3.3). I tested 100% of demanded energy to understand the cost and emissions of full reliability. In my description of energy demand met, there is no consideration of outages that may occur due to other system failings not related to energy deficits.

To understand how combining domestic and healthcare 'anchor' loads may impact the cost and emissions of these different off-grid systems I modelled the five demand scenarios, using the demand profiles created. These are provided in Table 4.2.

Table 4.2: The electricity demand scenarios modelled for each of the system types.

Demand Scenario	Description
Domestic	Representing only the domestic demand profile
Domestic + PHC	Representing the domestic combined with the PHC demand profile
Domestic + Pathlab	Representing the domestic combined with the Pathlab demand profile
PHC	Representing only the PHC demand profile
Pathlab	Representing only the pathlab demand profile

The optimisation criteria used for selecting the systems were the lowest unit cost (LCUE) and emissions of electricity used (EIUE). I describe these metrics in more detail in Section 3.2.4 in the methods chapter.

4.3.3 Comparison with the National Grid

To compare with the results of the use of mini-grids to meet domestic and health centre loads, I used monitored grid availability data to assess the reliability with which the national grid, as

it currently operates, could meet these loads. Grid data are publicly available from monitoring locations across India from the Prayas Energy Supply Monitoring Initiative (ESMI) database [239]. I examined the data availability and the grid service level for the months between November 2017 and March 2019, across nine monitoring locations ranging between 4 and 37 km from the Sarvantara mini-grid site. From these, I selected three monitoring locations with a range of service levels and settlement types that had adequate data availability to construct three one-year profiles using data from February 2018 to January 2019. The profiles represent high, medium and low availability from a city, a small town and a village, respectively (see Table 4.3). In this instance, availability is defined as the percentage of minutes during which electricity supply is available in the time period examined.

Table 4.3: The locations and their characteristics used for the grid analysis

Name	Grid Strength	Availability (% of time)	Availability (hours/ day)	Settlement type	Km to mini-grid site
Bhavaniyapur	Weak	52.2%	12.5	Village	4km
Nanpara	Moderate	71.0%	17.0	Small town	8km
Nazipura	Strong	95.5%	22.9	City	37km

The data for monitoring locations in the ESMI database has some hours where no data is recorded. This occurs when the recording device is turned off, for example, due to device maintenance [240]. Data availability ranges from 81 to 100% of hours depending on the month and location. There is a mean data availability of 95% of hours. To fill these gaps, I used the mean values for that hour of the day in the relevant month to give a complete profile. Whilst the averaging process for times with data unavailability may create a smoother grid reliability profile than would be seen in real data, due to the low number of hours it is assumed this will have a minimal impact on the results.

I used the grid profiles to compute the unmet energy for each of the demand scenarios. The load profiles are comprised of total energy demand (in kWh) for each hour $E^L(t)$; the demand is assumed constant within each time step. The unmet energy (in kWh) in each time-step $E^U(t)$ is the product of the fraction of the hour where supply is unavailable $U(t)$ and the demand in that hour $E^L(t)$:

$$E^U(t) = (U(t))(E^L(t)) \quad (4.1)$$

The unmet energy fraction E^{UF} calculated for each grid profile and demand scenario is therefore described as:

$$E^{UF} = \sum_{t=0}^{t_{max}} \frac{E^U(t)}{E^L(t)} \quad (4.2)$$

This method assumes that unmet energy demand caused by a blackout would not be shifted to other hours with higher availability. The unmet energy fraction E^{UF} calculated for each demand profile on an hourly basis from each grid location provides a truer indication of the suitability of the local grid to supply each demand scenario than assuming that the overall percentage of time where supply is unavailable is equal to the fraction of energy unmet. The results of this for each demand scenario are shown in Section 4.4.2.

4.4 Results of Analysis

4.4.1 Demand Profiles

The hourly energy demand curves for all the demand scenarios are shown in Figure 4.2. Table 4.4 gives the average total daily energy demand for all scenarios and the fraction of demand occurring at night. For the domestic demand, the average daily household consumption is 0.15 kWh, equating to just under tier 2 levels as defined by the ESMAP MTF [32].

For both health centres, the demand is highest during their respective opening hours. The required energy for the Pathlab is considerably higher than for the PHC as it uses more electrical equipment. Relatively low and constant demand is observed out of hours due to the use of devices requiring constant supply such as refrigerators. The domestic demand of the village shows a different demand pattern, with the demand lowest during the day, and increasing in the evening, remaining high overnight. The village residents keep their lights on overnight for security reasons, to protect their homes and livestock [241], meaning the domestic scenario has the greatest proportion of demand at night of any scenario. The domestic demand profile and the health centres' demand profiles are complementary and when combined, result in a more balanced demand pattern over the day.

Nonetheless, as shown in Table 4.4, the Domestic + PHC scenario still has most of its demand occurring out of the solar irradiation hours (at 59%, albeit lower than the 76% of the Domestic only scenario, as expected). For the Domestic + Pathlab scenario, the peak demand occurs in the evening hours after sunset which would be expected to increase the storage requirement of the system compared to the Pathlab only scenario.

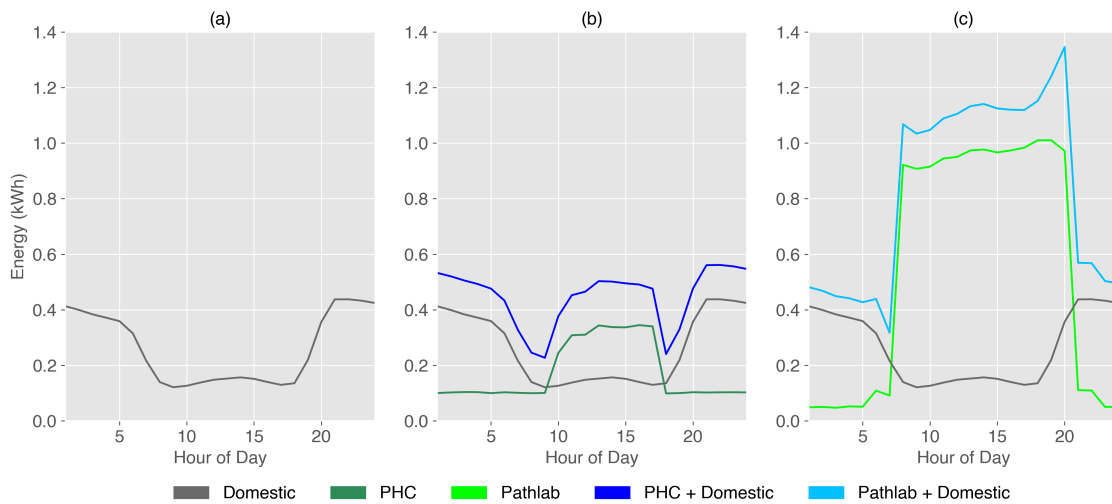


Figure 4.2: shows the shape of the daily demand profiles for all scenarios. Panel (a) shows the domestic load curve alone. Panel (b) shows the domestic, PHC and PHC + domestic curves. Panel (c) shows the domestic, Pathlab and Pathlab + domestic curves. Figure adapted from Beath et al.(2021) [18]

Table 4.4: Energy demand of the five scenarios. The night demand fraction is calculated by assuming that the average daylight hours are from 7:00 to 18:00.

Scenario	Average Daily Demand (kWh)	Average Night Demand Fraction
Domestic	6.9	76%
PHC	4.5	32%
Pathlab	14.4	21%
Domestic + PHC	11.4	59%
Domestic + Pathlab	21.3	39%

4.4.2 National Grid Unmet Energy Estimations

For comparison with the alternative electrification option of connection to the national grid, I estimated unmet energy that would arise for each of the demand scenarios from three grid locations close to the mini-grid site. The three locations represent a range of grid availability levels and are from different geographies (see Table 4.3) with the closest site and most representative of the grid locations being the ‘weak’ one, from the nearby village of Bhavaniyapur. Service unavailability has a late morning peak across all three locations, with further increases in the evening peaking again at 21:00 and 22:00 for the weak and moderate locations respectively.

For the Domestic demand scenario, for all three grid locations, the unmet energy fraction is lower compared to the percentage of minutes the grid is unavailable (see Table 4.5). This is due to the majority of demand happening overnight when the average availability of all three grid locations is higher (see Figure 4.3). Critically, for the PHC and Pathlab scenarios, the unmet energy fraction is substantially higher than the proportion of time the grid is unavailable overall, meaning that blackouts tend to occur during periods of higher demand during daytime hours when the clinics are open.

These results suggest that in rural locations the grid is inadequate, especially for the daytime uses of the health centres. The high levels of unmet demand estimated suggest that for a more reliable supply, in the absence of national grid improvements users may require other forms of generation such as mini-grids or stand-alone systems.

Table 4.5: showing the results of estimated unmet energy for each demand scenario across three grid profiles. The time unavailable for each grid profile can be compared with the amount of unmet energy expected for each demand profile.

		Bhavaniyapur (Weak)	Nanpara (Moderate)	Nazipura (Strong)
	% Time unavailable:	47.8%	29.0%	4.5%
Scenario	Night Demand	Unmet energy fraction		
Domestic	74%	46.1%	26.6%	4.3%
Domestic + PHC	58%	48.6%	28.9%	4.8%
Domestic + Pathlab	43%	50.7%	31.9%	5.3%
PHC	32%	52.2%	32.4%	5.5%
Pathlab	28%	52.9%	34.4%	5.7%

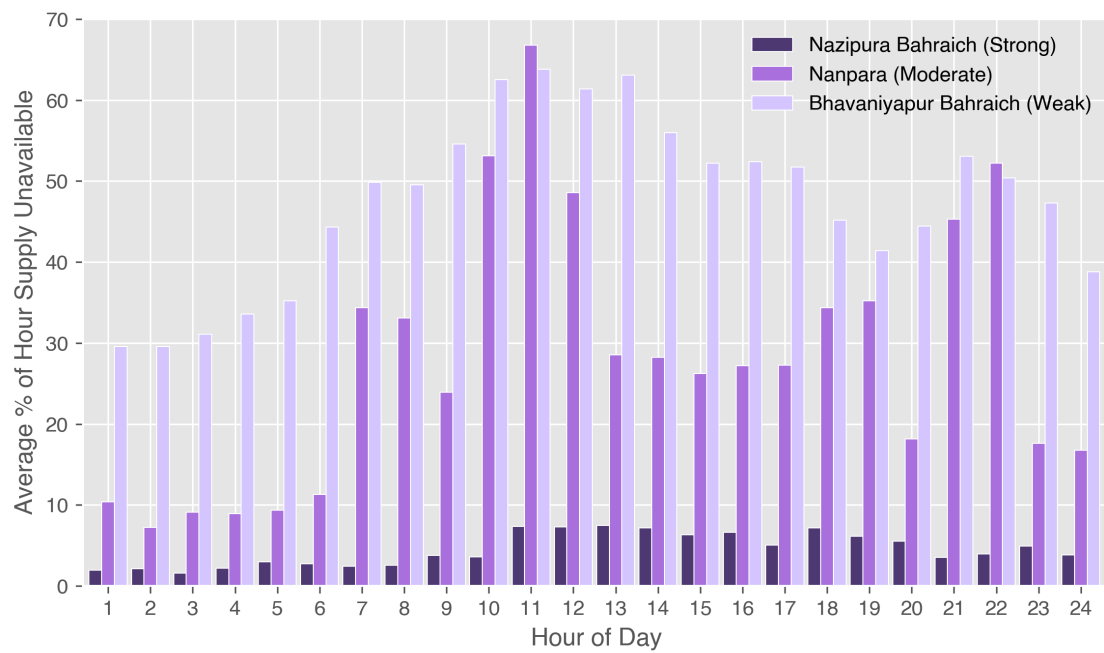


Figure 4.3: Average proportion of each hour of the day where electricity supply is unavailable for each of the three grid locations. For all three locations, a peak towards the middle of the day can be observed, with a further clear evening peak for the Moderate and Weak locations. Figure adapted from Beath et al.(2021) [18]

4.4.3 Cost and Emissions Implications of Health Centre Anchor Loads

This section outlines the findings on the cost and emissions of electricity for PV with battery systems as well as diesel only systems that meet 95% of energy demand.

PV and Battery Systems

Here I compare optimised mini-grid systems incorporating solar PV and batteries (without diesel backup) to meet domestic and health centre loads alone and in combination. Relative to the Domestic scenario, the Domestic + PHC and Domestic + Pathlab scenarios achieve a 30% and 53% reduction in LCUE, respectively. When comparing the individual health centre scenarios (PHC or Pathlab) to the scenarios combined with domestic demand (Domestic + PHC or Domestic + Pathlab), the results are mixed. There is a substantial reduction in the LCUE of 35% for the PHC, but a smaller reduction of 2.5% for the Pathlab (see Table 4.6). This may be explained by the increased fraction of dumped energy that the Domestic + Pathlab scenario has compared to the Pathlab scenario due to the higher fraction of night demand, which offsets some of the cost savings from the economies of scale of larger systems. Additionally, the health centre-only scenarios are considered without the distribution network for domestic users, and therefore have a reduced capital equipment cost, which subsequently reduces their LCUE.

The emissions intensity of energy is lower for the Domestic + PHC (by 5.6%) and Domestic + Pathlab (by 19.3%) scenarios compared with the Domestic only scenario. However, when comparing the respective combined scenarios with the PHC or Pathlab only scenarios, there are greater emissions intensities (of 21.9% and 5.3% respectively). There is an association of an increased fraction of dumped electricity with greater emissions intensity, which indicates why the health

centre-only scenarios have lower emissions intensities.

Table 4.6: Results summary for PV and Battery systems optimising for cost with 95% of energy demand met

	PV Size (kWp)	Battery Size (kWh)	Dumped energy %	LCUE (\$/kWh)	Emissions Intensity (gCO ₂ /kWh)
Domestic	2.1	12	15.0%	1.35	120.44
Domestic + PHC	3.3	17	11.7%	0.95	113.68
Domestic + Pathlab	6	20	10.5%	0.63	97.23
PHC	1.2	5	9.6%	1.45	93.23
Pathlab	3.9	10	9.1%	0.65	92.38

In addition to optimising for cost (LCUE), I also optimised the PV and battery systems for emissions intensity of used electricity (EIUE). As it is unlikely a system would be optimised for emissions in reality, I only discuss this briefly. For all but the PHC scenario, the results showed system configurations that have larger battery capacities, but identical or slightly smaller PV sizes than the systems optimised for cost. The emissions intensity of energy is reduced by 12-16% and is accompanied by increases in LCUE (of 1-4%), representing a reasonably low increase in cost for a more substantial reduction in emissions intensity. The larger battery sizes accommodate greater degradation caused by their usage whilst still being able to supply electricity throughout the considered time period, negating the need for the battery to be replaced during the project's lifetime. For the PHC scenario, the optimal system size for both cost and emissions is the same, likely due to the step size (resolution) of the PV panel and battery units iterated over during the modelling process and the smaller system size.

Mirroring the results from the cost-optimal systems, the combined scenarios have lower emissions when compared to the domestic only scenario; however, emissions increase for the health centres when their load is combined with the domestic mini-grid.

Diesel Systems

Table 4.7 shows the results of the different demand scenarios at 95% of demand met solely using diesel generation. Notably, when comparing with the cost-optimal PV and battery systems, the diesel-powered systems have 14.3 – 22.1% higher LCUE, and the emissions intensities are in the order of magnitude of ten times higher than the emissions optimal PV and battery systems (see Figure 4.4).

Table 4.7: Results summary for diesel-only systems optimising for cost with 95% of energy demand met

	Diesel Capacity (kW)	Avg. Capacity Factor	Fuel efficiency litres/kWh	LCUE (\$/kWh)	Emissions Intensity (gCO ₂ /kWh)
Domestic	2	18.5%	0.46	1.55	1259.96
Domestic + PHC	2	30.5%	0.32	0.97	858.92
Domestic + Pathlab	4	27.6%	0.34	0.75	917.55
PHC	2	15.1%	0.56	1.81	1539.47
Pathlab	3	39.9%	0.29	0.74	794.35

Apart from the Pathlab scenario, the same trend is observed with the PV and battery scenarios: lower costs for the combined scenarios (Domestic + PHC, and Domestic + Pathlab) than for the

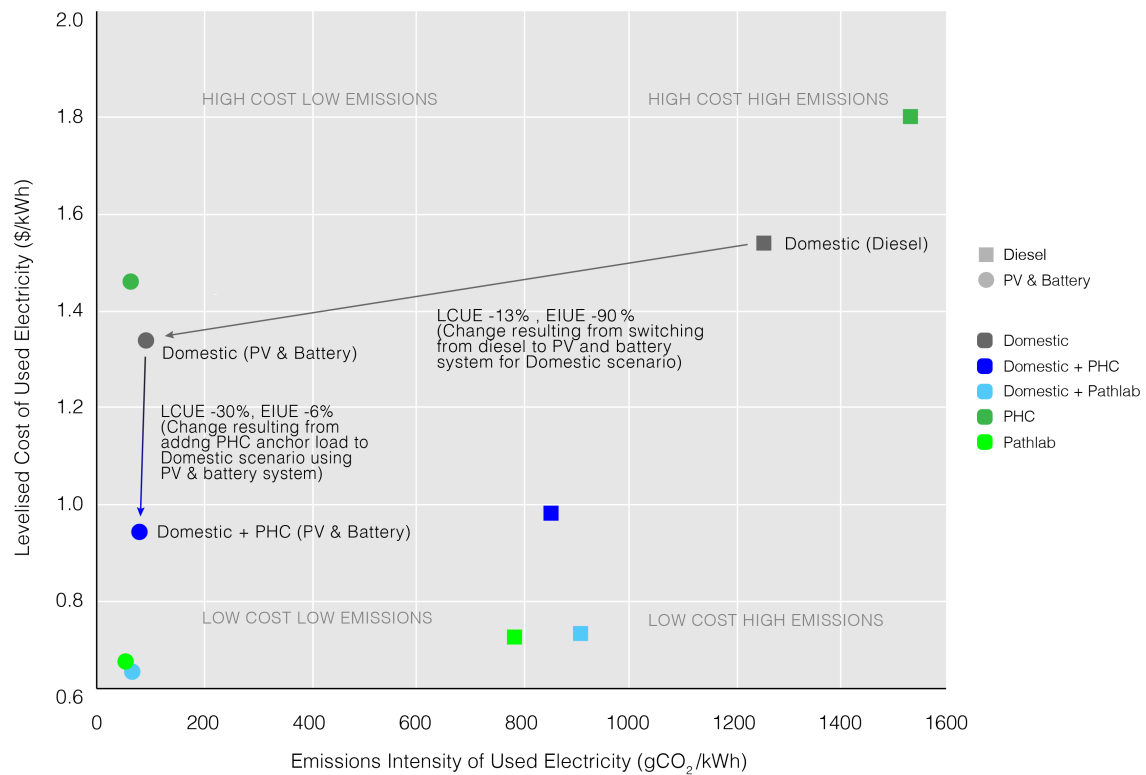


Figure 4.4: compares the levelised cost of used electricity (\$/kWh) and emissions intensity of used electricity for PV and diesel systems for each demand scenario at 95% of energy met. The figure highlights the change in LCUE and EIUE from switching from diesel to PV for the domestic scenario, and from integrating the PHC as an additional load to the domestic mini-grid. Figure adapted from Beath et al.(2021) [18].

individual scenarios (Domestic, PHC). An exception to this rule is the Domestic + Pathlab scenario, which is marginally more expensive (<1%) than the Pathlab alone. Broadly, for diesel systems, the scenarios with greater overall demand have both higher operating fuel efficiency and higher generator capacity utilisation, which are associated with lower LCUEs and EIUEs.

Diesel costs may change due to global oil price fluctuations therefore I tested how the cost competitiveness of diesel-only systems may shift with a doubling and halving of the diesel fuel price. Under a doubling of the fuel price, the diesel-only systems are 16 to 77% more expensive than PV-based ones, with the largest impact seen for the PHC scenario. With fuel prices halved, the picture is mixed. PV and battery systems remain at a lower cost for the PHC and Domestic scenarios (15 and 2% lower respectively). However, for the Pathlab, Domestic + Pathlab and Domestic + PHC scenarios diesel becomes lower cost (3%, 6% and 26% lower, respectively).

4.4.4 Sensitivity of Results to Increased Energy Demand Met

In a healthcare context, it is particularly important to provide a reliable electricity supply. This section presents the cost and emissions of systems that meet 100% of the energy demanded over the system lifetime (see Figure 4.5). These are compared to systems already presented meeting 95% of demanded energy.

PV and Battery Systems

Table 4.8: Results summary for PV & Battery systems optimising for cost with 100% energy demand met

	PV Size (kWp)	Battery Size (kWh)	Dumped energy %	LCUE (\$/kWh)	Emissions Intensity (gCO ₂ /kWh)
Domestic	3.9	18	42.6%	1.55	158.32
Domestic + PHC	6.3	26	41.2%	1.16	153.49
Domestic + Pathlab	9.9	40	37.2%	0.82	132.09
PHC	2.1	8	38.2%	1.60	142.71
Pathlab	6.3	22	35.7%	0.82	126.67

Optimising PV and battery systems for cost which could hypothetically meet 100% of energy demand led to significant increases in the installed capacity of both batteries and PV panels for all scenarios, and LCUE figures that were 10.7 to 30.2% higher than the cost-optimal systems meeting 95% of energy demand (see Figure 4.6).

Still, either health centre acting as an anchor load has beneficial cost impacts: the combined scenarios are cheaper in both cases than the domestic-only mini-grid, with lower emissions. The combined Domestic + PHC scenario has also a lower cost than the PHC alone although with higher emissions. However, the combined Domestic + Pathlab scenario sees marginally higher LCUE when compared to the Pathlab alone and with a higher EIUE.

As for the 95% reliable systems, I also optimised 100% reliable systems for emissions intensity rather than cost. In all five scenarios with 100% of energy demand met, the emissions optimal systems have smaller PV panels, larger batteries, and increased costs but lower emissions than the cost optimal systems. The increases in cost for all scenarios are relatively small (from 0.1 to 3.3%), with reductions in emissions of between 1.2 and 6.6% when compared to the cost-optimal PV and battery systems. This again highlights the potential for emissions reductions at low cost.

The results relating to anchor load benefits are the same as the cost-optimal findings, with reductions in emissions and cost for the combined scenarios when compared to the Domestic scenario. The PHC also benefits from integration into the mini-grid, with substantially lower costs but emissions increase marginally. Both cost and emissions increase for the Pathlab when integrated into the mini-grid, albeit by relatively small amounts: 1.3% and 5.3% respectively.

Diesel Systems

When using solely diesel generation at 100% of demand being met, the LCUE figures are almost the same as for the 95% case (see Figure 4.6). The combined scenarios have the lowest costs of all five scenarios suggesting anchor load benefits with diesel when meeting 100% of energy demand (see Table 8).

The combined scenarios see reductions in emissions against their respective individual scenarios except for a small increase of 3.1% for the combined Domestic + Pathlab scenario when compared to the Pathlab scenario.

The results are mixed when comparing the LCUE figures with the cost-optimal PV and Battery systems for 100% of energy being met (see Figure 4.5). For the Domestic-only scenario and the PHC scenario using solely diesel is more expensive, by 0.6% and 13.5%, than the PV mini-grid respectively. These scenarios have the poorest diesel-generating capacity utilisation and operating

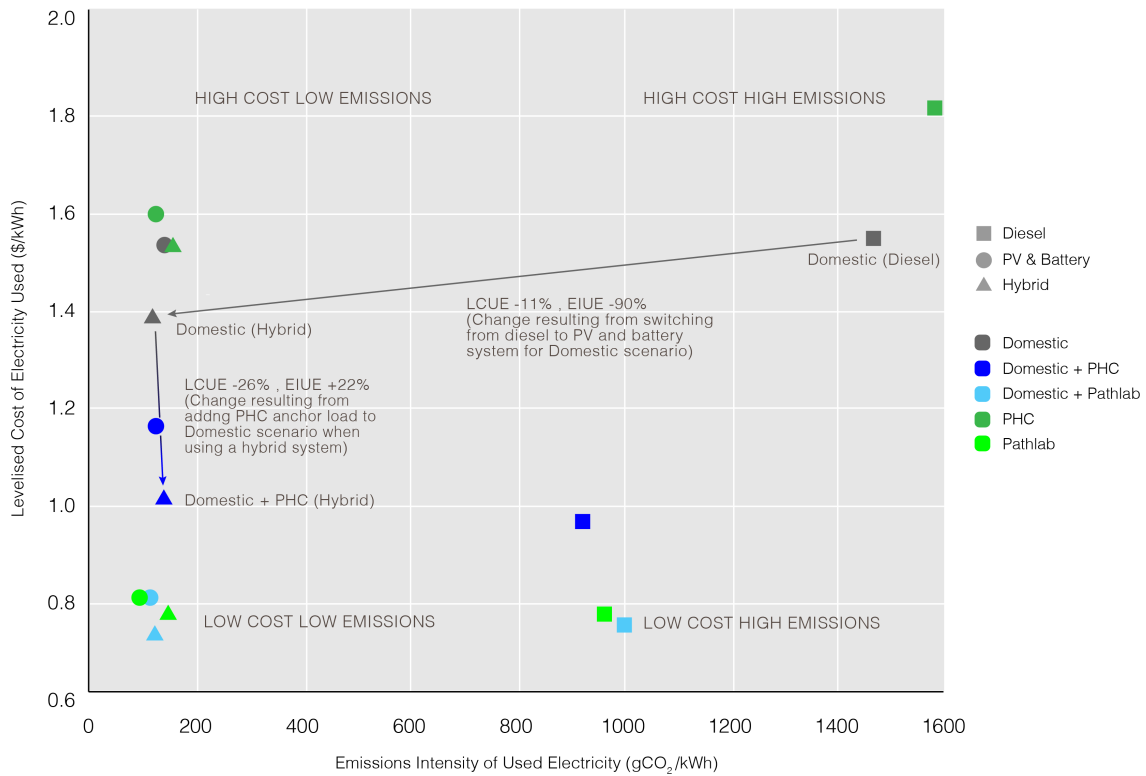


Figure 4.5: compares the levelised cost of used electricity (\$/kWh) and emissions intensity of used electricity for cost-optimal PV and battery, diesel and hybrid PV, battery and diesel systems for each demand scenario at 100% of energy met. The annotations highlight the change in LCUE and EIUE when hybrid technology is used over diesel for the domestic scenario; and when the PHC is added as an anchor load to the domestic scenario for a hybrid system. Figure adapted from Beath et al.(2021) [18]

efficiency. For the remaining scenarios, the diesel system is lower cost than the cost-optimal PV and battery-only systems at 100% energy satisfaction: 16.9% lower for the Domestic + PHC scenario, 7.9% lower for the Domestic + Pathlab scenario and finally a 4% reduction for the Pathlab scenario. As would be expected, the EIUE values are far greater for all systems using diesel only than PV and battery systems at 100% of energy being met.

Table 4.9: Results summary for diesel-only systems optimising for cost with 100% of energy demand met

	Diesel Capacity (kW)	Avg. Capacity Factor	Fuel efficiency litres/kWh	LCUE (\$/kWh)	Emissions Intensity (gCO₂/kWh)
Domestic	2	16.0%	0.53	1.56	1438.39
Domestic + PHC	2	27.5%	0.34	0.96	933.26
Domestic + Pathlab	4	25.1%	0.37	0.76	998.56
PHC	2	14.6%	0.58	1.82	1587.54
Pathlab	3	28.4%	0.36	0.78	968.77

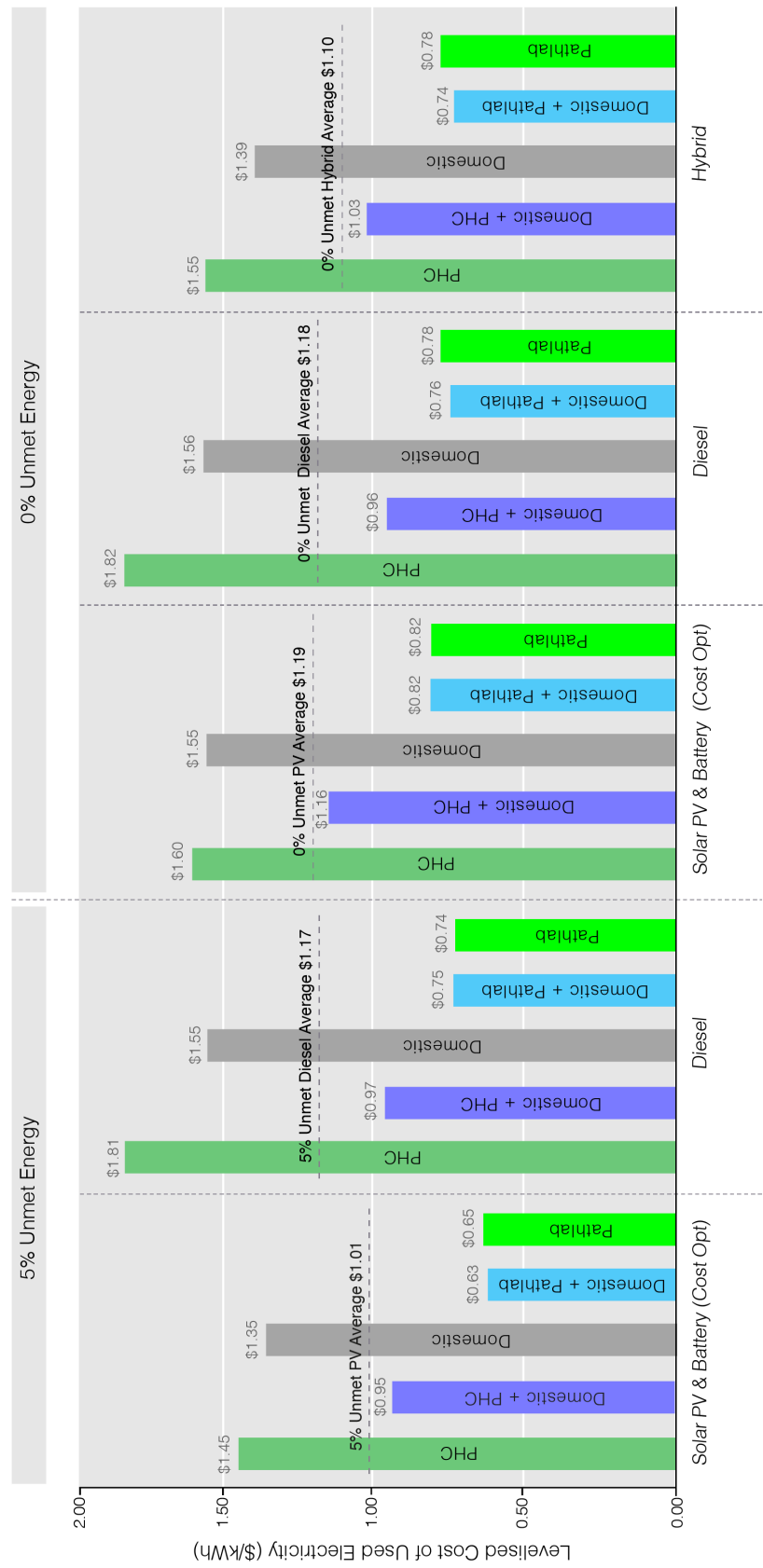


Figure 4.6: Compares the LCUE values for each demand scenario with each cost-optimised system configuration at both levels of reliability.

4.4.5 Cost and Emissions of 100% Reliable Hybrid PV, Battery and Diesel Systems

Finally, I investigated using diesel to meet the last 5% of the demand unmet. I did this by taking the cost-optimal PV and battery systems that already meet 95% of energy demand, as described in Section 4.4.3, and added diesel generation for the remaining demand to give fully reliable systems. The results show (see Figure 4.5) that for all scenarios the LCUE is lower than the exclusively PV and battery systems providing 100% of energy demand. For three scenarios, Domestic, Domestic + Pathlab and PHC the hybrid systems are cheaper than exclusively diesel systems for meeting 100% of demanded energy. For the Pathlab scenario, the LCUE figures are the same, with a higher figure returned for the Domestic + PHC scenario. Additionally, for the PHC, Domestic and Domestic + Pathlab scenarios the hybrid system is cheaper at 100% than a diesel-only system at 95% of energy demand met (see Figure 4.6).

In all cases, the combined scenarios are cheaper than the individual ones. The Domestic + PHC scenario is 26.2% and 33.8% cheaper than the Domestic and PHC ones respectively. The Domestic + Pathlab scenario is 46.7% than the Domestic scenario and 4.6% than the Pathlab scenario. The results for emissions intensity are mixed, with the hybrid systems having in most cases similar but slightly higher emissions of electricity used than solely the PV and battery systems. The emissions intensities are much lower than for the diesel-only systems.

Table 4.10: Results summary for hybrid systems optimising for cost with 100% of energy demand met

Scenario	PV Size (kWp)*	Battery Size (kWh)*	Diesel Capacity (kW)	Avg. Capacity Factor	Fuel efficiency (l/kWh)	Dumped energy fraction*	LCUE (\$/kWh)	Emissions Intensity (gCO ₂ /kWh)
Domestic	2.1	12.0	1	34.1%	0.33	15.1%	1.39	142.35
Domestic + PHC	3.3	17.0	2	24.4%	0.38	11.7%	1.03	150.96
Domestic + Pathlab	6.0	20.0	4	14.4%	0.59	10.5%	0.74	173.52
PHC	1.2	5.0	1	18.3%	0.47	9.6%	1.55	164.11
Pathlab	3.9	10.0	3	13.8%	0.63	9.1%	0.78	182.66

*These values are from the 95% energy met cost optimal PV systems.

4.5 Discussion of Findings

Integrating Anchor Loads to Domestic Mini-grids

At a 95% of demand met threshold the results presented in this chapter indicate that there are cost and emissions benefits for health centre anchor loads regardless of the generating technology used. The benefits are clearest when comparing a mini-grid meeting only domestic demand to one meeting both domestic and health centre demands: across all the assessed scenarios, LCUE reduces between 23% and 38% when adding the PHC, and between 46% and 53% when adding the Pathlab. When comparing health centre systems with the respective combined domestic and health centre mini-grid, there are significant reductions in LCUE for the PHC but no or only marginal gains for the Pathlab. In other words, the results suggest that whilst it is always beneficial for the domestic mini-grid to integrate with one of the two health centres, there is far less of a cost incentive for the larger health centre (Pathlab) to connect to surrounding domestic loads than the smaller health centre (PHC).

When using PV and batteries without backup generation, the emissions intensity of electricity

used across the demand scenarios sits at 92 – 120gCO₂/kWh for systems capable of meeting 95% of energy demand. This is substantially lower than that reported for the Indian national grid, which has an emissions intensity of approximately 700gCO₂/kWh [242], and diesel generation, which ranges from 794-1540gCO₂/kWh. Although it is more likely in practice that an off-grid PV system would be optimised for cost rather than emissions, my results suggest that optimising off-grid systems for emissions can achieve noteworthy reductions in the emissions intensity of electricity used (12-16%) with only small cost increases (1-4%), when compared to cost-optimal systems.

At 95% of demand met, my analysis finds diesel generation to be between 14.3% and 22.1% more expensive than PV across all scenarios, demonstrating that PV and battery technology can be a lower-cost option than diesel. The diesel price used was the average of one year of daily prices from Delhi, the nearest major city for which reliable price data over a long period of time was available. In reality, the cost of diesel at the mini-grid site may be higher when considering transport costs to a more remote location, therefore possibly making diesel even less competitive at 95% of energy demand met. The LCUE for PV systems are lower than diesel reflecting lower lifetime averaged costs; however, the upfront capital cost of equipment is often the primary challenge for installing off-grid energy infrastructure and this is still lower for diesel.

When testing the sensitivity of the results by changing the threshold to 100% of energy demand met, integrating a health centre remains broadly beneficial in terms of LCUE reductions, yet costs are 10.7 - 30.2% higher for solar PV and battery systems compared to systems meeting 95% of demand, indicating a significant premium for the final 5% of electricity demand. The dumped energy fraction of the 100% reliable PV systems is roughly three to four times greater than that of the 95% reliable ones, demonstrating the large quantity of poorly utilised capacity required to meet the last 5% of energy demand. A hybrid system where a diesel generator is used to meet the final 5% of demand that the PV and battery capacity does not meet is the cheapest way to reach 100% of energy demand met for all but one scenario investigated. This finding is unsurprising, with modern 3rd generation mini-grids typically utilising the same generation combination to achieve high levels of energy demand met and low costs [15]. Critically, when using a hybrid configuration, the anchor load effects are always beneficial in terms of cost, whether compared to the domestic or the health centre-only scenarios.

Comparing the Indian National Grid for Meeting Demand Scenarios

The national grid availability data and these results showing the estimated fraction of demand that would be unmet across different locations underline the fact that the grid in the region is far from adequate for supplying rural areas. In the ‘weak’ case, the closest and most representative of the case study site, the unmet energy proportion is at or above 50%, worsening for the scenarios including the health clinics. This demonstrates that to provide reliable high-quality electricity for both healthcare and domestic needs, either drastic improvements to the national grid or alternatives such as mini-grid systems are needed.

To improve the service level for rural grid customers to similar levels to those seen in the urban or ‘strong’ grid location, investment in improved infrastructure is needed either by strengthening the grid or through alternatives such as mini-grids, as is explored in this chapter. In 2019-20 the grid in Uttar Pradesh suffered from transmission and distribution losses of 22%, and an energy supply shortfall of 7%, up from 5% in the previous year. Improving service levels requires upgrading power lines as well as increasing generating capacity or implementing energy efficiency measures. The remote location of the case-study site and the fact that recent electricity demand increases in

the district of Bahraich are among the highest in the state, averaging 27% a year between 2016 and 2019, suggest that fairly significant investment is likely to be needed to improve service levels [243].

The grid electricity unit cost for metered customers is low, with a tariff at around \$0.11/kWh, varying somewhat for the type of consumer and the amount of electricity consumed [244]. However, electricity in India is subsidised: in Uttar Pradesh, there is around a 30% gap between revenue per unit and expenditure per unit, widening to around 50% for domestic customers [243], meaning that the cost of production is higher than the tariff rates paid by many consumers. Adding further complexity, more than half of customers in Uttar Pradesh do not have a meter and are therefore thought to pay a higher equivalent unit cost of up to \$1.13/kWh [14]. Whilst even the lowest cost mini-grid systems described here offer a unit cost of electricity considerably higher than the national grid, it is worth considering that the cost premium for more reliable electricity from the mini-grid compared to the grid may be outweighed by increased welfare that could result from the improved quality of electricity supply. For healthcare uses, previous research has demonstrated that a more reliable supply can reduce mortality [64] and boost health centre capabilities [10], potentially resulting in socioeconomic benefits to a host community.

Limitations and Further Research

This work is limited by the modelling framework used. For example, the hourly temporal resolution of the CLOVER model. As discussed in Section 4.3.1, some medical devices such as X-ray machines are used at high power in time periods shorter than an hour. Sub-hourly fluctuations in demand may have impacts on the system not accounted for by the CLOVER model and it is important to be aware of this. In reality, sub-hourly power requirements for such devices would need to be carefully considered for the design of any off-grid system. An additional limitation is with the model inputs. The prices of components and inputs were accurate at the time of completing the study; however, prices such as diesel prices fluctuate and may have changed substantially since the time of the study and thus the results could be different. For example, hybrid systems may no longer be the least cost for the highest levels of reliability demand due to increased diesel prices and lower PV and storage prices. The reader should be mindful of this limitation when considering the results. Finally, this work does not consider the practical requirements for integrating health centres into domestic mini-grids. Legal and regulatory issues, as well as community-specific factors, would likely present significant barriers to the implementation of health centres as anchor loads.

Further analysis should focus on the practicalities of implementing mini-grids with anchor loads, and the tariff structures required for feasible operation in different contexts. If the benefits identified in this chapter can be realised in practice, the application of such systems has the exciting potential to tackle poor levels of rural electricity access for both households and anchor users in India, as well as in other countries with electricity access challenges.

4.6 Summary

This chapter quantitatively assesses the impact of integrating health centres as anchor loads into a domestic mini-grid using solar, diesel and hybrid electricity generation. In addition, it explores the suitability of the national grid for meeting demand, to understand the necessity of off-grid systems for more reliable power.

Using monitored demand data, and detailed system and cost data, the modelling work presented

in this chapter demonstrates promising results for rural health centres to act as anchor loads in mini-grids. There are cost and emissions benefits at both levels of reliability investigated, and across generating technologies. Additionally, the modelling demonstrates that solar PV and battery technologies used in mini-grids are cost-competitive or cheaper than using diesel-only generators. At the highest level of service provision, hybrid systems whereby diesel supplies 5% of electricity are the lowest cost option, concurring with other previous studies. These findings indicate that for the study area, the national grid is not adequate for meeting either domestic or healthcare electricity demand, with poorer service levels for healthcare demand.

This chapter demonstrates the potential that anchor loads - in this case using the example of health centres - have in improving both the financial and environmental sustainability of mini-grids, through lower unit cost and greenhouse gas emissions intensity of electricity. Improving the economics of solar mini-grids will help to speed up their deployment and ensure they fulfil their potential role in providing high-quality, low-carbon access to electricity in countries still facing the persistent challenge of poor access.

In this work I have used a case study example of rural health centres acting as anchor loads in northern India; however, the findings have relevance beyond this case: other anchor loads with a similar pattern and share of demand within domestic solar mini-grids may produce similar benefits. As discussed previously, in Chapter 2, given the challenges faced by mini-grid operators: low demand, and low revenue per user, diversifying consumers e.g. to health centres may offer an opportunity for improved economics.

Chapter 5

Improving the Operational Performance of Hybrid Mini-grids

Acknowledgement of Contributions

This study was completed in conjunction with J Baranda Alonso and P Sandwell. J Baranda Alonso and P Sandwell completed the load profile construction and the tariff analysis presented in this chapter. J Baranda Alonso and P Sandwell wrote code for processing modelling results and plotting figures. J Baranda Alonso and P Sandwell produced written sections of the chapter's corresponding paper. R Mori advised on data inputs and study design. A Gambhir and J Nelson provided supervision and feedback for the study and suggested edits to the paper manuscript. The original CLOVER model (before modifications made) was written by P Sandwell.

Previous Presentation of this work

Hamish Beath, Javier Baranda Alonso, Richard Mori, Ajay Gambhir, Jenny Nelson, Philip Sandwell, Maximising the benefits of renewable energy infrastructure in displacement settings: Optimising the operation of a solar-hybrid mini-grid for institutional and business users in Mahama Refugee Camp, Rwanda, Renewable and Sustainable Energy Reviews, Volume 176, 2023, 113142, doi.org/10.1016/j.rser.2022.113142.

5.1 Overview

In off-grid contexts, diesel generation is commonly used to provide electricity. This has associated problems, such as high operating costs, and local and GHG emissions, as outlined in Section 2.5.4 of Chapter 2. As demonstrated in Chapter 4, solar PV and battery storage capacity working alongside diesel generators may be able to provide significant cost and emissions savings over operating diesel alone. A hybrid mini-grid - incorporating diesel, PV and batteries - can be configured in a variety of ways, depending on context-specific factors such as demand profile, and type of components used. The way the system is operated may impact costs via diesel usage or battery degradation and therefore ensuring that the operational strategy suits the context is important for ensuring a given system is economical.

Diesel generator-based electricity supply systems are widely used within humanitarian settings such as refugee camps. Humanitarian organisations typically rely on diesel generators to provide power for services in refugee camps, such as healthcare and camp administration, whilst camp residents often have no access to electricity. Increasingly, the use of solar and battery capacity used alongside diesel generators is being considered for refugee camps as a route to reducing expenditure on fuel and expanding electricity access within camps.

In this chapter, I use an adapted version of the CLOVER model (see section 3.3 in the methods chapter) and monitored electricity demand data to assess the savings made possible by the integration of solar PV (18.4 kWp) and battery (78 kWh) capacity into an existing 13kW diesel-powered mini-grid in Mahama Refugee Camp, Rwanda. Further, I assess alternative operating strategies from the one previously used at the camp to recommend an improved strategy with reduced fuel costs. Finally, I evaluate the impact of the extension of the existing users to include refugee businesses in the camp marketplace; to assess how additional demands (as was discussed in Chapter 4) may influence the operation of the system. This chapter then assesses how different tariff structures may recover additional costs incurred from higher demands and is presented here for reference.

The work presented in this chapter contributes novelty in several ways. Firstly, it presents a detailed case study of the use and optimisation of a hybrid mini-grid in a refugee camp setting, with usage data which is sparse in this context. Secondly, it recommends an approach to improving the usage of energy-generating resources in a refugee context and considers different demand components and how they may alter the outputs. Finally, as discussed in section 3.3, this work applies an open-source, customisable model that facilitates the modelling of hybrid mini-grids with dispatched diesel generation with applicability to other contexts where similar energy systems are in use.

5.2 Background

5.2.1 Operating hybrid mini-grids

For national electricity grids composed of many generators with different marginal costs of output, system operators will employ economic dispatch by meeting demand using the available generator(s) with the lowest marginal cost [245]. For a hybrid mini-grid with dispatched generation the aim will also be to meet demand at the lowest marginal cost but, in the absence of an active system operator, using a dispatch strategy with predetermined settings for how different electricity supply components are operated is typically used [246].

Dispatch strategies include “Load following” (LF), which is where the dispatched generator, typically diesel, is operated to provide the net remaining demand if the renewables and batteries are insufficient. Under this strategy, the generator is not used to charge the batteries, but only to meet the load directly [188]. Alternatively, “cycle charging” (CC) means that the generator is used to charge the batteries, and may also meet demand simultaneously. This can have operational benefits: high capacity factors, increased efficiency and therefore reduced fuel use per unit output, and the number of generator starts may also be lower, further reducing fuel consumption and maintenance costs [247]. CC can reduce the total hours that the generator runs, and can respect the desire for “quiet hours”, as is the case for the system described in this paper. However, CC increases the energy throughput of the battery, increasing degradation and representing a financial

burden if the batteries must be replaced sooner. Additionally, the strategy may result in renewable energy being dumped rather than stored if battery capacity is used to store dispatched generation instead [246]. Finally, a “combined dispatch” (CD) strategy, whereby the generator may operate either as load following and cycle charging under different conditions, is possible and aims to maximise possible benefits of both LF and CC strategies [188]. This requires a net load threshold to be calculated at which it is more costly to discharge the battery than run the diesel generator to meet net load. The system then switches between CC and LF modes during operation according to the electricity demand level.

For hybrid mini-grid operation that includes CC, important variables are the points at which the battery state of charge (SOC) is considered low enough that the dispatched generator will turn on, and high enough that it will turn off. These battery state of charge “set points” can significantly affect the lifetime cost of a system [248, 249].

Studies that investigate the improved operation of hybrid mini-grids are limited and have mixed findings. Arévalo-Cordero et al. [250] and Ramesh & Saini [251] conclude that a CD strategy offers the lowest cost of electricity, whilst others [188, 252–255] found the CC strategy to be the most economical. Das & Zaman [256] find almost no difference in the cost of energy between the three main dispatch strategies. Although Bernal-Agustín et al. [257] in their optimisation of size and operational strategy found the lowest unit cost of electricity to be from a CD strategy, the CC strategies investigated give the lowest diesel costs. Few studies adjust the battery SOC set points for operation that includes cycle charging, which limits their findings. For those that do [188, 246, 257], different optimal state of charge values are found, given the variation in storage technology used and the shape of the demand curve. These studies suggest the potential for a study into using a CC strategy to reduce diesel costs under a variety of SOC set points. They also suggest a dependence on the case study under investigation which given the lack of previous application of this research to displacement settings, highlights a research gap for this work.

5.2.2 Energy in Displacement Settings

More than 80 million people are estimated to be forcibly displaced for reasons such as conflict and violence in their place of origin, a number which has more than doubled from a decade ago [258]. If the number of people displaced globally continues to rise at the current rate then there will be more than a quarter of a billion people displaced by 2030 [259]. Extreme weather events caused by climate change are likely to trigger increased involuntary migration in future years [260], and more than 200 million people could be internally displaced by climate change alone by 2050 [261] causing increased pressure on humanitarian organisations. Protracted crises often lead to long-term displacement, averaging 10 years for refugees, with little change to this over the last three decades [262]. Consequently, temporary settlements can be occupied for many years and require appropriate infrastructure and services to function [263].

The vast majority of refugees are hosted in developing countries [258], and often in remote, geographically challenging environments where access to modern energy services is limited [264]. The Moving Energy Initiative estimated that as many as 20,000 displaced people die per year as a result of indoor air pollution caused by a lack of modern energy services, mainly for cooking [265]. A lack of electricity, meanwhile, has an impact on livelihoods as it limits the capacity of displaced people to carry out income-generating activities that may improve their quality of life [266, 267]. Coupled with the protracted nature of many displacement contexts, there is a need for long-term solutions to provide the benefits of clean energy access in these settings.

Recently, humanitarian organisations have developed targets to reduce their environmental footprint. UNHCR, for example, operates over 3,000 diesel generators powering vital institutional functions such as health clinics in refugee camps [268]: acknowledging the GHG emissions that result from this, UNHCR has introduced initiatives to increase the use of renewable energy and reduce its diesel consumption [269]. These high-level commitments can provide institutional recognition of the need for sustainable energy solutions, which can be vital in initiating implementation projects in the field.

Diesel usage in humanitarian settings is also a significant cost burden. Transporting diesel to geographically challenging locations means that the unit cost of electricity can be several times higher than the grid in a respective host county, and the humanitarian sector as a whole is estimated to spend \$400 million per year on diesel fuel for electricity generation [264, 270]. This expenditure is against a backdrop of funding difficulties for agencies working to support displaced people [271]. In 2020, for example, UNHCR faced a funding gap of 51% [272]. Assessing the potential cost savings of reducing the usage of diesel fuel can therefore provide evidence in support of implementing clean energy solutions.

Techno-economic studies offer insight into addressing the financial, environmental and humanitarian need for more sustainable electricity. Frack et al. [273] estimated a payback time of 5.4 years for a solar PV, wind and storage system that replaces diesel generation for a refugee camp in Juba, South Sudan. Neves et al. [274] calculated that incorporating solar PV and batteries with existing diesel yields a payback period of 3.4 years for Mantapala refugee camp in Zambia, with GHG emissions reduced by 55% compared with the baseline of diesel alone. Alonso et al. [221] estimated that completely displacing diesel usage at Nyabiheke camp in Rwanda would result in the initial investment in PV and battery equipment being paid back after 6.2 years. A hybrid diesel-solar system, with no storage, that reduces diesel usage by 32% was found to have a payback period of 0.9 years.

A recent study by Baldi et al. [275] applied a macro-level techno-economic modelling approach to assess the energy needs of displaced households, businesses, and institutions in 288 refugee settlements in Sub-Saharan Africa. Using this assumed electricity demand to design renewable energy solutions for each settlement, they found that the upper bound for the cumulative upfront cost of meeting the total demand would be \$1.34 billion and, by deploying PV mini-grids in place of diesel, could offset 2.86 MtCO_{2eq} over 20 years. The scale and scope of this study offers great insight into the cumulative potential of solar and hybrid mini-grids to provide power in displacement settings and, amongst its recommendations, stresses the importance of using reliable and representative data to inform future research.

Despite examples of PV reducing diesel usage in camps for displaced people, there are significant barriers to broader deployment. Most critically, short-term budgets and uncertainty over how long camps will exist impede the humanitarian sector in deploying more renewable generation [221, 265, 274]. Annual budget cycles prevent investment in the higher upfront costs of renewable energy technologies when compared to diesel, even when they could enable significant cost savings over the lifespan of a camp [274]. Involving the private sector can help to overcome some of the issues around long-term investments. There have been recent efforts to help to provide contractual certainty such as guarantee mechanisms to de-risk private sector investment [276, 277]. However, these are yet to be used at scale. Highlighting the opportunities to achieve greater profitability for private sector companies operating renewable energy systems in displacement settings may help to scale investment; these opportunities have not yet been well demonstrated.

A further barrier is the sparse data on the current energy sources and potential future energy demands of displaced people. This lack of data can inhibit the impact of energy interventions: a renewable energy system may be incorrectly sized, or connected only to a small number of key users to preserve reliability, but this could limit its benefits and potential to be extended to other users in the camp, for example, businesses or households [278].

5.3 Case Study: Rwanda and Mahama Refugee Camp

The Government of Rwanda has ambitious plans for achieving SDG 7. Its plan for universal electrification expects that around half of new connections are to be from off-grid solutions, including 10% from mini-grids, which have featured in its electrification strategies since 2004 [279]. Rwanda has a progressive attitude towards refugees: hosting around 127,000 displaced people as of August 2021 [280], the Government is an adopter of the New York Declaration for Refugees and the Comprehensive Refugee Response Framework (CRRF) [281, 282]. These international agreements acknowledge the protracted situations that many displaced people face and aim to integrate them into host community societies, offering a relatively conducive environment for long-term sustainable energy solutions.

In this Chapter, I assess the case of Mahama, which is the largest refugee camp in Rwanda. I investigate the mini-grid, which recently has solar and battery storage integrated into its system. Mahama Refugee Camp, located in Eastern Province, is home to approximately 46,000 refugees who have fled conflict in Burundi since 2015 [280]. The Ministry in Charge of Emergency Management (MINEMA) and UNHCR oversee the camp's administration and the protection of displaced people, whilst critical services are provided by NGO partners. The international non-profit humanitarian aid organisation Alight, formerly the American Refugee Committee (ARC), has operated in Rwanda since 1994 and is responsible for infrastructure in the camp which includes the provision of electricity services for institutional users, such as the health facility [283].

Before the upgrade of the electricity system, power was provided by diesel generators and was available to a restricted group of users including the camp administration offices and the health clinic, which provides primary healthcare services. The provision of electricity in the camp was the responsibility of Alight and the fuel costs were covered by UNHCR, with the users receiving electricity at no cost. MeshPower Ltd., a solar mini-grid company, was contracted by Alight to install PV and battery storage to hybridise the existing system. The capital expenditure was grant-funded by various donors with the intent of reducing fuel usage and costs. MeshPower operates more than 70 PV mini-grids around Rwanda serving domestic and commercial customers, and a smaller number of institutional users, via direct current (DC) connections for basic energy access and alternating current (AC) connections to support productive and more intensive electricity requirements [284]. In December 2019 MeshPower installed 18.4 kW_p of PV and 78 kWh of battery capacity at the Mahama health clinic to serve consumers via an AC distribution network (see Figure 5.1). For the present study, MeshPower Ltd. provided access to the data remotely monitored from the system, guidance on its technical and operational aspects and potential extension to refugee businesses, but had no influence on the results, conclusions or recommendations presented in this chapter. The electricity system is configured so that the solar capacity and diesel generator are used to meet the load, and also to charge the batteries. The diesel generator operates using a CC strategy, with the SOC set points dictating when the charging occurs depending on the time of day. Late at night, noise from the generator must be avoided where possible: during these times

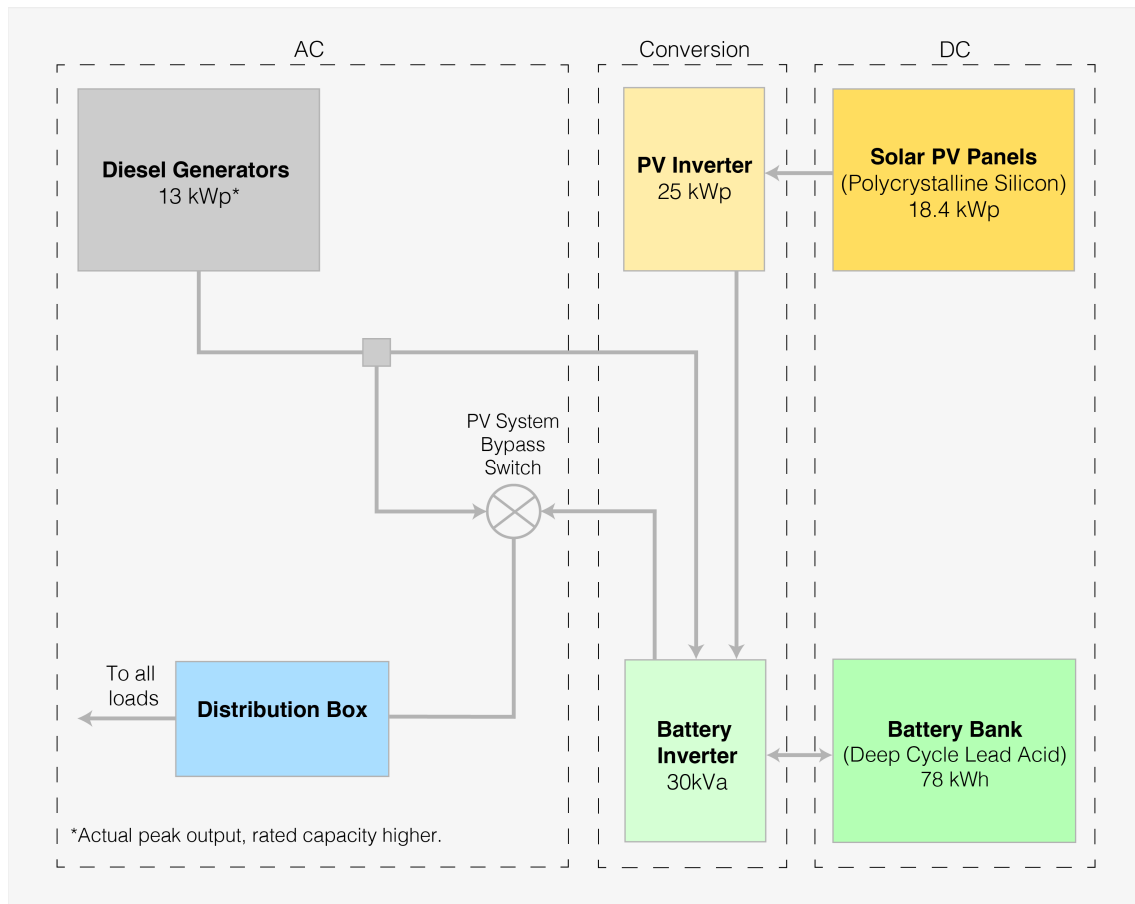


Figure 5.1: A simplified system diagram of the electricity system at Mahama Refugee Camp which shows the original diesel capacity (grey), and the solar PV (yellow) and battery storage (green) equipment installed by MeshPower Ltd. Arrows indicate the direction of energy flows.

the SOC set points are configured such that the generator is run only in exceptional circumstances when the state of charge of the battery is very low. Earlier in the evening, the settings ensure that (as required) the generator will charge the batteries to a higher SOC to provide electricity throughout the night. It is this requirement, to compress diesel generator operation into the earlier evening, that partly necessitates a CC strategy.

Since the integration of the solar and battery capacity into the system in December 2019 the electricity demand has increased to accommodate additional loads in the health centre (including a recently constructed maternity centre), connections for further offices and the camp police station, and new street lighting in the area. In addition, refugee businesses in the nearby marketplace will be connected to the system and supplied with AC power on a commercial basis under a tariff structure, with any additional system fuel costs being covered by MeshPower.

In this chapter, I consider the different objectives of each of the key stakeholders: reducing fuel costs and GHG emissions (Alight), ensuring system reliability and optimal performance (MeshPower), providing improved opportunities for livelihoods in the camp (UNHCR, MINEMA and refugee entrepreneurs), and learning how to improve the design or operation of hybrid electricity systems in other situations of displacement (academics and the wider practitioner community).

5.4 Methods

A summary of the methods used to produce the results in this chapter is listed below. The primary objective of the study was to reduce the fuel usage of the diesel generator (particularly overnight) by increasing the utilisation of the solar and battery storage capacity, whilst maintaining the adequate performance of the system. The secondary objective was to quantify the additional impacts of providing electricity for refugee businesses. A summary of the methodological process followed is shown in Figure 5.2 and the methods are covered in more detail in the following sections:

- Section 5.4.1: Description of how demand data collected from the Mahama Camp system was analysed and business demand data generated.
- Section 5.4.2: Summarises the CLOVER modifications made to emulate the system operation and describes how the performance, cost and emissions impacts were quantified
- Section 5.4.3 Describes the assessment of the impact of different numbers of refugee businesses connecting to the system.

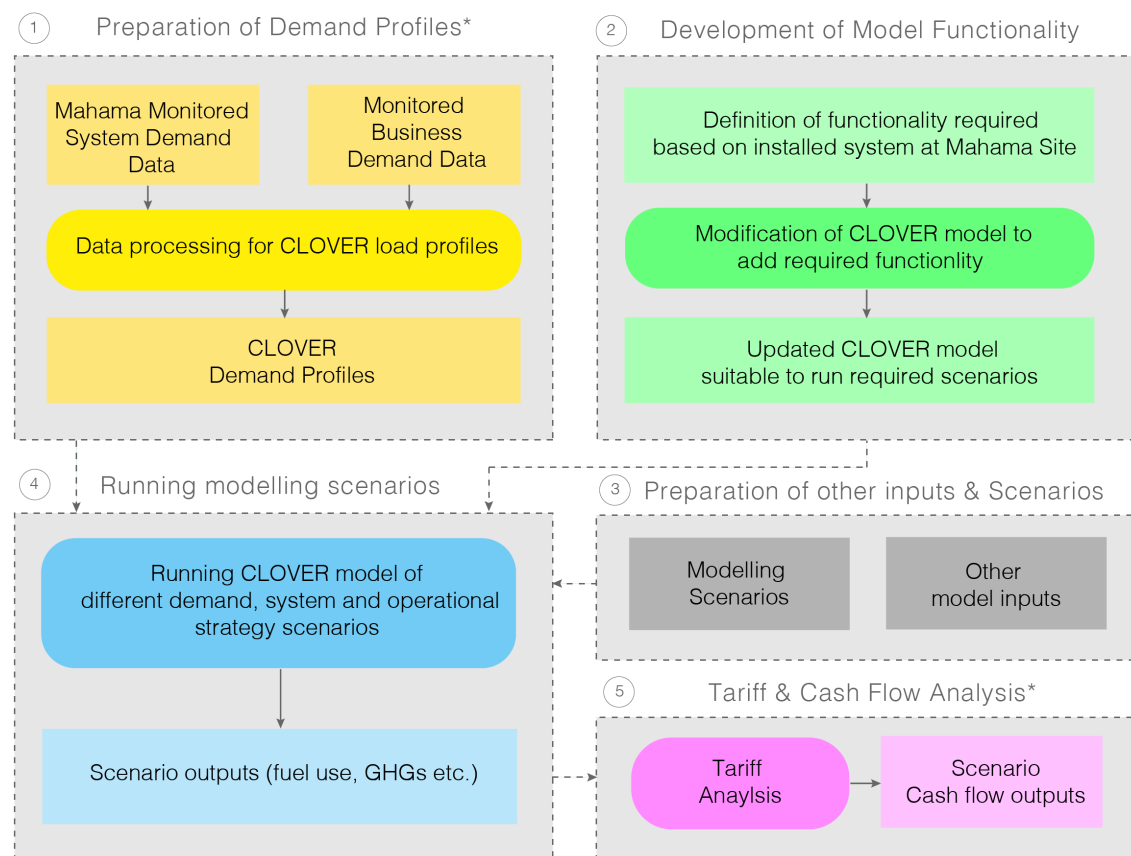


Figure 5.2: A high-level flow diagram explaining the main methodological components for the analysis in this chapter; * denotes parts that were undertaken by co-investigators on this project (see below)

5.4.1 Demand Data

Analysing data from the Mahama system

Electricity usage and operational data from the Mahama system are remotely monitored and stored via the Victron Remote Management (VRM) energy monitoring platform. This is a commercial platform compatible with the components used on the system such as the battery management system. Data is recorded at a minute or hour resolution, with the latter used in this analysis for congruence with the modelling approach. The data is segregated by the energy source (diesel generator, PV or battery storage) and destination (consumers or battery charging).

The profile of the institutional electricity demand was prepared by a co-investigator, Dr Philip Sandwell, who used the total load supplied by the system from all sources, which was monitored from 1 January 2020 to 31 December 2020, to synthesise the load profile for institutional users. Whilst this monitoring period was during the COVID-19 pandemic, this did not materially impact demand due to the self-contained nature of the camp and the institutional activities at the site [285]. First, he derived the mean and standard deviation of the monitored data for each hour of the day. To synthesise a new load profile as an input for the modelling process, he selected values from a normal distribution for the corresponding hour of the day for each hour of a five-year period. This yielded a profile which emulates the hourly variation of the original monitored data but is seasonally invariant. More detail of this process is found for reference within the Appendix Section B.1.

Load profiles of refugee businesses

In addition to the load profile of the institutional energy users in Mahama Camp, I also consider the additional loads of refugee-owned and -operated businesses in the nearby camp marketplace. I do this by considering several cumulative load profiles for a total of N business connections, which allows the evaluation of the impact of extending electricity access at different scales. This part of the load profile preparation was completed by Dr Philip Sandwell and Javier Baranda Alonso and a full description of how the business load profiles were constructed is available for reference in the appendix (B.1).

The above two co-investigators used a qualitative survey of 142 refugee businesses, conducted in Mahama in January 2019, to provide an indication of the types of businesses operating in the camp [286]. This allowed them to estimate the existing distribution of business types and those to be modelled in this work. Monitored electricity use data from refugee businesses in the camp was not available, as they did not have connections to monitor, and so to derive the load profiles of the businesses they used data from 19 businesses provided by MeshPower Ltd. These business customers are served by a mini-grid operated by MeshPower in the village of Gitaraga, situated in a rural area close to of the Rwandan capital Kigali. Businesses in this site are similar to those found in the camp and in other displacement situations [267, 287]. The energy usage has been monitored by MeshPower as part of their usual business operations and is considered representative of both refugee businesses and future MeshPower customers. Data from the Gitaraga site was processed similarly to that in Section 5.4.1 to calculate the hourly means and standard deviations of electricity use of each business type [204].

Table 5.1 gives a breakdown of the types of businesses operating in the Mahama marketplace found by the surveys (“Survey %”) and the simplified proportions used in the model to ensure integer numbers of businesses (“Modelling %”). The proportions of shops and money transfer agents were

reduced slightly in favour of an increase in the proportion of workshops (also including phone and device repairs), cinemas and tailors to better reflect the potential makeup of a marketplace with increased access to electricity.

Table 5.1: The breakdown of the businesses in the marketplace in Mahama Camp found by the surveys (“Survey %”), the simplified proportions used in this work (“Modelling %”), and a description of their operations.

Business Type	Survey %	Modelling %	Description
Bar/Restaurant	48	50	Food, refreshments and entertainment
Shop	24	15	Sells food and small items
Tailor	7	10	Clothes production and repair
Hair salon	8	10	Hairdressing and barber services
Device repair	4	–	Electronic and device repair services
Money transfer	7	–	Accounting and money transfer services
Cinema	2	5	Television, football matches and films
Workshop	–	10	Welding and device repair services

I consider the synthesised system load profile with low, medium and high levels of business connections, corresponding to $N \in \{20, 50, 100\}$. In alignment with the UNHCR goal of providing Tier 2 electricity access to displaced households, the magnitudes of the synthesised load profiles were scaled to provide 200 Wh per day for each business user whilst the relative temporal variation remained the same. This may overestimate the energy use of businesses, as those in Gitaraga regularly used less than this amount [204], but nonetheless represents the energy requirements necessary to meet this Tier 2 goal.

5.4.2 Modelling diesel generation

To ensure the usefulness and accuracy and reproducibility of results produced, software used in electricity access research should, 1) model real-life electricity systems faithfully; should 2) have an open-source code base available to others to replicate the study, and should 3) be able to optimise for desired parameters. Following these principles, I adapted the CLOVER model to add new functionality required for the results presented in this chapter. These modifications enabled me to satisfy the first criterion, with the latter two already met by the existing modelling framework. Making these changes allowed me to simulate and explore the impacts of many different operational strategies before implementing any changes in practice, avoiding causing potential outages or disruption on the real system.

The new model functionality means that the use of the diesel generator can be modelled using a CC dispatch strategy and that the specific SOC set points determining whether the generator will run can vary in different parts of the day. A detailed description of these changes is given in Section 3.3 of the methods chapter.

I modelled several scenarios to assess the impact of a) the inclusion of solar and battery capacity, b) the optimisation of the hybrid mini-grid and c) the inclusion of refugee businesses. Firstly, I modelled the previous diesel-only system to provide a baseline to compare cost and emissions reductions. Secondly, I modelled the inclusion of solar PV and battery storage capacity to observe expected reductions in fuel use. Next, I modelled a range of different operational strategies for the hybrid system to select an option with further reductions in fuel use. Finally, I modelled the inclusion of different numbers of refugee business connections under a range of operational strategies to see the impact of additional loads and which strategy is most advantageous.

Where possible I gathered model inputs that were for the actual system from MeshPower and for others representative inputs were taken from the literature. The technical, financial and emissions inputs are provided in Appendix B in tables B.1, B.2 and B.3 respectively.

System performance and impacts

The primary financial metric I use is the net present cost (NPC, in 2020 US dollars) and, in particular, the differences in NPC, C^T , as a result of differences in fuel costs in systems with different operational strategies. The NPC, C^T is the sum of all costs (in 2020 US dollars), C_n^j of type j (for example equipment, fuel, maintenance), occurring in year n of the considered lifetime of the system, N years, and subject to the discount rate r :

$$C^T = \sum_n^N \left(\sum_j \frac{C_n^j}{(1+r)^n} \right) \quad (5.1)$$

Similarly, the main environmental metric used is the total GHG emissions (kgCO_{2eq}/kWh) ξ^T given by the sum of the emissions from each component j in year n , defined by

$$\xi^T = \sum_n^N \left(\sum_j \xi_n^j \right) \quad (5.2)$$

Both C^T and ξ^T are primarily impacted by diesel fuel and operation and maintenance requirements. The equipment costs for the generation and storage capacity have already been covered before the considered lifetime begins, which offers a fair analysis given the consistent comparison to the same reference system.

In addition and consistent with the analysis in Chapter 4, I use the per-unit impacts of electricity with the levelised cost of used electricity (LCUE, \$/kWh). This measure is similar to the levelised cost of electricity (LCOE) but explicitly accounts only for electricity used by customers, rather than unused over-generation from PV [114, 205]. I also calculate the emissions intensity of used electricity (EIUE), measured in kgCO_{2eq}/kWh, please see section 3.2.4 of the methods chapter for a mathematical description of these metrics.

Battery degradation is assessed as the amount of energy in kWh that has been supplied by the battery as a proportion of the total that would be expected during its lifetime (dependent on the number of cycles at a given depth of discharge) [288–290]. As battery capacity degrades over time this measure, h , scales linearly from $h = 0\%$ when the battery is new to $h = 100\%$ when the battery has degraded to 80% of its original capacity, defined to be the end of its useful life. Finally, I assess the overall performance of the diesel generator via the average probability that it is in use during a given hour of the day $\rho^D(t)$ and its average load factor Γ . Mathematical descriptions of these metrics are provided in Appendix (B.2). The financial and GHG emissions inputs used for this study are also provided in the appendix, in Tables B.2 and B.3.

Identifying the optimum operational strategies

To compare a range of alternative strategies I altered two key criteria: the SOC set points at which the diesel generator is used, and the times at which these are in effect. I modelled the same system over the same five-year period but with five depths of discharge: in the evening these ranged from

a switch-on SOC of 50% to 70%, and in the daytime from 30% to 65%, each in 5% increments. For each set of SOC parameters, I varied the times of the more restrictive evening period, with start times from 17:00 to 21:00 and end times from 22:00 to 04:00 (presently 18:00-02:00 for the original strategy). Each combination of start and end times was modelled for each set of SOC parameters. Including the original SOC parameters (90% and 65%) this totalled 260 simulations.

From these simulations, I selected an optimum “alternative strategy” which both reduces costs and avoids frequently reaching a low SOC, defined as below 40% occurring more than once per week (14.3% of days), and never below 30%. This was informed by the operational advice provided by MeshPower Ltd.

5.4.3 Assessing the impacts of connecting business customers

Finally, I assessed the potential impacts on the system of connecting 20, 50 or 100 refugee businesses. I looked at the increase in costs and GHGs, relative to those for the institutional loads only, under the alternative operating strategy, the original strategy for comparison and finally under a third strategy adapted to suit the new load profile.

The final part of the analysis, regarding tariff structures for business consumers, was carried about by my co-investigators Dr Philip Sandwell and Javier Baranda Alonso. They applied each of four potential tariff structures which are assumed to be paid by the businesses:

1. The current grid tariff in Rwanda of 89 RWF/kWh (\$0.09/kWh) for consumers using 0-15 kWh/month,
2. A representative mini-grid tariff with a daily fee of 30 RWF (\$0.03) plus usage tariffs of 400 RWF/kWh (\$0.40/kWh) in the daytime and 600 RWF/kWh (\$0.60/kWh) at night,
3. A usage-based tariff that will break even against the increase in diesel costs resulting from the business connections,
4. A daily fee-based tariff that will break even against the increase in diesel costs.

Tariffs 1 and 2 would offer the greatest parity with the host community, depending on whether they are connected to the national grid or a privately operated mini-grid. Tariff 2 is similar to those used by the mini-grid operator MeshPower Ltd. in other rural communities and from which the demand profiles were constructed. Tariffs 3 and 4 are calculated to exactly offset the additional fuel costs incurred by the business users. These allow a simple consideration of the minimum tariff required for the operator to break even.

5.5 Results of Analysis

5.5.1 Input data and load profiles

Institutional and business loads

The energy demand monitored from the existing system serving institutional loads is shown in Figure 5.3 for the period from 1 January to 31 December 2020. The daily total energy demand is shown in Figure 5.3 a), with both the monitored sum (translucent) and the five-day moving average (solid). Two breaks in data collection occurred as a result of minor system faults during which time the system was non-operational and so data was not recorded, resulting in 1.9% downtime. Total day-to-day demand is relatively consistent throughout, at 70.1 ± 10.0 kWh per day, but with

a maximum daily demand of 96.9 kWh and a minimum of 46.7 kWh recorded during the one-year period.

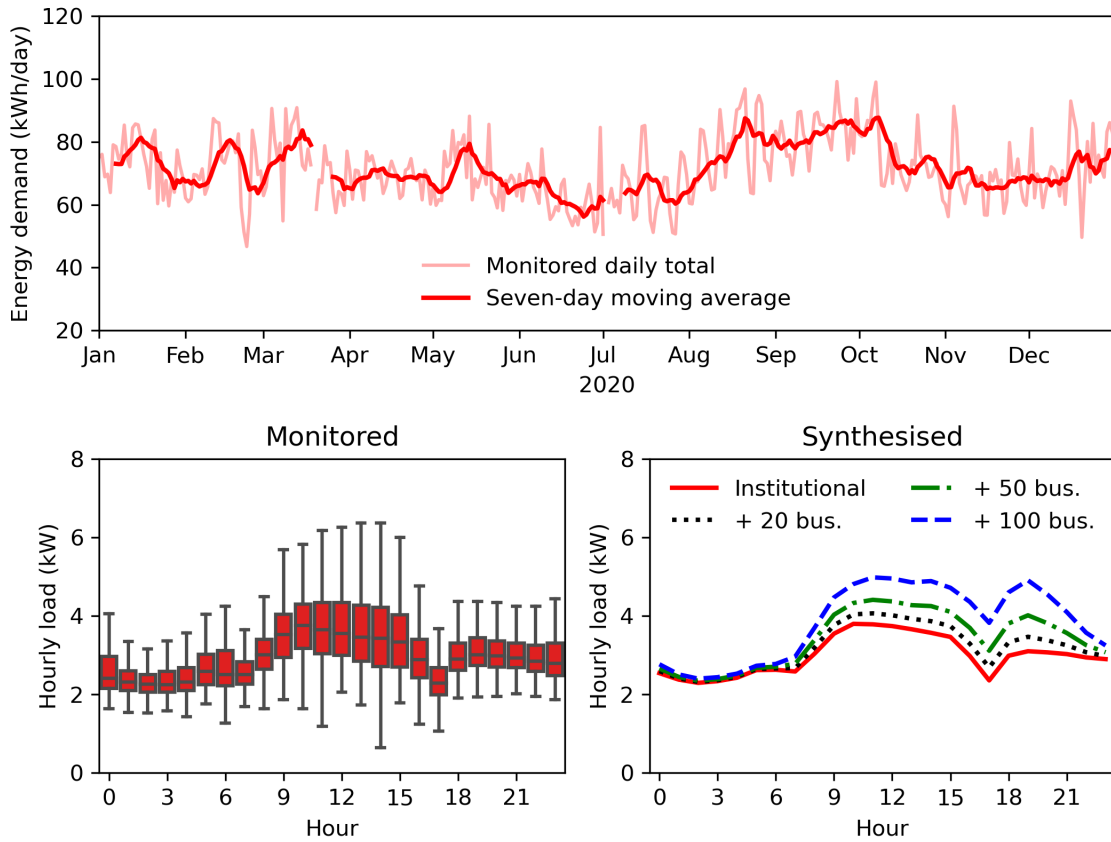


Figure 5.3: Monitored energy demand of the Mahama camp system, showing a) daily total energy demand and b) hourly energy demand distributions, and c) synthesised profiles of the institutional load and scenarios with 20, 50 and 100 business connections. Figure from Beath et al. (2023) and produced by P Sandwell [222]

Figure 5.3 b) shows the hourly load profile of the monitored data as a series of box and whisker plots. The lower, central and upper bars of the boxes correspond to the first, second (median) and third quartiles of the data whilst the whiskers show the extent of values outside of 1.5 times the interquartile range, with values beyond these limits not shown for clarity. Monitored electricity demand typically falls within a range of 2-3 kW during the nighttime and 3-4 kW during the daytime, with much greater variability during the day.

Figure 5.3 c) shows the mean synthesised daily load profile of the system (red solid line), which maintains the temporal profile of the monitored data to include higher demand during the daytime. Figure 5.3 c) shows the additional load of 20, 50 and 100 refugee business connections (black dotted, green dash-dotted and blue dashed lines respectively) in each scenario, using the breakdown of the types of businesses shown in Table 5.1. The average load profile for the connected businesses is concentrated during the daytime with demand amongst businesses peaking in the evening to prolong opening hours after sunset (around 18:00), after which it decreases overnight. The average of the peaks above the baseline existing load is around 350 W, 900 W and 1.8 kW for the low, medium and high connection scenarios. Electricity demand from businesses during the night is negligible. These load profiles are matched to the operating patterns of the businesses in the camp: while small shops might be open from early in the morning, other businesses with higher-power appliances such as hair salons typically operate around midday and the afternoon, whilst

bars and restaurants stay open into the evening.

Overnight energy use

Changing the way the diesel generator is operated could affect the usage of the battery storage. This may have a greater impact overnight when the storage cannot be charged by solar, possibly requiring the diesel generator to be run again and negating any fuel reduction benefits. In addition to the overall daily load profile, it is therefore important to assess whether typical energy use overnight could be satisfied via the storage alone, and to what extent the SOC limits may need to be altered to accommodate it.

Figure 5.4 shows the cumulative overnight electricity demand remaining until sunrise (blue) and cumulative demand since sunrise (green) for each day of the monitored data. When solar generation is unavailable (typically 18:00 to 6:00), around 25-40 kWh of electricity (equivalent to around 30-50% of the battery capacity) is required to meet the needs of the health centre and offices. A further 10-15 kWh (10-20% of the battery capacity) would be required to power the system until 10:00 the following morning, for example, if solar generation were lower than expected owing to cloudy weather. For the highest-demand evenings, an additional 10 kWh may be required to supply the system overnight.

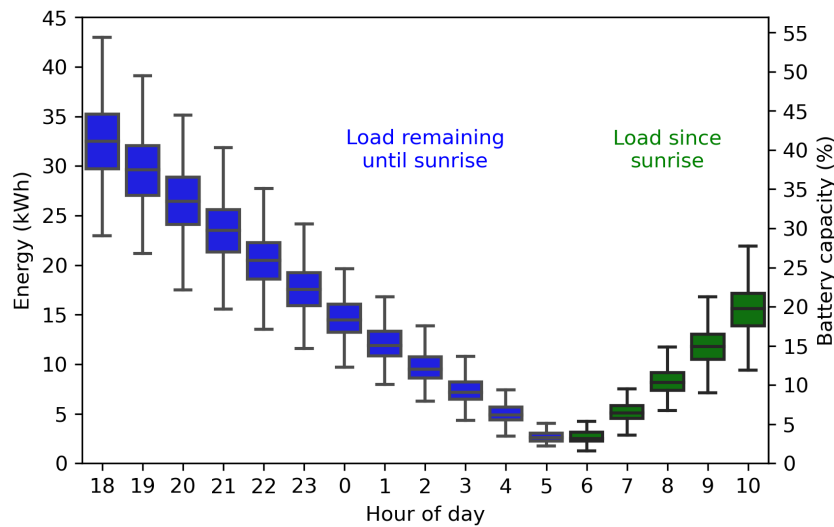


Figure 5.4: Monitored night energy demand remaining until sunrise (blue) and energy demand since sunrise (green) in the Mahama camp system. Energy demand is represented in kWh (left) and as a percentage of the battery capacity installed (right). Outliers are not shown for clarity.

Figure from Beath et al. (2023) and produced by Phillip Sandwell [222]

The storage capacity is technically capable of supplying this amount of energy but is not permitted to do so by the operating strategy. The current strategy used by the system initiates the usage of the diesel generator when the battery SOC falls below 90% between 18:00–02:00, or 65% thereafter, resulting in regular usage. As Figure 5.4 suggests, if these SOC set points could be lowered then it may be possible to use battery energy alone to meet the overnight demands of the system. This necessitates an evaluation of the potential operating strategies and SOC set points to assess the optimal configuration.

5.5.2 Investigation of diesel operation strategies

The diesel-only system and original strategy

To establish a baseline against which to assess the hybrid system and its possible operation strategies, I first evaluated the performance of the original diesel-only system under the synthesised institutional load profile. I found that using diesel generation only, over a five-year period, yielded a total diesel fuel cost of \$56,200. Considering the logistical issues in transporting diesel fuel to remote locations such as refugee camps [291] this may underestimate the true cost. I found the system has a high environmental impact with an emissions intensity of 1182 gCO_{2eq}/kWh and GHG emissions of 168 tCO_{2eq} over the five-year period. Over a 15-year time horizon, the emissions intensity remains the same and the GHG emissions increase proportionally to 504 tCO_{2eq}, as no new capital expenditure is assumed.

I then modelled the performance and impacts of the hybrid system – following the installation of the solar and battery equipment – under its original strategy. During the “diesel hours” (18.00–02.00) the batteries were permitted to discharge just 10% of their available capacity, leaving 90% of the stored energy in the batteries but not available to be used. Below this threshold, the diesel generator would be engaged to meet the loads and fully recharge the storage. During the remaining “quiet hours” the batteries were permitted to discharge a maximum of 35% of capacity, at which point the diesel charged them to 85% SOC. This strategy, designed to prevent excessive battery drainage overnight and to avoid potential damage from deep cycling, meant the generator would run almost exclusively during the diesel hours only. On average, this resulted in the diesel generator operating for 4.02 hours between 18:00–02:00, and just 0.06 hours during the quiet hours after 02:00. In the diesel-only system, the generator was required to run continuously to meet demand.

The hybrid system with the original operating strategy resulted in diesel costs of just \$14,700 over five years, a reduction of \$41,500 (74%) compared to the diesel-only system. The total GHG emissions from fuel use were reduced by the same percentage, as expected, to 44 tCO_{2eq}, and the emissions intensity fell by 45% to 651 gCO_{2eq}/kWh. The reduction is more modest than the reduction from emissions from diesel fuel due to the embedded emissions of the added renewable infrastructure (see Table B.3 in Appendix B). When considering a longer 15-year time period I find that the emissions intensity decreases substantially to 436 gCO_{2eq}/kWh.

This analysis focuses predominantly on fuel costs, and the potential of the system to reduce them using solar and battery equipment. It treats the initial investment in capital equipment as a sunk cost to directly focus on future savings, given that the equipment has already been installed. Calculating the LCUE for each system, however, offers an assessment of the overall costs of electricity inclusive of the costs of the grant-funded capital equipment, the prices of which are shown in Table B.2, for a total equipment cost at the time of installation of \$42,700. The diesel-only system was found to have an LCUE of \$0.60/kWh over five years, while the LCUE of the hybrid system was \$0.61/kWh. Considering instead a fifteen-year period the diesel LCUE is the same, as it is tied to the ongoing use of fuel, but the renewable equipment would generate more energy to pay off its initial investment, reducing the LCUE to \$0.42/kWh.

No additional equipment is considered to be installed during the 15-year period, reflective of the lifetime of many of the system assets, however, expected battery lifetimes are typically shorter. Whilst the battery does not degrade below its recommended minimum capacity for replacement in the simulations, the model of battery degradation does not take into account other factors

affecting battery performance. As opposed to my main focus on five-year time periods, the longer fifteen-year periods may require the replacement of batteries or other equipment which would lead to additional costs and environmental impacts.

Analysing alternative operating strategies

The simulations showed that the original operating strategy was conservative in its use of the batteries. Only 15% of the battery lifetime was consumed after five years, but at the expense of using more diesel than might be necessary to meet demand, and so modifying the SOC set points and altering the time periods could offer opportunities to further reduce diesel consumption.

Figure 5.5 shows the impacts of each modelled combination of SOC parameters (each categorised by colours) and timings over a five-year period. The horizontal axis displays the fuel cost savings relative to the original operating strategy and the vertical axis shows an estimate of the battery life consumed, as a percentage of the total expected life, h . Strategies which allow greater depths of discharge result in higher usage, and subsequent degradation, of the batteries. Within each strategy, the variations in timings offered fuel cost savings compared to the original strategy or, for some, increases.

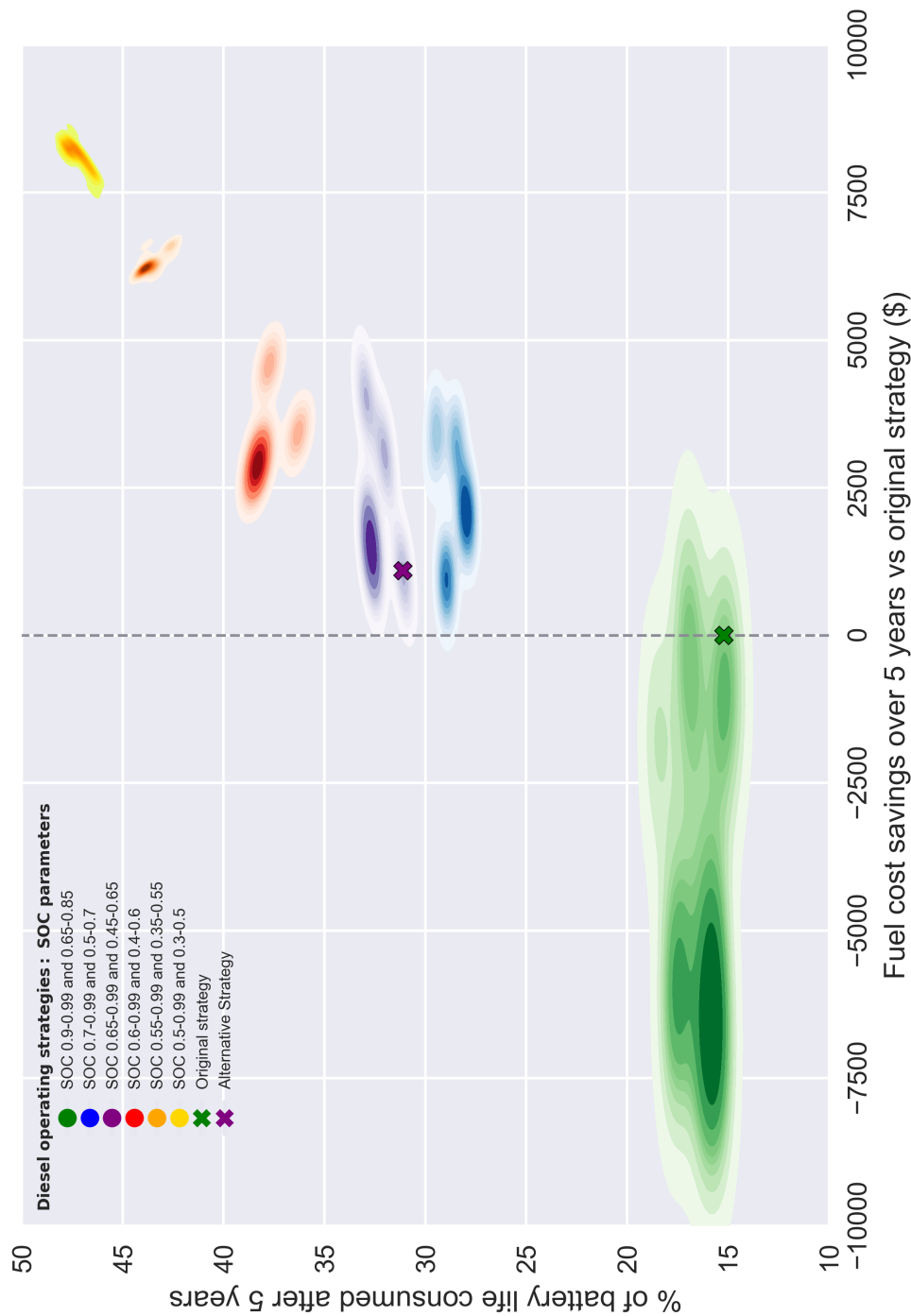


Figure 5.5: The savings in fuel costs compared to original strategy and % of battery lifetime consumed at different SOC (each categorised by colour) and timings over a 5-year period. SOC parameters are shown in the legend for “diesel hours” (left) and “quiet hours” (right). Darker shades of each colour represent a greater frequency of results. Crosses highlight the original (green) and alternative (purple) strategies. The original fuel cost savings (grey dashed line) is \$41,500 compared to the diesel-only system. Figure adapted from Beath et al. (2023) [222], figure code by J Barando Alonso

For the original SOC parameters (green) the original timings perform relatively well: the majority of timing combinations offer lower fuel cost savings, although some would offer more. All of the timings have a modest impact on the battery, consuming between 14.6–18.5% over five years, owing to the low depths of discharge. The most aggressive SOC parameters (yellow) offer the greatest utilisation of the battery storage and therefore have the lowest variations resulting from the timings of the diesel generator operation. These parameters use the highest amount of usable battery lifetime, up to $h = 48\%$, but also offer the greatest fuel savings, up to \$8,600 – a 21% increase beyond the original fuel savings from the initial installation of solar and battery equipment.

Whilst the additional potential fuel savings are relatively high, none of the SOC combinations produces additional savings over five years that are large enough to fully cover the cost of replacing batteries should they need replacing, at present storage prices. The price of batteries at the time of installation was \$200 per kWh of nominal capacity, with a cumulative cost of \$15,600 (equivalent to \$9,300, discounted at 10% over five years). For the cost of replacing all of the battery storage to match the maximum fuel savings (\$8,600), there would need to be a cost decrease of 1.6% per year. This rate is plausible but, in reality, the need to replace the batteries and its costs will depend on the actual technical performance of the storage capacity and prevailing prices. Further, batteries are complex components to maintain and manage, with the possibility of deeper cycling or temperature leading to unpredictable impacts on lifetime and potentially necessitating earlier replacement. This may lead to additional issues and costs relating to the responsible end-of-life management of the batteries, for example, to minimise their environmental impact, and for the sourcing of their replacement. System operators may therefore prefer to use a less extreme battery discharge strategy to balance fuel savings and battery lifetime.

Selecting an alternative strategy

An alternative strategy to improve on the current SOC and timings of the diesel generator operation should provide greater fuel cost savings whilst also respecting the additional constraint suggested by the mini-grid operator that the battery SOC should not drop below 40% on more than one day per week (14.3% of days) and never below 30%.

I identify the alternative strategy highlighted in Figure 5.5 (purple cross) as meeting these conditions. This strategy allows the SOC to drop to 65% between 18:00–00:00 and 45% during the rest of the day. The SOC drops below 40% on just 6.4% of days. The SOC dropping to this level is a result of the resolution of the computational model, in which the diesel generator is engaged on an hourly basis which may occur after the lower threshold has been crossed, but similarly could replicate any delays in switching on the generator in the real system.

Over a five-year time horizon, I find that this alternative strategy offers a further reduction in fuel costs of \$4,100 and 12.4 tCO_{2eq} of GHG emissions compared to the original strategy and consumes 33% of the battery lifetime; this equates to an economic cost of \$2,900, considering installed battery costs of \$200/kWh at a 10% discount rate (see Table B.2, in Appendix B). The LCUE decreases marginally by 3% to \$0.59/kWh, and the emissions intensity of electricity falls by 11% to 579 gCO_{2eq}/kWh. Over a 15-year time horizon, the LCUE falls to \$0.40/kWh and the emissions intensity to 340 gCO_{2eq}/kWh.

Impacts on battery usage

Figure 5.6 compares the distribution of SOC for the original (left) and alternative (right) operating strategies for each hour of the day, as well as the respective switch-on and switch-off SOC. Between 00:00 and 06:00 the original strategy has a relatively narrow distribution of SOC, which decrease until sunrise. This is a result of the batteries being fully charged in the evening before being depleted by overnight demands. The SOC distributions widen as the batteries charge in the morning, owing to variations in solar generation, and are almost always fully charged by 12:00. The batteries discharge to meet demand during the afternoon but the SOC is typically kept high by the solar generation, and by the diesel generator throughout the evening and night.

The alternative strategy, however, allows a greater depth of discharge and hence greater utilisation of the battery. Throughout all periods of the day, there is a wider distribution of SOC and deeper discharges, particularly between 23:00–11:00, but despite this, the battery is reliably charged to a high SOC by solar generation by 15:00. The greater discharging during the evening occasionally causes the battery to reach the switch-on SOC and initiates the generator to run, resulting in three distinct peaks in the early morning depending on if or when this occurs. As required by the system operator, this strategy rarely has a SOC below 40%.

Impacts on generator usage times and load factors

Figure 5.7 shows the effects of the two strategies on the times and the probability of diesel usage, $\rho^D(t)$, and load factor, Γ , of the diesel generator. I find that under the alternative strategy, the usage of the diesel generator decreases by 84% per night, to 1.06 hours, but it is sometimes now used between 05:00–08:00 to top up the batteries if the SOC is low (cf. Figure 5.6). Under this alternative strategy, the diesel generator is not used at all on 16% of days, while under the original strategy, it runs every night.

The load factor of the generator increases significantly by switching from the original strategy ($\Gamma = 55\%$) to the alternative strategy ($\Gamma = 97\%$). This could increase the fuel efficiency of the generator, which in my analysis is constant rather than linked to the load factor, and could yield additional fuel savings. The original strategy also has more instances of lower load factors near the minimum of 35%, which could cause operational issues and higher fuel expenditure per unit of electricity output. Meanwhile the fraction of electricity from renewable sources increases from 63% under the original strategy to 73% under the alternative.

In summary, I find that the alternative strategy more flexibly utilises the available battery capacity without exceeding the constraints recommended by the system operator. It also offers reductions in fuel costs, LCUE, GHGs and emissions intensity as well as reducing the duration of generator usage overnight. Table 5.2 summarises the impacts of the diesel-only system, original strategy and alternative strategy.

5.5.3 Impact of connecting business customers

Impacts on system costs and GHG emissions

I modelled the integration of the additional loads from 20, 50 and 100 refugee businesses in the camp marketplace, shown in Figure 5.3 c), and assessed their additional resultant cost and GHGs over the same five-year time frame under the alternative operating strategy.

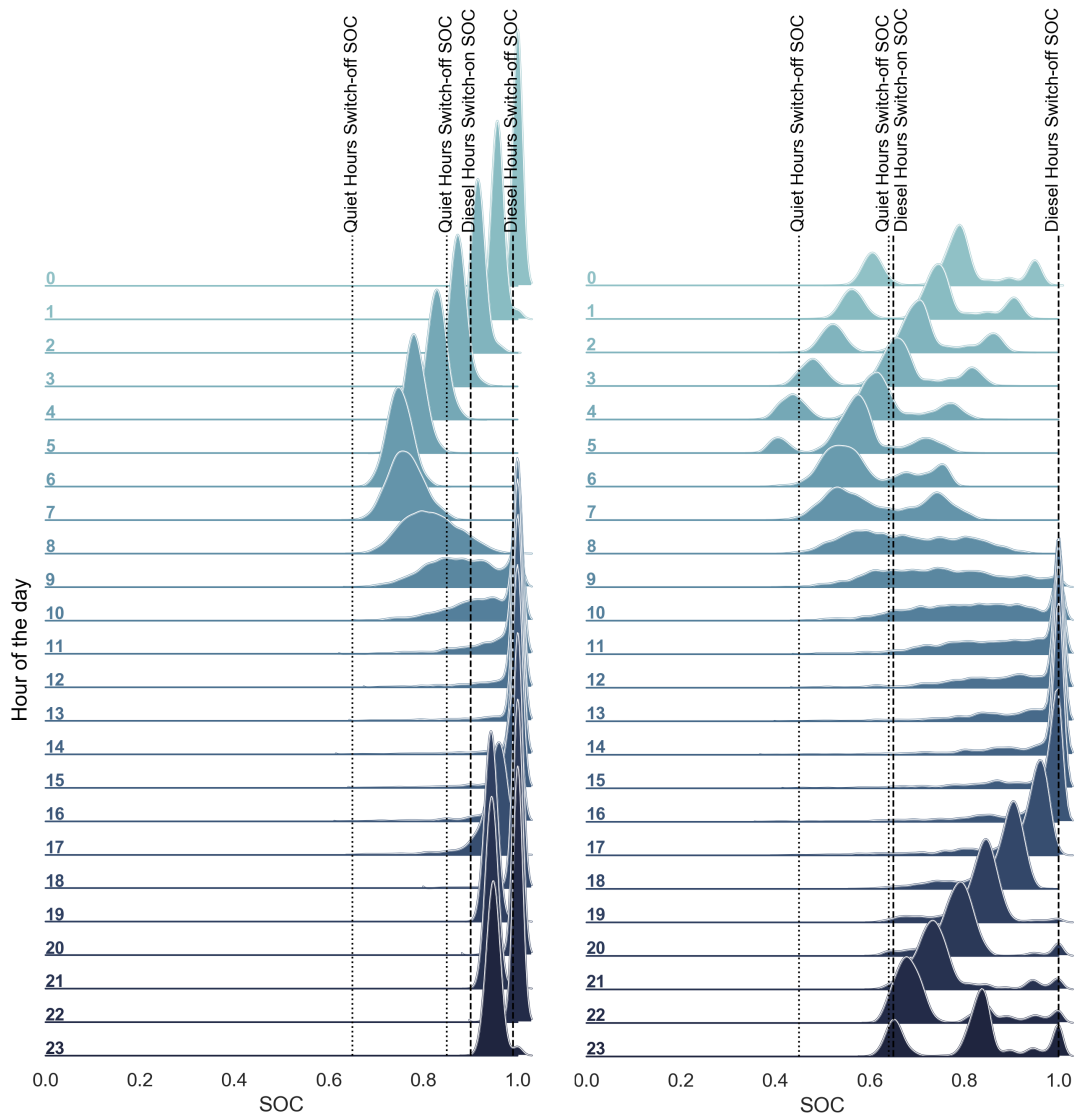


Figure 5.6: The hourly state of charge of battery storage for the original strategy (left) and the alternative strategy (right). State of charge thresholds for both cases are represented as vertical lines (quiet hours thresholds in dotted line, diesel hours thresholds in dashed line). Figure from Beath et al. (2023) and produced by J Barando Alonso [222].

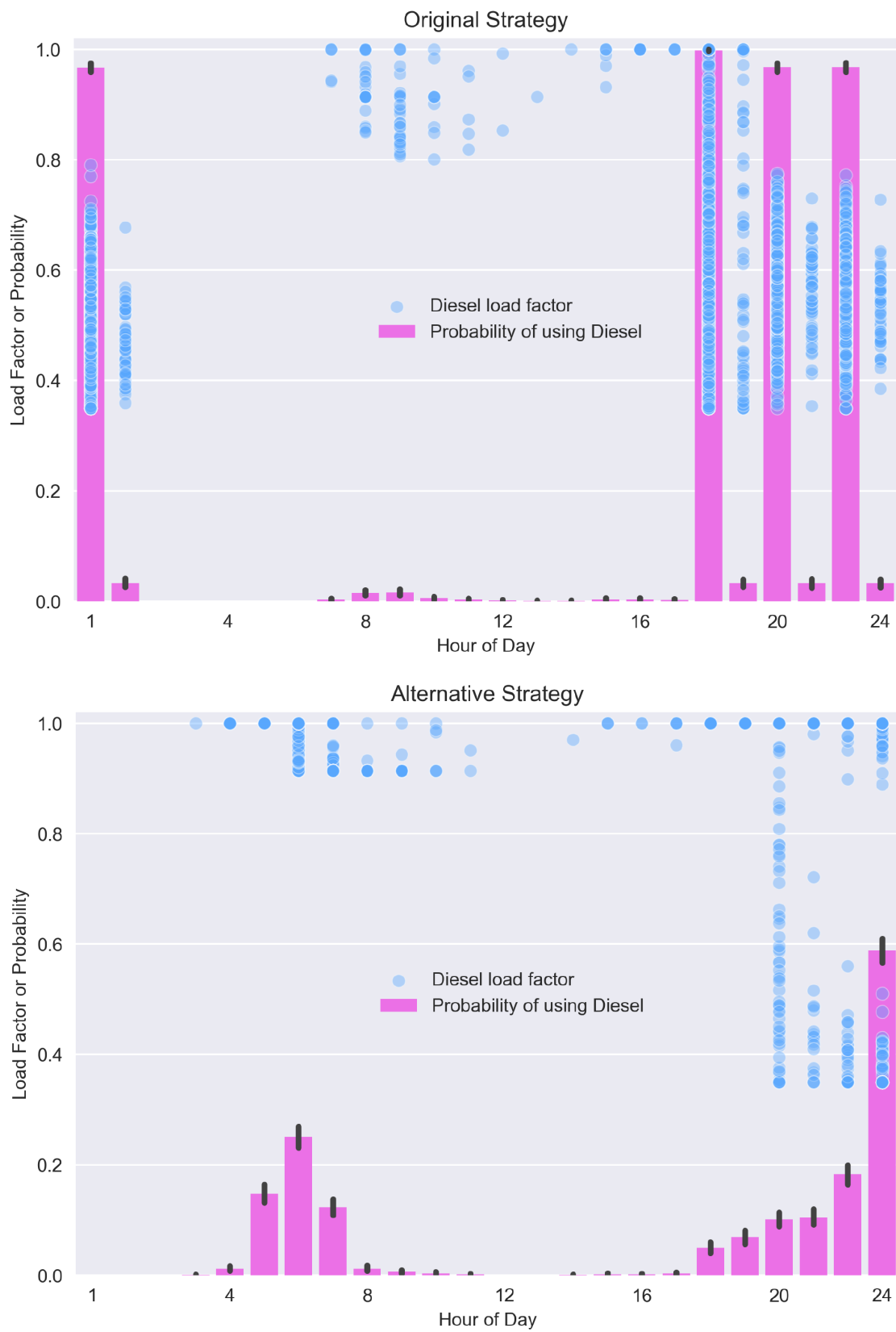


Figure 5.7: The probability of use (bars) and load factor (points) of the diesel generator under the original strategy (top) and alternative strategy (bottom). Each point represents the load factor during one instance of the generator being used. Figure adapted from Beath et al. (2023), original by J Barando Alonso [222].

Table 5.2: The fuel costs, fuel GHG emissions and technical performance of the diesel-only system, and the hybrid system under each of the original and alternative strategies over five years of operation.

Indicator	Diesel-only	Original	Alternative
Fuel cost (\$)	56,200	14,700	10,600
Diesel GHG emissions (tCO _{2eq})	168.0	44.0	31.6
Battery life consumed (%)	N/A	15	33
Diesel generator load factor (%)	25	55	97
Diesel usage (hrs/evening)	8	4.02	1.06
Diesel usage (hrs/day)	24	4.08	1.66
Nights using generator (%)	100	100	84

I found that connecting 20, 50 and 100 businesses increases fuel costs by \$266, \$3,170 and \$6,270 respectively, with corresponding GHG emissions increases of 0.9, 9.7 and 19 tCO_{2eq}. It is clear that connecting 20 businesses has relatively little impact on the system but at 50 and 100, the fuel expenditure increases substantially suggesting that at a lower level of additional usage, there is surplus solar electricity available. The cost increases for 50 and 100 businesses are comparable in magnitude to the fuel cost saving availed by switching from the original strategy to the alternative strategy (\$4,100). Switching to the alternative strategy could therefore unlock fuel savings which could be channelled, at least in part, to supporting business connections – minimising the overall financial impact but increasing opportunities for refugees.

By altering the timings of the alternative strategy (18:00-00:00), but keeping the SOC set points limits the same, I found that it is possible to reduce diesel costs and emissions. When connecting 50 businesses, diesel dispatch hours of 19:00–23:00 can achieve savings of around \$400 and 1.5 tCO_{2eq} over five years compared to the original timings for the alternative strategy. For 100 businesses, the same cost and emissions reductions are realised for dispatch times of 20:00–00:00. The technical performance recommendations of the operator remain unaffected. These results suggest that businesses can be connected to the system with moderate increases in diesel fuel usage and that further adjustments to the operating strategy can reduce the burden of additional connections.

Tariff structures for business customers

This part of the analysis was conducted by Philip Sandwell and Javier Barando Alonso and is presented here for reference as it is a component of the wider study.

The structure of the four tariff options, and the difference in their revenues against additional fuel costs resulting from powering the businesses, are shown in Table 5.3.

Connecting 20 businesses has a minimal impact on the system performance as this relatively modest increase in electricity demand can mostly be met by the excess solar generation during the day, which was previously unused. This results in tariff scenarios for 20 businesses being more favourable, owing to their lower costs. For 50 and 100 businesses, however, the increased energy demand requires greater usage of the diesel generator and hence the relative profitability of the same tariffs is lower.

Tariff 1, the national grid tariff, is able to recoup the additional fuel costs incurred in supplying 20 businesses but operates at a loss of around 60% for 50 or 100 businesses. For Tariff 2, informed by the mini-grid operator's experiences in rural areas of Rwanda, the system makes a profit taking

Table 5.3: The tariff structures for $N = 20, 50$ and 100 businesses and the percentage difference between the additional fuel costs and the revenue from customers (income minus fuel costs, divided by fuel costs). The mini-grid tariff is 400 RWF/kWh in the daytime, 600 RWF/kWh in the nighttime, plus a 30 RWF/kWh daily fee.

Tariff structure	Rate (RWF)			Difference (%)			
	<i>N</i>	20	50	100	20	50	100
1. National grid (/kWh)		89	89	89	+108	-59	-60
2. Mini-grid		Variable			+1322	+182	+179
3. Usage break even (/kWh)		43	218	223	–	–	–
4. Daily break even (/day)		10	45	44	–	–	–

into account additional fuel costs for all customer levels. The inclusion of the daily fee is influential by increasing the profitability of providing power to these businesses whose electricity consumption (and hence revenue when charged per kWh) is low.

The revenue for the mini-grid tariff assumes that businesses pay the daily fee, and consume energy 365 days per year, which would give \$3,900, \$9,300 and \$18,200 in revenue over five years for 20, 50 and 100 businesses respectively. This is in line with the current implementation of this tariff by the mini-grid operator in rural communities, but might be higher than reality. Instead, considering the mini-grid tariff with the same consumption but over 300 days per year, the revenues for 20, 50, and 100 businesses are \$3,800, \$8,900 and \$17,500 respectively. This tariff maintains high-profit margins, of 1,322%, 182% and 179%, compared to the increase in fuel use.

The rates of Tariffs 3 and 4 vary significantly between those for 20 and 100 businesses. This highlights the importance of both estimating the demand for connections before setting the tariff, and in potential economies of scale. Both tariffs, by definition, achieve the goal of breaking even on additional fuel costs. Tariff 4 – designed around a daily fee, rather than being consumption-based – may be favourable as it would allow more straightforward planning and financial forecasting for both the operator and businesses. It may, however, incentivise higher consumption.

Comparative effects of each stage of the investigation

Figure 5.8 summarises the overall financial impacts of each stage of the system investigation over the five-year period. Hybridising the system by first integrating solar and storage capacity yields the greatest single opportunity for savings (\$41,500) followed by further cost reductions (\$4,100) resulting from changing the state of charge and timing parameters to those of the recommended strategy. To contextualise these savings, the total equipment cost at the time of installation (\$42,700) is approximately the same as the five-year fuel savings (\$41,500 to \$45,600). Depending on the real-life reductions in fuel use actualised over the system lifetime, and other factors such as diesel prices and operation and maintenance costs, this suggests the investment in equipment would have a payback time of around five years.

Integrating businesses, shown here for the case of 100 connections, increases fuel costs (\$6,300) to satisfy their demand. These costs can be offset by introducing tariffs to earn revenue from the businesses: either partially offset under the grid tariff (\$2,500), fully offset by either a usage-based or daily fee-based break-even tariff, or more than offset (\$18,200) if the mini-grid tariff were applied. All considered, the total fuel costs over five years of operation are completely abated as a result of the mini-grid tariff.

These results suggest that the initial benefits of hybridising the system by introducing solar

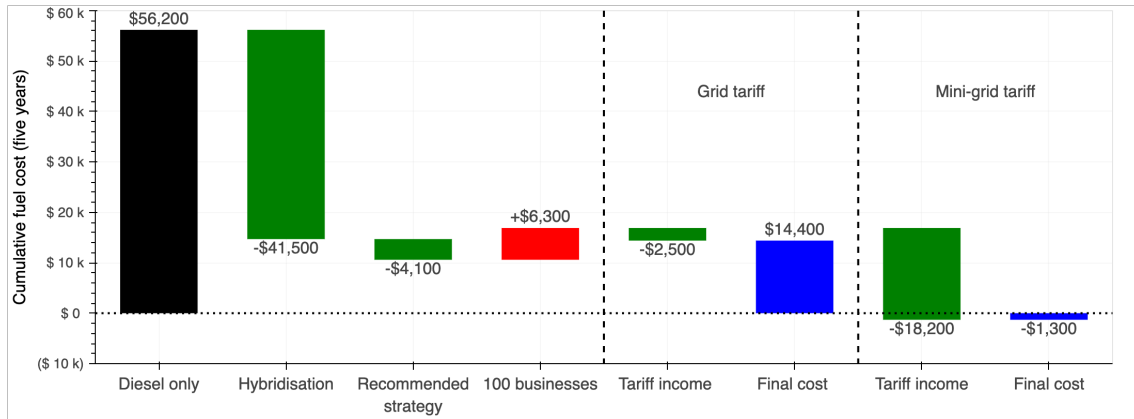


Figure 5.8: The five-year cumulative fuel costs of the original diesel-only system (black), potential savings (green) and costs (red), and to alternatives for the final cost (blue) of the system with 100 business connections when using either the grid tariff (left) or the mini-grid tariff (right). Figure from Beath et al. (2023) and produced by P Sandwell [222]

PV and battery capacity can be further extended by reassessing and optimising the operating strategy. They also suggest that businesses can be connected to the system in a manner that offsets the increased fuel costs or even provides a net income for the system operator. Matching the tariffs set by the subsidised national grid network would result in a loss (\$3,800 over five years) compared to the increased fuel costs, but these would be of a similar magnitude to the savings achievable through the optimisation of the diesel operational strategy (\$4,100). Permitting a mini-grid operator to replicate their tariffs from other systems would allow them to make a profit whilst potentially allowing both price parity with the host community and a sustainable financial return for the operator.

5.6 Conclusions

5.6.1 Designing hybrid systems to mitigate diesel usage

The results presented in this chapter reaffirm the potential for solar PV and battery storage to mitigate the usage of diesel generators in displacement settings and provide power for critical services [221, 273–275]. The analysis by my co-investigators of data monitored from the electricity system in Mahama Refugee Camp, Rwanda, found that the average load was 70.1 ± 10.0 kWh/day in 2020. Using this demand profile to model the baseline system, I found that the diesel-only system would have fuel costs of \$56,200 and emit 168 tCO_{2eq} over a five-year period.

Using the introduction of a cycle-charging diesel dispatch strategy into the CLOVER modelling framework, I estimate that the introduction of 18.4 kW_p of solar capacity and 78 kWh of battery capacity by the mini-grid developer significantly reduces costs to just \$14,700 and GHG emissions to 44 tCO_{2eq} over five years under the original conservative operating strategy for the generator. The 74% reductions in costs and GHGs are comparable to those found by previous studies. Considering the total equipment costs at the time of installation of roughly \$43,000 are similar to the fuel savings of \$41,500 to \$45,600, it indicates a payback period of around 5 years, also in line with other studies of this nature [221, 273, 274].

Considering the institutional loads only, I identified that overnight energy use currently equates to around 30-50% of the battery capacity and so altering the operational strategy of the generator

would gain greater utilisation of the storage capacity to reduce fuel usage. Under the original strategy degradation was found to be just 15% over five years; under other combinations of depths of discharge and generator usage times this could increase to up to 48%.

Using these results I suggest an alternative operating strategy which further reduces fuel costs by \$4,100 and GHGs by 12.4 tCO_{2eq} over five years. Under this strategy the storage SOC would drop below 40% on just 6.4% of days, meeting the technical performance requirements desired by the system operator. This achieves greater utilisation of the batteries, consuming 33% of the expected battery lifetime, and decreases generator usage by 84% per night whilst running at higher load factors. The economic cost of the battery lifetime used up equates to \$2,900, considering installed battery costs of \$200/kWh at a 10% discount rate (see Table B.2). The fuel savings outweigh this cost by a factor of 1.4, suggesting it is financially advantageous to cycle the batteries more deeply under this strategy.

These results suggest that system operators should explore options for increasing the utilisation of the batteries via more liberal depths of discharge and combinations of generator timings, both of which can decrease the overall fuel consumption, and that funders (such as donors, financing organisations or the companies themselves) should consider battery replacement costs as part of a long-term deployment strategy to allow system operators to fully utilise the batteries.

5.6.2 Supporting business customers

The final research question focused on the viability and potential profitability of connecting refugee businesses, whilst maintaining the overall reliability of the electricity system. I presented modelling scenarios in which 20, 50 or 100 enterprises in the local marketplace were connected to the electricity system and found that this additional energy use increased fuel consumption and costs, but alternative operating strategies could reduce these below those of the original institutional-only system.

The tariff structures and cash flow questions relating to the business customers were addressed by my co-investigators, Dr Philip Sandwell and Javier Barando Alonso: a summary is presented here. They found that the tariff charged and the number of business customers affect the overall profitability of connecting refugee entrepreneurs. For 20 businesses, which mostly rely on excess generation from solar that would otherwise have been wasted, the required tariffs to offset fuel usage were low. For 50 or 100 businesses, they found that matching the grid tariff (89 RWF/kWh) results in significant losses and would therefore need to be subsidised at that rate. Similarly, for 50 and 100 customers estimates for usage-based break-even tariffs (218 and 223 RWF/kWh) are several times higher than the national grid tariff, which could result in a lack of take-up of new connections due to a perception of high cost. Daily tariffs (10, 45 and 44 RWF/day for 20, 50 and 100 businesses) could however be received better by refugee businesses and be easier to administer and plan around for both customers and operators.

Not using a break-even-based tariff avoids a potential perception of discriminatory pricing under which refugee businesses would be paying for fuel usage only, whereas the institutional users would also be paying off the costs of the system. Charging the tariff used in the host community, however, could instead be perceived as favouring institutional users: refugee businesses would pay a commercial rate whilst institutional users have their electricity costs covered by Alight and UNHCR. The concerns for the latter could be mitigated by moving to a long-term power purchase agreement for the institutional users at a rate equal to or greater than the tariff for the

businesses [221, 292]. This would align the two payment structures and potentially offer simpler financial planning for the system operator and off-takers.

For better outcomes, systems operated by companies should be permitted to replicate the tariff structures used for their other sites in rural areas. In addition to parity with the host communities, they found that this tariff structure could provide a profit for the company. This could incentivise both greater numbers of connections and more sustainable business opportunities, satisfying the electricity demand for livelihood opportunities found in this study and others [267, 287]. This also aligns with the livelihoods objectives of humanitarian agencies and is desirable for long-term private sector engagement [281].

5.6.3 Limitations of the study

The use of this techno-economic modelling framework leads to methodological limitations relating primarily to its temporal resolution and inputs. The CLOVER model operates on an hourly resolution and therefore misses some of the dynamic behaviour that occurs at the sub-hourly level in a real-life system. Demand fluctuations are aggregated into hourly values and the modelled system is not required to respond to short, high-power events which could affect the system and may cause greater battery degradation than modelled. Reformatting the input data and model processes to accommodate a higher temporal resolution could help to solve some of these issues. However, this may increase uncertainty due to the greater level of granularity needed in model inputs such as demand or how batteries respond to different sub-hourly events. Additionally, modelling a higher temporal resolution over a multi-year time horizon increases computation time, and this could be a limiting factor in addressing research questions requiring high numbers of simulations and scenarios.

Where data availability permitted, the model inputs accurately represented the system installed at Mahama Camp. However, I have not used dynamic prices. This is a shortcoming as a change in fuel price, for example, may impact the results. Care should be taken when comparing the results of this study to other situations with significantly lower or higher fuel prices.

The model simplifies the treatment of fuel consumption by the diesel generator when compared to the installed system. Although fuel consumption per kWh supplied is typically lower at higher capacity factors [293], similarly to other studies [127, 275] this modelling framework assumes a constant rate of fuel consumption. The original and alternative strategies in my analysis have average capacity factors of 55% and 97% respectively and I can therefore reasonably assume the fuel savings in the alternative strategy would be higher, and therefore more favourable than outlined in this chapter. Future work should incorporate variable fuel consumption to assess this.

To evaluate business connections it was necessary to use proxy data based on customers elsewhere in the country. The size of daily demand was based on the aspirations of humanitarian agencies and the Government of Rwanda but the actual daily demand is likely to vary significantly between days and types of businesses with different needs, which could affect system utilisation [294]. The number of business connections could vary depending on the local demand, tariff structures, ability and willingness to pay, and the supporting environment. This study, therefore, offers different numbers of connections and types of tariff structures to inform on a range of eventualities.

Finally, aside from the specifics of electricity generation and use in Mahama Camp, Rwanda has both well-defined electricity access targets and a progressive environment for displaced people.

Rwanda offers a high potential for the implementation of hybrid mini-grid systems, which may not be representative of other countries. The technological and economic benefits of hybrid mini-grids would remain applicable in countries with similar prices and solar resources; however, the enabling environment is a critical component for implementation and should be considered when comparing this work to other countries.

5.7 Summary

This chapter uses energy system modelling and monitored data from a real system in a refugee camp to investigate how the hybridisation of diesel mini-grids can reduce fuel use and cost and emissions savings. Further, it highlights the importance of considering the way that hybrid systems - comprising PV, batteries and dispatched generation - are operated to ensure cost efficiency. There are crucial trade-offs between deeper battery cycling and increased fuel use depending on the operational strategy used, and these can be investigated through energy systems modelling.

As with the previous chapter, this work also considers different demand scenarios. In this case, I consider the impact on the system of adding additional users, and how the operational strategy may need to be adjusted in response to an additional demand. Further, whilst there may be some surplus generation (that would otherwise be dumped) to meet the electricity demand of some additional users, it is important to consider the costs either in additional fuel use or battery cycling that may arise from higher numbers of additional users, and whether these costs can be recouped through different tariff structures.

This chapter provides a highly detailed case study for a system in a displacement setting. The use of diesel generators in refugee camps is widespread and costly and there is broad applicability of this study to other similar camps. Considering other off-grid contexts, this research has the potential to give insights that are applicable when hybrid mini-grids are in use and configured in a similar way. When considering the need to meet SDG7, and provide sustainable, reliable and affordable electricity to all, hybrid mini-grids can play a crucial part; but considering the way they are operated to maximise benefits is crucial.

Chapter 6

Assessing Pathways to Universal Electricity Access using Geospatial Modelling

Acknowledgement of Contributions

Methodology for electricity demand modelling and profile construction was devised in conjunction with S Few and A Gambhir. S Few wrote code for processing the CREST model electricity demand profiles. S Mittal advised and supported on the use of geospatial data sets. S Mittal and A Gambhir advised on the design of the geospatial modelling tool used in this study. B Winchester wrote code used for running the CLOVER model on high-performance computers. P Sandwell and B Winchester wrote the version of the CLOVER model used in this study. A Gambhir, J Nelson and S Mittal gave supervision and guidance on the study design and development and gave feedback on the written manuscript.

6.1 Overview

The previous two Chapters 4 and 5 used energy systems modelling, demonstrating for case studies in India and Rwanda, the importance of demand and system reliability, and also the cost impacts of how hybrid mini-grids are operated. Whilst these case study chapters have broader applicability to other similar contexts, they do not give analysis at the country or regional level. As highlighted in section 2.7 of the literature review, research using modelling tools can be done both at the more local scale or in less detail but using tools suitable for regional or country-level studies. Tools for both types of research (local level and national/regional level) are important and useful and can be used in conjunction with one another, as I seek to demonstrate in this chapter.

In this work, I use a combination of modelling tools, including a new geospatial data model (outlined in Section 3.4 of the methods chapter), to explore pathways to universal household electricity access for 43 countries in Sub-Saharan Africa. I undertake, 1) energy demand modelling at the country level and for access tiers 1-4; 2) off-grid energy systems modelling at the country level and demand level and for a range of system types and at two reliability levels; and 3) integration of these models with geospatial data and data modelling to explore different scenarios for electricity

access. My scenarios examine how the shares of different types of access solution change at the regional and country levels, under different demand, reliability and carbon price conditions.

The work presented in this chapter contributes novelty in several ways. Firstly there is methodological novelty: this research uses multiple tools and approaches and presents a new open-source tool suitable for exploring least-cost scenarios of universal electricity access that could be adapted for other regions or to explore different questions. Secondly, in looking at the reliability of different options using this methodology. Whilst data on grid reliability is poor, there is novelty in looking at how differences in reliability between different electricity access options might be important for considering which pathways to take. Finally, this research addresses how carbon pricing might impact pathways to universal access, which has not yet been explored.

6.2 Background

Prior work that looks at the use of different approaches for universal access, e.g. centralised grid versus decentralised, and fossil versus renewables-based, has used both integrated assessment models (IAMs) [156, 157, 295] and geospatial data models and analysis [7, 24, 127, 167, 296, 297]. Although IAMs can help to investigate solutions to Africa's electricity supply at a high level [156, 295], they are not well suited to the research questions in this chapter that require more detail within countries, such as on electricity demand.

Work using geospatial analysis has explored pathways to universal access from a range of perspectives. Electricity demand level is an important consideration for understanding which solutions are most cost-effective: Mentis et al (2017) find that for Sub-Saharan Africa, higher tiers of electricity access are likely to have a least-cost pathway with a larger proportion of grid extension deployed [24]. Additionally, a further consideration in relation to demand is how it may be impacted by the demand for cooling, especially given scenarios of future global heating. In some parts of Sub-Saharan Africa, once this additional demand is factored in, a study found that national grids displace off-grid solutions [296]; although national grid reliability in this context has not been considered. Financing rates for different types of access solutions in different countries have been shown, using geospatial analysis, to have an impact on least-cost pathways to universal electricity access. When applying a universal discount rate, versus technology and country-specific ones, least-cost pathways in different areas will vary [297]. As discussed in previous chapters, diesel generation is commonly used in off-grid contexts; therefore diesel prices and subsidies have been demonstrated to have an impact on the relative cost-effectiveness of different access solutions [127]. Finally, geospatial analysis has been used to examine specific electricity access needs of both healthcare facilities [60] and refugee camps [275] globally.

As explored in Chapter 4, the reliability of supply of both grid and off-grid options is a factor that should be taken into account when considering electrification approaches. The level of reliability (in terms of the percentage of electricity demand met) is a key consideration impacting the costs of off-grid PV-based systems. Systems designed to meet higher levels of reliability cost more, with the final few fractions of demand adding to system cost disproportionately [298]. In low-income countries, national grids often deliver poor service levels [25] and lead to associated negative impacts such as reduced safety and security, reduced household income and additional expenditure on backup alternatives [4, 26, 42]. An equal comparison of the extension of unreliable grids versus off-grid infrastructure capable of meeting higher levels of demand may therefore lead to sub-optimal outcomes for households. Whilst Moner-Girona et al (2021) include grid reliabil-

ity as an indicator in a multi-dimensional analysis of decentralised access solutions for SSA [299], reliability of supply has not been included in least-cost modelling.

As countries and corporations seek to mitigate their greenhouse gas (GHG) emissions, and carbon pricing becomes more widespread it is important to consider how carbon pricing could impact pathways to universal household access. TVoluntary carbon markets have the potential to provide significant amounts of climate finance for African countries, allowing them to sell carbon credits and spend money on clean energy infrastructure, among other things [163]. The Africa Carbon Markets Initiative (ACMI) launched at COP27 aims to boost the number of voluntary carbon credits dramatically; currently, Africa only produces a small fraction of the carbon credits it could issue. Clean energy is an obvious recipient, providing financial incentives for the use of low-carbon options [164]. In the context of electricity access, another scheme, D-REC initiative [165], acknowledges the lower GHG emissions from off-grid PV systems and provides financial incentives, increasing their uptake. Understanding how carbon pricing could influence the relative cost and shares of different access modes is important for forward planning. The impact of carbon pricing on pathways to electricity access has not been explored in a meaningful way. Dangachew et al. (2018) include a carbon pricing scenario when looking at electricity access in SSA but do so using an IAM-based approach, only looking at electricity generation shares at the macro level. They do not give an analysis on the likely impact modal split i.e. grid vs off-grid [156].

As discussed in Section 2.7.2 in the literature review, there are existing geospatial tools to evaluate and plan electricity access. OnSSET, which has had the most widespread use, is a powerful tool with a high spatial resolution; which is also open-source and modular [24, 167]. However, the tool has a simplistic representation of electricity demand and temporal dynamics. Although the high spatial resolution is a strength when considering electrification strategies, it is computationally intensive to run at the multi-country level which makes it less suitable for analysis with a large number of scenarios.

6.3 Methods

This chapter combines electricity demand modelling, off-grid energy systems modelling, and geospatial data modelling with additional country-specific inputs to investigate the impact of demand, system reliability and carbon pricing on the shares of each mode in pathways to universal electricity access. A summary of the methodological process followed is shown in Figure ?? and is covered in more detail in the following parts of this thesis:

- Section 6.3.1 gives a description of the demand modelling process and the basis for growth in demand profiles
- Section 6.3.2 provides a description of how the CLOVER model was used for the research in this chapter
- Section 6.3.3 gives detail of the geospatial data sets used to create the inputs to the least-cost model
- Section 6.3.5 explains at a high level the geospatial least-cost model; with further details explained in Section 3.4 of Chapter 3.

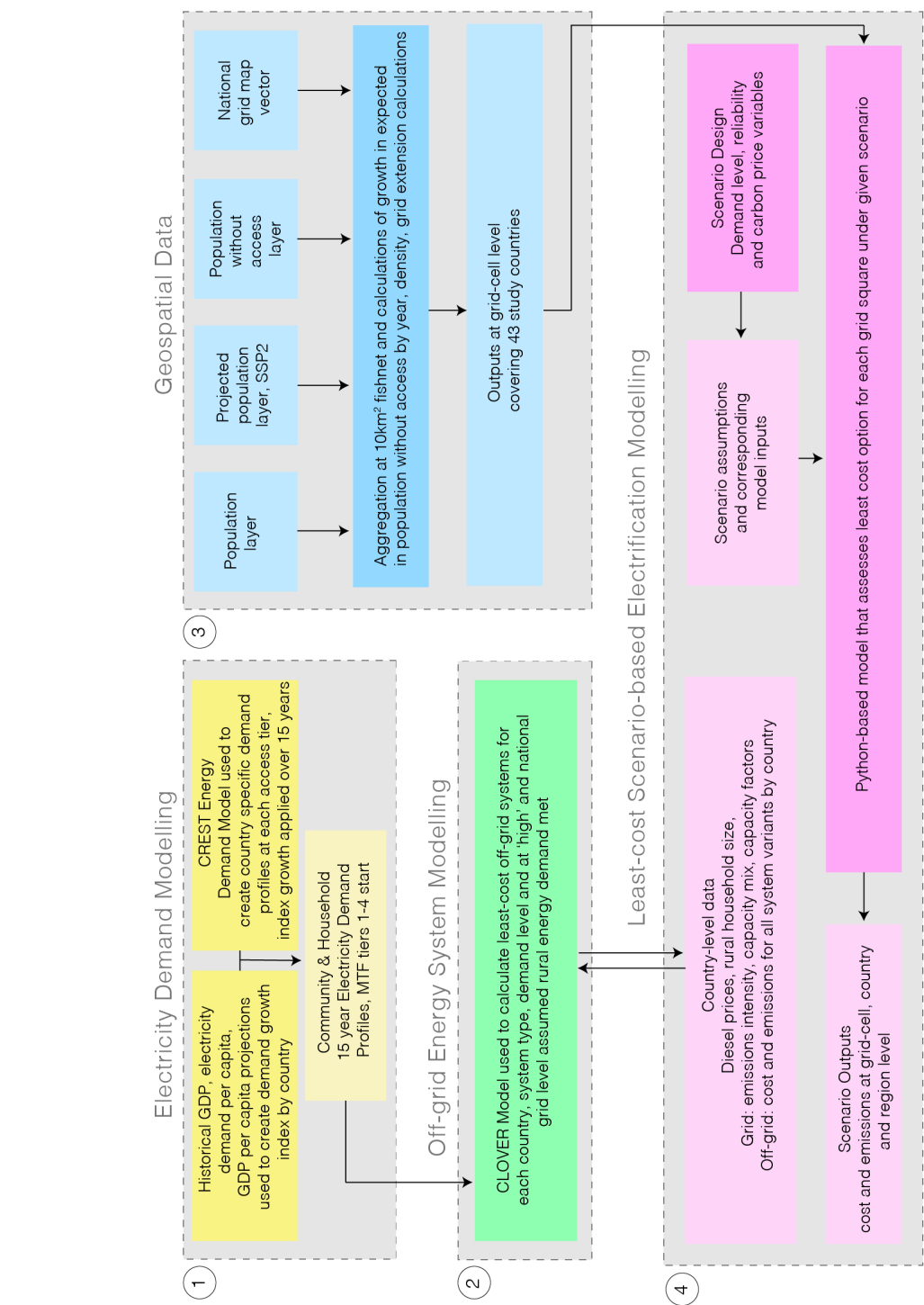


Figure 6.1: A high-level flow diagram explaining the main methodological components for the analysis in this chapter.

6.3.1 Electricity Demand

Household electricity demand levels are guided by the ESMAP MTF tiers of access [300] (see Table 6.3.1), but demand growth assumptions are underpinned by expected increases in GDP per capita growth, and the historical relationship between GDP per capita growth and electricity per capita growth, by country.

Table 6.1: MTF Annual energy consumption values per household from [300]

Tier	1	2	3	4
Annual Energy Demand per household (kWh)	4.5	73	365	1250

For the population and GDP per capita growth projections, I used Shared Socio-Economic Pathway (SSP) 2 as the basis for the main results. The SSPs are a set of global trajectories for development and growth that relate to challenges humanity will face with regard to climate change adaptation and mitigation. SSP2 represents the 'middle of the road' or central scenario [301]. I re-indexed long-range GDP per capita growth projections from SSP2 using 2020 figures of recorded GDP per capita (PPP) [301, 302], combined with IMF economic growth forecasts [303] that take into account the economic impact of the Covid-19 crisis. From 2025 onwards, I applied the SSP2 growth rates to the adjusted data for the immediate future. I used historical GDP per capita (PPP) and electricity demand per capita data [302], available in varying quantities between the years 1970 - 2014, to derive a linear relationship between both variables by country. For countries without adequate data availability, a country with a similar socioeconomic profile in the same region was selected as a proxy (see table C.5 within Appendix C).

I used the historical relationship calculated above for each country as the income elasticity of demand for electricity for each country and assumed it to be constant going forward. Using the elasticity of the GDP per capita projections I calculated an electricity demand growth index up to 2050, with 2020 as year 1; to guide the increases that should be used for the demand profiles. A range of other factors that may influence demand are not captured here; however, this approach rests on existing evidence suggesting income growth can lead to growth in electricity demand [304] including in low and middle-income countries [305, 306].

To create relevant country-specific demand profiles to be used in the off-grid modelling, I used the CREST open-source thermal-electrical demand model, which uses stochastic programming techniques to represent dwelling diversity and creates location-specific profiles that vary depending on temperature and daylight [208, 209]. I generated 25 household electricity demand profiles for each country for each of the tiers 1-4 of electricity access [204], as defined by the MTF.

The household profiles were then used to construct hourly, multi-year demand profiles that fit the annual future electricity demand estimates, derived from the GDP per capita projections for each country as explained above. For the mini-grid demand profiles, I assumed a community size of 100, and for each year took a proportional mix of demand profiles from the electricity access tier above and below the annual household consumption value in kWh, A for a given year; with the annual energy values in kWh represented by T^U and T^L for the upper and lower tiers respectively. I calculated the relevant percentages of the upper T_{Sh}^U and lower tier profiles T_{Sh}^L , with the upper given by

$$T_{Sh}^U = \frac{100(A - T^L)}{T^U - T^L} \quad (6.1)$$

and the lower tier share T_{Sh}^L the corresponding value that totals 100%.

Using these shares, I added the respective proportion of household demand profiles for each tier. I selected households (1-25) at random for each tier profile added during this process. I used the one-year profiles for each country for relevant years to form 15-year demand profiles for each location that track the predicted demand growth with start years for every year between 2020 and 2035. For the stand-alone systems, I used a proportional fraction (1/100th) of the community values to represent a single household. This approach risks a diversity benefit handed to stand-alone systems and thus potentially reducing the system size needed. This could then cause a lower-cost bias towards these systems. However, our modelling considers energy demand on an hourly basis (see below) and therefore does not consider sub-hourly spikes in power demand, reducing the impact. Additionally, demand profiles are uniform for PV and diesel-based systems and so any diversity benefit would be seen in the sizing of both technology types.

6.3.2 Off-grid Energy System Modelling

This chapter also uses the CLOVER energy system model, as described in the methods section 3.2. Here I will not describe the structure of the CLOVER model as this has been outlined in previous chapters. However, I will give an overview of how the model is applied. In this Chapter, my usage of the CLOVER model is different from in previous chapters as I used it to simulate different types of systems, at different demand levels, but over many locations (countries), rather than in a single location.

I used the CLOVER model to find the cost-optimal stand-alone systems for single households and mini-grid systems designed to power 100 households, for each country, using solar PV and battery systems, as well as diesel-powered systems. I do not consider other forms of off-grid generation e.g. micro-hydro or wind as other similar work found these sources to be the least-cost for relatively few people in Sub-Saharan Africa [24]. I assumed a 15-year project length, but re-optimising systems every five years to take into account the growing demand profiles and degrading components. In other words, throughout the system's lifetime at five-year intervals, the model adjusts the battery and PV capacity to account for increased demand or degraded capacity.

For the main reference scenarios, I set the sufficiency criterion as 90% of energy demand met. For simplicity, I assume consistency in the type of components used across system sizes, with them scaled as necessary by the model for the demand level; the technical inputs used are given in Table C.1 in Appendix C. For the location used for each country, I used the country centroid latitude and longitude as defined in the list provided in Google Developer Tools [307].

For solar inputs, the Renewables.ninja model, which integrates with CLOVER, requires tilt and orientation (azimuth towards north) as well as coordinates for a given location; the model has a spatial resolution of 50km² [217]. For panel tilt, I used the value of the latitude (multiplied by -1 for countries in the Southern Hemisphere) with a minimum value of 10 degrees reflecting the need for some angle for rainwater cleaning. For the orientation, I assume panels are placed horizontally tilting towards the equator, either Northwards in the Southern Hemisphere or Southwards in the Northern Hemisphere.

I optimised the off-grid systems for cost, given that this is most representative of how systems are designed in reality. As in Chapter 4, I used the levelised cost of used electricity (LCUE) as the optimisation objective and metric of unit cost; this has been discussed previously and a description is found in Section 3.2.4. Associated total and unit emissions for the system are given as outputs for each system. For a full breakdown of the cost, emissions and technical inputs used for the

off-grid modelling, see Tables C.2 and C.1 in Appendix C.

For each of the study countries, I used the model to find the cost-optimal mini-grid and stand-alone systems with starting electricity demand tiers (1-4), 15-year system lifetimes and start years of 2020, 2025, 2030 and 2035 to reflect growing demand and reduced component costs. Further, systems were modelled at 'high' reliability (90%) of electricity demand met, and 'grid' reliability; which equates to the estimated rural grid reliability in each country (see Table C.4 in Appendix C). The emissions and cost outputs of this process are used as inputs into the least-cost modelling process, explained below in Section 6.3.5. For a small number of countries, required inputs were not available and so a country in close geographical proximity with a similar socioeconomic profile was selected (see Table C.5).

6.3.3 Geospatial Data

I used geospatial data sets with coverage of the 43 Sub-Saharan African countries for this analysis (see Table C.4). These were for population, projected population growth, population without access to electricity and national grid infrastructure. Population data sets for the years 2014 and 2020 were taken from the LandScan Global data sets [308, 309]. Down-scaled population projections (SSP2) at ten-year intervals were used (2020, 2030 and 2040) for spatially dis-aggregated population trajectories [310, 311]. I used a gridded data set providing estimates for the population without access to electricity across Sub-Saharan Africa, based on remote sensing night-light data from Falchetta (2019) [40]. I used the data from the years 2014 and 2020 in this analysis [312]. Finally, I used vector maps of the African grid available from Moner-Girona & Georgia (2021) [313]. I aggregated these data and performed relevant calculations for my own fishnet (gridded map) at a 10km² resolution to be used in the modelling.

6.3.4 Calculating Population Trajectories

Estimation of baseline trend population without access

To calculate estimates of the population for the years 2020-35 I first used the historical geospatial electricity access population data (2014 and 2020) to produce estimated annual changes at the fishnet grid-cell level. In my forward projections, I assumed that population growth in each cell is divided by the 2020 share of access in each cell; so, population growth occurring in partially electrified cells is divided on this basis. I calculated annual estimations of the population without access at the cell level, up to 2035. This calculation considers assumed population growth amongst the population without access and any expected change in access. Given that in any given grid cell, improvements in access may mean that households in a cell would reach universal access; any surplus trend improvement that may exist is calculated annually and aggregated at the national level. In other words, trends in improvements in access are kept within national borders. This remaining trend amount is then used and allocated to connect households within other cells without access, starting with cells with the highest number of people without access first. This process is recorded by year for use in the next section. This logic rests on the assumption that an investor or government agency would seek to provide access to the areas with the highest numbers of people without access first. I assume that trend gain stays within national borders and once a country reaches universal access it is no longer considered in the process. The output of this process is a geospatial data set covering my 43 countries, with the numbers of people gaining access and the

absolute number remaining without access, at the map cell level, by year.

Defining pathways to universal access

To reach universal household access in all countries not expected to reach universal access by the target year, a pathway must be defined. This is done on a country-by-country basis, with the number of people connected for each country, P_X , in each year n defined by

$$P_X^n = P_R^n \left(\frac{n}{N} \right)^2 \quad (6.2)$$

where P_R^n is the remaining population for a given country in year n . These annual amounts result in an S-curve for each country, leading to most of the remaining connections above the baseline happening in the middle of the period [314]. The model then takes the approach of electrifying cells by the year in each country up to the amount P_X^n . Any growth in the non-electrified population in already 'connected' cells is also taken into account first, after which the next highest population density remaining non-connected cells are added, to reach the annual amount P_X^n ; following a similar logic to the baseline calculation described above. Cells that are connected under the baseline scenario (outlined above) are skipped, avoiding double counting. The output of this process is a geospatial data set that gives the estimated number of people gaining access at the fishnet cell level for each year n between $n = 1$ and the target year for universal electrification N . The trajectory of the population reaching universal access (by individual map cell) is then converted to the number of households by using the rural household size by the country for the most recent year available from the Global Data Lab average household size database [315]. See Table C.4 for the assumed household size for each country.

6.3.5 Least-cost Electrification Modelling

This section outlines the least-cost electrification model. This part of the methods is explained mathematically in the methods chapter, in Section 3.4. Below is a brief explanation given for clarity and to put in context with the chapter-specific methods. The code for the model used is available on [Github](#).

For grid extension costs, firstly, the distance from the nearest national grid line to the centroid of the selected cell, combined with an estimated cost per km of grid extension gives an estimate of the cost of the lines needed to connect the given cell. Secondly, per-household infrastructure and labour costs are added to represent connecting each household to the extended grid. The installation of equipment such as transformers or substations is omitted from this research (see limitations in discussion Section 6.5). In addition to the extension infrastructure, an additional national grid capacity cost is added. For the additional national grid capacity needed (in kW), I derived values for weighted average grid capacity factor (%) G_y and cost (in 2022 USD) C_G^I for units of capacity on the grid in each country κ using generation mixes per country [316, 317], and country-relevant capacity factors and costs [318]. With these values, the model estimates additional grid capacity, G^{cap} , required in each cell, ϕ , based on the average hourly electricity demand in kWh, E^L , in the final year N ; taking into account the assumed reliability of the rural grid G^R (as a percentage of energy demand met) in each country κ and is given by

$$G^{cap}(\phi) = \left(\frac{1}{G_y(\kappa)} \right) E_N^L (G^R(\kappa)) \quad (6.3)$$

The estimated additional grid capacity for each grid cell $G^{cap}(\phi)$ is then multiplied by the estimated cost per installed unit of capacity for the relevant country $C_G^I(\kappa)$. This work assumed the grid reliability to stay constant which is why the energy demand, E_N^L , is multiplied by the network reliability, G^R , for the country, κ . The generating and network costs are discounted according to the year of construction in the model, as guided by the above.

Data on grid reliability in Sub-Saharan Africa is sparse; however, some estimates of varying recentness for grid reliability by country are available from the World Bank Enterprise Survey [25]. These figures are more relevant for urban connections which are known to be considerably higher than those in rural locations [18, 319]. I used these data from the Enterprise Survey to give an estimated share of outage time in a year. I then assumed a 25% further penalty to reflect that national grids are typically lower for rural areas, as is explored in Chapter 4.

To calculate national grid greenhouse gas emissions, I used the demand estimations (as described in Section 6.3.1) by country and by electricity access tier, scaled to the relevant number of households in a given cell, ϕ . I multiplied the energy demand (in kWh) by the emissions intensity of the grid (in gCO₂e/kWh) to provide the total emissions resulting from grid operation in a given cell. Grid emissions intensity figures are from varying sources and are assumed to be constant (see Table C.4 in Appendix C). The assumed rural reliability, as described above, is applied to the demand totals before calculating emissions. This is then done by multiplying the reliability-reduced electricity demand by the relevant grid emissions factor. Additionally, I assume an emissions factor for each km of grid extension, this is added per grid cell based on the estimated number of grid extension km required to connect it (see Table C.3).

For the off-grid systems, emissions and investment figures are calculated for system types (stand-alone, mini-grid; pv and diesel) and country, using the off-grid modelling process described in Section 6.3.2. I use these figures to calculate estimates of the cost and emissions of electrification via each system type, for all the households requiring access in each cell; to be used in the final part of my calculations. The number of households requiring connection is multiplied by the investment estimate (per household) for each given system: diesel and PV mini-grid and stand-alone systems, sized for each country and at each start tier of access; and for two levels of reliability (high and national grid level). These numbers are then discounted (at 8% /year) accordingly with the respective year of connection [320].

For the final stage of the methods, I designed the model to provide estimates of which mode of access was the least cost for each cell, covering all countries requiring electrification to reach universal access by the target year for universal access (in most scenarios 2030). Cells are first assessed for population density, to establish whether stand-alone or mini-grid systems are more appropriate. Next, the grid and the relevant off-grid PV and diesel systems' total investment estimations are compared for each cell covering the entire gridded map. Each cell is connected using one mode only, and the system or capacity added is enough to meet all populations requiring access up to the target year N . Grid capacity is added in the year the cell is first connected, and off-grid capacity is added iteratively, by-year as additional households gain access. Using the estimated investment required for access via each mode, the least cost approach for each cell is selected for the given scenario. A more detailed description of this process is outlined in Section 3.4 of Chapter 3.

For scenarios applying a carbon price or unmet demand penalty, these are applied uniformly across all modes to tonnes of CO₂e and kWh of unmet demand respectively, and discounted, prior to finding the least-cost approach. Carbon prices are applied with a dynamic rate, with 2020,

2025 and 2030 values. The unmet demand penalty is a fixed value with no increase over time. For further detail on the carbon prices and unmet demand penalty, see the scenarios Section 6.3.5 below.

For each scenario run (see next section) this modelling process gives outputs of a geospatial dataset covering all countries with the estimated least cost mode of electrification, the associated investment requirements (in 2022 USD) and emissions. Additionally, the model provides summary outputs at the aggregate (all countries) and individual country levels.

Scenarios of Universal Household Electricity Access

In this chapter, I use the modelling approach described above to explore a range of scenarios that consider electricity demand, reliability of supply and carbon pricing. The scenarios (see Table 6.2) rest on population and electricity demand growth assumptions based on the shared-socioeconomic pathway (SSP) 2 [321].

In my baseline assessment of the population requiring access by 2030 and 2035, some countries are expected to reach universal household access. Therefore, although a total of 43 countries are included in the study and the baseline scenario, not all are included in the scenarios for universal access. In other words, the scenarios include only the additional population requiring access above what is expected to occur under the baseline scenario (see next section): the gap existing for meeting universal access. In line with SDG7, the target year for universal access is 2030, unless specified.

I use the ESMAP multi-tier framework (MTF) [300] of access as a guide for demand, in step with other studies of this nature [24, 297]. Although the Tiers of access span 1-5, this research looks at only tiers 1-4, with a particular focus on tiers 2-4. Tier 1, equating to annual household demand of 4.5kWh, is extremely low, and not representing a meaningful supply. Tier 5 is a high level of access, and largely unobtainable for households first gaining access who are typically poor, with a low ability to pay. Tiers 1, 2, 3 and 4 represent household demand of 4.5, 73, 365 and 1250 kWh per year respectively (see Table 6.1). The MTF tiers of electricity access are used as the starting point for household annual demand. However, household demand trajectories vary depending on expected GDP/capita growth by country, as in Section 6.3.1. I use the MTF to guide consumption only, not for blackouts or hours of service since I perform my own analysis on service reliability (unmet demand).

To explore scenarios of reliability, I take two approaches. The first is to compare the grid with off-grid systems that have been cost-optimally modelled to meet the same level of energy demand met as the national grid in each country (Rel_grid_all). Systems are sized according to the electricity demand met and therefore have lower investment costs at lower reliability levels (see Section 6.3.2). The second approach is to apply an unmet demand penalty universally for each unit of unmet demand (Rel_penalty); representing the cost that outages cause for households. This penalty is included in the total investment costs, and would hypothetically be incurred by an electricity supply company when blackouts occur.

The carbon prices used are values for 2020, 2025 and 2030 from the IPCC AR6 C1 and C2 scenarios, in 2022 USD, provided from the AR6 scenarios database [322]. C1 and C2 scenarios represent keeping global heating to 1.5 degrees, without and with overshoot, respectively. See Table 6.5 in Section 6.4.5 for the values used.

I perform sensitivity analysis on a range of carbon price and reliability penalty levels, to explore the values that are most crucial for initiating modal shifts (see Section 6.4.5). Including all variants and sensitivity analyses, this work presents results from over 100 scenarios.

Table 6.2: Describing the main scenarios described in the results section.

Scenario	Demand	Description
Baseline	N/A	Scenario that uses trend access and projected population growth to estimate the population remaining without access by target year (2030 or 2035).
Reference		
<i>Ref_central</i>	Tiers 1-4	Central reference scenario where all unmet population above the baseline gain access via calculated least-cost electrification mode by 2030.
<i>Ref_single_modes</i>	Tiers 3 & 4	Reference scenario where all unmet population above the baseline gain access via single modes of access e.g., all via grid/diesel etc.
<i>Ref_late</i>	Tiers 2-4	Central reference scenario where all unmet population above the baseline gain access via calculated least-cost electrification mode by 2035.
Reliability		
<i>Rel_grid_all</i>	Tiers 2-4	Reliability scenario where all unmet population above the baseline gain access by 2030 via calculated least-cost mode with the off-grid modes are sized at the reliability of the grid in for each respective country.
<i>Rel_penalty_0.50</i>	Tiers 2-4	Reliability scenario where all unmet population above the baseline gain access by 2030 via calculated least-cost mode with a \$0.5/kWh penalty for unmet demand applied across all modes.
Climate Policy		
<i>Ctax_median</i>	Tiers 2-4	Carbon price scenario where all unmet population above the baseline gain access by 2030 via calculated least-cost mode, with a carbon price scheme representing the median values from the AR6 database is applied across all modes.

6.4 Results of Analysis

This section presents the results of the analysis. First, it will describe the results in the reference scenarios at different demand levels. Secondly, it will discuss findings on the scenarios relating to reliability. Thirdly, it will cover carbon pricing, and finally, the sensitivity analysis will be described.

6.4.1 Baseline scenario

Based on calculations using spatially distributed historical population [308, 309] and electricity access data [312], and population projections (SSP2) [310, 311], I estimate that under the baseline scenario that while some countries reach or make significant progress towards universal access by 2030, many countries in Sub-Saharan Africa could see an increase in the number of people without access to electricity (see Figure 6.2). This is due to population growth occurring in regions with present low access and low historical trend progress, coupled with limited trend progress at the national level respectively. My baseline expectations for the total number without access in 2030 and 2035 for the 43 study countries are that just over 470 million people are expected to remain without access to electricity in 2030, with almost no further reduction by 2035 (see Table 6.3). This is in the context of population growth from 2020 of 240 million and 360 million in 2030 and 2035 respectively. Countries such as Ghana, Kenya, South Africa, and Cote d'Ivoire are expected to reach universal access which can be seen in areas of blue in Figure 6.2. Parts of the Sahel and East African countries, such as Ethiopia and Uganda are expected to see the greatest increases in the

number of people without access, shown in areas of red. Figure 6.3 (left), shows the distribution of the population without access in 2020 (based on dataset from Falechetta et al (2021) and in 2030 under my baseline scenario (right). Whilst other approaches factor in stated policies, my approach considers historical trends in changes in access and anticipated population growth only as explained above. These results are in-line with other work taking a similar approach [41].

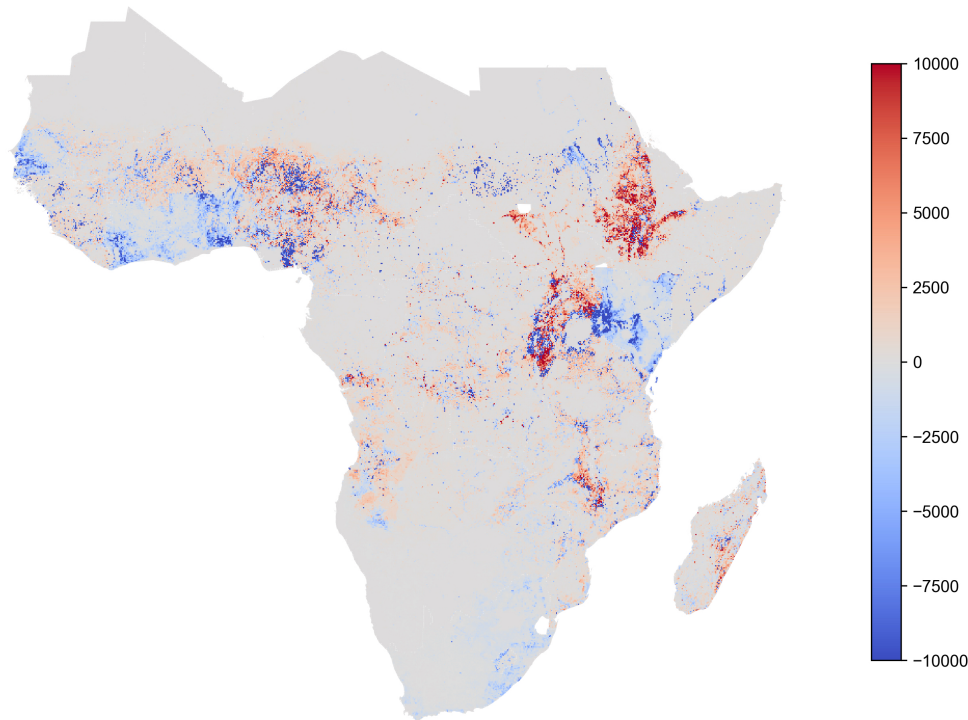


Figure 6.2: Change in population without access between 2020 and 2030 under the Baseline scenario. Areas coloured red represent areas where the population without access is expected to increase, with blue being areas where the population is expected to decrease. Spatial distribution at 10km² resolution. For more clarity in greyscale, instead see adapted version of Figure 6.3: Figure C.1 in Appendix C.

Table 6.3: Describing the population and population without access in 2020 and expected in 2030 and 2035 under the baseline scenario.

Year	Population Total Without Access	Net Change in No Access	Total population
2020	505 million		1.10 billion
2025	478 million	-27 million	1.21 billion
2030	472 million	-6 million	1.34 billion
2035	471 million	-1 million	1.46 billion

6.4.2 Pathways to universal household access

Conforming with previous work on this topic [24], my results indicate that the demand level is an important factor in determining the modal distribution. For the Ref_central scenario at the lowest demand level (tier 1), off-grid PV-based technologies dominate, estimated to be the least-cost option for 78% of the population requiring access. At tiers 2 and 3 (Figures 6.4a and 6.4b), grid connections, as well as off-grid diesel feature prominently, with diesel mini-grids in use in parts of Western and Eastern Africa, particularly in countries with lower fuel costs (see Figure 6.5a). At

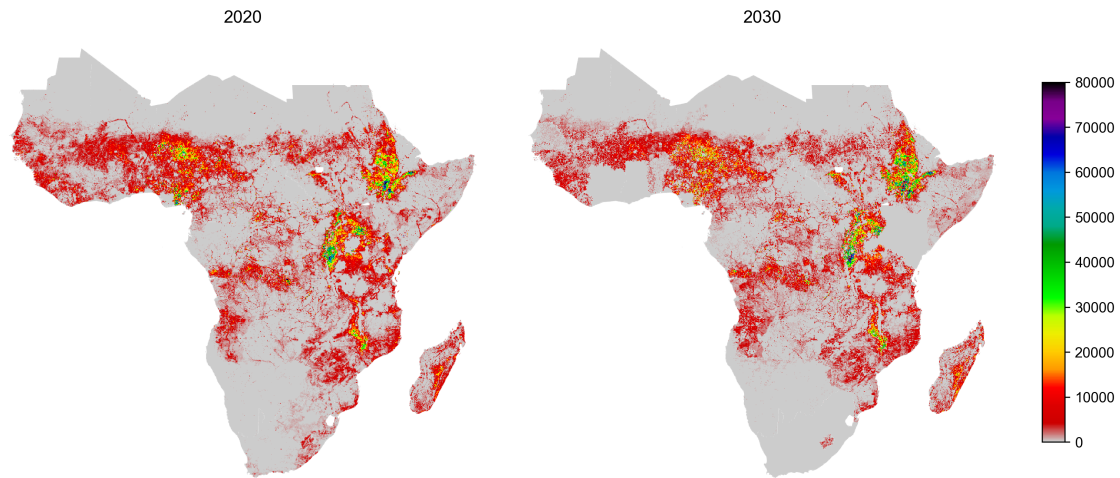


Figure 6.3: Showing distribution of population without access in 2020 (left) (aggregated dataset from Falchetta (2021), and predicted distribution of population without access in 2030 under baseline scenario (right). For adapted greyscale version see Figure C.1 in Appendix C

tier 3 level demand, the grid is the least-cost option for 49% of the population and at tier 4 level access (Figure 6.4c), the share of all off-grid technologies falls with the diesel share falling the most and with the grid share increasing to 63%. The share of off-grid PV falls from over 1/3 at tier 2, to 25% and 20% at tiers 3 and 4 respectively.

At lower tiers, off-grid PV systems, whether SHSs or mini-grids are relatively cheaper than other modes as they can be sized for low levels of consumption due to their modular nature. The costs of grid extension remain relatively high regardless of the consumption level, with only added capacity changing according to the demand tier. Diesel systems are more economical above tier 1 due to minimum generator sizes and output capacity factors.

I also give estimates for the investment and emissions implications of single-mode scenarios (Ref_single_modes), shown in Table 6.4. They indicate that there are pronounced gains from taking a mixed, least-cost approach rather than an electrification strategy that relies too heavily on one approach such as grid extension. The emissions contributions, even from the most polluting single-mode pathway (all diesel mini-grids, tier 4) are low when compared to global annual emissions. Although not captured by this study, there may be technology ‘lock-in’ from initial access modes and significant emissions growth in future years as electricity demand grows, which means that modal choice is still important for emissions [161].

A later universal access target of 2035 (Ref_late) has varying impacts on the share of both forms of off-grid PV when compared with a 2030 target; the cost of PV systems falls over time in the modelling (see Section 6.3.2). At tier 1 there is almost no difference: 78% with a 2030 target versus 79% with 2035. However, at tier 2, this shift is much more pronounced: 36% with 2030 year for universal access, increasing to 51% with 2035. At tiers 3 and 4, there are notable but smaller increases (25-31% and 20-27%), respectively in relation to 2030 shares. In general, these results suggest that later electrification will favour off-grid PV due to cost reductions and that the most sensitive to these reductions demand level is tier 2.

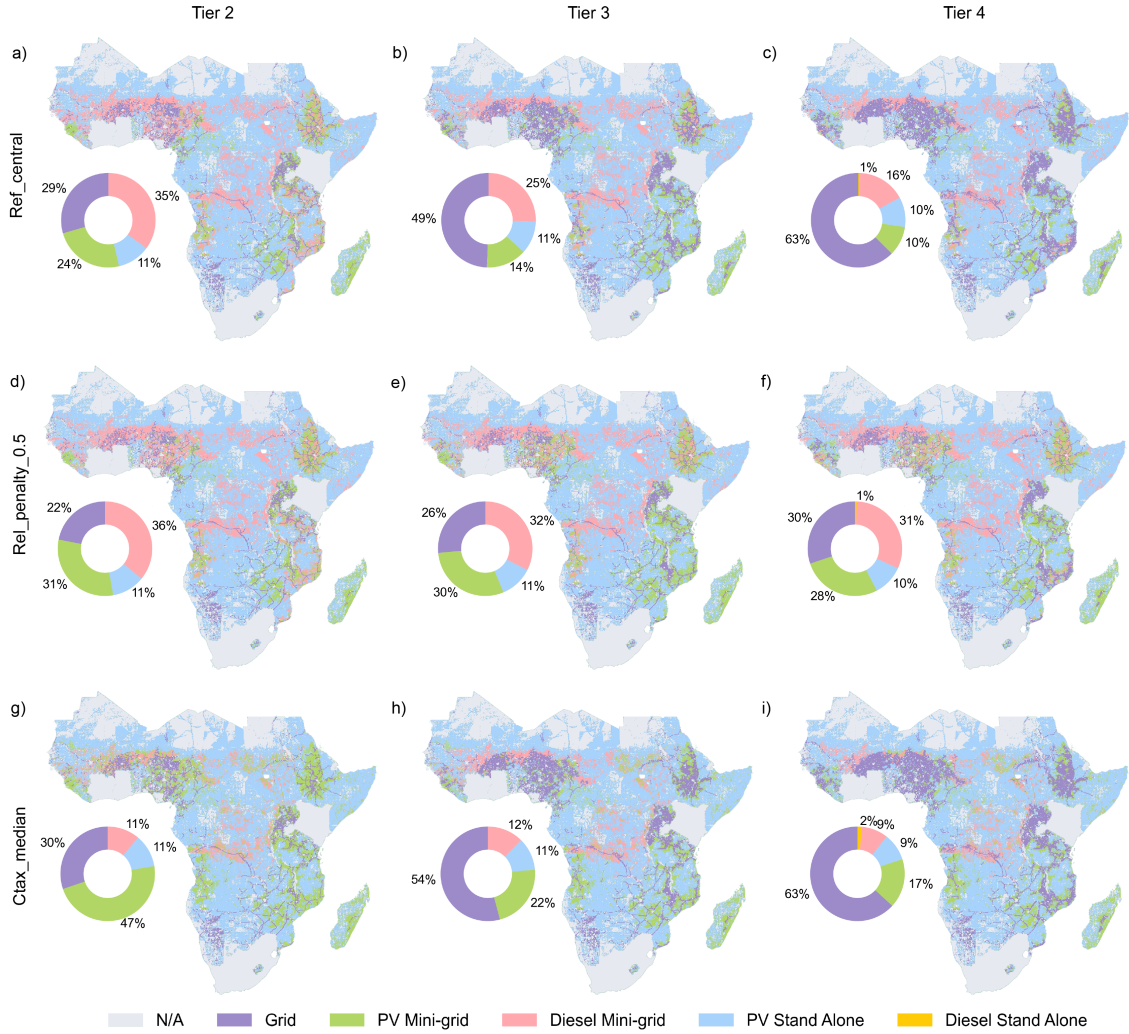


Figure 6.4: Spatially distributed least-cost electrification method for the Ref_central, Rel_penalty_0.5 and Ctax_median scenarios at demand tiers 2-4. Panels a-c represent the Ref_central results, with no carbon tax or unmet demand penalty applied, they show a declining role for off-grid technologies as demand increases. Panels d-f represent the Rel_penalty_0.5 scenario applying a 0.5\$/kWh of unmet demand penalty universally across all modes, this reduces the share of the grid at all demand tiers, with the largest difference at the highest demand tier. Panels g-i represent the Ctax_median scenario, equating to a carbon price scheme of around \$4 (2022 USD) in 2020, rising to \$146 by 2030, here diesel usage is reduced in all three tiers, with the largest difference with Ref_central seen at Tier 2. For adapted greyscale version see Figure C.2 in Appendix C.

6.4.3 Variation in Reliability of Electricity Supply

Treating all electrification modes equally regarding electricity demand met may be seen as short-sighted, as national grids provide low levels of reliability in some countries, such as Nigeria, the Central African Republic, Senegal and Zambia [25] (See Figure 6.5b). Here I explore how accounting for differences in assumed electricity demand met may impact the modal shares and their spatial distribution.

For the Ref_central scenario, the off-grid systems were sized to meet 'high' levels (90%) of the electricity demand profiles. In the Rel_grid_all scenario, I removed the assumption that off-grid means of access provide high levels of demand met and instead look at the cost-optimal outcome

Table 6.4: detailing Investment, emissions and reliability metrics for selected scenarios. These figures represent additional above the baseline, not representing all requiring access. See methods for further detail on what is included and omitted from investment and emissions values.

Scenario	Investment* 2020-2030	Total Emissions (Mt, CO ₂ e) 2020-2030	Demand-weighted Mean Reliability
Ref_central			
<i>Cost-optimal, Tier 3</i>	\$29 billion	98	0.73
<i>Cost-optimal, Tier 4</i>	\$41 billion	204	0.71
Ref_single_modes			
<i>PV mini-grids, Tier 3</i>	\$58 billion	57	0.90
<i>Off-grid Diesel Tier 3</i>	\$66 billion	286	0.90
<i>Grid, Tier 3</i>	\$235 billion	141	0.62
<i>PV mini-grids, Tier 4</i>	\$87 billion	113	0.90
<i>Diesel mini-grids, Tier 4</i>	\$100 billion	507	0.90
<i>Grid, Tier 4</i>	\$244 billion	207	0.63
Rel_Penalty_0.5			
<i>Tier 3 with 0.5 penalty</i>	\$52 billion	125	0.81
<i>Tier 4 with 0.5 penalty</i>	\$72 billion	215	0.81
Clim_ctax			
<i>Tier 3 with Carbon Tax</i>	\$44 billion	86	0.72
<i>Tier 4 with Carbon Tax</i>	\$64 billion	157	0.71

*Investment figures include reliability penalty/carbon tax incurred by this modal distribution.

at tiers 2-4 when the off-grid systems are sized to meet the same assumed reliability as the rural grid in each country (see Section 6.3.5). I find a significant shift away from the grid compared to the Ref_central scenario: 29 to 22% at tier 2, 49 to 29% at tier 3 and 63 to 34% at tier 4. At tier 2, this reduction in grid share is all met by an increased share off-grid PV; and the share of diesel also falls. At tiers 3 and 4, shares of both off-grid diesel and PV increase. At tier 3, the majority of the fall in grid share is met by PV, and at tier 4 the majority by diesel.

Further, I modelled scenarios that universally apply a financial penalty for unmet demand across all means of access. Although this is impossible to quantify consistently and will vary across countries, income groups and demand levels, the inclusion of a cost in the modelling of each lost kWh represents the economic cost for each lost unit of electricity demand for households [4, 42]. The results (Figure 6.4, panels d-f) represent the inclusion of a \$0.5/kWh penalty (Rel_penalty_0.5) applied universally and show that at all tiers there are shifts from the grid to off-grid modes, with the largest difference at tier 4, when compared to Ref_central.

Looking at the country level, for electricity access tier 3, countries such as Nigeria and Senegal with low-grid reliability see significant shifts to off-grid PV (Figures 6.5b and e). Other countries with low grid reliability such as Zambia and Central African Republic do not: this is due to them already having high levels of off-grid PV already and lower levels of the grid (Figures 6.5 b, d and e, and Figure 6.6). Countries with higher grid reliability e.g. Ethiopia and Tanzania also see shifts, with these countries having a higher initial share of the grid as the least-cost option (Figure 6.6 and Figure 6.5b and e).

As would be expected, applying a reliability penalty of \$0.5/kWh sees the overall demand-weighted mean reliability for the scenario increase at all electricity demand tiers, with more impact

at the higher two tiers: at tier 3, it increases from 0.73 in the Ref_central scenario to 0.81; at tier 4 it increases from 0.71 to 0.81. At the aggregate level, the results indicate a trade-off between reliability and carbon emissions: higher reliability applying a reliability penalty of 50 cents (Rel_penalty_0.5) sees higher overall emissions at tiers 3 and 4, a 28% and a 5% increase respectively (see Table 6.4). This increase in emissions for greater reliability is due to off-grid diesel being used in place of the grid, which typically has higher emissions intensity of electricity, but varies depending on the country (see Table C.4).

6.4.4 Inclusion of Carbon Pricing Scheme

Factoring in the implementation of a carbon pricing scheme (Ctax_median), of around \$4/tonne in 2020, rising to \$146/tonne in 2030 (2022 USD), causes notable shifts in the modal shares and distribution (Figure 6.4, panels g to i). As would be expected, there is a shift away from diesel generation as it is the most polluting form of generation and thus incurs the greatest carbon cost. At each of the three tiers (2-4), this shift occurs largely from off-grid diesel to off-grid PV, with the largest change at tier 2 demand. The overall shares of grid electricity remain stable at tiers 2 and 4 but increase at tiers 3 at the expense of diesel generation.

At the country level, the largest shifts to off-grid PV are seen in countries that are reliant on diesel generation or that have more carbon-intensive grids. At tier 3, countries such as Sudan, South Sudan and the DRC see the largest shifts to off-grid PV, and each has a high share of diesel generation under the Ref_central scenario; these countries also have fairly low levels of grid coverage (Figures 6.6, 6.5 d and f). Angola, which has a high grid emissions intensity sees shifts away from both diesel and the grid towards off-grid PV (Figures 6.6, 6.5 c and f).

However, At the whole continent level, there is little difference as the shifts from the grid to PV are counter-weighted by shifts from diesel to the grid occurring in other countries with lower grid emissions intensities. The modelling does not assume off-grid PV to be emissions-free: embedded GHG in PV and battery equipment leads to positive emissions resulting from their construction, and associated carbon pricing.

The application of carbon pricing sees reduced total emissions, saving 12Mt (12%) and 47Mt (23%) at tiers 3 and 4 respectively. The carbon tax scenarios have little or no impact on the overall reliability at the aggregate level: with a 1% reduction at tier 3 and the same demand-weighted mean reliability at tier 4 (see Table 6.4.)

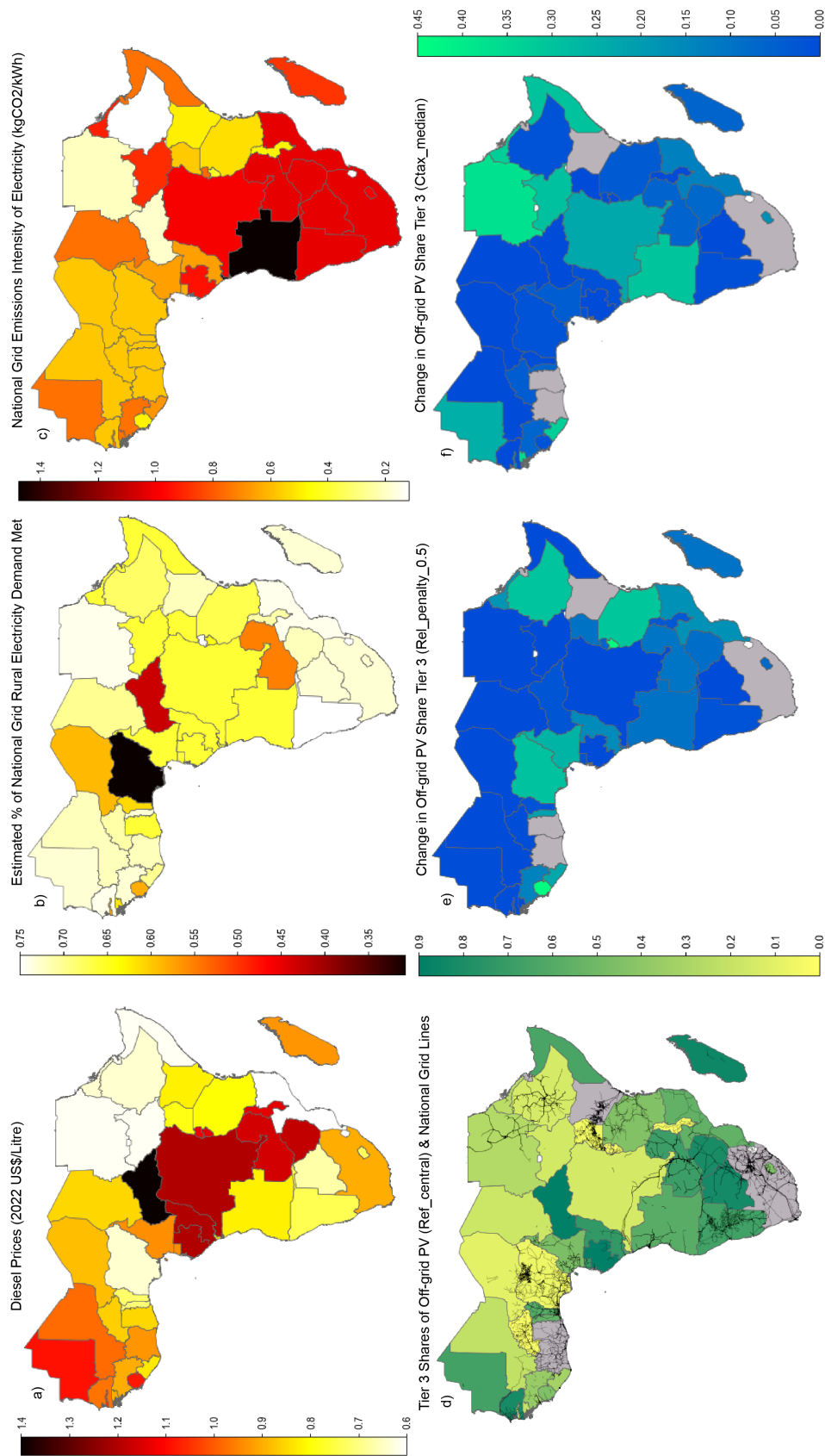


Figure 6.5: Showing diesel prices (top left), grid reliability (top centre), and national grid intensity of electricity (top right) by country; with the share of off-grid PV at tier 3 in the Ref_central scenario with mapped national grid lines (bottom left), and the change in the share of off-grid PV in the Rel_penalty_0.5 and Ctax_median scenarios (bottom centre and bottom right, respectively). These plots demonstrate that a range of country factors influence the initial shares and changes in shares of off-grid PV under reliability and carbon price scenarios. Greyed-out countries in the bottom panels are not included as these reach universal access under the baseline scenario. For country data used for diesel price, grid reliability and grid emissions intensity figures see Table C.4 in Appendix C.

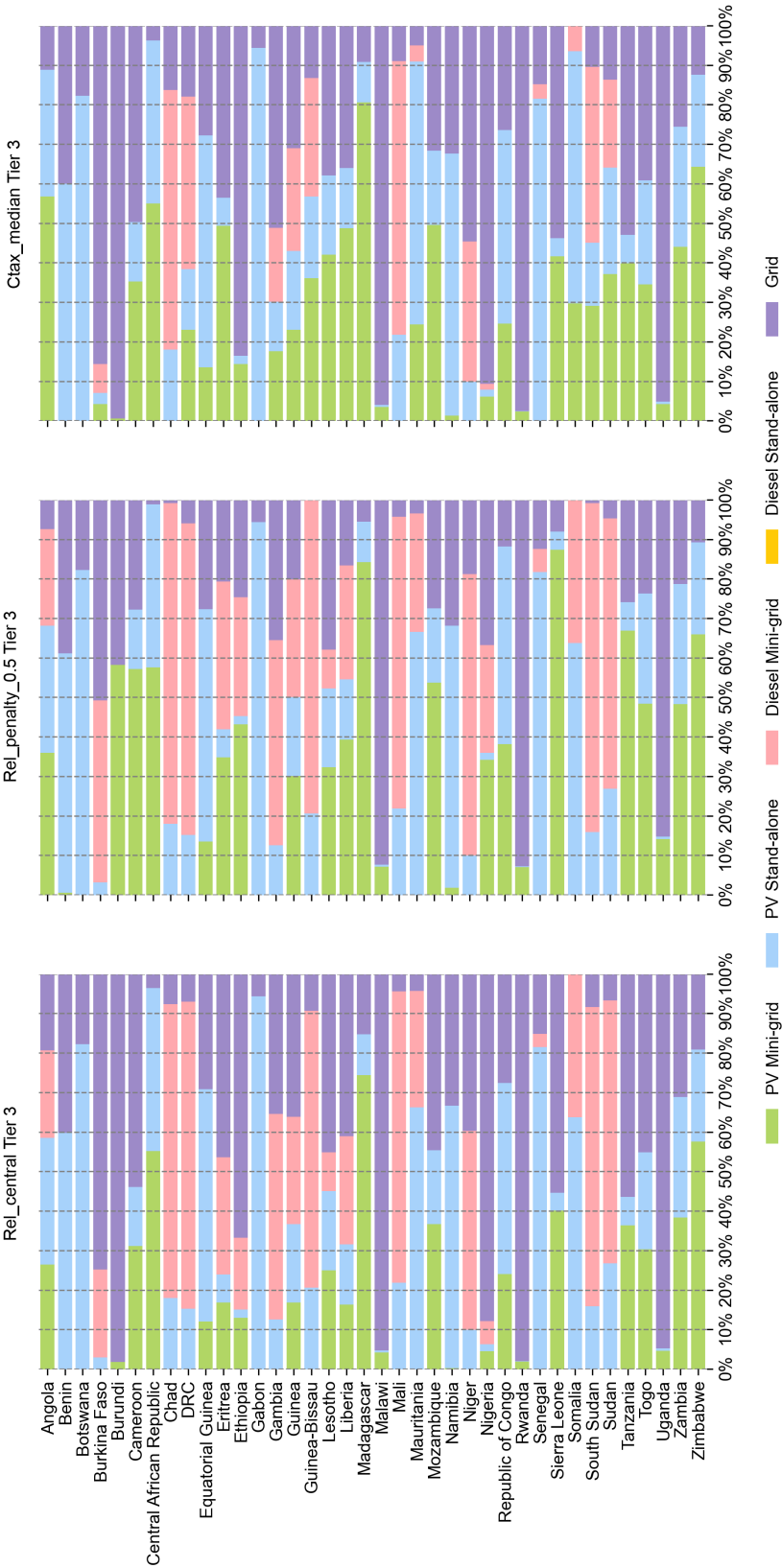


Figure 6.6: Showing stacked horizontal bars of each electrification mode by country (Tier 3) for the Ref_central (Left), Rel_penalty_0.5 (middle), and Ctax_median (right) scenarios. For adapted greyscale version see Figure C.3 in Appendix C.

6.4.5 Sensitivity Analysis

This section outlines the sensitivity analysis I completed for both the carbon pricing and reliability penalty. For reliability, I ran scenarios of penalties between \$0.05 and \$1.00 at 5-cent increments (see Figure 6.7 top panel). For the carbon pricing, I took percentile values of all the carbon prices in the AR6 database for C1 (1.5 degrees no overshoot) and C2 (1.5 degrees with overshoot) scenarios between the 10th and 90th percentile at 10% increments (Figure 6.7 bottom panel) [322]. For carbon prices in 2022 dollars see Table 6.5.

The analysis shows that for the reliability penalty, the results are most sensitive up to a 40-cent penalty, however it varies between tiers of access. At tier 2, the share of the grid drops from an already low level and plateaus after 20 cents. The plateau is due to the share of the population without access that is in direct proximity to or 'under' the grid meaning that the cost of connection is very low and even with a high unmet demand penalty, off-grid modes cannot compete (see Section 6.3.5). Tiers 3 and 4 require higher penalties to see more significant reductions in grid shares, with tiers 3 and 4 seeing the largest shift between 20 and 40 cents. The magnitude of the reductions is greater at tier 4 due to the larger grid shares. At tiers 3 and 4, lower levels of unmet demand penalty (up to 30 cents) see the grid displaced more by off-grid diesel, and at higher levels (30-50 cents) the grid is displaced by off-grid PV.

Table 6.5: Percentile values of carbon prices from the AR6 report for the C1 and C2 scenarios, in 2022 USD [322]

Percentile	2020	2025	2030	2035
10	0.00	18.44	20.75	98.91
20	0.00	28.28	44.05	135.54
30	0.32	39.80	67.94	183.78
40	2.45	64.96	98.30	213.16
50	3.69	89.11	148.01	272.51
60	5.86	120.65	191.40	320.46
70	9.53	161.66	248.40	364.02
80	10.91	200.48	288.77	445.20
90	13.94	271.80	381.10	694.04

For carbon pricing, the shift at the aggregate level is predominantly between off-grid PV and off-grid diesel, the least and most carbon-intensive modes. For tier 2 and 3 electricity demand, carbon prices have more impact at lower levels than for tier 4 demand but continue to increase the share of off-grid PV and reduce off-grid diesel as the carbon prices increase up to the 90th percentile value. For tier 4, the carbon pricing only becomes effective after the median values, after which there are reductions in both off-grid diesel and grid, with off-grid PV filling the gap. The results suggest that generally, at higher levels of demand, a higher carbon price would be required to see meaningful changes; this is logical as the overall investment levels are greater at higher tiers and thus the value of the carbon price will need to be greater to comprise the same share of costs.

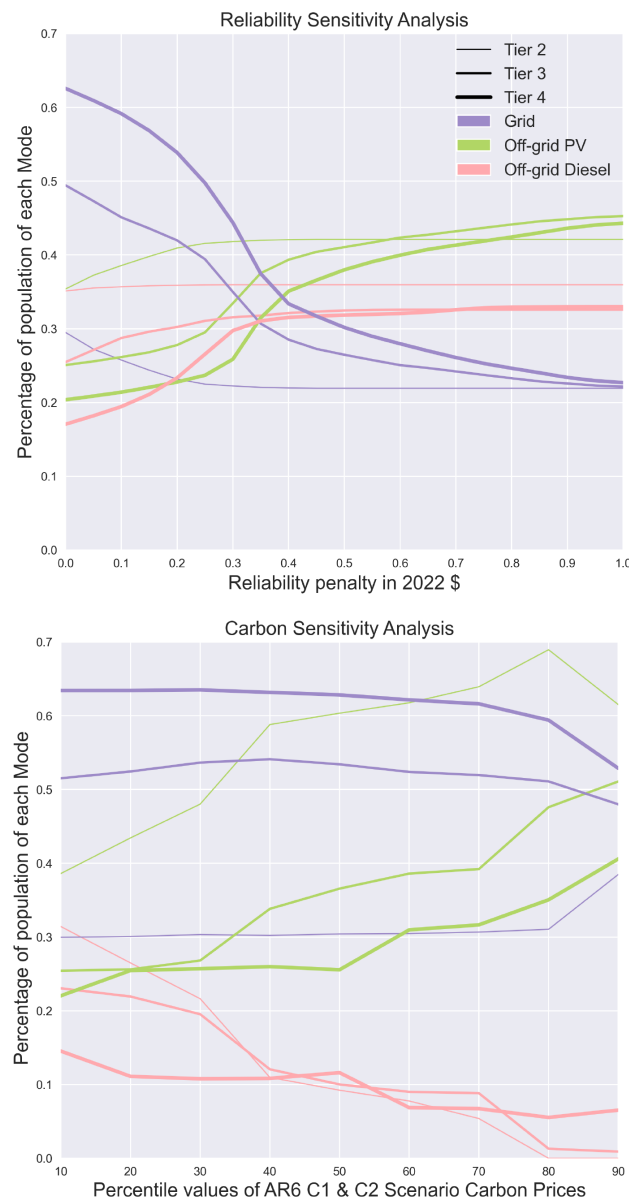


Figure 6.7: Showing the outputs of the reliability and carbon pricing sensitivity analysis. The lines represent the changing shares of each mode of access as the reliability penalty and carbon prices increase. The thickness of the lines represents the different electricity access demand levels (tier 2-4) and the colours of the lines represent the modes. The shares of population displayed are the total shares of all people gaining access under each of the reliability and carbon price scenarios from 2020-2030. For adapted greyscale version see Figure C.4 in Appendix C.

6.5 Discussion and Conclusions

The role of off-grid PV

The work presented in this chapter has used a novel modelling framework to investigate how demand levels, reliability and carbon pricing may influence the share of different electricity access modes, particularly considering the role of off-grid PV. My results suggest an important role for off-grid PV, at all tiers of access examined. At tier 2 level demand, in the reference scenario for over

a third of the population, off-grid PV is the least-cost solution. This equates to around 165 million people from the subset of the population considered in this study (see Section 6.3.4). Critically, this role for off-grid PV is increased when considering both a carbon price and a cost to unmet electricity demand. At tier 2, applying a 50-cent penalty (Rel_penalty_0.5) to each lost kWh of demand sees the number gaining access via off-grid PV increase to almost 200 million; and by applying a carbon price (Ctax_median) it increases further to just under 270 million people (also at tier 2). These findings rest on significant assumptions (discussed more below); however, they indicate that when designing pathways to universal electricity access, carbon pricing and the costs of unmet electricity demand are factors that may need consideration.

Demand Levels

Similar to other work that addresses this question [24], the results presented in this chapter indicate that demand level is crucial when considering pathways to access. This work differs from other research in that it includes growth in demand that varies at the country level; so the importance of the electricity demand level is confirmed but with a different approach. Considering the growth in demand, my results suggest that at higher demand (tier 4), grid extension is most economical for just under two-thirds of connections. For lower tiers of demand (tiers 2 and 3), off-grid solutions (both diesel and PV) are the least cost for more than half of the population that will require access above our baseline scenario estimates. When considering the level of access that policymakers, electrification authorities, NGOs or other actors may want to enable, thinking about demand and future demand growth is vital. Investing in systems most suited to electricity demand levels can ensure efficient use of capital.

Carbon Pricing

The level of demand also matters for carbon pricing. In my main Ctax_median scenario, there is a greater impact on the most polluting sources at a lower level of demand (tier 2). The sensitivity analysis reveals that for higher levels of demand, higher carbon prices would be needed to cause larger reductions in diesel usage. Most still living without a connection are poor households that typically have a low ability to pay and lower consumption levels likely to be closer to tier 2. Consequently, the median AR6 carbon price values may be effective at boosting the uptake of lower-carbon off-grid PV, or grid connections in some countries with low grid emissions intensity. Another consideration is whether the median AR6 prices for carbon would ever be achieved. Reflecting on that, for tier 2, the sensitivity analysis indicates that even at the lowest carbon prices investigated there is some impact in reductions in polluting generation sources, and so a lower carbon price is likely to be sufficient to see some incentivisation of lower-carbon options. Regardless, carbon pricing, whether through the existence of voluntary schemes to encourage carbon credits [163, 165]; or through more formal taxation is salient for considering the future uptake of off-grid low-carbon electricity access solutions. My results indicate that for higher demand levels when the infrastructure costs are higher, carbon prices will need to be higher to incentivise moves away from polluting options such as off-grid diesel or carbon-intensive national grids. In countries with a low grid emissions intensity of electricity, carbon pricing schemes will incentivise both off-grid PV and grid extension over diesel.

Reliability

On the reliability of electricity supply, applying a financial penalty for unmet units of demand to represent the cost to households of outages [4, 42] has significant effects. As with carbon pricing, lower financial penalties are effective at lower demand tiers. For tier 2, 20 cents per kWh unmet and below is effective for shifting to more reliable means (see Figure 6.7). At higher tiers (3 and 4), lower demand penalties see shifts from the grid to diesel first, then PV at higher levels. This is something policymakers should have an awareness of if their intention is to stimulate the uptake of low-carbon technology rather than diesel. In reality, an unmet demand penalty would be challenging to implement: with demand not met only possible to quantify based on historical trends of consumption and hourly outage data. In reality, fines could more easily be applied for lost hours of service, or large frequency fluctuations; both beyond the capability of this modelling framework to consider in its current form. However, albeit with limited data (see below), this work highlights the importance of factoring in the reliability of different modes of access when modelling pathways to universal access; this has so far been overlooked in the literature.

Limitations and Further Work

This study uses up-to-date information regarding costs, and where appropriate assigns costs at the country level. Further, it takes into account anticipated cost reductions in low-carbon technologies (PV and battery systems). However, other costs used are static such as diesel prices and the price of additional grid capacity. In reality, these would change over time. Additionally, the grid emissions intensity of electricity is likely to change over time, impacting the shifts between modes when carbon prices are applied. Whilst these could be incorporated within the modelling framework, I chose not to include them due to the uncertainty in their trajectory; and the high number of other variables this work considers.

This work only considers households needing access and household electricity demand. In reality, there are additional users such as businesses and institutions that need electricity access. These other users are critical and likely make up a significant portion of the currently unmet demand for electricity. This work did not consider them due to poorer data availability and the added complexity of doing so. Further work should incorporate these demands to give a fuller picture of the electricity resource needs in different places.

This study uses synthetic demand profiles whilst modelling different energy systems. Whilst the methodology considers differences in demand dynamics that may occur at different tiers and in different locations, there is likely to be demand diversity within countries that would impact the least-cost supply option. Further work is needed to better understand electricity demand at a high resolution and to calibrate and validate demand model predictions against real data.

This work suggests that different access solutions may be appropriate at different demand levels for some households depending on where they are. This work also takes into account electricity demand growth up to the target year of universal electrification (typically 2030). However, the question of what may happen when users gain access from one solution e.g. solar mini-grids and demand levels grow past 2030, is not addressed in this work. When considering future growth in the very long term it may be better to invest in other solutions e.g. grid extension. Future work should seek to look at initial investment in access, as well as investment in upgrading access for connected users as incomes and electricity demand grow in the much longer term. Doing so could help to avoid stranded assets (e.g. stand-alone systems or mini-grids) and help to consider how

different access solutions can be used alongside one another.

The modelling framework used has strengths: the representation of demand and off-grid systems allows the questions above to be investigated. Further, the least-cost model is relatively fast to run and is therefore well suited to investigating multiple scenarios with an annual temporal resolution. However, there are key weaknesses. Firstly, the spatial resolution (10km²) misses some detail. There may be households in a cell categorised as having grid extension as the least cost solution but are far enough away from the centroid of the cell that an off-grid solution would be the most economical. Further, a higher spatial resolution would permit more accurate distinctions between households most suited for stand-alone systems versus mini-grids.

Secondly, the representation of grid extension in this modelling framework is simplistic. The model does not incorporate a grid-routing algorithm. Instead, it uses the distance from the grid as a determinant of the cost of grid extension. In reality, many other factors come into play, e.g. the route taken (via other settlements) and the terrain the grid travels over. The OnSSET model incorporates a sophisticated grid routing algorithm that takes account of these factors [24, 167]; however, this approach would be less suited to the analysis presented in this chapter, which requires many scenarios to be addressed, without significant increases in the available computing power. Additionally, distinguishing between low-, medium- and high-voltage lines, and the positioning and cost of transformers and other equipment would further strengthen the analysis.

Finally, whilst there is novelty in this work in the fact that it attempts to take into account reliability in the least-cost modelling, the available data on grid reliability for Sub-Saharan Africa is extremely poor, limiting the depth of the analysis. In reality, and as demonstrated in Chapter 4, the reliability of the grid in low-income countries can vary greatly. However, the vast majority of households without access, exist in areas outside urban centres [1, 41], and this is also where grid reliability is lower in Sub-Saharan African countries [319]. It is therefore logical to assume reliability is lower for rural households than for enterprises in urban centres [25]. A more sophisticated representation of reliability would permit greater accuracy from the results. As yet, no geospatial data set exists that approximates grid reliability over space. Additionally, this work does not account for improvements in grid reliability and a comparison of the cost of grid strengthening versus investments in off-grid technologies under the unmet demand penalty scenarios.

Further work should seek to address these key weaknesses: 1) Improving the spatial resolution whilst keeping the model as computationally light as possible, 2) incorporating a more sophisticated representation of grid extension, that still facilitates annual temporal resolution and a scenario-based approach; and 3) including a spatially dis-aggregated representation of grid reliability and considering the possibility and costs of grid strengthening.

6.6 Summary

This chapter combines several methodological approaches to investigate how demand, carbon pricing and reliability may have an impact on pathways to universal household electricity access in Sub-Saharan Africa. It highlights the importance of considering all three factors in decision-making around improving electricity access, and their impact on each other.

Differing from the previous two chapters, this chapter focuses on a region-wide study, rather than a case-specific one. However, the findings in this chapter can inform case-specific studies. The availability of carbon credits in future may enable electrification at the community level via

the use of off-grid PV. Additionally, off-grid systems designed to meet high levels of reliability can be seen as advantageous over weaker grids, that may be near or present at a given case study site.

While this chapter applied the methods and modelling framework to 43 countries in Sub-Saharan Africa, the approach could be adapted and new data used, to look in-depth at a single country or district, or at another region. Improved data such as on national grid reliability would provide greater granularity and would deepen the possibilities for further analysis. This chapter has demonstrated that using geospatial data models, combined with energy system modelling can allow complex questions to be answered and both sets of tools should be considered when asking research questions relating to improving electricity access in low-income countries.

Chapter 7

Conclusions

7.1 Overview

This thesis has developed and used modelling tools to address research questions relating to improving electricity access with off-grid, renewables-based electricity systems. It has done so by investigating a range of contexts and, where possible, by using the most up-to-date and detailed system-level or country-level data available for the studies to ensure the relevance of the results. This section will present a summary of each chapter, the key results of the thesis, the implications of the results, the limitations of the work and recommendations for future work.

Chapter 3 presented the modelling tools that were used and developed for the work in this thesis and included my two main methodological contributions. Firstly, it presented new functionality within the open-source CLOVER energy system model that facilitates the modelling of dispatched generation within hybrid diesel, PV and battery mini-grids. The changes made allow the modelling of cycle-charging capabilities, whereby diesel generators are used to meet demand and charge the batteries. This updated version of the CLOVER model was used for the results in Chapter 5. Secondly, in Chapter 3 I also presented a new open-source geospatial energy model that is designed to consider least-cost approaches to electricity access under scenarios that apply carbon pricing and penalties for unmet electricity demand. This was developed to produce the results in Chapter 6.

Chapter 4 demonstrated the potential that diversifying the electricity consumer base beyond domestic users has for improving the cost of electricity from solar-based mini-grids in India. Anchor loads - in this case using the example of rural health centres - have the potential to be complementary electricity consumers to domestic users, improving both the financial and environmental sustainability of solar-PV and battery-based mini-grids through a lower unit cost and greenhouse gas emissions intensity of electricity. Improving the economics of solar mini-grids can help to speed up their deployment and ensure they fulfil their potential role in providing high-quality, low-carbon access to electricity in countries still facing the persistent challenge of poor electricity access. Additionally, Chapter 4 explored a range of off-grid system types and reliability levels and used monitored grid reliability data to determine whether the grid could meet the demand profiles of the study. For the highest levels of reliability (0% unmet demand), my results indicate that hybrid mini-grids offer the lowest cost of electricity. The rural grid offers poor reliability and is unable to meet the electricity demands of households or health centres. The reliability of the grid is lower further away from urban centres.

Chapter 5 highlighted the importance of considering the way that hybrid mini-grid systems - comprising PV, batteries and dispatched generation - are operated to ensure they are done so in the most cost-efficient way. There are crucial trade-offs between deeper battery cycling and increased fuel use depending on the operational strategy selected. Cycling the batteries too much could result in higher costs from battery degradation, whereas cycling batteries too little will result in higher fuel costs. For the system detailed in my study, the alternative strategy identified by my modelling saw deeper battery cycling unlock fuel savings that outweighed the cost of battery degradation. However, this was not the case for all strategies assessed and operators of hybrid mini-grids with a similar configuration should be mindful of the trade-offs between battery cycling and fuel use. Chapter 5 also highlighted the potential for the use of solar and battery technology to drastically reduce fuel costs in humanitarian settings. High expenditure on fuel for diesel generators in such settings means that installing solar and battery capacity can have relatively short payback times. For the study presented in Chapter 5 the payback time was estimated to be around five years. Chapter 5 also highlighted the potential to expand the electricity consumer base. At different levels of additional consumers, in this case, refugee businesses, there were different technical impacts on the electricity system. A low number of additional users had a relatively low technical impact, but a higher number resulted in increased fuel burn. Depending on the tariffs charged to these consumers, the additional costs could be more than recouped.

Chapter 6 demonstrated the potential importance of electricity demand levels, carbon pricing schemes and reliability levels for determining least-cost pathways to universal household electricity access in Sub-Saharan Africa. The analysis used a combination of modelling tools to provide outputs at the grid-cell, country and whole region levels. The work demonstrated that demand level is critical: low levels of demand lend better to off-grid, PV-based systems, while higher demand levels see increased levels of usage of off-grid diesel systems and national grid extension. Carbon pricing reduces the use of diesel generators and the impact on the share of the grid, although this varies by country. Applying a penalty for unmet demand causes a shift away from national grid extension towards both off-grid PV and diesel systems. The sensitivity analysis in Chapter 6 also highlighted the need for higher carbon pricing and unmet demand penalties to cause shifts between different modes at higher levels of demand. This work highlighted how country-specific factors such as grid reliability, grid emission intensity and diesel pricing have an impact on the shares of each mode in each country and how they change under different scenarios.

7.2 Key Results of Thesis

- When considering adding health centre anchor loads to domestic PV and battery mini-grids, there are substantial cost and emissions reductions with anchor loads included, relative to domestic-only demand profiles. The magnitude of the reductions depends on the type of health clinic and the level of reliability, but the savings in the unit cost of electricity range from 25 to 53% for solely PV and battery-based systems.
- I find that for the mini-grid site in northern India, at 95% of demand met, PV and battery systems are 14-22% cheaper than diesel-only systems, with 10 times lower carbon intensity. The national grid has poor availability, at just over 50% of minutes at the most comparable monitoring site to the mini-grid location. I find that the national grid would meet a lower fraction of the health centre anchor load demand profiles than the domestic-only ones. In other words, blackouts have a greater coincidence with health centre demand than domestic

demands.

- For the hybrid mini-grid site in Rwanda, installing solar PV and battery capacity reduces fuel expenditure and fuel-related emissions by 74%, with further cost savings of 7% by optimising the operational strategy of the system. The fuel savings of the alternative operating strategy outweighed additional battery degradation costs. Based on system costs and reduced fuel usage, the payback period for the equipment is around 5 years.
- The hybrid mini-grid in Rwanda could permit a lower number (20) business customers to be connected with little additional fuel burn due to excess solar-generated electricity available. Adding greater numbers of businesses (50 or 100) resulted in much higher levels of additional fuel burn, with it only possible to recoup costs under the higher 'mini-grid' tariff.
- In considering pathways to universal household access, the level of demand matters: in the reference case at lower demand (tier 2) off-grid PV amounts to 35% of connections, at higher demand (tier 4) this share drops to just 20%, with the grid the dominant mode used. The share of off-grid diesel used declines at higher levels of demand and is greatest at tier 2 demand.
- When applying a penalty of 50 cents per kWh, the share of off-grid PV used increases across all tiers, with increases of 35 to 42%, 24 to 41% and 20 to 36% at tiers 2, 3 and 4 respectively. Carbon pricing sees substantial increases in the share of off-grid PV used, but the impact declines at higher demand (tier 4). Sensitivity analysis implies that a higher carbon price scheme is needed to have a larger impact on reducing diesel usage at greater demand levels.

7.3 Implications of Research

Here I will briefly detail some relevant implications for different groups of the research presented in this thesis.

Firstly, for mini-grid developers, the results in Chapter 4 of this thesis quantify the financial up-sides to connecting larger, 'anchor' loads to mini-grids serving households. This thesis presents results for the first time exploring rural health clinics as an anchor load. The results suggest that the load profiles may be complementary, providing a considerable unit cost of electricity saving. Primary health clinics in more remote parts of lower-income countries are often without adequate power, and there are considerable benefits from energising healthcare facilities. The inclusion of rural health clinics may offer a route to enhanced financial sustainability for solar-based mini-grids. Mini-grid companies may wish to seek partnerships with local authorities operating health clinics and stress the potential mutual benefits of their inclusion within mini-grids. Additionally, of relevance to mini-grid developers and operators, the results in Chapter 5 quantify how the operating strategy of hybrid mini-grids may impact financial sustainability. Given the high cost of both diesel fuel and battery replacement, it is important that hybrid mini-grids are operated in a way that considers the lifetime costs of each. The use of open-source modelling tools (more below) can be an effective way of exploring the trade-offs between deeper battery cycling and diesel fuel costs for a given existing or future system.

For humanitarian organisations that may have considerable annual energy costs and are typically reliant on polluting diesel generators, the results in Chapter 5 of this thesis offer insights into the benefits of hybridising diesel systems. Investing in solar and PV battery capacity to complement existing diesel generators allows humanitarian organisations to save money on energy costs over the medium term, with relatively short payback times from reduced fuel expenditure. Additionally, this thesis presents results for the first time exploring the feasibility of connecting refugee business owners to camp energy systems. These results indicate that humanitarian organisations that hybridise energy systems may wish to explore connecting other user groups such as business owners to use surplus energy. However, depending on the scale and shape of demand profiles they should be mindful of additional fuel usage and the tariff requirements to break even.

For electricity planning authorities and policymakers, the results in Chapter 6 of this thesis highlight the importance of considering reliability levels of access solutions when devising electricity access plans. Poor grid reliability results in costs for households. When including a financial penalty for unmet electricity demand in least-cost modelling, a greater proportion of households gain access via off-grid systems. This finding is particularly important for relevant actors in countries with weaker grids. Factoring in a financial penalty for lost demand within such models offers a new way to consider reliability in models (a key tenet of SDG7) that can ensure it is better considered when conducting electricity access research.

Additionally, the results in Chapter 6 also emphasise the impact of the electricity demand level. Planners and policymakers should consider existing and predicted levels of future electricity demand level and consider cheaper off-grid systems for lower levels of electricity demand. However, there is considerable spatial heterogeneity and modelling tools (more below) may be useful for exploring different modes of access in different places and at varying demand levels.

With implications for organisations working within the voluntary carbon market, the results in Chapter 6 highlight the impact on the shares of each technology under a carbon pricing scenario. Identifying areas where carbon pricing may have an impact on a shift between a higher carbon and lower carbon mode of access is needed to properly target interventions to where they are most effective.

Finally, and most importantly, it is important to highlight the potential impact of the two open-source models this thesis describes in Chapter 3. Much of the modelling of off-grid electricity systems is undertaken using 'black box' commercial software leading to issues around transparency and reproducibility of results. Paid software also may create a financial barrier for some user groups. This thesis presents the development of, and detailed description of, new functionality in the open-source CLOVER model that enables the modelling of hybridised diesel mini-grids that are increasingly found in off-grid settings. The changes to the model were undertaken with consultation from a mini-grid owner-operator, and with the consideration of a real system in the hope that real-world usefulness is improved. The updated model gives those who wish to model hybrid mini-grids a free, open-source alternative to other tools that they can adapt and apply to future settings as needed.

In Chapter 3 this thesis also presents a new, open-source and flexible modelling framework for considering different electricity access technologies across countries or regions that has implications for planners or researchers who need to conduct relevant work. Whilst other open-source tools exist, they are computationally intensive and typically are best designed for snapshot analysis. The tool outlined in this thesis offers a new approach. It is designed with an annual temporal time horizon and allows users to easily consider different time horizons for universal electricity access. It permits

users to run many scenarios and sensitivities easily and includes emissions accounting of different modes, facilitating users to test the impacts of carbon prices. This alternative to existing tools can be used and adapted by other researchers to be applied in other contexts.

7.4 Limitations of Work

In this section, I will discuss the broad limitations of the work presented in this thesis. There are more detailed, chapter-specific limitations relating to the respective methods used that are not discussed here but can be found in the discussion sections of each chapter. Here I cover limitations relating to modelling-based work, data inputs, important non-technical and cost factors, and finally applicability of studies to other contexts.

Models are flawed and provide a coarse representation of reality. They have limitations both in their scope and with the outputs they provide. Both the models used in this thesis have some fundamental limitations that should be highlighted. The CLOVER model, used in all results Chapters (4-6), has characteristics in its design that create limitations for the research. The hourly resolution has drawbacks both on the demand and the supply side. For demand, some devices, such as for medical uses (as discussed in Chapter 4), have a usage profile that demands high levels of power for short periods, e.g. minutes or seconds. The CLOVER model does not represent this and therefore misses the peak demands placed on the system, and possible blackouts that could occur. For the supply side, these sub-hourly peak demands may, for example, place additional strain on battery capacity and cause additional battery degradation. These dynamics are missed and exemplify the oversimplification of my modelling approach. There are trade-offs between representing greater complexity and requiring additional data and computation. Likewise, specifically for the work in Chapter 6 using the least-cost geospatial tool, a range of oversimplifications in the modelling limit the work. Firstly, using a higher spatial model resolution for the analysis would permit greater accuracy in the results, but would require more intensive computation and would limit the scenarios and sensitivities looked at. The work uses country-level data for demand, reliability and other variables such as grid capacity costs. In reality, there are high levels of intra-country differences not captured. This would require new high-resolution datasets or downscaling, using other available data. It would also require more intensive computation. Future work (see below) should seek to address this gap. This is particularly important for electricity demand. Whilst the study outlined in Chapter 6 explores different levels of demand, with variation between countries, policymakers and planners need to better understand what demand is likely to be, so as to effectively plan electricity systems.

To enhance model accuracy and usefulness, and to gain an understanding of how it compares to similar models, models should undergo a validation process where outputs are tested against reality, and other model outputs. The CLOVER model has been applied to many real-world settings, and developed in conjunction with organisations engaged in real-world projects with the aim of ensuring it provides useful outputs. It has however never undergone a structured validation process whereby the outputs are tested against real-world conditions or other models. This is a limitation of the research. With regards to Chapter 6 given the nature of the research, there is also a limitation around the validation of the geospatial model. Whilst many of the data inputs e.g. the population without access have undergone validation, the framework used and developed in Chapter 6 has not. To improve the accuracy of geospatial tools and data, testing output values against ground truth data or other model outputs at given points is important. For example, in the

case of the least-cost geospatial model, testing against real-world data e.g. via specific electricity access plans, existing infrastructure or investor attractiveness for a given technology. This process requires time and availability of data, which is not always readily available in low-income countries. Model validation is important and should be addressed if the tools presented in this thesis are to have enhanced usefulness and robustness.

Modelling analysis requires data inputs and assumptions to be made about the subject matter. Efforts have been made in this research to use the most relevant and up-to-date data available and to use assumptions that most accurately represent reality. One of the research gaps that this thesis addresses is to provide detailed studies with data from real electricity access projects to answer research questions on how to improve economic performance. However, there are limitations arising from the data and assumptions used, this is the case for all three results Chapters 4 - 6. Costs change over time, often due to factors that cannot be predicted. Diesel prices, for example, are used as an input for all three studies presented here. Unexpected global events, such as Russia's 2022 invasion of Ukraine, can have large and lasting impacts on fossil fuel prices, and have significant consequences on the accuracy of modelling results. Likewise, I have cost reduction assumptions for the low-carbon technologies included in my work. One can have a high degree of certainty that there is a downward direction due to learning rates of modular low-carbon technologies; however, unpredictable factors such as supply-chain disruption and heightened global demand for commodities can cause disruption to expected cost reduction trends. These uncertainties exist beyond cost data: assumptions for the spatial distribution of population and population growth, or forward projections of electricity demand growth may prove quickly to be incorrect. The data and assumptions for this work were up to date at the time of completion of the analysis but the work has been carried out during a period of great uncertainty: a global pandemic, war and associated economic shocks. It is important to consider this when reading the results and this uncertainty is a significant limitation of research of this nature.

The work presented in this thesis has focused on technologies and their costs for electricity access under different conditions, using models for analysis. The technical and economic aspects are important considerations for electricity access and project implementation. However, there are other elements that have importance. For example, when implementing projects such as renewables-based off-grid electricity systems, there are critical social, political, and cultural aspects that need to be taken into account. These may include willingness and ability to pay, community-specific needs for electricity, laws and regulations surrounding electricity generation and supply, and social structures within communities and how these impact technology use. While this is beyond the scope of the research presented in this thesis, it is relevant to consider. For the work in Chapter 4, connecting health centres as anchor loads would have community-specific practical challenges, and would also require engagement and support from local health authorities and other bodies. For the work presented in Chapter 5, connecting to refugee businesses in the camp would have a range of hurdles beyond the technical and financial impact on the system. These may include administering connections and billing, allocating connections fairly or obtaining institutional approval. Finally, the work in Chapter 6 which examines technology choice based on cost under different scenarios for all of Sub-Saharan Africa clearly misses details at the country or community level. This limitation must be acknowledged for the work presented in this thesis. Decision-makers should refer to research on electricity using a range of methodological approaches and areas of focus to ensure a fuller picture beyond technology costs and performance is understood.

A further point, related to the above, is the applicability of the case studies presented in Chapters 4 and 5. It is important to acknowledge the findings and consider their relevance to

other situations. For example, in Chapter 4 my results suggest that PV is cost-competitive with diesel and that for higher reliability, hybrid systems may be the most cost-effective. It is likely that these findings are relevant to other contexts given their dependence on prices of components and fuel, which apply beyond a single community. That said, community-specific factors e.g. the demand profiles, that may be related to the economic activity or income level also play a part. The case study outlined in Chapter 5 describes a system in a refugee camp in Rwanda, the configuration and setting of which will have similarities and differences with other refugee camps that must be taken into account when drawing insights. This is an obvious limitation of the work; however, important to highlight that there will be context-specific factors that will have an impact on the cost, technical performance and feasibility of implementing off-grid electricity systems.

7.5 Future Work

There is important further work that relates to the research in each of the Chapters 3 to 6 and I have outlined some of these chapter-specific points already. Here I will briefly summarise what I believe to be the most pressing future work that has resulted from the outcomes of this thesis.

Firstly, although the work in this thesis in Chapters 4 and 6 addressed the reliability of the supply of different electricity access options, this is an area that needs urgent further work. Data availability on the reliability of national grids, and also of off-grid systems, is poor. As decisions are made on pathways to connecting remaining populations without electricity access, without having a clear understanding of the relative service level of different options, resources may be allocated poorly. In addition, there are an estimated 3.5 billion people who are thought to have an electricity supply that is not reliable [3], but our understanding of where the problem is worst and who is most affected is also poor. Improved data would allow better planning of different resources in different places. For example, resources may be more efficiently used building a mini-grid, rather than adding new grid connections at the end of a grid line with very poor service levels. Further work is needed to build up databases on the reliability of national grids and off-grid systems in different locations to enable better planning of electricity systems so that more people can exist with high-quality, reliable connections that are taken for granted in high-income countries.

This thesis presented two in-depth case studies that look at ways to reduce costs for off-grid solar-based electricity systems. Both of these examples demonstrated the potential of adding additional components of electricity demand to these off-grid systems. Although the research used representative monitored electricity demand data where possible, this modelling work was exploratory and its findings are the result of running simulations. An important next step is to test these results on real systems. In the process, future work should look at the practical barriers to implementation such as tariff schemes, willingness to pay for electricity, regulatory hurdles, and how systems perform in real life over time with different user groups. For example, this work has demonstrated the potential for health centres to act as consistent consumers of electricity, and to potentially cross-subsidise domestic users. However, practical implementation of such an arrangement would require regulatory permissions and the involvement of local health authorities. Future work to address implementation will help to build on the work presented in this thesis and ensure that barriers are reduced for achieving lower-cost, sustainable electricity for those who need it.

Finally, there is important further work needed on the use, development, maintenance and promotion of open-source modelling tools for electricity access. There is a growing ecosystem

of different tools available, each with strengths and weaknesses and appropriate use cases. The tools presented in this thesis and their application to different research questions demonstrate the potential that they have to give insights for electricity access researchers. Additionally, this thesis has highlighted many shortcomings of the models used, some of which could be addressed with further development, or use in conjunction with other existing tools. Future work should explore the expansion of these tools and how they can be better linked with other existing modelling tools, techniques and data sets to avoid duplication of work. Additionally, it is important that modelling tools are well-maintained, and that efforts are made to provide appropriate materials and resources for other potential users to make good use of them to ask other research questions.

7.6 Thesis Summary and Final Considerations

This thesis has demonstrated how open-source modelling tools, and high-resolution and real-world data can be used for addressing research questions focused on electricity access. It has outlined new ways of thinking; developed new tools and applied them to new settings. The results in each chapter exemplify the ways in which modelling can demonstrate ways in which cost optimisation of off-grid electricity access can be addressed, at the single system or regional level. Whilst techno-economic modelling is a useful method for electricity access research, it is only part of the picture. Electricity access is a complex challenge that goes beyond technology and technology costs. To come within touching distance of reaching universal electricity access this decade, a multi-faceted approach is needed, considering many other equally important challenges such as the right policy support, viable business models and addressing community-level barriers.

Appendix A

Supplementary Tables and Information Chapter 4

Table A.1: Grid of probabilities used for a single device to generate a profile. This example shows the grid for fans used in the Pathlab.

	Month											
	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
Hour	1	0	0	0	0	0	0	0	0	0	0	0
	2	0	0	0	0	0	0	0	0	0	0	0
	3	0	0	0	0	0	0	0	0	0	0	0
	4	0	0	0	0	0	0	0	0	0	0	0
	5	0	0	0	0	0	0	0	0	0	0	0
	6	0	0	0	0	0	0	0	0	0	0	0
	7	0	0	0	0	0	0	0	0	0	0	0
	8	0	0	0	0	0	0	0	0	0	0	0
	9	0	0	0	0	0	0	0	0	0	0	0
	10	0	0	0	0	0	0	0	0	0	0	0
	11	0	0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0	0
	12	0	0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0	0
	13	0	0	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0	0
	14	0	0	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0	0
	15	0	0	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0	0
	16	0	0	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0	0
	17	0	0	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0	0
	18	0	0	0.75	0.75	0.75	0.75	0.75	0.75	0.75	0	0
	19	0	0	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0	0
	20	0	0	0	0	0	0	0	0	0	0	0
	21	0	0	0	0	0	0	0	0	0	0	0
	22	0	0	0	0	0	0	0	0	0	0	0
	23	0	0	0	0	0	0	0	0	0	0	0
	24	0	0	0	0	0	0	0	0	0	0	0

Table A.2: Technical system modelling inputs used for Chapter 4.

Item	Value	Unit	Source
Battery Maximum Charge	0.85	State of charge (0.00-1.00)	[323]
Battery Minimum Charge	0.15	State of charge (0.00-1.00)	[323]
Battery Leakage	0.004	Fractional leakage per hour	[144]
Battery Conversion in	0.93	Conversion efficiency (0.00-1.00)	[144]
Battery Conversion out	0.93	Conversion efficiency (0.00-1.00)	[144]
Battery Cycle Lifetime	2500	Expected number of cycles over the lifetime	[324]
Battery Lifetime Loss	0.30	Fractional loss over the lifetime (0.00-1.00)	[323]
Battery C Rate	0.20	Charge/Discharge rate	[324]
Battery Replacement Capacity	0.70	Minimum level of capacity before replacement (0.0-1.0)	
Transmission efficiency DC	0.96	Efficiency of distribution network (0.00-1.00)	[325]
Transmission efficiency AC	0.92	Efficiency of distribution network (0.00-1.00)	[325]
DC to AC conversion	0.97	Conversion efficiency (0.00-1.00)	[326]
DC to DC conversion	0.95	Conversion efficiency (0.00-1.00)	[325]
AC to DC conversion	0.90	Conversion efficiency (0.00-1.00)	[327]
AC to AC conversion	0.98	Conversion efficiency (0.00-1.00)	[326]
PV panel tilt	29	Degrees above horizontal	
PV panel azimuth	180	Degrees from North	
PV panel lifetime	20	years	
PV panel degradation	0.01	Fractional degradation per year	
Diesel generator fuel consumption	0.28	Litres per kW capacity per year	[328]
Diesel minimum load	0.30	Minimum capacity factor (0.00 – 1.00)	

Table A.3: Emissions modelling inputs used for Chapter 4.

Item	Value	Unit	Comments	Source
PV Panels	1520	kgCO ₂ /kWp	Based on multi-crystalline silicon PV manufactured in China	[116]
Battery	110	kgCO ₂ /kWh		[329]
Diesel Generator	476	kgCO ₂ -eq/kW		[330]
Diesel Fuel	2.68	kgCO ₂ /litre		[331]
Balance of System	134	kgCO ₂ /kW	Based on the sum of figures of BOS	[332]
Inverter	124	kgCO ₂ /kW	Based on manufacture in China, 0.89 kW inverter/kWp	[116]

Table A.4: Financial modelling inputs used for Chapter 4.

Item	Value	Unit	Comments	Source
PV Panels	392.40	\$/kWp	17% efficiency, 300-watt panel	[333]
Discount Rate	10	%		[328]
PV O&M	7.85	\$/kWp p.a.		[334]
Battery Cost	325.61	\$/kWh	Lithium-ion Phosphate battery 1.02kWh	[335]
Battery O&M	3.26	\$/kWh p.a.		[334]
Battery cost reduction	7.5	% p.a.		[336]
Diesel generator cost	426.92	\$/kW		[337]
Diesel fuel cost	0.97	\$/litre	Based on one year of daily diesel prices, Delhi, India	[338]
Diesel O&M	42.69	\$/kW p.a.	10% of capital cost p.a.	[328]
Balance of System	348.69	\$/kWp	Cables, Racks, Charge controllers	[339] & Private correspondence, technical employee Oorja Solutions
Connection & Distribution Network	5627.95	\$	Estimate for 45 Households	Private correspondence with Oorja Solutions and [328]
Connection & Distribution Network (Community + Health centre)	5798.72	\$	Estimate for 45 Households + Health centre	Private correspondence with Oorja Solutions and [328]
Inverter	170.46	\$/kW	SMA inverter	[340]
Misc. Costs	1458.76	\$/kW		Private correspondence, technical employee Oorja Solutions
General O&M	1785	\$ p.a.	Costs of spare parts and technician	Private correspondence, technical employee Oorja Solutions

Appendix B

Supplementary Tables, Figures, and Information Chapter 5

B.1 Constructing load profiles for Mahama Camp

Institutional profile

This part of the load profile generation was undertaken by Dr Philip Sandwell and is described here for reference as it comprises part of the study presented in Chapter 5.

The total load monitored in the hour t , $L^M(t)$, in kWh, is taken to be the sum of the energy supplied to the consumers by each source i , $S^i(t)$, for $i \in \{\text{PV, Diesel, Battery}\}$, (each in kWh), such that

$$L^M(t) = \sum_i S^i(t) \quad (\text{B.1})$$

The total load $L^M(t)$ is then reformatted into a matrix L_{dh}^M of dimensions $d \times 24$ for each day $d \in Z^+$ recorded over a total of d^T days, and at the hour $t \in \{0, 1, 2, \dots, 23\}$. The mean μ_t^M and standard deviation σ_t^M of the monitored data in each hour t are given respectively by the usual formulae

$$\mu_t = \frac{1}{d^T} \sum_d L_{dt}^M \quad (\text{B.2})$$

and

$$\sigma_t^M = \sqrt{\frac{\sum_d (L_{dt}^M - \mu_h^M)^2}{d^T}} \quad (\text{B.3})$$

These are used as inputs to generate a synthesised load profile given in kWh, $L^S(t)$, which can be taken as an input to the modelling process. A new matrix L_{lt}^M of dimensions $l \times 24$, for each day $l \in Z^+$ in the time frame considered, in our case $l \in \{1, 2, \dots, 1825\}$ for five years, is constructed. Each element of L_{lt}^S is set to the value of Q_{lt} drawn at random from the normal distribution

$\mathcal{N}(\mu_h^M, \sigma_h^{M2})$, for its corresponding hour t , and subject to the condition

$$L_{lt}^S = \begin{cases} Q_{lt} & \text{for } Q_{lt} \geq 0 \\ 0 & \text{for } Q_{lt} < 0 \end{cases} \quad (\text{B.4})$$

This condition avoids unphysical negative electricity demands. As this process does not allow values less than zero, as $\mathcal{N}(\mu_h, \sigma_h^2)$ does, this results in the mean of the synthesised data, μ_h^S , being greater than the monitored data, i.e. $\mu_h^S \geq \mu_h$ by definition. In practice, the difference between the two values is negligibly small and $\mathcal{N}(\mu_h^M, \sigma_h^{M2}) \sim \mathcal{N}(\mu_h^S, \sigma_h^{S2})$ as desired by this framework. The synthesised matrix L_{lh}^S is then converted into a load profile $L^S(t)$ by sequentially ordering its elements by day and hour.

Business load profiles

This part of the load profile construction was completed by Dr Philip Sandwell and Javier Barando Alonso and is described here for reference as it comprises part of the study presented in Chapter 5.

We assumed that each business of type $b \in B$, where B is the set of all business types in the camp (for example shops, restaurants, hair salons, phone chargers and others), has an associated daily load profile L_t^b . This is assumed to be seasonally invariant and has an hourly mean and standard deviation of μ_t^b and σ_t^b . Using a process similar to that described in the previous section, each individual business is assigned a randomly allocated load profile $L_n^b(t)$ for each of n_b businesses of type b , which sum to the total number in the marketplace N .

The number $n_b \in \mathbb{Z}^+$ of each business type b was informed by a survey of 142 businesses in the Mahama Camp marketplace conducted in January 2019. The proportion of businesses of each type in the survey is converted into an integer n_b and is constant throughout the considered time frame, conforming to

$$N = \sum_{b \in B} n_b \quad (\text{B.5})$$

The load profile of an individual business is considered independently with each being assigned an index $n \in \{1, 2, \dots, n_b\}$ and its associated generic load profile $L_n^b(t)$. The cumulative load profile of all N businesses in the marketplace is therefore given by

$$L_N^B(t) = \sum_{b \in B} \left(\sum_{n \in n_b} L_n^b(t) \right) \quad (\text{B.6})$$

and therefore the total load profile, considering both the institutional loads and business customers, is given by

$$L_N^T(t) = L^S(t) + L_N^B(t) \quad (\text{B.7})$$

Input data for business loads

Monitored electricity usage data for refugee businesses was not available and so they used data measured from 19 business customers connected to a MeshPower system in the village of Gitaraga,

Rwanda which is similar to those in Mahama Camp and other displacement situations. Electricity usage data from the Gitaraga site was taken from a previous study which made these data available [204].

From these data, they categorised businesses according to type (bars, shops, etc.) and calculated the mean and standard deviation of electricity use of each business type for each hour of the day. They then scaled the daily usage for each business type to the target value of 200 Wh per day. These new load profiles were used for L_t^b as described above. The hourly load profiles for each business type and the cumulative load profile of each scenario are shown in Figure B.1 (a) and (b) respectively. The load profiles for individual businesses and each scenario used in this work are publicly available [341].

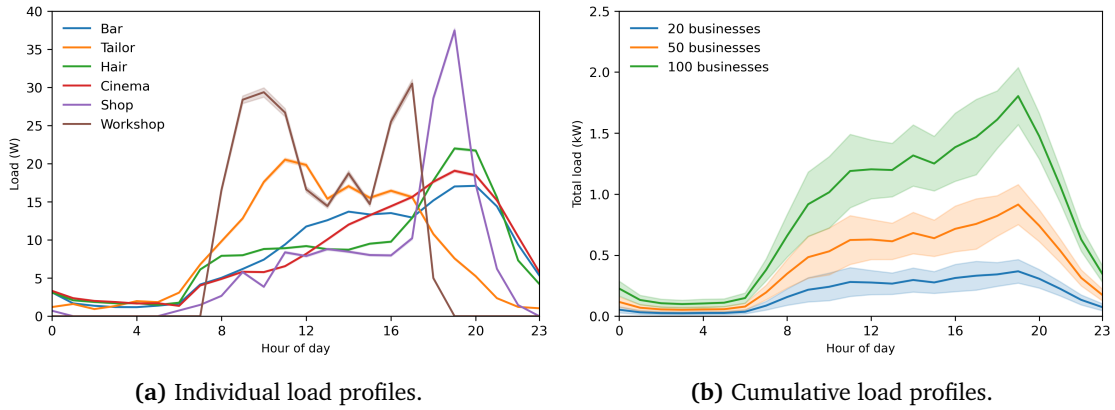


Figure B.1: The load profiles of (a) individual businesses of each type and (b) the scenarios of 20, 50 and 100 businesses cumulatively. Figure from Beath et al. (2023), produced by Javier Barando Alonso [222]

Most business types demonstrate higher demand throughout the afternoon and evening periods, with the exception of the tailor and workshop which have peaks of demand in the morning. When considered cumulatively, the total load demand is high throughout the morning and afternoon with a peak in the evening at 19:00. The addition of the three business scenarios to those of the institutional loads, to give the total system load, is shown in Figure 5.3.

In this work, the same composition of business types in each scenario is used and so the cumulative load scales linearly with the number of connections. Changing the relative proportions of businesses would affect the timings and magnitude of demand throughout the day which would have subsequent effects on the system performance and impacts. As the modelled business demand is relatively high in the evening, if more demand were instead concentrated in the daytime then this might be able to be met through solar power and therefore potentially reduce battery and/or fuel usage.

B.2 Mathematical Description of Appraisal Metrics

The lifetime usage of the battery storage is assessed via the proportion of the total expected battery usage over its lifetime, E_T^{Bat} , that has been consumed by time t [288–290]. This measure, $h(t)$, assumes a linear degradation in capacity and is taken to be the cumulative energy supplied by the battery in kWh, $E_{Cum}^{Bat}(t)$, divided by E_T^{Bat} . E_T^{Bat} , and therefore $h(t)$, will be dependent on the battery usage strategy k which will allow for an expected number of charging and discharging

cycles γ_k at a maximum depth of discharge (% of battery capacity) δ_k . This gives us

$$h_k(t) = \frac{E_{Cum}^{Bat}(t)}{B \gamma_k \delta_k} \equiv \frac{E_{Cum}^{Bat}(t)}{E_{T,k}^{Bat}} \quad (\text{B.8})$$

for an installed battery storage capacity of B kWh. $h(t)$ is expressed as a percentage of the total expected lifetime cumulative energy throughput that the battery storage has experienced. For example, for a battery which is judged to have reached the end of its life ($h = 100\%$) once its ability to store energy degrades to 80% of its original capacity, it is defined to have $h = 25\%$ when operating at 95% of its original storage capacity, $h = 50\%$ at 90% of its original capacity, and so on.

The usage and technical performance of the diesel generator are assessed via two metrics. The first, $\rho^D(t)$, is the probability that the generator is in use in during a given hour of the day t and is given by

$$\rho^D(t) = \frac{1}{T} \sum_t^T D^{on}(t) \quad (\text{B.9})$$

where

$$D^{on}(t) = \begin{cases} 1 & \text{for } E_{Gen}^D(t) > 0 \\ 0 & \text{otherwise} \end{cases} \quad (\text{B.10})$$

Electricity supplied by the generator, $E_{Gen}^D(t)$, of any amount triggers $D_{dt}^{on} = 1$ for that hour and day. This therefore represents the probability that the generator is in use at any point during the hour, rather than the total usage time, as the latter would require a sub-hourly resolution which is not possible in this modelling framework.

The load factor of the generator, $\Gamma(t)$, is the proportion of the maximum available power output of the generator, D , being used in that hour. It has a lower bound set by the designated minimum operating load factor for the generator, $\Gamma_{\min}$ described in Section 3.3, and is given by

$$\Gamma(t) = \frac{1}{D} E_{Gen}^D(t) \quad (\text{B.11})$$

which results in $\Gamma(t) \in [\Gamma_{\min}, 1]$.

B.3 Model Inputs

Table B.1: Model inputs used for the system technical specification.

Item	Value	Unit	Source
PV capacity	18.4	kW _p	
PV tilt	15	Degrees above horizontal	
PV azimuth	0	Degrees from North	
PV lifetime	20	Years	
Diesel capacity	13.0	kW	
Diesel fuel consumption	0.31	Litres per kW per hour	[127, 342]
Diesel minimum capacity factor	35	%	[343]
Battery capacity	78.0	kWh	
Battery maximum SOC	Varies	%	[344]
Battery minimum SOC	Varies	%	[344]
Battery leakage	0.13	% per hour	[344]
Battery round-trip efficiency	92	%	
Battery cycle lifetime	2000	cycles	[344]
Battery lifetime capacity loss	20	%	[344]
Battery C rate discharging	0.2	/hour	[344]
Battery C rate charging	0.13	/hour	[344]
Inverter lifetime	10	Years	
Transmission efficiency (DC)	96	%	[325]
Transmission efficiency (AC)	92	%	[325]
DC to AC conversion efficiency	97	%	[326]
DC to DC conversion efficiency	95	%	[325]
AC to DC conversion efficiency	90	%	[327]
AC to AC conversion efficiency	98	%	[326]

Table B.2: Model inputs used for the economic assessment.

Item	Value	Unit	Source
Discount rate	10	%	
PV cost	400	\$/kW _p	[345]
Storage cost	200	\$/kWh	[345]
Diesel fuel cost	1.13	\$/litre	[345]
Diesel O&M	43	\$/kW p.a.	[328]
BOS cost	100	\$/kW.	[345]
PV inverter cost	163	\$/kW	[345]
Battery inverter Cost	10000	\$	[345]
General O&M	2400	\$ p.a.	[345]

Table B.3: Model inputs used for the environmental assessment for the analysis.

Item	Value	Unit	Source	Comments
PV	1498	kgCO _{2eq} /kWp	[346]	Manufacturing in China
Storage	233	kgCO _{2eq} /kWh	[347]	
Diesel generator	0	kgCO _{2eq} /kW		Already installed
Diesel fuel	2.68	kgCO _{2eq} /litre	[18]	
Balance of systems	134	kgCO _{2eq} /kW	[332]	
Inverters	124	kgCO _{2eq} /kW	[346]	Manufacturing in China

Appendix C

Supplementary Tables, Figures, and Information Chapter 6

Table C.1: Model inputs used for technical specification of off-grid systems.

Item	Value	Unit	Reference	Notes
Battery Depth of Discharge	70	%	[323]	LFP battery chemistry
Battery Leakage	0.004	% per hour	[144]	LFP battery chemistry
Battery Cycle lifetime	2500	Cycles	[348]	LFP battery chemistry
Battery Roundtrip efficiency	86	%	[144]	
Battery Lifetime loss	80	%	[348]	
Battery C-rate in	1			
Battery C-rate out	1			
Transmission efficiency AC	92	%	[325]	
Transmission efficiency DC	96	%	[325]	
DC to AC conversion	97	%	[326]	
DC to DC conversion	95	%	[325]	
AC to DC conversion	90	%	[327]	
AC to AC conversion	98	%	[326]	
PV Inverter lifetime	10	Years		
Battery Inverter lifetime	15	Years		
Diesel consumption	0.28	Litres / kW / hour	[328]	
Diesel minimum load factor	0.35	%		
Solar PV Lifetime	20	Years		

Table C.2: Model inputs used for the economic and GHG assessment of off-grid systems.

Item	Cost/Value	Unit	Ref	GHGs	Unit	Ref
Solar PV Panels	400	\$/kWp	[345]	790*	kgCO2/kWp	[118]
Solar PV Cost Decrease	3	% p.a	[110]			
Solar PV O&M	1	% of CAPEX p.a	[328]			
PV Inverter	163	\$/kW	[345]	124*	kgCO2/kWp	[346]
PV Inverter Cost Decrease	3	% p.a	[110]			
Battery Storage	300	\$/kWh	[345]	110	kgCO2/kWh	[329]
Battery Storage Cost Decrease	7.5	% p.a	[336]			
Battery O&M	1	% of CAPEX p.a	[328]			
Balance of System	100	\$/kWh	[345]	134	kgCO2/kWp	[346]
Diesel Generator	400	\$/kWp	[349]	476	kgCO2/kW	[330]
Diesel Fuel	See table A.5			2.67	kgCO2/litre	[350]
Diesel O&M	10	% of CAPEX p.a	[328]			
Discount Rate	10	% per annum	[298]			
Distribution Cost	125	\$ per household	[18]			

*Based on manufacture in China.

Table C.3: Non-country specific model inputs used for the least-cost electrification modelling in Chapter 6

Item	Value	Unit	Source
Grid Cost	16000	\$/km	[320]
Meter Cost	90	\$ per household	[320]
Install cost	250	\$ per household	[345]
Discount Rate	8	%	[320]
Density Threshold	200	people per sq km	[345]
Grid emissions factor	3000	kg CO2e per km	

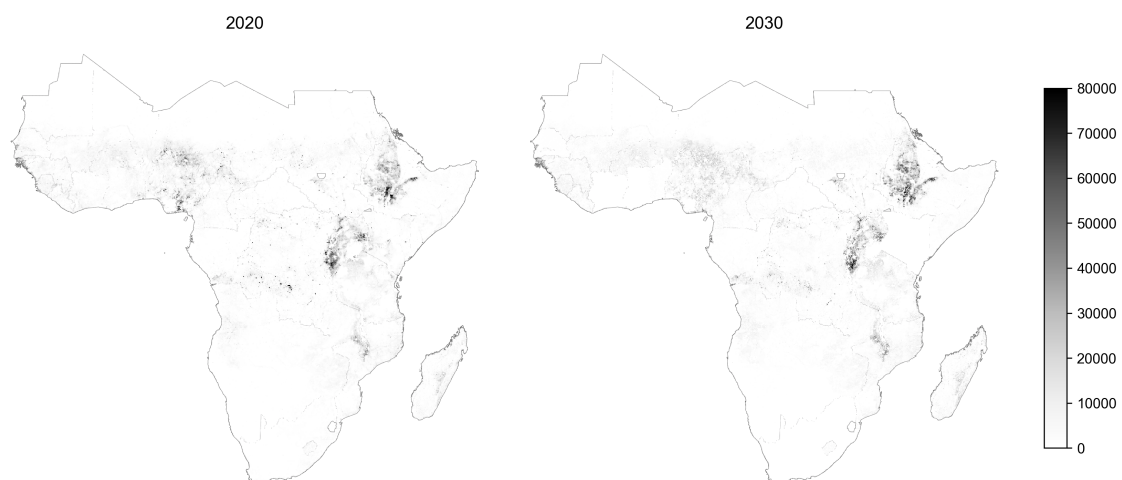


Figure C.1: Greyscale version of Figure 6.3 showing the distribution of population without access in 2020 (left) (aggregated dataset from Falchetta (2021)), and predicted distribution of population without access in 2030 under baseline scenario (right).

Table C.4: Country-level data used for the least-cost electricity access modelling in Chapter 6. Diesel prices are taken from World Bank data for the most recent value for each country [302]. Rural electricity demand met is based on data from the World Bank Enterprise Survey [25]. Rural household size by country is taken from Global Data Lab database [315]. Grid capacity costs and capacity factors are calculated using estimated shares from each country (see Section 6.3.5 of Chapter 6) with cost and capacity data by country from Allington (2022) [318]. Grid emissions intensity data is taken from various sources [351–353]

Country	Diesel Prices	Rural Grid Electricity Demand Met	Household Size	Grid Capacity Cost (\$/kWh)	Grid Capacity Factor	Grid Emissions Intensity (kgCO ₂ /kWh)
Angola	0.82	66%	5.94	2129	0.61	1.48
Benin	0.72	61%	8.3	1241	0.82	0.58
Botswana	0.67	73%	5.63	2500	0.85	1.03
Burkina Faso	0.86	71%	7.2	1447	0.75	0.58
Burundi	1.14	64%	5.94	2128	0.54	0.41
Cameroon	0.94	66%	8.66	2018	0.69	0.66
Central African Republic	1.4	43%	6.96	2452	0.70	0.19
Chad	0.85	70%	7.84	1269	0.82	0.75
Cote d'Ivoire	0.93	72%	7.02	1642	0.67	0.58
Djibouti	0.61	66%	6.18	1300	0.80	0.75
DRC	1.2	66%	6.21	2475	0.34	1.00
Equatorial Guinea	0.94	66%	8.45	1651	0.62	0.63
Eritrea	0.64	66%	6.37	1304	0.78	0.92
Ethiopia	0.64	68%	5.96	2460	0.40	0.12
Gabon	1.2	66%	6.11	1754	0.71	0.95
Gambia	1.06	58%	15.6	1300	0.80	0.71
Ghana	0.85	66%	7.97	1760	0.73	0.58
Guinea	0.9	73%	7.82	1980	0.60	0.75
Guinea-Bissau	0.9	62%	9.42	1300	0.80	0.75
Kenya	0.82	72%	5.83	2987	0.65	0.50
Lesotho	0.73	73%	5.5	2500	0.69	1.00
Liberia	0.83	70%	6.74	1912	0.70	0.68
Madagascar	0.93	73%	5.83	2068	0.76	0.88
Malawi	1.14	71%	5.49	2199	0.59	0.49
Mali	0.98	72%	7.41	2068	0.63	0.58
Mauritania	1.09	73%	7.95	1323	0.73	0.75
Mozambique	0.6	74%	5.46	2253	0.72	1.03
Namibia	0.74	75%	6.94	2443	0.60	1.03
Niger	0.88	59%	7.46	1691	0.80	0.58
Nigeria	0.64	23%	7.28	1461	0.74	0.58
Republic of Congo	1.2	66%	5.66	1504	0.76	0.66
Rwanda	1.13	75%	5.16	1748	0.65	0.77
Senegal	0.97	74%	12	1535	0.78	0.58
Sierra Leone	1.08	58%	7.23	2137	0.49	0.49
Somalia	0.61	66%	7.06	1300	0.80	0.75
South Africa	0.9	73%	6.09	2604	0.81	1.03
South Sudan	0.61	66%	6.76	1301	0.79	0.89
Sudan	0.61	74%	7.25	2020	0.58	0.21
Tanzania	0.8	67%	7.15	1721	0.69	0.53
Togo	0.71	73%	5.15	1632	0.64	0.58
Uganda	0.79	66%	6.28	2431	0.54	0.57
Zambia	1.16	55%	6.62	2441	0.66	1.03
Zimbabwe	1.18	72%	5.33	2488	0.77	1.03

Table C.5: Showing the substitutions used for electricity demand, diesel prices or CLOVER modelling outputs when the required data inputs were unavailable. Country substitutions were selected based on countries with a similar socio-economic profile in the closest geographical proximity

Country	Electricity Demand	Diesel	CLOVER
Angola			
Benin			
Botswana			
Burkina Faso	Benin		
Burundi	DRC		
Cameroon			
Central African Republic	DRC		
Chad			Niger
Cote d'Ivoire			
Djibouti			Sudan
DRC			
Equatorial Guinea			Cameroon
Eritrea			Ethiopia
Ethiopia			
Gabon			Republic of Congo
Gambia	Senegal		
Ghana			
Guinea	Senegal		
Guinea-Bissau	Senegal	Guinea	
Kenya			
Lesotho	Botswana		
Liberia	Ghana		
Madagascar	DRC		
Malawi	Zambia		
Mali	Niger		
Mauritania	Senegal		
Mozambique			
Namibia			
Niger			
Nigeria			
Republic of Congo		DRC	
Rwanda	Tanzania		
Senegal			
Sierra Leone	Senegal		
Somalia			Sudan
South Africa			
South Sudan			Sudan
Sudan			
Tanzania			
Togo			
Uganda	Tanzania		
Zambia	Zimbabwe		
Zimbabwe			

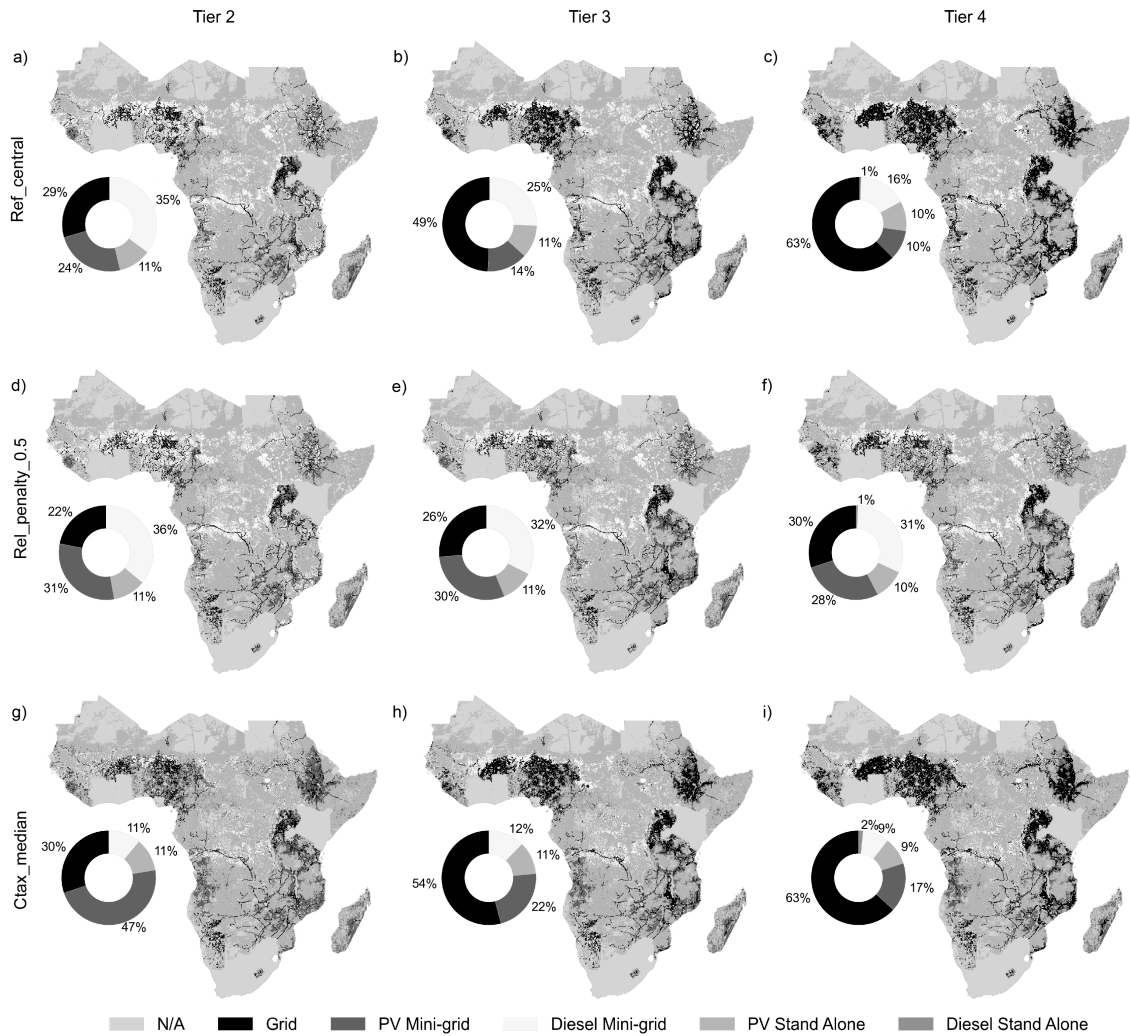


Figure C.2: Greyscale version of Figure 6.4, spatially distributed least-cost electrification method for the Ref_central, Rel_penalty_0.5 and Ctax_median scenarios at demand tiers 2-4. Panels a-c represent the Ref_central results, with no carbon tax or unmet demand penalty applied, they show a declining role for off-grid technologies as demand increases. Panels d-f represent the Rel_penalty_0.5 scenario applying a 0.5\$/kWh of unmet demand penalty universally across all modes, this reduces the share of the grid at all demand tiers, with the largest difference at the highest demand tier. Panels g-i represent the Ctax_median scenario, equating to a carbon price scheme of around \$4 (2022 USD) in 2020, rising to \$146 by 2030, here diesel usage is reduced in all three tiers, with the largest difference with Ref_central seen at Tier 2.

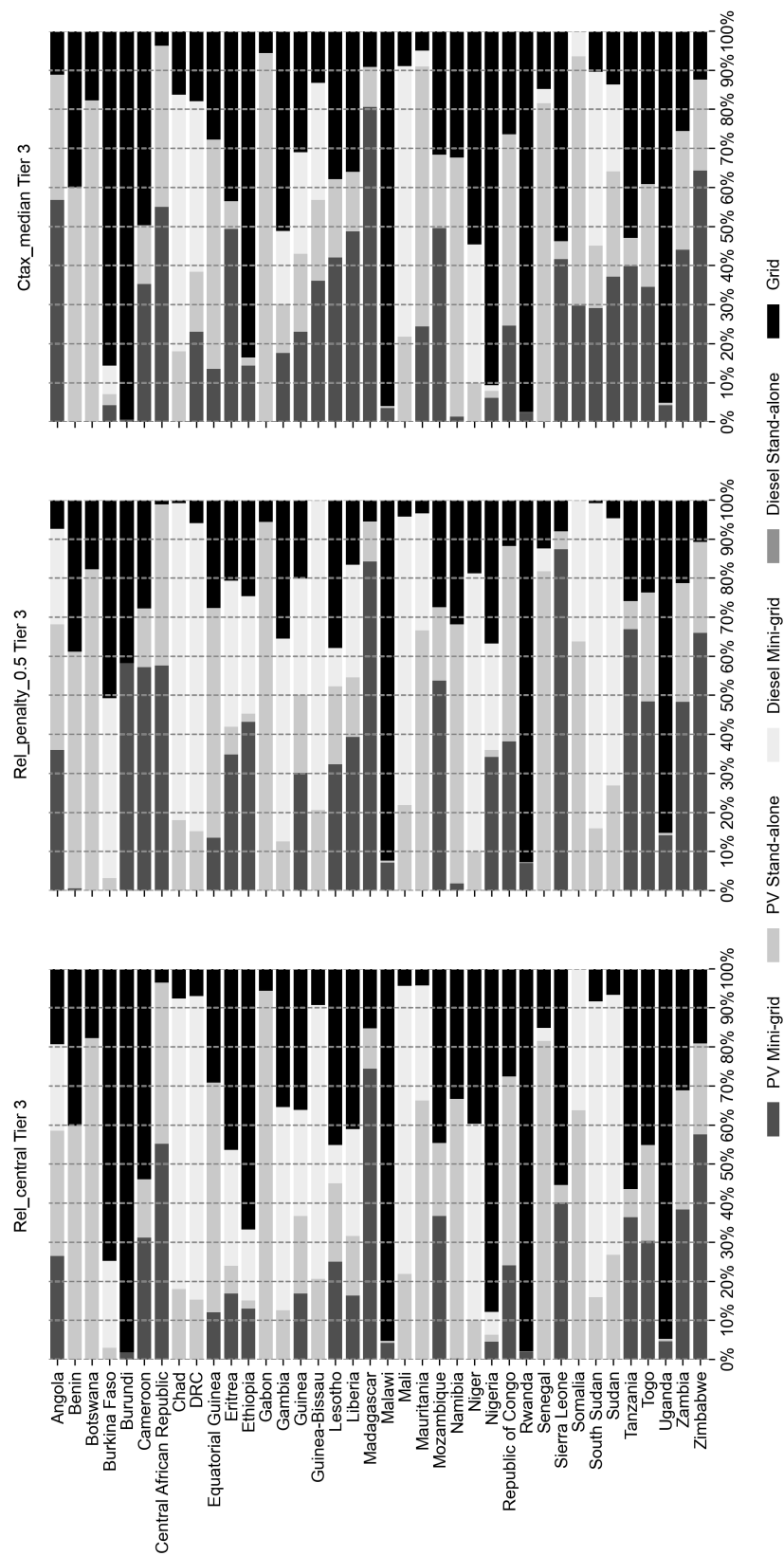


Figure C.3: Greyscale version of Figure 6.6 showing stacked horizontal bars of each electrification mode by country (Tier 3) for the Ref_central (Left), Rel_penalty_0.5 (middle), and Ctax_median (right) scenarios.

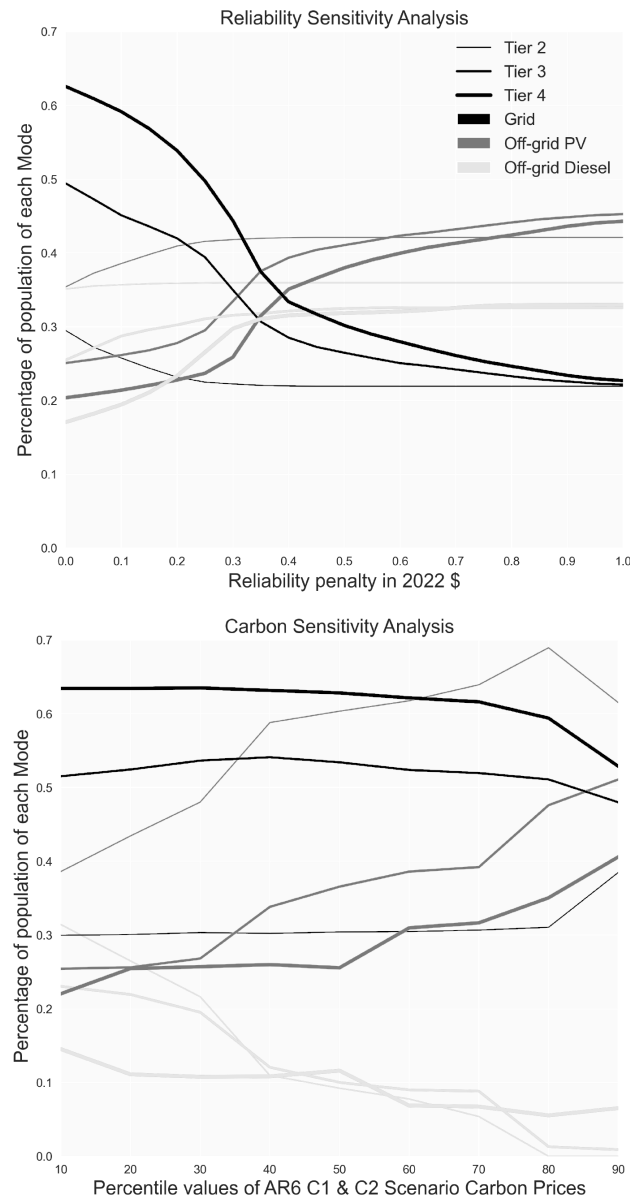


Figure C.4: Greyscale version of Figure 6.7, showing the outputs of the reliability and carbon pricing sensitivity analysis. The lines represent the changing shares of each mode of access as the reliability penalty and carbon prices increase. The thickness of the lines represents the different electricity access demand levels (tier 2-4) and the colours of the lines represent the modes. The shares of population displayed are the total shares of all people gaining access under each of the reliability and carbon price scenarios from 2020-2030.

References

1. IEA. *SDG7: Data and Projections* (2022).
2. IEA. *SDG7: Data and Projections – Analysis – (2023)* (2023). <https://www.iea.org/commentaries/for-the-first-time-in-decades-the-number-of-people-without-access-to-electricity-is-set-to-increase-in-2022>.
3. Ayaburi, J., Bazilian, M., Kincer, J. & Moss, T. Measuring “Reasonably Reliable” access to electricity services. *The Electricity Journal* **33**, 106828. ISSN: 1040-6190 (7 Aug. 2020).
4. Meles, T. H. Impact of power outages on households in developing countries: Evidence from Ethiopia. *Energy Economics* **91**, 104882. ISSN: 0140-9883 (Sept. 2020).
5. Carlsson, F., Demeke, E., Martinsson, P. & Tesemma, T. Cost of power outages for manufacturing firms in Ethiopia: A stated preference study. *Energy Economics* **88**, 104753. ISSN: 0140-9883 (May 2020).
6. Andersen, T. B. & Dalgaard, C. J. Power outages and economic growth in Africa. *Energy Economics* **38**, 19–23. ISSN: 0140-9883 (July 2013).
7. Blechinger, P., Köhler, M., Juette, C., Berendes, S. & Nettersheim, C. *Off-Grid Renewable Energy for Climate Action – Pathways for change, published by Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ)* (Deutsche Gesellschaft für Internationale Zusammenarbeit (GIZ), 2019).
8. Nerini, F. F. *et al.* Mapping synergies and trade-offs between energy and the Sustainable Development Goals. *Nature Energy*. ISSN: 2058-7546. <http://www.nature.com/articles/s41560-017-0036-5> (2017).
9. Chen, Y. J., Chindarkar, N. & Xiao, Y. Effect of reliable electricity on health facilities, health information, and child and maternal health services utilization: evidence from rural Gujarat, India. *Journal of health, population, and nutrition* **38**, 7. ISSN: 20721315. <https://jhpn.biomedcentral.com/articles/10.1186/s41043-019-0164-6> (1 Feb. 2019).
10. Suhlrie, L. *et al.* The role of energy in health facilities: A conceptual framework and complementary data assessment in Malawi. *PLoS ONE* **13**, e0200261. ISSN: 19326203. <https://doi.org/10.1371/journal.pone.0200261> (7 July 2018).
11. Gustavsson, M. Educational benefits from solar technology—Access to solar electric services and changes in children’s study routines, experiences from eastern province Zambia. *Energy Policy* **35**, 1292–1299. ISSN: 0301-4215. <https://www.sciencedirect.com/science/article/pii/S0301421506001492?via%5C%3Dihub> (2 Feb. 2007).

12. Electricity access and rural development: Review of complex socio-economic dynamics and causal diagrams for more appropriate energy modelling. *Energy for Sustainable Development* **43**, 203–223. ISSN: 0973-0826. <https://www.sciencedirect.com/science/article/pii/S097308261731390X?via%5C%3Dihub> (Apr. 2018).
13. Agency, I. R. E. Off-grid renewable energy solutions to expand electricity access: An opportunity not to be missed. www.irena.org (2019).
14. Agrawal, S., Bali, N. & Urpelainen, J. *Rural Electrification in India: Customer Behaviour and Demand* (2019). <https://www.rockefellerfoundation.org/report/rural-electrification-india-customer-behaviour-demand/>.
15. ESMAP. *Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers* (2019). <https://openknowledge.worldbank.org/handle/10986/31926>.
16. ESMAP. *Mini Grids for Half a Billion People: Market Outlook and Handbook for Decision Makers*. <https://openknowledge.worldbank.org/handle/10986/38082> (Sept. 2022).
17. Ortega-Arriaga, P., Babacan, O., Nelson, J. & Gambhir, A. Grid versus off-grid electricity access options: A review on the economic and environmental impacts. *Renewable and Sustainable Energy Reviews* **143**, 110864. ISSN: 1364-0321 (June 2021).
18. Beath, H. *et al.* The cost and emissions advantages of incorporating anchor loads into solar mini-grids in India. *Renewable and Sustainable Energy Transition* **1**, 100003. ISSN: 2667-095X (Aug. 2021).
19. Ma, S. & Urpelainen, J. Distributed power generation in national rural electrification plans: An international and comparative evaluation. *Energy Research Social Science* **44**, 1–5. ISSN: 2214-6296 (Oct. 2018).
20. IFC. Benchmarking Mini-Grid DESCOs 2017 Update Summary of Findings (May 2018 2018).
21. Mandelli, S., Brivio, C., Colombo, E. & Merlo, M. Effect of load profile uncertainty on the optimum sizing of off-grid PV systems for rural electrification. *Sustainable Energy Technologies and Assessments* **18**, 34–47. ISSN: 2213-1388. <https://www.sciencedirect.com/science/article/pii/S2213138816300637> (Dec. 2016).
22. Blodgett, C., Moder, E., Kickham, L. & Leaf, H. Achieving Mini-Grid Scalability in Kenya: Financing the Future of Rural Electrification, 1–30. http://www.vulcan.com/MediaLibraries/Vulcan/Documents/Financing-the-Future-Whitepaper-Oct-2017.pdf?email_sent1 (2017).
23. Hazelton, J., Bruce, A. & MacGill, I. A review of the potential benefits and risks of photovoltaic hybrid mini-grid systems. *Renewable Energy* **67**, 222–229. <https://www.sciencedirect.com/science/article/pii/S0960148113006046> (July 2014).
24. Mentis, D. *et al.* Lighting the World: the first application of an open source, spatial electrification tool (OnSSET) on Sub-Saharan Africa. *Environmental Research Letters* **12**, 085003. ISSN: 1748-9326. <https://iopscience.iop.org/article/10.1088/1748-9326/aa7b29/20https://iopscience.iop.org/article/10.1088/1748-9326/aa7b29/meta> (8 July 2017).
25. Bank, T. W. *Enterprise Surveys Indicators Data - World Bank Group* 2022. <https://www.enterprisesurveys.org/en/data>.
26. Farquharson, D. V., Jaramillo, P. & Samaras, C. Sustainability implications of electricity outages in sub-Saharan Africa. *Nature Sustainability* **2018 1:10** **1**, 589–597. ISSN: 2398-9629. <https://www.nature.com/articles/s41893-018-0151-8> (10 Oct. 2018).

27. Straeten, J. V. D. *et al.* From dualism to convergence – A research agenda for energy access. https://www.tu-berlin.de/fileadmin/f27/PDFs/Discussion%5C%5C_Papers%5C%5C_neu/discussion%5C%5C_paper%5C%5C_Nr%5C%5C_%5C%5C_38%5C%5C_17.pdf (2017).
28. Agency, I. E. WEO Energy Access Outlook 2017, Methodology for Energy Access Analysis, 1–9. www.iea.org/energyaccess. (2017).
29. For Global Development, C. *More than a Lightbulb: Five Recommendations to Make Modern Energy Access Meaningful for People and Prosperity* (Center for Global Development, 2016), 37. www.cgdev.org/node/3124016%20http://www.cgdev.org/publication/more-than-lightbulb-recommendations-modern-energy-meaningful.
30. Pachauri, S. & Brew-Hammond, A. Energy Access for Development -19. *Global Energy Assessment*, 1401–1458. http://www.iiasa.ac.at/web/home/research/Flagship-Projects/Global-Energy-Assessment/GEA%5C_Chapter19%5C_energyaccess%5C_hires.pdf http://www.homeofgeography.org/uk/news%5C_2009/EarTrend%5C_May09.pdf (2012).
31. Banerjee, S. *et al.* Global Tracking Framework. *Sustainable Energy for All*, The World Bank. **3**, 289. ISSN: 18766102. http://www.iea.org/publications/freepublications/publication/Global%5C_Tracking%5C_Framework.pdf <http://documents.worldbank.org/curated/en/2013/05/%2017765643/global-tracking-framework-vol-3-3-main-report> (Washington D.C. 2013).
32. Bhatia, M. & Angelou, N. Beyond Connections Energy Access Redefined, Technical Report 008/15, 1–24. http://www.worldbank.org/content/dam/Worldbank/Topics/Energy%20and%20Extract/Beyond%5C_Connections%5C_Energy%5C_Access%5C_Redefined%5C_Exec%5C_ESMAP%5C_2015.pdf (2015).
33. Pelz, S., Pachauri, S. & Groh, S. A critical review of modern approaches for multidimensional energy poverty measurement. *Wiley Interdisciplinary Reviews: Energy and Environment* **7**, e304. ISSN: 2041-840X. <https://onlinelibrary.wiley.com/doi/full/10.1002/wene.304> <https://onlinelibrary.wiley.com/doi/abs/10.1002/wene.304> <https://wires.onlinelibrary.wiley.com/doi/10.1002/wene.304> (6 Nov. 2018).
34. What are we measuring? An empirical analysis of household electricity access metrics in rural Bangladesh. *Energy for Sustainable Development* **30**, 21–31. ISSN: 23524669. <https://www.researchgate.net/publication/283351391> (2016).
35. Action, P. *Poor people's energy outlook 2012* ISBN: 9781853397318 (2012).
36. Action, P. *Poor people's energy outlook 2010* http://www.ecowrex.org/fr/system/files/documents/2013%5C_poor-people-energy-outlook%5C%5C_practical-action.pdf www.practicalaction.org.uk (2010).
37. Doll, C. N. & Pachauri, S. Estimating rural populations without access to electricity in developing countries through night-time light satellite imagery. *Energy Policy* **38**, 5661–5670. ISSN: 0301-4215 (10 Oct. 2010).
38. Li, X., Zhou, Y., Zhao, M. & Zhao, X. A harmonized global nighttime light dataset 1992–2018. *Scientific Data* **2020 7:1** **7**, 1–9. ISSN: 2052-4463. <https://www.nature.com/articles/s41597-020-0510-y> (1 June 2020).

39. Min, B., Gaba, K. M., Sarr, O. F. & Agalassou, A. Detection of rural electrification in Africa using DMSP-OLS night lights imagery. <http://dx.doi.org/10.1080/01431161.2013.833358> **34**, 8118–8141. ISSN: 13665901. <https://www.tandfonline.com/doi/abs/10.1080/01431161.2013.833358> (22 2013).
40. Falchetta, G., Pachauri, S., Parkinson, S. & Byers, E. A high-resolution gridded dataset to assess electrification in sub-Saharan Africa. *Scientific Data* **2019** 6:1 **6**, 1–9. ISSN: 2052-4463. <https://www.nature.com/articles/s41597-019-0122-6> (1 July 2019).
41. Falchetta, G., Pachauri, S., Byers, E., Danylo, O. & Parkinson, S. C. Satellite Observations Reveal Inequalities in the Progress and Effectiveness of Recent Electrification in Sub-Saharan Africa. *One Earth* **2**, 364–379. ISSN: 2590-3322 (4 Apr. 2020).
42. Impacts of Electricity Outages in Urban Households in Developing Countries: A Case of Accra, Ghana. *Energies* **2021**, Vol. 14, Page 3676 **14**, 3676. ISSN: 1996-1073. <https://www.mdpi.com/1996-1073/14/12/3676/htm%20https://www.mdpi.com/1996-1073/14/12/3676> (12 June 2021).
43. Oseni, M. O. & Pollitt, M. G. A firm-level analysis of outage loss differentials and self-generation: Evidence from African business enterprises. *Energy Economics* **52**, 277–286. ISSN: 0140-9883 (Nov. 2015).
44. Longa, F. D. & van der Zwaan, B. Heart of light: an assessment of enhanced electricity access in Africa. *Renewable and Sustainable Energy Reviews* **136**, 110399. ISSN: 1364-0321 (Feb. 2021).
45. Bos, K., Chaplin, D. & Mamun, A. Benefits and challenges of expanding grid electricity in Africa: A review of rigorous evidence on household impacts in developing countries. *Energy for Sustainable Development* **44**, 64–77. ISSN: 0973-0826 (June 2018).
46. Saing, C. H. Rural electrification in Cambodia: does it improve the welfare of households? *Oxford Development Studies* **46**, 147–163. ISSN: 1360-0818. <https://www.tandfonline.com/doi/full/10.1080/13600818.2017.1340443> (2 Apr. 2018).
47. Gibson, J. & Olivia, S. The effect of infrastructure access and quality on non-farm enterprises in rural Indonesia. <https://researchcommons.waikato.ac.nz/handle/10289/4136> (2008).
48. Bastakoti, B. P. The electricity-livelihood nexus: some highlights from the Andhikhola Hydroelectric and Rural Electrification Centre (AHREC). *Energy for Sustainable Development* **10**, 26–35. ISSN: 0973-0826. <https://www.sciencedirect.com/science/article/pii/S0973082608605414> (3 Sept. 2006).
49. Khandker, S. R., Barnes, D. F. & Samad, H. A. Welfare Impacts of Rural Electrification: A Panel Data Analysis from Vietnam. *Source: Economic Development and Cultural Change* **61**, 659–692. <https://www.jstor.org/stable/pdf/10.1086/669262.pdf?refreqid=excelsior%5C%3A37ab21057e78e1bb1178723cd8441032> (3 2013).
50. Vernet, A., Khayesi, J. N., George, V., George, G. & Bahaj, A. S. How does energy matter? Rural electrification, entrepreneurship, and community development in Kenya. *Energy Policy* **126**, 88–98. ISSN: 0301-4215 (Mar. 2019).
51. Wassie, Y. T. & Adaramola, M. S. Socio-economic and environmental impacts of rural electrification with Solar Photovoltaic systems: Evidence from southern Ethiopia. *Energy for Sustainable Development* **60**, 52–66. ISSN: 0973-0826 (Feb. 2021).

52. Azimoh, C. L., Klintenberg, P., Wallin, F. & Karlsson, B. Illuminated but not electrified: An assessment of the impact of Solar Home System on rural households in South Africa. *Applied Energy* **155**, 354–364. ISSN: 0306-2619. <https://www.sciencedirect.com/science/article/pii/S030626191500759X?via%5C%3Dihub> (Oct. 2015).
53. Aklin, M., Bayer, P., Harish, S. P. & Urpelainen, J. Does basic energy access generate socio-economic benefits? A field experiment with off-grid solar power in India. *Science Advances* **3**, e1602153. ISSN: 2375-2548. <http://advances.sciencemag.org/lookup/doi/10.1126/sciadv.1602153> (5 May 2017).
54. Terrapon-Pfaff, J., Gröne, M.-C., Dienst, C. & Ortiz, W. Productive use of energy – Pathway to development? Reviewing the outcomes and impacts of small-scale energy projects in the global south. *Renewable and Sustainable Energy Reviews* **96**, 198–209. ISSN: 1364-0321. <https://www.sciencedirect.com/science/article/pii/S1364032118305252> (Nov. 2018).
55. Pueyo, A. & DeMartino, S. The impact of solar mini-grids on Kenya's rural enterprises. *Energy for Sustainable Development* **45**, 28–37. ISSN: 0973-0826. <https://www.sciencedirect.com/science/article/pii/S0973082617310359?via%5C%3Dihub> (Aug. 2018).
56. Quiroga, J. F. *Experiences to help build futures* ISBN: 9789585615779 (2017).
57. Bensch, G., Kluve, J. & Peters, J. Impacts of rural electrification in Rwanda. *Journal of Development Effectiveness* **3**, 567–588. ISSN: 1943-9342. <http://www.tandfonline.com/doi/abs/10.1080/19439342.2011.621025> (4 Dec. 2011).
58. Dholakia, H. H. Solar powered healthcare in developing countries. *Nature Energy*, 1. ISSN: 2058-7546. <http://www.nature.com/articles/s41560-018-0205-1> (July 2018).
59. Cronk, R. & Bartram, J. Environmental conditions in health care facilities in low- and middle-income countries: Coverage and inequalities. *International Journal of Hygiene and Environmental Health* **221**, 409–422. ISSN: 1618131X (3 Apr. 2018).
60. Moner-Girona, M., Kakoulaki, G., Falchetta, G., Weiss, D. J. & Taylor, N. Achieving universal electrification of rural healthcare facilities in sub-Saharan Africa with decentralized renewable energy technologies. *Joule* **5**, 2687–2714. ISSN: 2542-4785. [http://www.cell.com/article/S2542435121004384/fulltext%20http://www.cell.com/article/S2542435121004384/abstract%20https://www.cell.com/joule/abstract/S2542-4351\(21\)00438-4](http://www.cell.com/article/S2542435121004384/fulltext%20http://www.cell.com/article/S2542435121004384/abstract%20https://www.cell.com/joule/abstract/S2542-4351(21)00438-4) (10 Oct. 2021).
61. Adair-Rohani, H. *et al.* Limited electricity access in health facilities of sub-Saharan Africa: a systematic review of data on electricity access, sources, and reliability. *Global health, science and practice* **1**, 249–61. ISSN: 2169-575X. <http://www.ncbi.nlm.nih.gov/pubmed/25276537%20http://www.pubmedcentral.nih.gov/articlerender.fcgi?artid=PMC4168575> (2 Aug. 2013).
62. *Modern Energy Services for Health Facilities in Resource-Constrained Settings* ISBN: 978 92 4 150764 6. [www.who.int%20http://apps.who.int/iris/bitstream/10665/156847/1/9789241507646%5C%5C_eng.pdf](http://apps.who.int/iris/bitstream/10665/156847/1/9789241507646%5C%5C_eng.pdf) (2014).
63. Irwin, B. R., Hoxha, K. & Grépin, K. A. Conceptualising the effect of access to electricity on health in low- and middle-income countries: A systematic review. *Global Public Health* **15**, 452–473. ISSN: 17441706. <https://www.tandfonline.com/doi/abs/10.1080/17441692.2019.1695873> (3 Mar. 2020).

64. Apenteng, B. A., Opoku, S. T., Ansong, D., Akowuah, E. A. & Afriyie-Gyawu, E. The effect of power outages on in-facility mortality in healthcare facilities: Evidence from Ghana. *Global Public Health* **13**, 545–555. ISSN: 1744-1692. <https://www.tandfonline.com/doi/full/10.1080/17441692.2016.1217031> (5 May 2018).
65. Otiangala, D. *et al.* Oxygen insecurity and mortality in resource-constrained healthcare facilities in rural Kenya. *Pediatric Pulmonology* **55**, 1043–1049. ISSN: 10990496. <https://onlinelibrary.wiley.com/doi/10.1002/ppul.24679> (4 Apr. 2020).
66. Action, P. *Poor People's Energy Outlook 2013 - Practical Action* (2013). <https://practicalaction.org/knowledge-centre/resources/poor-peoples-energy-outlook-2013/>.
67. The imperative for stronger vaccine supply and logistics systems. *Vaccine* **31**, B73–B80. ISSN: 0264-410X. <https://www.sciencedirect.com/science/article/pii/S0264410X12016295#sec0055> (Apr. 2013).
68. Javadi, D. *et al.* Implementation research on sustainable electrification of rural primary care facilities in Ghana and Uganda. *Health Policy and Planning* **35**, ii124–ii136. ISSN: 0268-1080. https://academic.oup.com/heapol/article/35/Supplement%5C%5C_2/ii124/5959266 (Supplement 2 Nov. 2020).
69. SE4ALL. *Challenges in Vaccine Rollout Linked to Cooling | Sustainable Energy for All 2023*. <https://www.seforall.org/chilling-prospects-2021/impact-of-covid-19/challenges-in-vaccine-rollout-linked-to-cooling#18>.
70. Bruce, N., Perez-Padilla, R. & Albalak, R. The health effects of indoor air pollution exposure in developing countries. Geneva: World Health Organization, Report WHO/SDE/OEH/02.05, 1–40. http://apps.who.int/iris/bitstream/handle/10665/67496/WHO%5C%5C_SDE%5C%5C_OEH%5C%5C_02.05.pdf?sequence=1&isAllowed=y (2002).
71. Nazaroff, W. W. Illumination, lighting technologies, and indoor environmental quality. *Indoor Air* **24**, 225–226. ISSN: 09056947. <http://doi.wiley.com/10.1111/ina.12113> (3 June 2014).
72. Muller, E., Diab, R., Binedell, M. & Hounsome, R. Health risk assessment of kerosene usage in an informal settlement in Durban, South Africa. *Atmospheric Environment* **37**, 2015–2022. ISSN: 1352-2310. <https://www.sciencedirect.com/science/article/pii/S1352231003001250?via%5C%3Dihub> (15 May 2003).
73. Muyanja, D. *et al.* Kerosene lighting contributes to household air pollution in rural Uganda. *Indoor Air* **27**, 1022–1029. ISSN: 09056947. <http://doi.wiley.com/10.1111/ina.12377> (5 Sept. 2017).
74. Apple, J. *et al.* Characterization of particulate matter size distributions and indoor concentrations from kerosene and diesel lamps. *Indoor Air* **20**, 399–411. ISSN: 09056947. <http://doi.wiley.com/10.1111/j.1600-0668.2010.00664.x> (5 Oct. 2010).
75. Obeng, G. Y., Akuffo, F., Braimah, I., Evers, H.-D. & Mensah, E. Impact of solar photovoltaic lighting on indoor air smoke in off-grid rural Ghana. *Energy for Sustainable Development* **12**, 55–61. ISSN: 0973-0826. <https://www.sciencedirect.com/science/article/pii/S0973082608604196?via%5C%3Dihub> (1 Mar. 2008).
76. Lam, N. L. *et al.* Exposure reductions associated with introduction of solar lamps to kerosene lamp-using households in Busia County, Kenya. *Indoor Air* **28**, 218–227. ISSN: 09056947. <http://doi.wiley.com/10.1111/ina.12433> (2 Mar. 2018).

77. Barron, M. & Torero, M. Household electrification and indoor air pollution. *Journal of Environmental Economics and Management* **86**, 81–92. ISSN: 0095-0696. <https://www.sciencedirect.com/science/article/pii/S0095069617304825?via%5C%3Dihub> (Nov. 2017).
78. Bank, T. W. *The Welfare Impact of Rural Electrification* 178. ISBN: 978-0-8213-7367-5. http://www.worldbank.org/ieg%20https://ieg.worldbankgroup.org/Data/reports/rural%5C%5C_elec%5C%5C_full%5C%5C_eval.pdf (2008).
79. The geography of energy and education: Leaders, laggards, and lessons for achieving primary and secondary school electrification. *Renewable and Sustainable Energy Reviews* **58**, 107–123. ISSN: 1364-0321. <https://www.sciencedirect.com/science/article/pii/S1364032115016020?via%5C%3Dihub> (May 2016).
80. UNDESA. Electricity and education : The benefits , barriers , and recommendations for achieving the electrification of primary and secondary schools. *United Nations Department of Economic and Social Affairs*, 1–36. <https://sustainabledevelopment.un.org/content/documents/1608Electricity%20and%20Education.pdf%20//sustainabledevelopment.un.org/index.php?page=view&type=400&nr=160> (December 2014).
81. Aguirre, J. *The Impact of Rural Electrification on Education: A Case Study from Peru* (2017), 91–108. <https://search-proquest-com.gate3.library.lse.ac.uk/docview/1919458745/fulltextPDF/9513533566174D64PQ/1?accountid=9630>.
82. Bank, W. Rural Electrification and Development in the Philippines: Measuring the Social and Economic Benefits. *UNDP/Energy Sector Management Assistance Program (ESMAP) Report No. 255/02*, 1. <http://documents.worldbank.org/curated/en/765461468776759888/Rural-electrification-and-development-in-the-Philippines-measuring-the-social-and-economic-benefits> (May May 2002).
83. Kanagawa, M. & Nakata, T. Assessment of access to electricity and the socio-economic impacts in rural areas of developing countries. *Energy Policy* **36**, 2016–2029. ISSN: 0301-4215. <https://www.sciencedirect.com/science/article/pii/S0301421508000608#bib38> (6 June 2008).
84. Da Silveira Bezerra, P. B. *et al.* The power of light: socio-economic and environmental implications of a rural electrification program in Brazil. *Environmental Research Letters* **12**, 095004. ISSN: 1748-9326. <http://stacks.iop.org/1748-9326/12/i=9/a=095004?key=crossref.6c3113a7d31a2f8d495107634a49cac4> (9 Sept. 2017).
85. Pereira, M. G., Sena, J. A., Freitas, M. A. V. & Silva, N. F. D. Evaluation of the impact of access to electricity: A comparative analysis of South Africa, China, India and Brazil. *Renewable and Sustainable Energy Reviews* **15**, 1427–1441. ISSN: 1364-0321 (3 Apr. 2011).
86. Greco, G. & Ojong, N. Fostering Human Wellbeing in Africa through Solar Home Systems: A Systematic and a Critical Review. *Sustainability* **2022**, Vol. 14, Page 8382 **14**, 8382. ISSN: 2071-1050. <https://www.mdpi.com/2071-1050/14/14/8382/htm%20https://www.mdpi.com/2071-1050/14/14/8382> (14 July 2022).
87. Kollamparambil, U. Happiness, Happiness Inequality and Income Dynamics in South Africa. *Journal of Happiness Studies* **21**, 201–222. ISSN: 15737780. <https://link.springer.com/article/10.1007/s10902-019-00075-0> (1 Jan. 2020).

88. Nasrudin, R., Quarina, Q. & Dartanto, T. Revisiting the Energy-Happiness Paradox: A Quasi-Experimental Evidence of Electricity Access in Indonesia. *Journal of Happiness Studies* **23**, 3549–3576. ISSN: 15737780. <https://link.springer.com/article/10.1007/s10902-022-00567-6> (7 Oct. 2022).
89. IEA. *Energy Access Outlook 2017: From poverty to prosperity* (2017), 144. https://www.iea.org/publications/freepublications/publication/WE02017SpecialReport%5C_EnergyAccessOutlook.pdf.
90. Dfid. Support Study for DFID Low Carbon Mini Grids quot; Identifying the gaps and building the evidence base on low carbon mini-grids quot; Final Report. https://assets.publishing.service.gov.uk/government/uploads/system/uploads/attachment%5C_data/file/278021/IED-green-min-grids-support-study1.pdf (2013).
91. Ahmed, A. A. *et al.* OFF-GRID RENEWABLE ENERGY SYSTEMS: STATUS AND METHODOLOGICAL ISSUES About IRENA (2015). http://www.irena.org/documentdownloads/publications/irena_off-grid%5C_renewable%5C_systems%5C_wp%5C_2015.pdf.
92. Ton, D. T. & Smith, M. A. The U.S. Department of Energy's Microgrid Initiative. *The Electricity Journal*. <https://www.energy.gov/sites/prod/files/2016/06/f32/The%20US%20Department%20of%20Energy%5C%27s%20Microgrid%20Initiative.pdf> (2012).
93. Of New, M. & Renewable Energy, I. *National Policy for Renewable Energy based Micro and Mini Grids* (Ministry of New and Renewable Energy, 2012). https://mnre.gov.in/file-manager/UserFiles/draft-national-Mini_Micro-Grid-Policy.pdf.
94. Nordman, B. & Christensen, K. *Local power distribution with nanogrids* June 2013. <http://ieeexplore.ieee.org/document/6604464/>.
95. Mandelli, S., Barbieri, J., Mereu, R. & Colombo, E. Off-grid systems for rural electrification in developing countries: Definitions, classification and a comprehensive literature review. *Renewable and Sustainable Energy Reviews* **58**, 1621–1646. ISSN: 1364-0321. <https://www.sciencedirect.com/science/article/pii/S1364032115017219> (May 2016).
96. Avila, N., Carvallo, J. P., Shaw, B. & Kammen, D. M. *The energy challenge in sub-Saharan Africa: A guide for advocates and policy makers* (2017), 1–79. <https://www.oxfamamerica.org/static/media/files/oxfam-RAEL-energySSA-pt2.pdf>.
97. EU. *Africa-EU Renewable Energy Cooperation Programme Creating opportunities for renewable energy Fact File THE MINI-GRID POLICY TOOLKIT* (2014). www.africa-eu-renewables.org.
98. Ortega-Arriaga, P., Babacan, O., Nelson, J. & Gambhir, A. Grid versus off-grid electricity access options: A review on the economic and environmental impacts. *Renewable and Sustainable Energy Reviews* **143**, 110864. ISSN: 1364-0321 (June 2021).
99. Bajo-Buenestado, R. The effect of blackouts on household electrification status: Evidence from Kenya. *Energy Economics* **94**, 105067. ISSN: 0140-9883. <https://linkinghub.elsevier.com/retrieve/pii/S0140988320304072> (Feb. 2021).
100. Of Rwanda, G. *REPUBLIC OF RWANDA MINISTRY OF INFRASTRUCTURE RURAL ELECTRIFICATION STRATEGY* (2016).
101. Sergi, B. *et al.* Institutional influence on power sector investments: A case study of on- and off-grid energy in Kenya and Tanzania. *Energy Research Social Science* **41**, 59–70. ISSN: 2214-6296 (July 2018).

102. Bank, T. W., Funds, C. I. & ESMAP. MINI-GRIDS GENDER EQUALITY: INCLUSIVE DESIGN, BETTER DEVELOPMENT OUTCOMES KEY ISSUES, AND POTENTIAL ACTIONS (2017).
103. Markvart, T. *Solar electricity* 280. ISBN: 9780471988526 (Wiley, 2000).
104. IEA. *Solar PV – Analysis - IEA* <https://www.iea.org/reports/solar-pv>.
105. Institute, F. *Photovoltaics Report - Fraunhofer ISE* <https://www.ise.fraunhofer.de/en/publications/studies/photovoltaics-report.html>.
106. Institute, F. *Photovoltaics report* 2018. www.ise.fraunhofer.de.
107. IRENA. *Renewable Power Generation Costs in 2021* (2021). <https://www.irena.org/publications/2022/Jul/Renewable-Power-Generation-Costs-in-2021>.
108. IEA. *World Energy Outlook 2004. Electronics and Power* **23**, 577. ISSN: 00135127. <https://www.iea.org/media/weowebiste/2008-1994/WE02004.pdf> (4 2004).
109. IEA. *World Energy Outlook 2010* (2010). <https://www.iea.org/reports/world-energy-outlook-2010>.
110. DNV. *Energy Transition Outlook: A global and regional forecast to 2050* (2021).
111. IEA. *Renewables 2021 – Analysis - IEA* (2021). <https://www.iea.org/reports/renewables-2021>.
112. Hsu, D. D. *et al.* Life Cycle Greenhouse Gas Emissions of Crystalline Silicon Photovoltaic Electricity Generation. *Journal of Industrial Ecology* **16**, S122–S135. ISSN: 10881980. <http://doi.wiley.com/10.1111/j.1530-9290.2011.00439.x> (Apr. 2012).
113. Salavei, A. *et al.* Solar Resources and Carbon Footprint of Photovoltaic Power in Different Regions in Europe. *Proceedings of the EU PVSEC 2014 (Amsterdam)*, 1430–1432. <https://publications.jrc.ec.europa.eu/repository/handle/JRC89270> (Nov. 2014).
114. Sandwell, P. *et al.* Off-grid solar photovoltaic systems for rural electrification and emissions mitigation in India. *Solar Energy Materials Solar Cells* **156**, 147–156. ISSN: 0927-0248. <https://www.sciencedirect.com/science/article/pii/S0927024816300447?via%5C%3Dihub#f0010%20https://spiral.imperial.ac.uk/bitstream/10044/1/32892/4/Revised%20Manuscript%20ff-grid%20PV.pdf> (Nov. 2016).
115. Antonanzas-Torres, F. *et al.* Environmental Impact of Solar Home Systems in Sub-Saharan Africa. *Sustainability* 2021, Vol. 13, Page 9708 **13**, 9708. ISSN: 2071-1050. <https://www.mdpi.com/2071-1050/13/17/9708/htm%20https://www.mdpi.com/2071-1050/13/17/9708> (17 Aug. 2021).
116. De Wild-Scholten, M. (Energy payback time and carbon footprint of commercial photovoltaic systems. *Solar Energy Materials and Solar Cells* **119**, 296–305. ISSN: 0927-0248. <https://www.sciencedirect.com/science/article/pii/S0927024813004455> (Dec. 2013).
117. IEA. *Solar PV Global Supply Chains* (IEA, 2022). <https://www.iea.org/reports/solar-pv-global-supply-chains/executive-summary>.
118. Elementa & Dixon, W. *Whole life carbon of photovoltaic installations - Technical Report - February 2022* (2022). <https://www.renewableenergyhub.co.uk/main/solar-panels/solar-panels-carbon-analysis/>.
119. IRENA. *Solar PV in Africa: Costs and Markets* (2016), 84. www.irena.org%20https://www.irena.org/DocumentDownloads/Publications/IRENA_Solar_PV_Costs_Africa_2016.pdf.

120. Šúri, M., Huld, T. A., Dunlop, E. D. & Ossenbrink, H. A. Potential of solar electricity generation in the European Union member states and candidate countries. *Solar Energy* **81**, 1295–1305. ISSN: 0038092X. <http://linkinghub.elsevier.com/retrieve/pii/S0038092X07000229> (10 Oct. 2007).
121. Global/ESMAP, L., GOGLA, for Access, E. & Advisors, O. C. *Off-Grid Solar Market Trends Report 2022: State of the Sector* (World Bank, 2022). <https://documents1.worldbank.org/curated/en/099235110062231022/pdf/P175150063801e0860928f00e7131b132de.pdf>.
122. Phadke, A., Jacobson, A. A., Park, A. & Young, W. *Lawrence Berkeley National Laboratory Recent Work Title Powering a Home with Just 25 Watts of Solar PV: Super-Efficient Appliances Can Enable Expanded Off-Grid Energy Service Using Small Solar Power Systems: Permalink* <https://escholarship.org/uc/item/3vv7m0x7> Publication Date (2017). <https://escholarship.org/uc/item/3vv7m0x7>.
123. Sharif, I. & Mithila, M. Rural Electrification using PV: the Success Story of Bangladesh. *Energy Procedia* **33**, 343–354. ISSN: 1876-6102. <https://www.sciencedirect.com/science/article/pii/S1876610213000854> (Jan. 2013).
124. IRENA. OFF-GRID RENEWABLE ENERGY STATISTICS 2021. www.irena.org (2021).
125. ESMAP. Energy Sector Management Assistance Program Energy Sector Management Assistance Program Technical and Economic Assessment of Off-grid, Mini-grid and Grid Electrification Technologies Energy Sector Management Assistance Program. <http://documents.worldbank.org/curated/en/634581468333897517/pdf/430990ESMOREVISED01public1.pdf%20www.esmap.org> (2007).
126. Casillas, C. E. & Kammen, D. M. The energy-poverty-climate nexus. *Science*. ISSN: 00368075 (2010).
127. Szabó, S., Bódis, K., Huld, T. & Moner-Girona, M. Energy solutions in rural Africa: Mapping electrification costs of distributed solar and diesel generation versus grid extension. *Environmental Research Letters* **6**, 034002. ISSN: 17489326. <http://stacks.iop.org/1748-9326/6/i=3/a=034002?key=crossref.89590e74fed5b2bdbff442f147cb9d62> (3 July 2011).
128. IFC. *The Dirty Footprint of the Broken Grid The Impacts of Fossil Fuel Back-up Generators in Developing Countries* (2019), 1–52. www.ifc.org%20https://www.ifc.org/wps/wcm/connect/2cd3d83d-4f00-4d42-9bdc-4afdc2f5dbc7/20190919-Full-Report-The-Dirty-Footprint-of-the-Broken-Grid.pdf?MOD=AJPERES&CVID=mR9UpXC.
129. Valencia, A. M. Analysis of Decision-Making for Off-Grid Rural Electrification in Colombia, 257–265. http://link.springer.com/10.1007/978-3-319-15964-5_23 (2015).
130. Meyers, S., Marnay, C., Schumacher, K., Sathaye, J. & Lawrence, E. O. *Estimating Carbon Emissions Avoided by Electricity Generation and Efficiency Projects: A Standardized Method (MAGPWR)* (2000). <http://eta-publications.lbl.gov/sites/default/files/lbnl-46063-english.pdf>.
131. UNDP. Standardized Baseline Assessment for Rural Off-Grid-Electrification in Sub-Saharan Africa A standardization tool to streamline and simplify the CDM project cycle, 53. www.mdgcarbon.org (2013).

132. Sahu, S., Schultz, M. & Beig, G. Critical pollutant emissions from the Indian telecom network. *Atmospheric Environment* **103**, 34–42. ISSN: 1352-2310. <https://www.sciencedirect.com/science/article/pii/S135223101400973X?via%5C%3Dihub> (Feb. 2015).
133. Monitor, E. *Weekly data: Africa's diesel generation boom - Energy Monitor* <https://www.energymonitor.ai/sectors/power/weekly-data-africas-diesel-generation-boom/>.
134. Lucas, P. L. *et al.* Future energy system challenges for Africa: Insights from Integrated Assessment Models. *Energy Policy* **86**, 705–717. ISSN: 0301-4215. <https://www.sciencedirect.com/science/article/pii/S0301421515300616?via%5C%3Dihub#bib49> (Nov. 2015).
135. López-González, A., Domenech, B. & Ferrer-Martí, L. Lifetime, cost and fuel efficiency in diesel projects for rural electrification in Venezuela. *Energy Policy* **121**, 152–161. ISSN: 0301-4215. <https://www.sciencedirect.com/science/article/pii/S0301421518304105?via%5C%3Dihub#bib3> (Oct. 2018).
136. BNEF. *3Q 2017 Frontier Power Market Outlook, 31 July 2017* 2017. <https://about.bnef.com/blog/3q-2017-frontier-power-market-outlook/>.
137. IEA. *Grid-Scale Storage – Analysis* (IEA, 2022). <https://www.iea.org/reports/grid-scale-storage>.
138. Energy storage deployment and innovation for the clean energy transition. *Nature Energy* **2**, 17125. ISSN: 2058-7546. <http://www.nature.com/articles/nenergy2017125> (9 July 2017).
139. Grantham, J. GMO White Paper: The race of our lives. <https://www.gmo.com/docs/default-source/research-and-commentary/strategies/asset-allocation/the-race-of-our-lives-revisited.pdf> (2018).
140. BNEF. *2022 Battery Price Survey* (BNEF, 2022). <https://about.bnef.com/blog/lithium-ion-battery-pack-prices-rise-for-first-time-to-an-average-of-151-kwh/>.
141. Mauler, L., Duffner, F., Zeier, W. G. & Leker, J. Battery cost forecasting: a review of methods and results with an outlook to 2050. *Energy Environmental Science* **14**, 4712–4739. ISSN: 17545706. <https://pubs.rsc.org/en/content/articlehtml/2021/ee/d1ee01530c%20https://pubs.rsc.org/en/content/articlelanding/2021/ee/d1ee01530c> (9 Sept. 2021).
142. A comparison of lead-acid and lithium-based battery behavior and capacity fade in off-grid renewable charging applications. *Energy* **60**, 492–500. ISSN: 0360-5442. <https://www.sciencedirect.com/science/article/pii/S0360544213007044> (Oct. 2013).
143. -Shell, E. S. Energy storage trends for off-grid services in emerging markets Insights from social enterprises (2018).
144. IRENA. *Electricity storage and renewables: Costs and markets to 2030* 132. ISBN: 9789292600389. [www.irena.org%20http://www.irena.org/publications/2017/Oct/Electricity-storage-and-renewables-costs-and-markets](http://www.irena.org/publications/2017/Oct/Electricity-storage-and-renewables-costs-and-markets) (2017).
145. Few, S., Schmidt, O. & Gambhir, A. Energy access through electricity storage: Insights from technology providers and market enablers. *Energy for Sustainable Development* **48**, 1–10. ISSN: 0973-0826 (Feb. 2019).
146. Thomas, L. V., Schmidt, O., Gambhir, A., Few, S. & Staffell, I. Comparative life cycle assessment of lithium-ion battery chemistries for residential storage. *Journal of Energy Storage* **28**, 101230. ISSN: 2352-152X (Apr. 2020).

147. Yudhistira, R., Khatiwada, D. & Sanchez, F. A comparative life cycle assessment of lithium-ion and lead-acid batteries for grid energy storage. *Journal of Cleaner Production* **358**, 131999. ISSN: 0959-6526 (July 2022).
148. For All, P. *Power for All Research Summary: US\$220 billion needed to build mini-grids for half a billion people* (2019), 0-1.
149. Allee, A., Williams, N. J., Davis, A. & Jaramillo, P. Predicting initial electricity demand in off-grid Tanzanian communities using customer survey data and machine learning models. *Energy for Sustainable Development* **62**, 56–66. ISSN: 0973-0826 (June 2021).
150. Crossboundary. *Innovation Insight: Reducing tariff leads to large and sustained increases in demand on mini-grids* (2022). www.crossboundary.com/labs..
151. Ramchandran, N., Pai, R. & Parihar, A. K. S. Feasibility assessment of Anchor-Business-Community model for off-grid rural electrification in India. *Renewable Energy* **97**, 197–209. ISSN: 0960-1481. <https://www.sciencedirect.com/science/article/pii/S0960148116304475> (Nov. 2016).
152. BNEF. *2017 Q4 Frontier Power Market Outlook* (2017). https://data.bloomberglp.com/bnef/sites/14/2017/11/Master_-_Frontier-Power-MO_EL_IO_EL_IO.pdf.
153. Lemaire, X. Off-grid electrification with solar home systems: The experience of a fee-for-service concession in South Africa. *Energy for Sustainable Development* **15**, 277–283. ISSN: 09730826. <http://dx.doi.org/10.1016/j.esd.2011.07.005> (3 2011).
154. Bisaga, I., Puźniak-Holford, N., Grealish, A., Baker-Brian, C. & Parikh, P. Scalable off-grid energy services enabled by IoT: A case study of BBOX SMART Solar. *Energy Policy* **109**, 199–207. ISSN: 0301-4215. <https://www.sciencedirect.com/science/article/pii/S0301421517304378> <http://www.sciencedirect.com/science/article/pii/S0301421517304378> (Supplement C Oct. 2017).
155. Crossboundary. *Innovation Insight: Appliance Financing 1.0 - Offering household appliances on credit did not significantly increase mini-grid revenues* (2021).
156. Dagnachew, A. G., Lucas, P. L., Hof, A. F. & van Vuuren, D. P. Trade-offs and synergies between universal electricity access and climate change mitigation in Sub-Saharan Africa. *Energy Policy* **114**, 355–366. ISSN: 03014215 (Mar. 2018).
157. Pathways to achieve universal household access to modern energy by 2030. *Environmental Research Letters* **8**, 24015–7. ISSN: 17489326. <http://iopscience.iop.org/article/10.1088/1748-9326/8/2/024015/pdf> (2 2013).
158. Chakravarty, S. & Tavoni, M. Energy poverty alleviation and climate change mitigation: Is there a trade off? *Energy Economics* **40**, S67–S73. ISSN: 0140-9883. <https://www.sciencedirect.com/science/article/pii/S014098831300217X> (Dec. 2013).
159. IPCC. *Special Report on 1.5 degrees: Summary for Policymakers* (2018). https://www.ipcc.ch/site/assets/uploads/sites/2/2019/05/SR15_SPM_version_report_LR.pdf.
160. Unruh, G. C. Understanding carbon lock-in. *Energy Policy* **28**, 817–830. ISSN: 0301-4215 (12 Oct. 2000).
161. Bertram, C. *et al.* Carbon lock-in through capital stock inertia associated with weak near-term climate policies. *Technological Forecasting and Social Change* **90**, 62–72. ISSN: 0040-1625. <https://www.sciencedirect.com/science/article/pii/S004016251300259X?via%5C%3Dihub> (Jan. 2015).

162. Pachauri, S. Household electricity access a trivial contributor to CO₂ emissions growth in India. *Nature Climate Change* **4**, 1073–1076. ISSN: 17586798 (12 Jan. 2014).
163. Initiative, A. C. M. *Africa Carbon Markets Initiative: Roadmap Report | Sustainable Energy for All* (2022). <https://www.seforall.org/publications/africa-carbon-markets-initiative-roadmap-report>.
164. UNFCCC. *Africa Carbon Markets Initiative launched to dramatically expand Africa's participation in voluntary carbon market - Climate Champions* 2022. <https://climatechampions.unfccc.int/africa-carbon-markets-initiative/>.
165. Southpole & Partners, P. C. *The D-REC Initiative* 2022. <https://www.d-recs.energy/wp-content/uploads/2022/02/Overview-of-the-D-REC-Initiative.pdf>.
166. Bahramara, S., Moghaddam, M. P. & Haghifam, M. R. Optimal planning of hybrid renewable energy systems using HOMER: A review. *Renewable and Sustainable Energy Reviews* **62**, 609–620. ISSN: 1364-0321 (Sept. 2016).
167. Korkovelos, A., Khavari, B., Sahlberg, A., Howells, M. & Arderne, C. The Role of Open Access Data in Geospatial Electrification Planning and the Achievement of SDG7. An OnSSET-Based Case Study for Malawi. *Energies* 2019, Vol. 12, Page 1395 **12**, 1395. ISSN: 1996-1073. <https://www.mdpi.com/1996-1073/12/7/1395/htm%20https://www.mdpi.com/1996-1073/12/7/1395> (7 Apr. 2019).
168. Ouedraogo, N. S. A GIS approach to electrification planning in Cameroon. *Energy Strategy Reviews* **45**, 101020. ISSN: 2211-467X (Jan. 2023).
169. Photovoltaic-software.com. *HOMER Legacy v2.68 - hybrid solar design* <https://photovoltaic-software.com/pv-softwares-calculators/free-photovoltaic-software/homer-legacy-v2-68>.
170. Baring-gould, E. I. *Hybrid2 The Hybrid System Simulation Model Users Manual* 1998.
171. Lilienthal, P. & Gilman, P. *HOMER Modelling Software* (2004). <http://www.nrel.gov/homer>.
172. Lilienthal, P., Lambert, T. & Gilman, P. Computer Modeling of Renewable Power Systems. *Encyclopedia of Energy*, 633–647. <https://www.sciencedirect.com/science/article/pii/B012176480X005222> (Jan. 2004).
173. Lambert, T., Gilman, P. & Lilienthal, P. *Micropower System Modeling with Homer* (2006), 379–418. <https://www.homerenergy.com/documents/MicropowerSystemModelingWithHOMER.pdf>.
174. HOMER. *HOMER - Hybrid Renewable and Distributed Generation System Design Software* 2023. <https://www.homerenergy.com/index.html>.
175. Nfah, E. M., Ngundam, J. M., Vandenbergh, M. & Schmid, J. Simulation of off-grid generation options for remote villages in Cameroon. *Renewable Energy* **33**. Hypothetical place. No real load data.No multi-year technology modelling, 1064–1072. ISSN: 0960-1481 (5 May 2008).
176. Adaramola, M. S., Agelin-Chaab, M. & Paul, S. S. Analysis of hybrid energy systems for application in southern Ghana. *Energy Conversion and Management* **88**. No monitored load. Hypothetical settlementNo multi-year technology optimisation, 284–295. ISSN: 0196-8904 (Dec. 2014).

177. Nandi, S. K. & Ghosh, H. R. Techno-economical analysis of off-grid hybrid systems at Kutubdia Island, Bangladesh. *Energy Policy* **38**. Hypothetical placeNo multi-year optimisationNo load data/site-relevant cost data, 976–980. ISSN: 0301-4215 (2 Feb. 2010).
178. Olatomiwa, L., Mekhilef, S. & Ohunakin, O. S. Hybrid renewable power supply for rural health clinics (RHC) in six geo-political zones of Nigeria. *Sustainable Energy Technologies and Assessments* **13**. hyptothetical places. Load profiles.Lack of detailed analysis of results e.g. multi-years, 1–12. ISSN: 2213-1388 (Feb. 2016).
179. Khan, E. U. & Martin, A. R. Optimization of hybrid renewable energy polygeneration system with membrane distillation for rural households in Bangladesh. *Energy* **93**. No multi-year optimisationLack of clarity re: the load profiles., 1116–1127. ISSN: 0360-5442 (Dec. 2015).
180. Keddar, S., Strachan, S., Soltowski, B. & Galloway, S. An Overview of the Technical Challenges Facing the Deployment of Electric Cooking on Hybrid PV/Diesel Mini-Grid in Rural Tanzania—A Case Study Simulation. *Energies* **2021**, Vol. 14, Page 3761 **14**. Uses homer within larger set of modelNo consideration of multi-year analysis., 3761. ISSN: 1996-1073. <https://www.mdpi.com/1996-1073/14/13/3761/htm%20https://www.mdpi.com/1996-1073/14/13/3761> (13 June 2021).
181. Chang, K. C. *et al.* Standalone and Minigrid-Connected Solar Energy Systems for Rural Application in Rwanda: An in Situ Study. *International Journal of Photoenergy* **2021**. no multi year consideration. ISSN: 1687529X (2021).
182. Bhattacharyya, S. C. Mini-grid based electrification in Bangladesh: Technical configuration and business analysis. *Renewable Energy* **75**. No multi-year technology optimisation. No consideration of how batteries degrade etc., 745–761. ISSN: 0960-1481 (Mar. 2015).
183. Kamel, S. & Dahl, C. The economics of hybrid power systems for sustainable desert agriculture in Egypt. *Energy* **30**. No multi-year optimisation of technology., 1271–1281. ISSN: 0360-5442 (8 June 2005).
184. Bekele, G. & Palm, B. Feasibility study for a standalone solar–wind-based hybrid energy system for application in Ethiopia. *Applied Energy* **87**. Lack of multi-year analysis., 487–495. ISSN: 0306-2619 (2 Feb. 2010).
185. Kolhe, M. L., Ranaweera, K. M. U. & Gunawardana, A. G. Techno-economic sizing of off-grid hybrid renewable energy system for rural electrification in Sri Lanka. *Sustainable Energy Technologies and Assessments* **11**. No monitored load profile or financial dataMulti-year analysis but lack of clarity with how it is done., 53–64. ISSN: 2213-1388 (Sept. 2015).
186. Sen, R. & Bhattacharyya, S. C. Off-grid electricity generation with renewable energy technologies inIndia: An application of HOMER. *Renewable Energy* **62**. - highlights the challenges in cost / viable business model.]- uses pre homer analysis (detailed demand building)- uses post homer - assessment of business models and tariffs, 388–398. ISSN: 09601481. <http://dx.doi.org/10.1016/j.renene.2013.07.028> (2014).
187. Zebra, E. I. C., van der Windt, H. J., Olubayo, B., Nhumaio, G. & Faaij, A. P. Scaling up the electricity access and addressing best strategies for a sustainable operation of an existing solar PV mini-grid: A case study of Mavumira village in Mozambique. *Energy for Sustainable Development* **72**. Considers more than single yeardetailed consideration and description of load generation etc., 58–82. ISSN: 0973-0826 (Feb. 2023).
188. Dufo-López, R. & Bernal-Agustín, J. L. Design and control strategies of PV-diesel systems using genetic algorithms. *Solar Energy* **79**, 33–46. ISSN: 0038092X (1 July 2005).

189. López, R. D. *iHOGA User's manual* (2021). <https://ihoga.unizar.es/en/>.
190. García-Vera, Y. E., Dufo-López, R. & Bernal-Agustín, J. L. Techno-Economic Feasibility Analysis through Optimization Strategies and Load Shifting in Isolated Hybrid Microgrids with Renewable Energy for the Non-Interconnected Zone (NIZ) of Colombia. *Energies* 2020, Vol. 13, Page 6146 **13**, 6146. ISSN: 1996-1073. <https://www.mdpi.com/1996-1073/13/22/6146/htm%20https://www.mdpi.com/1996-1073/13/22/6146> (22 Nov. 2020).
191. Dufo-López, R. *et al.* Multi-objective optimization minimizing cost and life cycle emissions of stand-alone PV–wind–diesel systems with batteries storage. *Applied Energy* **88**, 4033–4041. ISSN: 0306-2619 (11 Nov. 2011).
192. Zubi, G., Dufo-López, R., Pasaoglu, G. & Pardo, N. Techno-economic assessment of an off-grid PV system for developing regions to provide electricity for basic domestic needs: A 2020-2040 scenario. *Applied Energy* **176**, 309–319. ISSN: 03062619. <https://www.sciencedirect.com/science/article/pii/S0306261916306043?via%5C%3Dihub> (Aug. 2016).
193. Pfenninger, S. *et al.* Opening the black box of energy modelling: Strategies and lessons learned. *Energy Strategy Reviews* **19**, 63–71. <http://creativecommons.org/licenses/by/4.0/> (2018).
194. Morrison, R. Energy system modeling: Public transparency, scientific reproducibility, and open development. *Energy Strategy Reviews* **20**, 49–63. ISSN: 2211-467X. <https://www.sciencedirect.com/science/article/pii/S2211467X17300949> (Apr. 2018).
195. Berendes, S., Bertheau, P. & Blechinger, P. *Sizing and Optimization of Hybrid Mini-Grids with micrOgridS-an Open-Source Modelling Tool* (2018). http://hybridpowersystems.org/wp-content/uploads/sites/9/2018/05/2B_1_TENE18_062_paper_Berendes_Sarah.pdf.
196. Hilpert, S., Kaldemeyer, C. & Krien, U. The Open Energy Modelling Framework (oemof) - A novel approach in energy system modelling. **49**, 1–24 (June 2017).
197. Hoffmann, M. M. *Optimizing the Design of Off-Grid Micro Grids Facing Interconnection with an Unreliable Central Grid Utilizing an Open-Source Simulation Tool* (June 2019). <https://zenodo.org/record/5174632>.
198. Hoffmann, M. M., Pelz, S., Monés-Pederzini, Ò., Andreottola, M. & Blechinger, P. Overcoming the Bottleneck of Unreliable Grids: Increasing Reliability of Household Supply with Decentralized Backup Systems. *Journal of Sustainability Research* **2**. ISSN: 26326582 (1 Jan. 2020).
199. Bertheau, P. Supplying not electrified islands with 100% renewable energy based micro grids: A geospatial and techno-economic analysis for the Philippines. *Energy* **202**, 117670. ISSN: 0360-5442 (July 2020).
200. Blair, N. *et al.* System Advisor Model (SAM) General Description (Version 2017.9.5). www.nrel.gov/publications. (2018).
201. NREL. *Home - System Advisor Model - SAM*. 2023. <https://sam.nrel.gov/>.
202. Sandwell, P., Winchester, B., Beath, H. & Nelson, J. CLOVER: A modelling framework for sustainable community-scale energy systems. *Journal of Open Source Software* **8**, 4799. ISSN: 2475-9066. <https://joss.theoj.org/papers/10.21105/joss.04799> (82 Feb. 2023).
203. Sandwell, P., Ekins-Daukes, N. & Nelson, J. What are the greatest opportunities for PV to contribute to rural development? *Energy Procedia* **130**, 139–146. ISSN: 18766102. <https://www.sciencedirect.com/science/article/pii/S1876610217345101> (Sept. 2017).

204. Few, S. *et al.* Electricity demand in populations gaining access: Impact of rurality and climatic conditions, and implications for microgrid design. *Energy for Sustainable Development* **66**, 151–164. ISSN: 0973-0826 (Feb. 2022).
205. Sandwell, P. R. The greenhouse gas mitigation potential of photovoltaic technologies (December 2017).
206. Boait, P. *et al.* EScobX: A Set of Tools for Mini-Grid Sustainability in the Developing World. *World. Sustainability* **9**, 1–15. <http://doi.org/10.3390/su9050738> (5 2017).
207. Boait, P., Advani, V. & Gammon, R. Estimation of demand diversity and daily demand profile for off-grid electrification in developing countries. *Energy for Sustainable Development* **29**, 135–141. ISSN: 0973-0826. <http://www.sciencedirect.com/science/article/pii/S0973082615001040> (Dec. 2015).
208. McKenna, E. & Thomson, M. High-resolution stochastic integrated thermal–electrical domestic demand model. *Applied Energy* **165**, 445–461. ISSN: 0306-2619 (Mar. 2016).
209. Barton, J., Thomson, M., Sandwell, P. & Mellor, A. *A Domestic Demand Model for India* 743–753. ISBN: 9789811526664. http://dx.doi.org/10.1007/978-981-15-2666-4_70 (Springer Singapore, 2020).
210. McKenna, E., Thomson, M. & Barton, J. *CREST Demand Model. Loughborough University. Dataset*. 2015. <https://doi.org/10.17028/rd.lboro.2001129.v8>.
211. A GIS-based decision support tool for renewable energy management and planning in semi-arid rural environments of northeast of Brazil. *Renewable Energy* **35**, 2921–2932. ISSN: 0960-1481 (12 Dec. 2010).
212. Isihak, S. R. *Achieving universal electricity access in line with SDG7 using GIS-based Model: An application of OnSSET for rural electrification planning in Nigeria* Jan. 2023.
213. Ciller, P. *et al.* Optimal Electrification Planning Incorporating On- And Off-Grid Technologies- And Reference Electrification Model (REM). *Proceedings of the IEEE* **107**, 1872–1905. ISSN: 15582256 (9 Sept. 2019).
214. Milosavljević, D. D., Kevkić, T. S. & Jovanović, S. J. Review and validation of photovoltaic solar simulation tools/software based on case study. *Open Physics* **20**, 431–451. ISSN: 23915471. <https://www.degruyter.com/document/doi/10.1515/phys-2022-0042/html> (1 Jan. 2022).
215. Huld, T., Müller, R. & Gambardella, A. A new solar radiation database for estimating PV performance in Europe and Africa. *Solar Energy* **86**, 1803–1815. ISSN: 0038-092X (6 June 2012).
216. Commission, E. *PVGIS Online Tool* 2023. https://joint-research-centre.ec.europa.eu/pvgis-online-tool_en.
217. Pfenninger, S. & Staffell, I. Long-term patterns of European PV output using 30 years of validated hourly reanalysis and satellite data. *Energy* **114**, 1251–1265. ISSN: 0360-5442. <https://www.sciencedirect.com/science/article/pii/S0360544216311744?via%5C%3Dihub> (Nov. 2016).
218. Staffell, I. & Pfenninger, S. Using bias-corrected reanalysis to simulate current and future wind power output. *Energy* **114**, 1224–1239. ISSN: 0360-5442. <https://linkinghub.elsevier.com/retrieve/pii/S0360544216311811> (Nov. 2016).

219. Staffell, I., Brett, D. J., Brandon, N. P. & Hawkes, A. D. Domestic Microgeneration: Renewable and distributed energy technologies, policies and economics. *Domestic Microgeneration: Renewable and Distributed Energy Technologies, Policies and Economics*, 1–354 (June 2015).
220. Sayani, R. *et al.* Sizing solar-based mini-grids for growing electricity demand: Insights from rural India. *Journal of Physics: Energy* **5**, 014004. ISSN: 2515-7655. <https://iopscience.iop.org/article/10.1088/2515-7655/ac9dc0><https://iopscience.iop.org/article/10.1088/2515-7655/ac9dc0/meta> (1 Nov. 2022).
221. Alonso, J. B., Sandwell, P. & Nelson, J. The potential for solar-diesel hybrid mini-grids in refugee camps: A case study of Nyabiheke camp, Rwanda. *Sustainable Energy Technologies and Assessments* **44**, 101095. ISSN: 2213-1388 (Apr. 2021).
222. Beath, H. *et al.* Maximising the benefits of renewable energy infrastructure in displacement settings: Optimising the operation of a solar-hybrid mini-grid for institutional and business users in Mahama Refugee Camp, Rwanda. *Renewable and Sustainable Energy Reviews* (2023).
223. Bhattacharyya, S. C. & Palit, D. Mini-grid based off-grid electrification to enhance electricity access in developing countries: What policies may be required? *Energy Policy* **94**, 166–178. ISSN: 03014215 (July 2016).
224. Robert, F. C., Sisodia, G. S. & Gopalan, S. The Critical Role of Anchor Customers in Rural Microgrids. *2017 International Conference on Computation of Power, Energy, Information and Communication (ICCPEIC)*, 398–403 (2017).
225. Welland, A. *Electrification of health clinics in rural areas: Challenges and opportunities Smart Villages* (2017). www.e4sv.org/5C%7Cinfo@e4sv.org/5C%7C@e4SmartVillages.
226. Mukherjee, S. & Symington, R. *Beyond Household Lighting: Promoting Anchor Load Agri-Business to Build Mini-Grids in India* (2018), 2–5. <http://energyaccess.org/news/recent-news/mlinda/>.
227. Nations, U. *Transforming our world: The 2030 agenda for sustainable development* (2015), 1–5. <https://sustainabledevelopment.un.org/content/documents/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf>.
228. Karp, C. L. *et al.* Evaluating the value proposition for improving vaccine thermostability to increase vaccine impact in low and middle-income countries. *Vaccine* **33**, 3471–3479. ISSN: 18732518 (30 July 2015).
229. Ouedraogo, N. S. & Schimanski, C. Energy poverty in healthcare facilities: a "silent barrier" to improved healthcare in sub-Saharan Africa. *Journal of Public Health Policy* **39**, 358–371. <https://doi.org/10.1057/s41271-018-0136-x> (2018).
230. Essendi, H. *et al.* *Reproductive Health*. <https://reproductive-health-journal.biomedcentral.com/track/pdf/10.1186/s12978-015-0078-8> (2015).
231. Forrester, J. A. *et al.* Impact of Surgical Lighting on Intraoperative Safety in Low-Resource Settings: A Cross-Sectional Survey of Surgical Providers. *World Journal of Surgery* **41**, 3055–3065. <http://www.lifebox.org> (2017).
232. Mani, S., Patnaik, S. & Dholakia, H. H. *The State of Electricity Access for Primary Health Centres in India Insights from the District Level Household and Facility Survey (DLHS-3 and DLHS-4) Issue Brief* | (2019). https://www.ceew.in/sites/default/files/CEEW-The-State-of-Electricity-Access-for-Primary_0.pdf.

233. Yadav, P., Davies, P. J. & Palit, D. Distributed solar photovoltaics landscape in Uttar Pradesh, India: Lessons for transition to decentralised rural electrification. *Energy Strategy Reviews* **26**, 100392. ISSN: 2211-467X. <https://www.sciencedirect.com/science/article/pii/S2211467X19300859> (2019).
234. Solutions, V. MK 144 Icelined Refridgerator Product Information (2018).
235. Ansons. Haier BD-206H Chest Freezer Product Information 2018. <https://ansons.ph/product/haier-bd-206h/>.
236. Meditech. Operating light Product Information 2018. <https://www.meditech-india.com/halogenoperation-lights.html>.
237. COCIR. Status Report 2016. Tech. rep. September. European Coordination Committee of the Radiological, Electromedical and Healthcare IT Industry (2017). www.cocir.org%20http://www.eceee.org/ecodesign/products/medical_imaging_equipment/draft_va_medical_imaging_equipment.
238. Twomey, J., Overcash, M. & Soltani, S. *Life cycle for engineering the healthcare service delivery of imaging* in (Springer-Verlag Berlin Heidelberg, 2012), 131–136. ISBN: 9783642290688.
239. Prayas. *Electricity Monitoring Initiative* 2019.
240. Prayas. *Private correspondence with Prayas* 29.03.19 (2019).
241. Kumar, P. *Personal Communication* 19 March 2019 2019.
242. IEA. WEO 2018. ISSN: 15493636. www.iea.org/t&c/%20https://www.iea.org/publications/freepublications/publication/WE02014_AfricaEnergyOutlook.pdf (2018).
243. Limited, U. P. P. C. *Statistics at a Glance 2019-20* (2020).
244. Commission, U. P. E. R. *Tariff for FY 2018-9* 2018.
245. Avi-Itzhak, B. Operating cost calculation of an electric power generating system under incremental loading procedure. *IEEE Transactions on Power Apparatus and Systems* **96**, 285–292. ISSN: 00189510 (1 1977).
246. Barley, C. D., Winn, C. B., Flowers, L. & Green, H. J. Optimal Control of Remote Hybrid Power Systems Part 1: Simplified Model. *Windpower '95*, 13 (1995).
247. Barley, C. D. & Winn, C. B. *Optimal dispatch strategy in remote hybrid power systems* in *Solar Energy* **58** (Pergamon, Oct. 1996), 165–179.
248. Muselli, M., Notton, G. & Louche, A. Design of hybrid-photovoltaic power generator, with optimization of energy management. *Solar Energy* **65**, 143–157. ISSN: 0038092X (Feb. 1999).
249. Ashari, M. & Nayar, C. V. An optimum dispatch strategy using set points for a photovoltaic (PV)-diesel-battery hybrid power system. *Solar Energy* **66**, 1–9. ISSN: 0038092X (May 1999).
250. Arévalo-Cordero, P. *et al.* Optimal energy management strategies to reduce diesel consumption for a hybrid off-grid system. *Revista Facultad de Ingeniería Universidad de Antioquia* **98**, 47–58. ISSN: 0120-6230. http://www.scielo.org.co/scielo.php?script=sci%7B%5C_%7Darttext%7B%5C%7Dpid=S0120-62302021000100047%7B%5C%7Dlng=en%7B%5C%7Dnrm=iso%7B%5C%7Dtlng=en%20http://www.scielo.org.co/scielo.php?script=sci%7B%5C_%7Dabstract%7B%5C%7Dpid=S0120-62302021000100047%7B%5C%7Dlng=en%7B%5C%7Dnrm=iso%7B%5C%7Dtlng=en (2021).

251. Ramesh, M. & Saini, R. P. Dispatch strategies based performance analysis of a hybrid renewable energy system for a remote rural area in India. *Journal of Cleaner Production* **259**, 120697. ISSN: 0959-6526 (June 2020).
252. Sopian, K. *et al.* Optimal operational strategy for hybrid renewable energy system using genetic algorithms. *WSEAS Transactions on Mathematics* **7**, 130–140. ISSN: 11092769. <http://pkukmweb.ukm.my/%7B~%7DSERI/indexhttp://www.utem.edu.my/> (2008).
253. Chauhan, A., Upadhyay, S., Khan, M. T., Hussain, S. M. S. & Ustun, T. S. Performance Investigation of a Solar Photovoltaic/Diesel Generator Based Hybrid System with Cycle Charging Strategy Using BBO Algorithm. *Sustainability* **2021**, Vol. 13, Page 8048 **13**, 8048. <https://www.mdpi.com/2071-1050/13/14/8048/htm%20https://www.mdpi.com/2071-1050/13/14/8048> (July 2021).
254. Katsigiannis, Y. A. & Georgilakis, P. S. *Optimal sizing of small isolated hybrid power systems using tabu search* in *Journal of Optoelectronics and Advanced Materials* **10** (2008), 1241–1245. <https://www.researchgate.net/publication/228877741>.
255. Upadhyay, S. & Sharma, M. P. Selection of a suitable energy management strategy for a hybrid energy system in a remote rural area of India. *Energy* **94**, 352–366. ISSN: 0360-5442 (Jan. 2016).
256. Das, B. K. & Zaman, F. Performance analysis of a PV/Diesel hybrid system for a remote area in Bangladesh: Effects of dispatch strategies, batteries, and generator selection. *Energy* **169**, 263–276. ISSN: 03605442 (Feb. 2019).
257. Bernal-Aguatín, J. L., Dufo-López, R. & Rivas-Ascaso, D. M. Design of isolated hybrid systems minimizing costs and pollutant emissions. *Renewable Energy* **31**, 2227–2244. ISSN: 09601481 (Nov. 2006).
258. UNHCR. *Global Trends Forced Displacement in 2020* UNHCR. 2021. <https://www.unhcr.org/60b638e37/unhcr-global-trends-2020%20http://www.unhcr.org/statistics/unhcrstats/576408cd7/unhcr-global-trends-2015.html>.
259. Save the Children & Boston Consulting Group. *Predicting Displacement* 2018.
260. World Economic Forum. *The Global Risks Report 2020 Insight Report 15th Edition* 2020. <https://www.weforum.org/reports/the-global-risks-report-2019>.
261. Clement, V. *et al.* *Groundswell Part 2: Acting on Internal Climate Migration* The World Bank. Washington, DC, 2021.
262. Devictor, X. *2019 update: How long do refugees stay in exile? To find out, beware of averages* 2021. <https://blogs.worldbank.org/dev4peace/2019-update-how-long-do-refugees-stay-exile-find-out-beware-averages> (2021).
263. Alshoubaki, H. The Temporary City: the Transformation of Refugee Camps from fields of Tents to Permanent Cities. *Housing Policies and Urban Economics* **7**, 5–15. <https://www.researchgate.net/publication/335161295> (2017).
264. Grafham, O. & Lahn, G. *The Costs of Fuelling Humanitarian Aid* Chatman House, Moving Energy Initiative. 2018. <https://www.chathamhouse.org/sites/default/files/publications/research/2018-12-10-Costs-Humanitarian-Aid2.pdf>.
265. Lahn, G. & Grafham, O. *Heat, light and power for refugees* Chatham House, Moving Energy Initiative. 2015.

266. Jeffries, E. In the dark. *Nature Energy* **2**, 905–907. ISSN: 20587546. <http://www.nature.com/articles/s41560-017-0034-7> (Dec. 2017).
267. Van Hove, E. & Johnson, N. G. Refugee settlements in transition: Energy access and development challenges in Northern Uganda. *Energy Research & Social Science*, 102103 (2021).
268. UNHCR. *Global Strategy for Sustainable Energy* 2019. www.facebook.com/UNHCREnv/.
269. UNHCR. *Strategic Framework for Climate Action* 2021. <https://www.unhcr.org/604a26d84/strategic-framework-for-climate-action>.
270. Lahn, G. *Energy in situations of displacement* Chatham House. 2018.
271. High Level Panel on Humanitarian Financing. High-Level Panel on Humanitarian Financing Report to the Secretary-General. *Too important to fail-addressing the humanitarian financing gap*, 1–31. <https://reliefweb.int/sites/reliefweb.int/files/resources/%7B%5C%7D5BHLF%20Report%7B%5C%7D5D%20Too%20important%20to%20fail%E2%80%9494addressing%20the%20humanitarian%20financing%20gap.pdf> (2016).
272. UNHCR. *Consequences of Underfunding in 2020* 2020. <https://www.unhcr.org/underfunding-2020/wp-content/uploads/sites/107/2020/09/Underfunding-2020-Full-Report.pdf%7B%5C%7D%7B%5C%7Dga=2.237341568.1004303944.1600906398-993617769.1600906398>.
273. Frack, P. F., Mercado, P. E., Sarriegui, G. & De Doncker, R. W. Renewable energy supply of refugee camps to decrease fuel consumption and CO2 emissions. *2015 IEEE 6th International Symposium on Power Electronics for Distributed Generation Systems, PEDG 2015* (Aug. 2015).
274. Neves, D., Baptista, P. & Pires, J. M. Sustainable and inclusive energy solutions in refugee camps: Developing a modelling approach for energy demand and alternative renewable power supply. *Journal of Cleaner Production* **298**, 126745. ISSN: 0959-6526 (May 2021).
275. Baldi, D., Moner-Girona, M., Fumagalli, E. & Fahl, F. Planning sustainable electricity solutions for refugee settlements in sub-Saharan Africa. *Nature Energy*, 369–379 (4 2022).
276. Thomson, L. & Caskie, D. *Feasibility and Applicability of a Global Guarantee Mechanism in Humanitarian Energy Contracts* Energy Market and Regulatory Consultants Limited. 2020.
277. Fouquet, D., Traum, Y. & Skehan, R. *Identification and analysis of standard clauses of PPA and leasing agreements for energy provision in the humanitarian sector* Becker Büttner Held. 2020.
278. Grafham, O. & Sandwell, P. Harness better data to improve provision of humanitarian energy. *Nature Energy* 2019 4:12 4, 993–996. ISSN: 2058-7546. <https://www.nature.com/articles/s41560-019-0518-8> (Dec. 2019).
279. IRENA. *Policies and regulations for renewable mini-grids* International Renewable Energy Agency. 2018. https://irena.org/-/media/Files/IRENA/Agency/Publication/2018/Oct/IRENA_mini-grid_policies_2018.pdf.
280. Rwanda, U. *Monthly Population Statistics* Aug. 2021. <https://reporting.unhcr.org/sites/default/files/Rwanda-Monthly%5C%20population%5C%20statistics-August%5C%202021.pdf>.
281. UNHCR. *New York Declaration for Refugees and Migrants* 2018.
282. UNHCR. *Comprehensive Refugee Response Framework* 2018.
283. Alight. *Our Humanitarian Work: Rwanda* 2021. <https://wearealight.org/our-work/rwanda/> (2021).

284. Ltd., M. *MeshPower: Connecting communities with clean energy* 2021. <https://meshpower.co.rw/> (2021).
285. MeshPower. *Private correspondence with MeshPower Rwanda Limited* 2021.
286. *Solar micro-grids for productive enterprise uses in humanitarian settings: Case study of Mahama 1 Refugee Camp, Rwanda* Strathmore Energy Research Centre. 2020.
287. Sandwell, P. *et al.* *Ensuring refugee camps in Rwanda have access to sustainable energy* Practical Action. 2020. https://infohub.practicalaction.org/bitstream/handle/11283/622633/Access%5C%20to%5C%20energy%5C%20in%5C%20refugee%5C%20camps%5C%20in%5C%20Rwanda_web%5C%20updated.pdf?sequence=5.
288. Gitizadeh, M. & Fakharzadegan, H. Battery capacity determination with respect to optimized energy dispatch schedule in grid-connected photovoltaic (PV) systems. *Energy* **65**, 665–674. ISSN: 360-5442 (2014).
289. Bordin, C. *et al.* A linear programming approach for battery degradation analysis and optimization in offgrid power systems with solar energy integration. *Renewable Energy* **101**, 417–430. ISSN: 0960-1481 (2017).
290. Guo, Y., Sheng, S., Anglani, N. & Lehman, B. *Optimal Power Management for Grid-Connected Microgrid Considering Modelling of Different Electricity Cost and Battery Degradation Cost in 20th Workshop on Control and Modeling for Power Electronics (COMPEL)* (2019), 1–7.
291. Tunge, T. & Whitehouse, K. *The power of data: Assessing operational energy use in camps to inform the design of alternative renewable energy* Practical Action. 2020.
292. Baranda Alonso, J. & Sandwell, P. Sustainable mini-grid systems in refugee camps: A case study of Rwanda. *Grantham Institute Briefing Note 12* (2020).
293. Manwell, J. F., Stein, W. A., Rogers, A. & McGowan, J. G. An investigation of variable speed operation of diesel generators in hybrid energy systems. *Renewable Energy* **2**, 563–571 (6 1992).
294. Nixon, J., Bhargava, K., Halford, A. & Gaura, E. The challenges of community-based solar energy interventions: Lessons from two Rwandan Refugee Camps. *Energy for Sustainable Development* **65**, 175–184. ISSN: 0973-0826 (2021).
295. Pappis, I. Strategic low-cost energy investment opportunities and challenges towards achieving universal electricity access (SDG7) in forty-eight African nations. *Environmental Research: Infrastructure and Sustainability* **2**, 035005. ISSN: 2634-4505. <https://iopscience.iop.org/article/10.1088/2634-4505/ac7900><https://iopscience.iop.org/article/10.1088/2634-4505/ac7900/meta> (3 July 2022).
296. Falchetta, G. & Mistry, M. N. The role of residential air circulation and cooling demand for electrification planning: Implications of climate change in sub-Saharan Africa. *Energy Economics* **99**, 105307. ISSN: 0140-9883 (July 2021).
297. Agutu, C., Egli, F., Williams, N. J., Schmidt, T. S. & Steffen, B. Accounting for finance in electrification models for sub-Saharan Africa. *Nature Energy* 2022. The are assuming that the private sector will build mini-grids and thus cost of capital much higher.

Would be fair to make the assumption that the, 1–11. ISSN: 2058-7546. <https://www.nature.com/articles/s41560-022-01041-6> (June 2022).
298. Lee, J. T. & Callaway, D. S. The cost of reliability in decentralized solar power systems in sub-Saharan Africa. *Nature Energy* **3**, 960–968. ISSN: 2058-7546. <http://www.nature.com/articles/s41560-018-0240-y> (11 Nov. 2018).

299. Moner-Girona, M. *et al.* A multidimensional high-resolution assessment approach to boost decentralised energy investments in Sub-Saharan Africa. *Renewable and Sustainable Energy Reviews* **148**, 111282. ISSN: 1364-0321 (Sept. 2021).
300. Bhatia, M. & Angelou, N. *Beyond Connections: Energy Access Redefined* ESMAP, World Bank, Washington, DC. Available from <https://openknowledge.worldbank.org/handle/10986/24368>.
301. Riahi, K. *et al.* The Shared Socioeconomic Pathways and their energy, land use, and greenhouse gas emissions implications: An overview. *Global Environmental Change* **42**, 153–168. ISSN: 09593780. <http://dx.doi.org/10.1016/j.gloenvcha.2016.05.009> 0959-3780/%C3%A3%20http://creativecommons.org/licenses/by/4.0/ (Jan. 2017).
302. Bank, T. W. *World Bank Open Data | Data 2022*. <https://data.worldbank.org/>.
303. IMF. *World Economic Outlook, April 2022: War Sets Back The Global Recovery* (). <https://www.imf.org/en/Publications/WE0/Issues/2022/04/19/world-economic-outlook-april-2022>.
304. Kraft, J. & Kraft, A. On the Relationship Between Energy and GNP. *Source: The Journal of Energy and Development* **3**, 401–403 (2 1978).
305. Nyasha, S., Gwenhure, Y. & Odhiambo, N. M. Energy consumption and economic growth in Ethiopia: A dynamic causal linkage: <https://doi.org/10.1177/0958305X18779574> **29**, 1393–1412. ISSN: 20484070. <https://journals.sagepub.com/doi/full/10.1177/0958305X18779574?journalCode=eaea> (8 June 2018).
306. Akinlo, A. E. Energy consumption and economic growth: Evidence from 11 Sub-Saharan African countries. *Energy Economics* **30**, 2391–2400. ISSN: 01409883 (5 Sept. 2008).
307. Tools, G. D. *countries.csv | Dataset Publishing Language | Google Developers* https://developers.google.com/public-data/docs/canonical/countries_csv.
308. E., B., A., R. & Urban, M. *LandScan Global 2014 [Data set]* 2015.
309. A., R. *et al.* *LandScan Global 2020 [Data set]* 2021.
310. Jones, B. & O'Neill, B. C. Spatially explicit global population scenarios consistent with the Shared Socioeconomic Pathways. *Environmental Research Letters* **11**, 084003. ISSN: 1748-9326. <https://iopscience.iop.org/article/10.1088/1748-9326/11/8/084003> 20<https://iopscience.iop.org/article/10.1088/1748-9326/11/8/084003/meta> (8 July 2016).
311. Gao, J. Downscaling Global Spatial Population Projections from 1/8-degree to 1-km Grid Cells. <https://opensky.ucar.edu/islandora/object/technotes%5C%3A553/> (2017).
312. Falchetta, G., Pachauri, S., Parkinson, S. & Byers, E. Data from: A High-Resolution Gridded Dataset to Assess Electrification in Sub-Saharan Africa. **6** (2021).
313. Moner-Girona, M. & KAKOULAKI, G. *Electricity grid Africa. European Commission, Joint Research Centre (JRC) [Dataset]* 2020. <https://data.jrc.ec.europa.eu/dataset/624c6e71-3b9c-4f48-8c67-645911798d41>.
314. Rao, N. D. & Pachauri, S. Energy access and living standards: some observations on recent trends. *Environmental Research Letters* **12**, 025011. ISSN: 1748-9326. <https://iopscience.iop.org/article/10.1088/1748-9326/aa5b0d> 20<https://iopscience.iop.org/article/10.1088/1748-9326/aa5b0d/meta> (2 Feb. 2017).

315. Lab, G. D. *Average household size - Area Database - Table - Global Data Lab* 2022. <https://globaldatalab.org/areadata/table/hhsize/>.
316. IEA. *World Energy Balances - Data product - IEA* (2022). <https://www.iea.org/data-and-statistics/data-product/world-energy-balances>.
317. IRENA. *IRENASTAT Online Data Query Tool: Electricity generation (GWh) by Country/area, Technology, Grid connection and Year* https://pxweb.irena.org/pxweb/en/IRENASTAT/IRENASTAT_-_Power%5C%20Capacity%5C%20and%5C%20Generation/ELECGEN_2022_cycle2.px/.
318. Allington, L. Selected 'Starter kit' energy system modelling data for selected countries in Africa, East Asia, and South America. <https://doi.org/10.21203/rs.3.rs-1178306/v1> (2022).
319. Lee, H. E., Kim, W. Y., Kang, H. & Han, K. Still lacking reliable electricity from the grid, many Africans turn to other sources. *Dispatch, Afrobarometer* (514 2022).
320. Institute, R. M. *Electrifying Economies: Comparing Costs of an Integrated Approach versus Pure Grid Extension* (2020).
321. O'Neill, B. C. *et al.* A new scenario framework for climate change research: The concept of shared socioeconomic pathways. *Climatic Change* **122**, 387–400. ISSN: 01650009. <https://link.springer.com/article/10.1007/s10584-013-0905-2> (3 Oct. 2014).
322. Byers, E. *et al.* AR6 Scenarios Database. <https://zenodo.org/record/7197970> (Nov. 2022).
323. GL, D. *Battery Energy Storage Study for the 2017 IRP*, 40 (2017).
324. Energy, I. *Inverted Energy Lithium Plus Batteries Datasheet* 2019.
325. Pang, H., Lo, E. & Pong, B. *DC Electrical Distribution Systems in Buildings in 2006 2nd International Conference on Power Electronics Systems and Applications* (IEEE, Nov. 2006), 115–119. ISBN: 962-367-544-5. <http://ieeexplore.ieee.org/document/4147795/>.
326. Starke, M., Tolbert, L. M. & Ozpineci, B. *AC vs. DC distribution: A loss comparison in 2008 IEEE/PES Transmission and Distribution Conference and Exposition* (IEEE, Apr. 2008), 1–7. ISBN: 978-1-4244-1903-6. <http://ieeexplore.ieee.org/document/4517256/>.
327. Mikhaylov, K., Tervonen, J. & Fadeev, D. *Development of Energy Efficiency Aware Applications Using Commercial Low Power Embedded Systems* ISBN: 978-953-51-0167-3 (InTechOpen, 2012).
328. Tongia, R. *Microgrids in India Myths, Misunderstandings, and the Need for Proper Accounting Acknowledgements* 2018. https://www.brookings.edu/wp-content/uploads/2018/02/impact_series_microgrid-folder_feb10-2.pdf.
329. Peters, J. F., Baumann, M., Zimmermann, B., Braun, J. & Weil, M. The environmental impact of Li-Ion batteries and the role of key parameters – A review. *Renewable and Sustainable Energy Reviews* **67**, 491–506. ISSN: 18790690 (Jan. 2017).
330. Smith, C. *et al.* Comparative Life Cycle Assessment of a Thai Island's diesel/PV/wind hybrid microgrid. *Renewable Energy* **80**, 85–100. ISSN: 18790682 (Aug. 2015).
331. Centre, B. E. *Carbon emissions of different fuels* (2011), 1. http://www.biomassenergycentre.org.uk/portal/page?_pageid=75,163182&_dad=portal&_schema=PORTAL.

332. Mason, J. E., Fthenakis, V. M., Hansen, T. & Kim, H. C. Energy payback and life-cycle CO₂ emissions of the BOS in an optimized 3.5 MW PV installation. *Progress in Photovoltaics: Research and Applications* **14**, 179–190. ISSN: 1062-7995. <http://doi.wiley.com/10.1002/pip.652> (2 Mar. 2006).
333. Amazon. *PowerHouse 300W 30V Polycrystalline Solar Panel: Amazon.in: Garden Outdoors* 2019.
334. Techno-economic assessment of biomass gasification-based mini-grids for productive energy applications: The case of rural India. *Renewable Energy* **154**, 432–444. ISSN: 18790682 (July 2020).
335. Indiamart. *Lithium Plus Battery, Capacity: 80 Ah, Rs 18000 /piece, Inverted Energy* | ID: 20551129330 2019. <https://www.indiamart.com/proddetail/lithium-plus-battery-20551129330.html>.
336. Schmidt, O., Hawkes, A., Gambhir, A. & Staffell, I. The future cost of electrical energy storage based on experience rates. *Nature Energy* **2**, 17110. ISSN: 2058-7546. <http://www.nature.com/articles/nenergy2017110> (8 July 2017).
337. Indiamart. *Kirloskar 10 KVa Koel Chhota Chilli Diesel Genset, Rs 200000 /unit* | ID: 20326816897 2019. <https://www.indiamart.com/proddetail/10-kva-koel-chhota-chilli-diesel-genset-20326816897.html>.
338. Oil, I. *Indian Oil Historical Diesel Prices* 2019. https://www.iocl.com/Product_PreviousPrice/DieselPreviousPriceDynamic.aspx.
339. Indiamart. *Pre Galvanized Fixed Type Mounting Structures, Thickness: 1.5 Mm To 4.0 Mm, Rs 2300 /kilowatt* | ID: 6865551512 2019. <https://www.indiamart.com/proddetail/fixed-type-mounting-structures-6865551512.html>.
340. Indiamart. *140 V SMA Solar Inverter, 7 A, Rs 48000 /piece, Sarth Solar Solutions Private Limited* | ID: 19053602488 2019. <https://www.indiamart.com/proddetail/sma-solar-inverter-19053602488.html>.
341. Baranda Alonso, J., Sandwell, P. & Beath, H. *Business load profiles used in “Maximising the benefits of renewable energy infrastructure in displacement settings: Optimising the operation of a solar-hybrid mini-grid for institutional and business users in Mahama Refugee Camp, Rwanda”* doi: 10.5281/zenodo.6759406. 2022.
342. Giles, P., Ross, M. & Sumanik, S. *Modelling Prime Diesel Electric Generator Fuel Consumption across Genset Sizings in 2020 IEEE Electric Power and Energy Conference (EPEC)* (2020), 1–8.
343. Jabeck, B. Available online October 2013. 2013. https://www.cat.com/en_GB/by-industry/electric-power/Articles/White-papers/the-impact-of-generator-set-underloading.html.
344. Company, T. B. *Data Sheet SOLAR SIND 04 2145* 2021. https://www.trojanbattery.com/pdf/datasheets/SIND_04_2145_DS.pdf.
345. MeshPower. *Private Correspondence with MeshPower Limited* Personal communication. 2021.
346. De Wild-Scholten, M. (Energy payback time and carbon footprint of commercial photovoltaic systems. *Solar Energy Materials and Solar Cells* **119**, 296–305. ISSN: 0927-0248. <https://www.sciencedirect.com/science/article/pii/S0927024813004455> (Dec. 2013).

347. Das, J., Abraham, A. P., Ghosh, P. C. & Banerjee, R. Life cycle energy and carbon footprint analysis of photovoltaic battery microgrid system in India. *Clean Technologies and Environmental Policy* **20**, 65–80. ISSN: 16189558 (1 2018).
348. Ioakimidis, C. S., Murillo-Marrodán, A., Bagheri, A., Thomas, D. & Genikomsakis, K. N. Life cycle assessment of a lithium iron phosphate (LFP) electric vehicle battery in second life application scenarios. *Sustainability (Switzerland)* **11**. ISSN: 20711050 (9 May 2019).
349. Färber, R., Guillod, T., Krismer, F., Kolar, J. W. & Franck, C. M. Endurance of polymeric insulation foil exposed to DC-biased medium-frequency rectangular pulse voltage stress. *Energies* **13**. ISSN: 19961073 (1 Dec. 2019).
350. For European Environmental Policy (IEEP), I. *Environmentally Harmful Subsidies: Identification and Assessment* - Annex 5: Subsidy level indicators for the case studies (2009). http://ec.europa.eu/energy/publications/statistics/doc/2009_energy_transport_figures.pdf.
351. UNFCCC. *Harmonized Grid Emission factor data set* | UNFCCC (). <https://unfccc.int/documents/198197>.
352. Takahashi, K. & Louhisuo, M. IGES List of Grid Emission Factors. **11.0**, 5. <https://www.iges.or.jp/en/pub/list-grid-emission-factor/en> (2022).
353. Brander, M., Sood, A., Wylie, C., Haughton, A. & Lovell, J. Electricity-specific emission factors for grid electricity (2011).